

Gestalten

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Use of generative AI

This document has been written with significant involvement of generative AI:

- The notes taken live during the seminar were turned into proper TeX by Claude Opus. This process also corrected various typos / unfinished sentences but otherwise did not greatly affect the mathematical contents.
- Certain material from [Sch26] not covered during the seminar has been added to the notes by Claude Opus or Codex. This happened for the following chapters:
 - The end of Chapter 6;
 - The material at the end of Chapter 8;
 - The last three sections of Chapter 10;
 - The material in Chapter 12.
- The background material on six-functor formalisms in Appendix B was originally added by Claude Opus, taken primarily from [HM24; CLL25; CLL24]. It has subsequently been edited.
- The content of Appendix A was mostly written by Codex, though with significant prompting and manual editing.

1 Presentable categories

Talk given by Sil.

1.1 Preliminaries

Concepts we should know:

- (-2) *Regular cardinal κ* : any union of sets of cardinality $< \kappa$ indexed by a set of cardinality $< \kappa$ has cardinality $< \kappa$.
- (-1) *Essentially κ -small (∞ -)category C* : (If κ uncountable) The set of isomorphism classes of C has cardinality $< \kappa$, and for all $x, y \in C$ the homotopy groups $\pi_i \operatorname{hom}_C(x, y)$ have cardinality $< \kappa$ (For $\kappa = \aleph_0$ one would take the smallest subcategory of $\operatorname{Cat}_\infty$ generated under finite colimits by $[0]$ and $[1]$);
- (0) *κ -filtered category I* : Every κ -small diagram in I admits a cocone;
- (1) *κ -compact object $X \in C$* : $\operatorname{Hom}_C(X, -): C \rightarrow \mathbf{An}$ preserves κ -filtered colimits;
- (2) *κ -presentable categories*: A category C which has small colimits, for which the subcategory C^κ of κ -compact objects is essentially small, and such that every object is a κ -filtered colimit of objects of C^κ ;
- (3) *Presentable categories*: A category C which is κ -presentable for some κ .

Remark 1.1. C is presentable iff it “admits a presentation:” There is a small category S such that $C \simeq \langle S \rangle[R^{-1}]$, where $\langle S \rangle = \operatorname{PSh}(S) = \operatorname{Fun}(S^{\operatorname{op}}, \mathbf{An})$ is the free cocompletion of S , where R is a (small) set of morphisms in $\langle S \rangle$, and where $\langle S \rangle[R^{-1}] \subseteq \langle S \rangle$ is the full subcategory spanned by those $X \in \langle S \rangle$ such that $\operatorname{Hom}_{\langle S \rangle}(f, X): \operatorname{Hom}_{\langle S \rangle}(B, X) \rightarrow \operatorname{Hom}_{\langle S \rangle}(A, X)$ is an isomorphism for every morphism $f: A \rightarrow B$ in R .

Similarly, C is κ -presentable if we can choose S and R above such that R consists of maps between κ -compact objects.

Remark 1.2. The description of $\langle S \rangle[R^{-1}]$ as a full subcategory of $\langle S \rangle$ is an instance of the principle that “free constructions in $\operatorname{Pr}^{\operatorname{L}}$ correspond to cofree constructions in Cat , via passing to right adjoints.” Indeed, the localization functor $\langle S \rangle \rightarrow \langle S \rangle[R^{-1}]$ is a morphism in $\operatorname{Pr}^{\operatorname{L}}$, hence preserves colimits and admits a right adjoint; this right adjoint is automatically fully faithful and identifies $\langle S \rangle[R^{-1}]$ with the subcategory described above. See [Lur09, Proposition 5.5.4.20].

Definition 1.3. We denote by $\mathbf{Pr}^L$ the category of presentable categories and cocontinuous functors. We let $\mathbf{Pr}_\kappa^L \subseteq \mathbf{Pr}^L$ denote the (non-full) subcategory of κ -presentable categories and those cocontinuous functors which preserve κ -compact objects.

Example 1.4. The categories $\mathbf{An}$, $\mathbf{Sp}$, and $D(R)$ are all in $\mathbf{Pr}_{\aleph_0}^L$. If X is a second-countable locally compact Hausdorff space, then $\mathbf{Shv}(X) \in \mathbf{Pr}_{\aleph_1}^L$. Categories of almost modules arising from standard countably presented almost setups are also $\aleph_1$ -presentable. Without such countability hypotheses, these last two examples are presentable but need not be $\aleph_1$ -presentable.

Lemma 1.5. *Let κ be an uncountable regular cardinal. If D is κ -presentable and I is a κ -small category, then also $\mathbf{Fun}(I, D)$ is κ -presentable. Its κ -compact generators are given by $\mathbf{Fun}(I, D^\kappa)$.*

Proof. This follows from the argument given in [Lur09, Proposition 5.4.4.3]; note that no enlargement of the regular cardinal is needed: since I is κ -small, the argument only uses that κ -filtered colimits of anima commute with κ -small limits. This commutation holds for every regular cardinal; we assume κ uncountable only to stay within our standing conventions, under which the κ -compact objects of $\mathbf{Fun}(I, D)$ are genuine κ -small colimits of objects of $\mathbf{Fun}(I, D^\kappa)$ rather than mere retracts thereof, so that no separate idempotent-completion step (as would be required for $\kappa = \aleph_0$) intervenes. $\square$

Definition 1.6. Let $\mathbf{Rex}_\kappa$ be the category of essentially small categories with κ -small colimits and functors preserving them. In the case $\kappa = \aleph_0$, we additionally require the categories to be idempotent complete.

Proposition 1.7. *Let C be a κ -presentable category. Then the subcategory $C^\kappa \subseteq C$ is essentially small, and the inclusion $C^\kappa \hookrightarrow C$ uniquely extends to an equivalence $\mathbf{Ind}_\kappa(C^\kappa) \xrightarrow{\sim} C$. Moreover, the assignments $C \mapsto C^\kappa$ and $D \mapsto \mathbf{Ind}_\kappa(D)$ define an equivalence*

$$\mathbf{Rex}_\kappa \rightleftarrows \mathbf{Pr}_\kappa^L.$$

Theorem 1.8. *The category $\mathbf{Pr}_\kappa^L$ is κ -presentable. (In other words: $\mathbf{Pr}_\kappa^L \in \mathbf{Pr}_\kappa^L$.)*

Proof. By the previous proposition, $\mathbf{Pr}_\kappa^L$ is equivalent to $\mathbf{Rex}_\kappa$, so we need to show $\mathbf{Rex}_\kappa$ is κ -presentable. To this end, we consider the free-forgetful adjunction

$$\langle - \rangle^\kappa : \mathbf{Cat} \rightleftarrows \mathbf{Rex}_\kappa : \mathbf{fgt}.$$

We will first show that this adjunction is monadic. By the Barr–Beck theorem, we need to show:

- The functor $\mathbf{fgt}$ is conservative. This is clear.
- The category $\mathbf{Rex}_\kappa$ admits $\mathbf{fgt}$ -split simplicial colimits, and $\mathbf{fgt} : \mathbf{Rex}_\kappa \rightarrow \mathbf{Cat}$ preserves these. Here we say that a functor $C_\bullet : \Delta^{\text{op}} \rightarrow \mathbf{Rex}_\kappa$ is *fgt-split* if its image $\mathbf{fgt}(C_\bullet) : \Delta^{\text{op}} \rightarrow \mathbf{Cat}$ admits a splitting.

For the second claim, consider such a diagram $C_\bullet$, and let $C_{-1} := \text{colim}_{[n] \in \Delta^{\text{op}}} C_n$ be its colimit in $\mathbf{Cat}$. Since split simplicial colimits are absolute, the colimit functors for κ -small diagrams in the C_n descend to colimit functors in C_{-1} . Thus C_{-1} admits κ -small colimits, and each structure map

$\iota_n: C_n \rightarrow C_{-1}$ preserves them. We claim that the augmented diagram is a colimit in Rex_κ ; that is, for every $D \in \text{Rex}_\kappa$ the canonical map

$$\text{Hom}_{\text{Rex}_\kappa}(C_{-1}, D) \rightarrow \lim_{[n] \in \Delta} \text{Hom}_{\text{Rex}_\kappa}(C_n, D)$$

is an equivalence. Since C_{-1} is the colimit in Cat , we have

$$\text{Fun}(C_{-1}, D) \xrightarrow{\sim} \lim_{[n] \in \Delta} \text{Fun}(C_n, D).$$

It remains only to identify the colimit-preserving functors on both sides. This property is detected after precomposition with ι_0 : one direction uses that ι_0 preserves κ -small colimits, while the converse uses a splitting $s: C_{-1} \rightarrow C_0$ with $\iota_0 s \simeq \text{id}_{C_{-1}}$ to write every κ -small diagram in C_{-1} as the image under ι_0 of a diagram in C_0 . Hence the displayed equivalence restricts to the desired equivalence on the full subcategories of κ -small-colimit-preserving functors.

We now know that Rex_κ is monadic over Cat . Moreover, Cat is $\aleph_0$ -presentable, as it is a Bousfield localization of $\text{sAn} = \text{Fun}(\Delta^{\text{op}}, \text{An})$ at a small set of morphisms between compact objects. The free κ -cocompletion monad is κ -accessible, so Rex_κ admits all colimits and is generated by the objects $\langle [n] \rangle^\kappa$. These generators are κ -compact, because $[n]$ is κ -compact in Cat and a κ -small colimit in a κ -filtered colimit of κ -cocomplete categories is computed after some stage; equivalently, $\text{fgt}: \text{Rex}_\kappa \rightarrow \text{Cat}$ preserves κ -filtered colimits. $\square$

Proposition 1.9. *The object $\text{Pr}_\kappa^{\text{L}}$ of $\text{Pr}_\kappa^{\text{L}}$ is κ -compact if and only if κ is not a limit cardinal. (In particular, this applies to $\kappa = \aleph_1$.)*

Sketch. This is [Sch26, Proposition 2.11]. If κ is an uncountable regular limit cardinal, then

$$\text{Pr}_\kappa^{\text{L}} \simeq \text{colim}_{\substack{\lambda < \kappa \\ \lambda \text{ regular}}} \text{Pr}_\lambda^{\text{L}}$$

is a κ -filtered colimit in $\text{Pr}_\kappa^{\text{L}}$, but $\text{Pr}_\kappa^{\text{L}}$ is not a retract of any stage, so it is not κ -compact. If $\kappa = \lambda^+$, the data of κ -small colimits can be encoded by an initial object, pushouts, and coproducts indexed by sets of cardinality at most λ , subject to fewer than κ compatibilities. This presents $\text{Rex}_\kappa \simeq \text{Pr}_\kappa^{\text{L}}$ as a κ -compact object of itself. The case $\kappa = \aleph_0$ requires a separate argument. $\square$

Lemma 1.10. *Let $F: C \rightarrow D$ be a morphism in $\text{Pr}_\kappa^{\text{L}}$, and let G denote its right adjoint. Then the following are equivalent:*

- (1) *The functor F is fully faithful;*
- (2) *The induced functor*

$$F^\kappa: C^\kappa \rightarrow D^\kappa$$

is fully faithful;

- (3) *The unit $\eta_X: X \rightarrow GFX$ is an isomorphism for every $X \in C^\kappa$.*

Proof. The implication (1) $\Rightarrow$ (3) is the usual adjunction criterion for a functor to be fully faithful. The implication (3) $\Rightarrow$ (2) follows from the adjunction: for $X, Y \in C^\kappa$, we have

$$\text{Hom}_D(FX, FY) \simeq \text{Hom}_C(X, GFY) \simeq \text{Hom}_C(X, Y).$$

Conversely, assume that F^κ is fully faithful. For $Y \in C^\kappa$, the map $\eta_Y : Y \rightarrow GFY$ induces, for every $X \in C^\kappa$, the map

$$\mathrm{Hom}_C(X, Y) \rightarrow \mathrm{Hom}_C(X, GFY) \simeq \mathrm{Hom}_D(FX, FY),$$

which is an isomorphism by assumption. Since the κ -compact objects generate C under colimits, this implies that η_Y is an isomorphism. Thus (2) $\Rightarrow$ (3).

It remains to see that (2) implies (1). Consider the subcategory $C' \subseteq C$ consisting of those objects X such that the map $\mathrm{Hom}_C(X, Y) \rightarrow \mathrm{Hom}_D(FX, FY)$ is an isomorphism for each $Y \in C$. Note that we have $C^\kappa \subseteq C'$: if X is κ -compact, then both $\mathrm{Hom}_C(X, -)$ and $\mathrm{Hom}_D(FX, F(-))$ preserve κ -filtered colimits, and every $Y \in C$ may be written as a κ -filtered colimit of κ -compact objects. Also note that C' is closed under arbitrary colimits, since both $\mathrm{Hom}_C(-, Y)$ and $\mathrm{Hom}_D(F(-), FY)$ send colimits in X to limits. It follows that $C' = C$, showing that F is fully faithful. $\square$

1.2 Basic properties of $\mathbf{Pr}^L$

Theorem 1.11. *Let $F : C \rightarrow D$ be a functor between $C, D \in \mathbf{Pr}^L$.*

- (1) *Then F admits a right adjoint if and only if it preserves colimits.*
- (2) *Moreover, F admits a left adjoint if and only if it preserves limits and it is κ -accessible for some regular cardinal κ (i.e. it preserves κ -filtered colimits).*

Corollary 1.12. *There is an equivalence $(\mathbf{Pr}^L)^{\mathrm{op}} \simeq \mathbf{Pr}^R$.*

Proposition 1.13. (1) *The category $\mathbf{Pr}^L$ admits small limits, which are preserved by $\mathrm{fgt} : \mathbf{Pr}^L \rightarrow \mathbf{Cat}$.*

- (2) *The category $\mathbf{Pr}_\kappa^L$ admits limits, preserved by the composite*

$$\mathbf{Pr}_\kappa^L \simeq \mathbf{Rex}_\kappa \rightarrow \mathbf{Cat}.$$

- (3) *If κ is uncountable, then the inclusion $\mathbf{Pr}_\kappa^L \hookrightarrow \mathbf{Pr}^L$ preserves κ -small limits.*

Proof. For (1) and (3) we refer to Lurie's HTT. For (2), one needs to use that $\mathbf{Rex}_\kappa \rightarrow \mathbf{Cat}$ is monadic. $\square$

Proposition 1.14. *The categories $\mathbf{Pr}^R$ and $\mathbf{Pr}_\kappa^R$ admit small limits, and the functors $\mathbf{Pr}_\kappa^R \rightarrow \mathbf{Pr}^R \rightarrow \mathbf{Cat}$ preserve small limits.*

Corollary 1.15. *The categories $\mathbf{Pr}^L$ and $\mathbf{Pr}_\kappa^L$ admit colimits (which are computed by passing to adjoints and computing the limit in $\mathbf{Pr}^R$ resp. $\mathbf{Pr}_\kappa^R$.)*

Example 1.16. Let I be a set.

- (1) The product $\prod_{i \in I} C_i$ in $\mathbf{Pr}^L$ is just the product of underlying categories. But the product in $\mathbf{Pr}_\kappa^L$ would be $\mathrm{Ind}_\kappa(\prod_{i \in I} C_i^\kappa)$.
- (2) The coproduct $\coprod_{i \in I} C_i$ in $\mathbf{Pr}^L$ is also the product $\prod_{i \in I} C_i$ in $\mathbf{Cat}$. So is the coproduct in $\mathbf{Pr}_\kappa^L$.

- (3) The pushout of $*/H$ and $*/G$ over $*$ is $*/(G \star H)$, where $G \star H$ is the free product of G and H , and where $*/G$ denotes the classifying anima of G . But the pushout $\langle */H \rangle \sqcup_{\langle * \rangle} \langle */G \rangle$ is given by the pullback $\langle */H \rangle \times_{\text{An}} \langle */G \rangle$ in Cat .

Remark 1.17. Item (2) of the previous example is striking: the natural map

$$\coprod_{i \in I} C_i \rightarrow \prod_{i \in I} C_i$$

in Pr^{L} is an isomorphism for *arbitrary* indexing sets I . Adapting the notion of *semiadditivity* from [Lur17, Definition 6.1.6.13] to arbitrary cardinals, this says that Pr^{L} is κ -semiadditive for every cardinal κ . This is an extremely unusual property of Pr^{L} .

The analogous statement in $\text{Pr}_{\kappa}^{\text{L}}$ holds whenever $|I| < \kappa$: the coproduct $\coprod_{i \in I}^{\text{Pr}_{\kappa}^{\text{L}}} C_i$ is always the naive product $\prod_i C_i$, whereas the product $\prod_{i \in I}^{\text{Pr}_{\kappa}^{\text{L}}} C_i = \text{Ind}_{\kappa}(\prod_i C_i^{\kappa})$ generally differs from it. The two agree when $|I| < \kappa$, but may also agree for larger I in degenerate cases.

1.3 Tensor products in Pr^{L}

Proposition 1.18. *Let $C, D \in \text{Pr}^{\text{L}}$. There exists a universal presentable category $C \otimes D$ with a functor $C \times D \rightarrow C \otimes D$ which preserves colimits in both variables. If $C, D \in \text{Pr}_{\kappa}^{\text{L}}$, then $C \otimes D \in \text{Pr}_{\kappa}^{\text{L}}$, and the functor $C \times D \rightarrow C \otimes D$ preserves κ -compact objects.*

Proof. We may set $\langle S \rangle \otimes \langle T \rangle = \langle S \times T \rangle$. Moreover, we may set $\langle S \rangle [R^{-1}] \otimes \langle T \rangle [Q^{-1}] = \langle S \times T \rangle [W^{-1}]$, where W consists of the morphisms $f \otimes \text{id}_t$ for $f \in R$ and $t \in T$, together with $\text{id}_s \otimes g$ for $g \in Q$ and $s \in S$ (here s and t range over the representables). $\square$

Corollary 1.19. *The categories Pr^{L} and $\text{Pr}_{\kappa}^{\text{L}}$ admit symmetric monoidal structures $\text{Pr}^{L, \otimes}, \text{Pr}_{\kappa}^{L, \otimes}$.*

Proof. We define these as suboperads of $\text{CAT}^{\times}$. $\square$

Example 1.20. (1) For $C = \langle S \rangle$, we get $C \otimes D \simeq \text{Fun}(S^{\text{op}}, D)$.

(2) For $C \in \text{Pr}_{\kappa}^{\text{L}}$ and $D \in \text{Pr}^{\text{L}}$, we have $C \otimes D \simeq \text{Fun}^{\kappa\text{-lex}}((C^{\kappa})^{\text{op}}, D)$.

(3) Let A be a (static) ring. Then $D_{\geq 0}(A)$ has compact projective generators A^n for $n \geq 0$, so $D_{\geq 0}(A) \simeq \text{Fun}^{\times}(\{A^n\}^{\text{op}}, \text{An})$. It follows that

$$D_{\geq 0}(A) \otimes C \simeq \text{Fun}^{\times}(\{A^n\}^{\text{op}}, C).$$

(Objects of the RHS are also known as *A-module objects in C*.)

Proposition 1.21. *The symmetric monoidal category $\text{Pr}^{L, \otimes}$ is closed monoidal, with internal hom given by*

$$[C, D] = \text{Fun}^{\text{L}}(C, D),$$

the full subcategory of $\text{Fun}(C, D)$ spanned by the colimit-preserving functors.

Moreover, also $\text{Pr}_{\kappa}^{L, \otimes}$ is closed monoidal, with internal hom

$$[C, D] = \text{Ind}_{\kappa}(\text{Fun}^{\kappa\text{-rex}}(C^{\kappa}, D^{\kappa})).$$

Proof. By [Lur09, Proposition 5.5.3.8], the category $\text{Fun}^{\text{L}}(C, D)$ is presentable. A colimit-preserving functor $E \rightarrow \text{Fun}^{\text{L}}(C, D)$ is the same as a functor $E \times C \rightarrow D$ which preserves colimits separately in both variables, hence, by the universal property of the tensor product, the same as a colimit-preserving functor $E \otimes C \rightarrow D$. This proves the first assertion.

For the second assertion, Gabriel–Ulmer duality identifies $\text{Pr}_\kappa^{\text{L}}$ with Rex_κ . The internal hom in Rex_κ is $\text{Fun}^{\kappa\text{-rex}}(C^\kappa, D^\kappa)$, since a functor into it is precisely a functor of two variables preserving κ -small colimits separately. Applying Ind_κ gives the displayed internal hom in $\text{Pr}_\kappa^{\text{L}}$. $\square$

Proposition 1.22. *We have $(\text{Pr}_\kappa^{\text{L}})^\otimes \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$.*

Proof. Since $(\text{Pr}_\kappa^{\text{L}})^\otimes \simeq \text{Rex}_\kappa^\otimes$, it suffices to prove the claim for the latter. The only thing that remains is to show that the functor

$$- \otimes -: \text{Rex}_\kappa \times \text{Rex}_\kappa \rightarrow \text{Rex}_\kappa$$

preserves κ -compact objects. The monoidal unit $\text{An} = \langle [0] \rangle^\kappa$ is κ -compact. Moreover, the κ -compact objects of Rex_κ are retracts of κ -small colimits of the generators $\langle [n] \rangle^\kappa$. Since the tensor product preserves κ -small colimits separately and preserves retracts, it is enough to observe that

$$\langle [n] \rangle^\kappa \otimes \langle [m] \rangle^\kappa \simeq \langle [n] \times [m] \rangle^\kappa,$$

which is κ -compact. $\square$

1.4 C-linear presentable categories

Consider $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$. For $A \in \text{CAlg}(C)$, we may consider the module category $\text{Mod}_A(C) \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$. Moreover, the forgetful functor $\text{Mod}_A(C) \rightarrow C$ admits a left adjoint

$$\text{free}: C \rightarrow \text{Mod}_A(C), \quad X \mapsto A \otimes X,$$

which defines a functor in $\text{CAlg}(\text{Pr}_\kappa^{\text{L}})$.

Given two A -modules M, M' , we obtain a *relative tensor product over A*

$$M \otimes_A M' = \text{colim}_{\Delta^{\text{op}}} \left(\dots \begin{array}{c} \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \end{array} M \otimes A \otimes A \otimes M' \begin{array}{c} \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \end{array} M \otimes A \otimes M' \rightrightarrows M \otimes M' \right).$$

Definition 1.23. Let $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$. Then we may form the module category $\text{Mod}_C(\text{Pr}_\kappa^{\text{L}})$ (where we are using that $\text{Pr}_\kappa^{\text{L}} \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$). We refer to the objects of $\text{Mod}_C(\text{Pr}_\kappa^{\text{L}})$ as *C-linear κ -presentable categories* and to its morphisms as *C-linear functors*.

Given $D \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$ and $B \in \text{CAlg}(D)$, there is always an equivalence

$$\text{CAlg}(D)_{B/} \simeq \text{CAlg}(\text{Mod}_B(D)).$$

Theorem 1.24. *Given $A \in \text{CAlg}(C)$ as before, the relative tensor product defines a symmetric monoidal structure on $\text{Mod}_A(C)$, turning the module category into an object*

$$\text{Mod}_A(C) \in \text{CAlg}(\text{Mod}_C(\text{Pr}_\kappa^{\text{L}})) \simeq \text{CAlg}(\text{Pr}_\kappa^{\text{L}})_{C/}.$$

Moreover, the assignment $A \mapsto \text{Mod}_A(C)$ defines a fully faithful functor

$$\text{Mod}_{(-)}(C) : \text{CAlg}(C) \hookrightarrow \text{CAlg}(\text{Mod}_C(\text{Pr}_\kappa^{\text{L}}))$$

which is moreover a morphism in $\text{Pr}_\kappa^{\text{L}}$. Its right adjoint

$$\text{End}_{(-)}(\mathbb{1}) : \text{CAlg}(\text{Mod}_C(\text{Pr}_\kappa^{\text{L}})) \rightarrow \text{CAlg}(C)$$

sends an object $D \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})_{C/}$ to the endomorphism object of its monoidal unit. This functor preserves κ -filtered colimits.

Proof. Full faithfulness and the description of the right adjoint are [Lur17, Corollary 4.8.5.21 and Remark 4.8.5.12]. The displayed functor is a morphism in $\text{Pr}_\kappa^{\text{L}}$ precisely because its right adjoint preserves κ -filtered colimits; this final assertion is [Sch26, Theorem 3.11]. $\square$

Notation 1.25. For $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$, we refer to the adjunction of Theorem 1.24 as the *module-endomorphism adjunction*

$$\text{CAlg}(C) \begin{array}{c} \xrightarrow{\text{Mod}_{(-)}(C)} \\ \xleftarrow{\text{End}_{(-)}(\mathbb{1})} \end{array} \text{CAlg}(\text{Mod}_C(\text{Pr}_\kappa^{\text{L}})),$$

Thus $R \in \text{CAlg}(C)$ is sent to $\text{Mod}_R(C)$, and the right adjoint sends $D \in \text{CAlg}(\text{Mod}_C(\text{Pr}_\kappa^{\text{L}}))$ to $\text{End}_D(\mathbb{1}_D) \in \text{CAlg}(C)$.

Convention 1.26. From now on, we fix our cutoff cardinal to be $\kappa = \aleph_1$. This choice is somewhat arbitrary: we could have worked with any uncountable regular cardinal. The reason why we need uncountability is to make sure that various countable colimits of compact objects remain compact. For example, given two modules M and M' over an algebra A , the relative tensor product $M \otimes_A M'$ defined above is computed as the geometric realization of the bar complex, hence as a *countable* colimit. Moreover, while κ -compact objects in a presheaf category are generally *retracts* of κ -small colimits of representables, for κ uncountable we may absorb the retract into the colimit.

For these reasons, working with an uncountable cutoff cardinal κ gives better behavior of the κ -compact objects. We will record some basic facts about κ -compact objects in Appendix A.

1.5 Stefanich rings

We are now ready to define the main object of study in these notes.

Definition 1.27. Set

$$0\text{Pr} := \text{An}, \quad 1\text{Pr} := \text{Pr}_{\aleph_1}^{\text{L}}.$$

For $n \geq 1$, define inductively

$$(n+1)\text{Pr} := \text{Mod}_{n\text{Pr}}(1\text{Pr}) \in \text{CAlg}(1\text{Pr}).$$

Notation 1.28. For every $n \geq 1$, the module-endomorphism adjunction gives an adjunction

$$\text{Mod}_n^{\mathbb{P}} : \text{CAlg}((n-1)\text{Pr}) \rightleftarrows \text{CAlg}(n\text{Pr}) : \text{End}_n^{\mathbb{P}},$$

where

$$\text{Mod}_n^{\mathbb{P}}(R) := \text{Mod}_R((n-1)\text{Pr}), \quad \text{End}_n^{\mathbb{P}}(D) := \text{End}_D(\mathbb{1}_D).$$

Definition 1.29. We define the category of *Stefanich rings* as the colimit

$$\text{StRing} := \text{colim}_{1\text{Pr}}(\text{CAlg}(\text{An}) \xrightarrow{\text{Mod}_1^{\mathbb{P}}} \text{CAlg}(1\text{Pr}) \xrightarrow{\text{Mod}_2^{\mathbb{P}}} \text{CAlg}(2\text{Pr}) \xrightarrow{\text{Mod}_3^{\mathbb{P}}} \dots).$$

Equivalently, passing to the right adjoints gives

$$\text{StRing} \simeq \text{lim}_{\text{CAT}}(\text{CAlg}(\text{An}) \xleftarrow{\text{End}_1^{\mathbb{P}}} \text{CAlg}(1\text{Pr}) \xleftarrow{\text{End}_2^{\mathbb{P}}} \text{CAlg}(2\text{Pr}) \xleftarrow{\text{End}_3^{\mathbb{P}}} \dots).$$

The first stage $\text{CAlg}(\text{An})$ is redundant: the inclusion of the tail beginning with $\text{CAlg}(1\text{Pr})$ is cofinal, and in the limit description the degree-zero term is uniquely determined as the endomorphism algebra of the unit in degree one. We retain it to make the degree-zero component of a Stefanich ring explicit.

Notation 1.30. The initial Stefanich ring is denoted by

$$\mathbb{P} := (*, \text{An}, 1\text{Pr}, 2\text{Pr}, \dots).$$

The notations $\text{Mod}_n^{\mathbb{P}}$ and $\text{End}_n^{\mathbb{P}}$ are used in anticipation of later generalizations for arbitrary Stefanich rings in place of $\mathbb{P}$.

2 Quasicoherent sheaves of higher categories I

Talk given by Christoph.

The goal of the next three talks is to construct higher analogues of quasicoherent sheaves on fpqc-stacks. In this talk we treat the affine case, meaning stacks of the form $\mathrm{Spec}(A)$ for a commutative ring spectrum $A \in \mathrm{CAlg}(\mathrm{Sp})$. We write

$$\mathrm{Aff}_{\mathbb{S}} := \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{op}}$$

for the category of *affine spectral schemes*, and denote by $\mathrm{Spec}(A)$ the affine spectral scheme corresponding to A . A map $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ in $\mathrm{Aff}_{\mathbb{S}}$ is the same thing as a map $A \rightarrow B$ in $\mathrm{CAlg}(\mathrm{Sp})$.

For each such affine $\mathrm{Spec}(A)$ we will define presentable categories

$$0\mathrm{Pr}_A, 1\mathrm{Pr}_A, 2\mathrm{Pr}_A, \dots$$

which should be thought of as higher versions of $\mathrm{QCoh}(\mathrm{Spec}(A))$. These categories assemble into a Stefanich ring $\mathcal{O}(\mathrm{Spec}(A))$, resulting in a functor

$$\mathcal{O}: \mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \hookrightarrow \mathrm{StRing}.$$

The main goals of the talk are then the following:

- We show that the resulting assignments $\mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$ can be upgraded to a six-functor formalism

$$\mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}, \text{all, countably presented maps}) \rightarrow (n+1)\mathrm{Pr}, \quad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A.$$

- We show that these assignments satisfy *ambidexterity* in sufficiently high degrees.

2.1 Affine Stefanich rings

We start by constructing the categories $n\mathrm{Pr}_A$.

Definition 2.1. Consider a commutative ring spectrum $A \in \mathrm{CAlg}(\mathrm{Sp})$. We denote its *derived category* of modules by $\mathrm{D}(A)$, or by

$$0\mathrm{Pr}_A := \mathrm{D}(A) := \mathrm{Mod}_A(\mathrm{Sp}) \in \mathrm{CAlg}(1\mathrm{Pr})$$

in the indexing introduced below. For $n \geq 1$, we inductively define the presentable category $n\text{Pr}_A$ by

$$n\text{Pr}_A := \text{Mod}_{(n-1)\text{Pr}_A}(1\text{Pr}) \in \text{CAlg}(1\text{Pr}).$$

Observe that $n\text{Pr}_A$ comes equipped with a canonical functor $n\text{Pr} \rightarrow n\text{Pr}_A$. This is defined inductively: for $n = 0$, this is the unique symmetric monoidal left adjoint $\text{An} \rightarrow \text{Mod}_A(\text{Sp}) = 0\text{Pr}_A$, while for $n \geq 1$ it is defined as the base change functor

$$n\text{Pr} = \text{Mod}_{(n-1)\text{Pr}}(1\text{Pr}) \xrightarrow{(n-1)\text{Pr}_A \otimes_{(n-1)\text{Pr}} -} \text{Mod}_{(n-1)\text{Pr}_A}(1\text{Pr}) = n\text{Pr}_A.$$

We may thus regard $n\text{Pr}_A$ as an object

$$n\text{Pr}_A \in \text{CAlg}(1\text{Pr})_{n\text{Pr}/} \simeq \text{CAlg}(\text{Mod}_{n\text{Pr}}(1\text{Pr})) \simeq \text{CAlg}((n+1)\text{Pr}).$$

Observation 2.2. Recall from Notation 1.28 the fully faithful functor

$$\text{Mod}_n^{\mathbb{P}}: \text{CAlg}((n-1)\text{Pr}) \hookrightarrow \text{CAlg}(n\text{Pr}), \quad R \mapsto \text{Mod}_R((n-1)\text{Pr}),$$

with right adjoint $\text{End}_n^{\mathbb{P}}(D) := \text{End}_D(\mathbb{1}_D)$. Note that for $A \in \text{CAlg}(\text{Sp})$, there is a canonical equivalence

$$n\text{Pr}_A = \text{Mod}_{(n-1)\text{Pr}_A}(1\text{Pr}) \simeq \text{Mod}_{(n-1)\text{Pr}_A}(\text{Mod}_{(n-1)\text{Pr}}(1\text{Pr})) = \text{Mod}_{(n-1)\text{Pr}_A}(n\text{Pr}) = \text{Mod}_{n+1}^{\mathbb{P}}((n-1)\text{Pr}_A).$$

In particular, since $\text{Mod}_{n+1}^{\mathbb{P}}$ is fully faithful, there is a canonical equivalence

$$\text{End}_{n+1}^{\mathbb{P}}(n\text{Pr}_A) \simeq (n-1)\text{Pr}_A. \quad (2.1)$$

Definition 2.3. Given a commutative ring spectrum A , we define the associated Stefanich ring

$$\mathcal{O}(\text{Spec}(A)) \in \text{StRing} = \lim_{\text{CAT}} (\text{CAlg}(\text{An}) \xleftarrow{\text{End}_1^{\mathbb{P}}} \text{CAlg}(1\text{Pr}) \xleftarrow{\text{End}_2^{\mathbb{P}}} \text{CAlg}(2\text{Pr}) \xleftarrow{\text{End}_3^{\mathbb{P}}} \dots)$$

to be the sequence

$$\mathcal{O}(\text{Spec}(A)) := (\Omega^\infty(A), D(A), 1\text{Pr}_A, 2\text{Pr}_A, \dots),$$

equipped with the structure maps from (2.1).

Lemma 2.4. Let $(C_n)_{n \geq n_0}$ be a sequence of objects of 1Pr , together with fully faithful left adjoints $F_n: C_n \rightarrow C_{n+1}$ with right adjoints $G_n: C_{n+1} \rightarrow C_n$. Let

$$C := \text{colim}_{1\text{Pr}} (C_{n_0} \xrightarrow{F_{n_0}} C_{n_0+1} \xrightarrow{F_{n_0+1}} \dots).$$

Then for every $n \geq n_0$ the canonical functor $\iota_n: C_n \rightarrow C$ is fully faithful.

Proof. Since the F_n are left adjoints, the colimit defining C may equivalently be computed as the limit along the right adjoints,

$$C \simeq \lim_{\text{CAT}} (\dots \xrightarrow{G_{n+1}} C_{n+1} \xrightarrow{G_n} C_n),$$

and under this identification ι_n is left adjoint to the projection $q_n: C \rightarrow C_n$. We must show that the unit map $R \rightarrow q_n \iota_n(R)$ is an isomorphism for each $R \in C_n$. Since the functors F_m are fully faithful, the sequence of objects

$$X_j = F_{j-1} F_{j-2} \dots F_n(R) \in C_j$$

for $j \geq n$ defines an object $X := (X_j)_{j \geq n}$ in this limit. We claim that the identification $R = X_n = q_n(X)$ exhibits X as a left-adjoint object to R under q_n (that is, it induces an isomorphism $X \cong \iota_n(R)$). Indeed, for any object $B = (B_j)_{j \geq n}$ of the limit, we have

$$\begin{aligned} \mathrm{Hom}_C(X, B) &\simeq \lim_{j \geq n} \mathrm{Hom}_{C_j}(X_j, B_j) \\ &\simeq \lim_{j \geq n} \mathrm{Hom}_{C_n}(R, G_n \cdots G_{j-1}(B_j)) \\ &\simeq \mathrm{Hom}_{C_n}(R, B_n). \end{aligned}$$

Here the second equivalence uses the adjunctions $F_m \dashv G_m$, and the last equivalence uses the compatibility equivalences $G_i(B_{i+1}) \simeq B_i$ defining the object B of the limit. This shows that the unit of the adjunction $\iota_n \dashv q_n$ at $R \in C_n$ is the identity map $R \rightarrow R$, hence is an isomorphism, showing that ι_n is fully faithful. $\square$

Lemma 2.5. *The assignment $\mathrm{Spec}(A) \mapsto \mathcal{O}(\mathrm{Spec}(A))$ defines a fully faithful functor*

$$\mathcal{O}: \mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \hookrightarrow \mathrm{StRing}.$$

Proof. Since we have $n\mathrm{Pr}_A = \mathrm{Mod}_{n+1}^{\mathbb{P}}((n-1)\mathrm{Pr}_A)$ for all n , this assignment is equivalent to the following composite:

$$\mathrm{CAlg}(\mathrm{Sp}) \xrightarrow{\mathrm{Mod}_{(-)}(\mathrm{Sp})} \mathrm{CAlg}(1\mathrm{Pr}) \hookrightarrow \mathrm{StRing},$$

where the second functor is fully faithful by Lemma 2.4, being a colimit over the fully faithful left adjoints

$$\mathrm{Mod}_n^{\mathbb{P}}: \mathrm{CAlg}((n-1)\mathrm{Pr}) \hookrightarrow \mathrm{CAlg}(n\mathrm{Pr}).$$

But also the first functor is fully faithful, as it factors as

$$\mathrm{CAlg}(\mathrm{Sp}) \xrightarrow{\mathrm{Mod}_{(-)}(\mathrm{Sp})} \mathrm{CAlg}(\mathrm{Mod}_{\mathrm{Sp}}(1\mathrm{Pr})) \hookrightarrow \mathrm{CAlg}(1\mathrm{Pr}),$$

where we use that $\mathrm{Sp} \in \mathrm{CAlg}(1\mathrm{Pr})$ is an idempotent algebra. $\square$

Notation 2.6. Consider a map $f: A \rightarrow B$ in $\mathrm{CAlg}(\mathrm{Sp})$. By the functoriality of the associated Stefanich ring, this results in a map of Stefanich rings

$$\mathcal{O}(\mathrm{Spec}(f)): \mathcal{O}(\mathrm{Spec}(A)) \rightarrow \mathcal{O}(\mathrm{Spec}(B)).$$

The n -th component is a symmetric monoidal functor of the form

$$f_n^*: n\mathrm{Pr}_A \rightarrow n\mathrm{Pr}_B,$$

which we refer to as the n -th *pullback functor*. Concretely, $f_0^*: \mathrm{Mod}_A(\mathrm{Sp}) \rightarrow \mathrm{Mod}_B(\mathrm{Sp})$ is extension of scalars along f , and for $n > 0$ the functor $f_n^*: \mathrm{Mod}_{(n-1)\mathrm{Pr}_A}(1\mathrm{Pr}) \rightarrow \mathrm{Mod}_{(n-1)\mathrm{Pr}_B}(1\mathrm{Pr})$ is extension of scalars along

$$f_{n-1}^*: (n-1)\mathrm{Pr}_A \rightarrow (n-1)\mathrm{Pr}_B.$$

2.2 Extension and restriction of scalars

In preparation for the construction of the six-functor formalism $\text{Spec}(A) \mapsto n\text{Pr}_A$, we recall some basic properties of extension and restriction of scalars in presentably symmetric monoidal categories.

Consider some $\mathcal{V} \in \text{CAlg}(\text{Pr}^{\text{L}})$. Given a map $g: R \rightarrow S$ in $\text{CAlg}(\mathcal{V})$, we obtain an adjunction

$$g^*: \text{Mod}_R(\mathcal{V}) \rightleftarrows \text{Mod}_S(\mathcal{V}): g_*$$

given by extension/restriction of scalars. The functor g^* is given on underlying objects in $\mathcal{V}$ by $M \mapsto M \otimes_R S$. It is symmetric monoidal.

Remark 2.7. We recall some basic properties of this adjunction:

- (a) The functor $g_*: \text{Mod}_S(\mathcal{V}) \rightarrow \text{Mod}_R(\mathcal{V})$ commutes with the forgetful functors to $\mathcal{V}$. It is functorial in g : we have $(g'g)_* \simeq g_*g'_*$. Consequently, g^* is functorial in g .
- (b) The functor g^* preserves free modules. Recall that an R -module is called *free* if it lies in the image of extension of scalars $\mathcal{V} \rightarrow \text{Mod}_R(\mathcal{V})$, $M \mapsto R \otimes M$ along the unit map $\mathbb{1} \rightarrow R$, and similarly for S -modules. Since $g^*(R \otimes M) \simeq S \otimes M$ and the composite $\mathbb{1} \rightarrow R \xrightarrow{g} S$ is the unit map of S , the functor g^* sends the free R -module $R \otimes M$ to the free S -module $S \otimes M$.
- (c) The functor $g_*: \text{Mod}_S(\mathcal{V}) \rightarrow \text{Mod}_R(\mathcal{V})$ preserves colimits. This holds because the map $\text{Mod}_R(\mathcal{V}) \rightarrow \mathcal{V}$ creates colimits.
- (d) The functor g_* satisfies the **projection formula**: for $N \in \text{Mod}_S(\mathcal{V})$ and $M \in \text{Mod}_R(\mathcal{V})$, the map

$$M \otimes_R g_* N \rightarrow g_*(g^* M \otimes_S N)$$

adjoint to $g^*(M \otimes_R g_* N) \simeq g^* M \otimes_S g^* g_* N \rightarrow g^* M \otimes_S N$ is an isomorphism. Since both sides preserve colimits in M , we may reduce the claim to the case of a free module $M = f^* M' = R \otimes M'$, in which case we see that both sides reduce to $M' \otimes g_* N$ by an explicit computation of the formula for the relative tensor product.

- (e) The functors g_* satisfy **base change**: Consider a pushout square in $\text{CAlg}(\mathcal{V})$ of the form

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ T & \longrightarrow & U, \end{array}$$

meaning that $U \simeq T \otimes_R S$ as commutative algebras. By passing to extension of scalars functors, we obtain a commutative diagram in $\text{CAlg}(\text{1Pr})$ of the form

$$\begin{array}{ccc} \text{Mod}_R(\mathcal{V}) & \longrightarrow & \text{Mod}_S(\mathcal{V}) \\ \downarrow & & \downarrow \\ \text{Mod}_T(\mathcal{V}) & \longrightarrow & \text{Mod}_U(\mathcal{V}). \end{array}$$

The horizontal functors have right adjoints, and we need to show that both composites $\text{Mod}_S(\mathcal{V}) \rightarrow \text{Mod}_T(\mathcal{V})$ agree, along the canonical map. Computing both sides, we get $T \otimes_R M$ versus $(T \otimes_R S) \otimes_S M$, and these agree. (Details omitted.)

(f) Finally, assume that $\mathcal{V} \in \text{CAlg}(\text{1Pr})$, and let $g: R \rightarrow S$ be a countably presented map, meaning that S is $\aleph_1$ -compact when regarded as an object of $\text{CAlg}_R(\mathcal{V})$. Then the functor

$$g_*: \text{Mod}_S(\mathcal{V}) \rightarrow \text{Mod}_R(\mathcal{V})$$

preserves $\aleph_1$ -compact objects by Corollary A.3, applied with $\kappa = \aleph_1$ and $C = \mathcal{V}$. In combination with (c), we conclude that g_* is a right adjoint to g^* internal to the 2-category 1Pr .

Lemma 2.8. *Let $\mathcal{V} \in \text{CAlg}(\text{1Pr})$, and let $g: R \rightarrow S$ be a countably presented map in $\text{CAlg}(\mathcal{V})$. Then also the induced functor*

$$g^*: \text{Mod}_R(\mathcal{V}) \rightarrow \text{Mod}_S(\mathcal{V})$$

is countably presented in $\text{CAlg}(\text{Mod}_{\mathcal{V}}(\text{1Pr})) \simeq \text{CAlg}_{\mathcal{V}}(\text{1Pr})$.

Proof. Recall from Theorem 1.24 that the functor

$$\text{CAlg}(\text{Mod}_{\mathcal{V}}(\text{1Pr})) \rightarrow \text{CAlg}(\mathcal{V}), \quad D \mapsto \text{End}_D(\mathbb{1}_D)$$

preserves $\aleph_1$ -filtered colimits. It follows that its left adjoint

$$\text{CAlg}(\mathcal{V}) \hookrightarrow \text{CAlg}(\text{Mod}_{\mathcal{V}}(\text{1Pr})), \quad R \mapsto \text{Mod}_R(\mathcal{V})$$

preserves $\aleph_1$ -compact objects. Replacing $\mathcal{V}$ by $\text{Mod}_R(\mathcal{V})$ then gives the claim. $\square$

Corollary 2.9. *Let $g: A \rightarrow B$ be a countably presented morphism of commutative ring spectra. Then for each $n \geq 0$, the functor*

$$g_n^*: n\text{Pr}_A \rightarrow n\text{Pr}_B$$

is countably presented as a map in $\text{CAlg}((n+1)\text{Pr})$.

Proof. By induction this is immediate from Lemma 2.8. $\square$

Corollary 2.10. *Let $f: A \rightarrow B$ be a countably presented map in $\text{CAlg}(\text{Sp})$. Then, for every $n \geq 0$, the object $n\text{Pr}_B \in (n+1)\text{Pr}_A \simeq \text{Mod}_{n\text{Pr}_A}(\text{1Pr})$ is $\aleph_1$ -compact.*

Proof. By the previous corollary, $n\text{Pr}_B$ is $\aleph_1$ -compact as an object of $\text{CAlg}_{n\text{Pr}_A}(\text{1Pr})$. By Lemma A.1 this is equivalent to it being $\aleph_1$ -compact as an object of $\text{Mod}_{n\text{Pr}_A}(\text{1Pr})$. $\square$

2.3 Construction of the six-functor formalism

We will now construct the desired six-functor formalism on $\text{Aff}_{\mathbb{S}}$ we are after.

Definition 2.11. A map $\text{Spec}(B) \rightarrow \text{Spec}(A)$ in $\text{Aff}_{\mathbb{S}}$ is called *countably presented* if B is countably presented as an A -algebra.

This class of maps satisfies the usual closure properties.

Lemma 2.12. *Let $A \in \text{CAlg}(\text{Sp})$.*

- (i) *Countably presented A -algebras are closed under countable colimits, retracts, and finite colimits in $\text{CAlg}_A(\text{Sp})$.*

- (ii) Countably presented maps of affine spectral schemes are stable under base change and composition.
- (iii) If $A \rightarrow B$ and $A \rightarrow C$ are countably presented and $B \rightarrow C$ is a map over A , then $B \rightarrow C$ is countably presented.
- (iv) If $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is countably presented and admits a section $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$, then the section is countably presented.

Proof. Part (i) is Lemma A.4(1), applied with $\kappa = \aleph_1$ and $C = \mathrm{Sp}$.

For base change in (ii), suppose $A \rightarrow B$ is countably presented and $A \rightarrow A'$ is any map. The base change is $\mathrm{Spec}(A' \otimes_A B) \rightarrow \mathrm{Spec}(A')$, and $A' \rightarrow A' \otimes_A B$ is countably presented by Lemma A.4(2). For composition, if $A \rightarrow B$ and $B \rightarrow C$ are countably presented, then $A \rightarrow C$ is countably presented by Lemma A.4(3).

Part (iii) is Lemma A.4(4). Finally, a section $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ corresponds to an A -algebra retraction $B \rightarrow A$, so (iv) is exactly Lemma A.5. $\square$

We now come to the main result of this talk:

Theorem 2.13. *For every $n \geq 0$, there exists a six-functor formalism*

$$\begin{array}{lll} \mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}, \text{all, countably presented maps}) & \rightarrow & (n+1)\mathrm{Pr} \\ \mathrm{Spec}(A) & \mapsto & n\mathrm{Pr}_A \\ \mathrm{Spec}(A) \xleftarrow{f} \mathrm{Spec}(B) \xrightarrow{g} \mathrm{Spec}(C) & \mapsto & n\mathrm{Pr}_A \xrightarrow{f_n^*} n\mathrm{Pr}_B \xrightarrow{g_{n,*}} n\mathrm{Pr}_C, \end{array}$$

where $g_{n,*}$ is right adjoint to g_n^* .

Proof. We start with the case $n = 0$. The functoriality of module categories provides a functor

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \rightarrow 1\mathrm{Pr}, \quad \mathrm{Spec}(A) \mapsto \mathrm{Mod}_A(\mathrm{Sp}), \quad f \mapsto f^*.$$

To extend this to the span category via the right adjoints, it suffices to check the following properties:

- For every countably presented map $g: \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$, the functor $g^*: \mathrm{Mod}_B(\mathrm{Sp}) \rightarrow \mathrm{Mod}_A(\mathrm{Sp})$ admits a right adjoint g_* internal to the 2-category $1\mathrm{Pr}$. This was verified in Remark 2.7(f).
- These right adjoints g_* satisfy base change. This was verified in Remark 2.7(e).
- For every such f , the right adjoint f_* satisfies the projection formula. This was verified in Remark 2.7(d).

The proof for $n > 0$ is essentially identical. Functoriality of the assignment $A \mapsto n\mathrm{Pr}_A$ provides a functor

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \rightarrow (n+1)\mathrm{Pr}, \quad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A, \quad f \mapsto f_n^*.$$

If $g: \mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ is countably presented, then by Corollary 2.9 so is the functor

$$g_{n-1}^*: (n-1)\mathrm{Pr}_A \rightarrow (n-1)\mathrm{Pr}_B$$

as a morphism in $\mathbf{CAlg}(n\mathbf{Pr})$. It then follows from Remark 2.7(f) that the associated restriction of scalars functor

$$g_{n,*} = (g_{n-1}^*)_* : n\mathbf{Pr}_B \rightarrow n\mathbf{Pr}_A$$

is an internal right adjoint to g_n^* in $(n+1)\mathbf{Pr}$. Moreover, the other parts of Remark 2.7 show that these right adjoints satisfy the required base change and projection formula conditions. $\square$

Remark 2.14. If we drop the countable presentation hypotheses, we still obtain a presentable six-functor formalism:

$$\mathrm{Span}(\mathrm{Aff}_{\mathbb{S}}) \rightarrow \mathbf{Pr}^{\mathrm{L}}, \quad \mathrm{Spec}(A) \mapsto n\mathbf{Pr}_A$$

2.4 Ambidexterity

It turns out that for $n \geq 1$ the symmetric monoidal pullback functor f_n^* admits not only a well-behaved right adjoint $f_{n,*}$, but also a left adjoint $f_{n,\sharp}$, and the two coincide. This phenomenon, of a functor having identical left and right adjoints, is known as *ambidexterity*, a term introduced by Hopkins–Lurie [HL13]. Before applying this to our functors f_n^* , we set up the general framework.

Definition 2.15. Let C be a 2-category, and let $F : X \rightarrow Y$ be a 1-morphism in C .

- We say F is a *left adjoint* if there exist a 1-morphism $F^R : Y \rightarrow X$ together with 2-morphisms

$$\eta : \mathrm{id}_X \rightarrow F^R F \quad \text{in } \mathrm{End}_C(X), \quad \varepsilon : F F^R \rightarrow \mathrm{id}_Y \quad \text{in } \mathrm{End}_C(Y),$$

such that the composites

$$F \xrightarrow{F\eta} F F^R F \xrightarrow{\varepsilon F} F \quad \text{and} \quad F^R \xrightarrow{\eta F^R} F^R F F^R \xrightarrow{F^R \varepsilon} F^R$$

are equivalent to the identity.

- Dually, F is a *right adjoint* if there exist a 1-morphism $F^L : Y \rightarrow X$ and 2-morphisms $\eta : \mathrm{id}_Y \rightarrow F F^L$, $\varepsilon : F^L F \rightarrow \mathrm{id}_X$ satisfying the analogous triangle identities.

For the present discussion the reader may take C to be an object of $2\mathbf{Pr}$, though the definition makes sense for any reasonable notion of $(\infty, 2)$ -category. If F is a left adjoint, its right adjoint F^R is unique, and the left adjoint of F^R is again F (with the same η and ε).

Remark 2.16. In our formalism of presentable (∞, n) -categories, there are forgetful functors $n\mathbf{Pr} \rightarrow k\mathbf{Pr}$ for any $k \leq n$. Indeed, the symmetric monoidal functor $\mathbf{An} \rightarrow 1\mathbf{Pr}$ (since $\mathbf{An}$ is the unit object) yields, by passing to modules, symmetric monoidal functors $n\mathbf{Pr} \rightarrow (n+1)\mathbf{Pr}$ inductively, whose right adjoints are the desired forgetful functors. In particular, Definition 2.15 applies to 1-morphisms in $n\mathbf{Pr}$ for any $n \geq 2$.

We can now state the general ambidexterity result.

Proposition 2.17. *Let C be an $(\infty, 3)$ -category, and let $F : X \rightarrow Y$ be a 1-morphism in C . The following are equivalent.*

(i) The morphism F admits a right adjoint $F^R: Y \rightarrow X$, and both

$$\eta: \text{id}_X \rightarrow F^R F \quad \text{in } \text{End}_C(X) \quad \text{and} \quad \varepsilon: F F^R \rightarrow \text{id}_Y \quad \text{in } \text{End}_C(Y)$$

admit right adjoints (in the respective $(\infty, 2)$ -categories).

(ii) The morphism F admits a left adjoint $F^L: Y \rightarrow X$, and both

$$\eta: \text{id}_Y \rightarrow F F^L \quad \text{in } \text{End}_C(Y) \quad \text{and} \quad \varepsilon: F^L F \rightarrow \text{id}_X \quad \text{in } \text{End}_C(X)$$

admit left adjoints.

Moreover, when these conditions hold, there is a canonical identification $F^L \simeq F^R$.

Remark 2.18. There is a dual version of Proposition 2.17, in which the words “right” and “left” are swapped in (i) and (ii) respectively; it again yields a canonical identification $F^L \simeq F^R$. However, in any given application, η and ε must be assumed to admit the *same* type of adjoint for the result to apply.

Proof. We show that (i) implies (ii) with $F^L = F^R$; the dual implication is similar. Given (F^R, η, ε) as in (i), define

$$\eta': \text{id}_Y \rightarrow F F^R \quad \text{and} \quad \varepsilon': F^R F \rightarrow \text{id}_X$$

by $\eta' := \varepsilon^R$ and $\varepsilon' := \eta^R$. The triangle identities for $(F, F^R, \eta', \varepsilon')$ follow from those of $(F, F^R, \eta, \varepsilon)$ together with the fact that forming right adjoints is compatible with composition. By construction, η' and ε' admit left adjoints (namely ε and η themselves), giving (ii). $\square$

We now establish an ambidexterity property for the functors $f_{n,*}$.

Proposition 2.19. Let $n \geq 1$, and let $f: A \rightarrow B$ be a countably presented map in $\text{CAlg}(\text{Sp})$.

- The pullback functor $f_n^*: n\text{Pr}_A \rightarrow n\text{Pr}_B$ admits a left adjoint $f_{n,\sharp}: n\text{Pr}_B \rightarrow n\text{Pr}_A$, which the canonical comparison identifies with the right adjoint, $f_{n,\sharp} \simeq f_{n,*}$. As a left adjoint, $f_{n,\sharp}$ satisfies base change and the projection formula.
- The category $n\text{Pr}_B$ is self-dual as an object of $(n+1)\text{Pr}_A \simeq \text{Mod}_{n\text{Pr}_A}(1\text{Pr})$.

Proof. Since we have the adjunction $f_n^* \dashv f_{n,*}$, we obtain a unit and counit

$$\eta: \text{id} \rightarrow f_{n,*} f_n^* \in \text{End}_{(n+1)\text{Pr}_A}(n\text{Pr}_A) \quad \varepsilon: f_n^* f_{n,*} \rightarrow \text{id} \in \text{End}_{(n+1)\text{Pr}_A}(n\text{Pr}_B).$$

We will deduce the proposition from Proposition 2.17, applied in the $(\infty, 3)$ -category $\mathcal{C} = (n+1)\text{Pr}_A$ (which lies in $(n+2)\text{Pr}$ and hence in 3Pr , since $n \geq 1$); concretely, we need to verify that η and ε admit right adjoints. For η , note that evaluation at the unit induces an equivalence

$$\text{End}_{(n+1)\text{Pr}_A}(n\text{Pr}_A) \xrightarrow{\sim} n\text{Pr}_A.$$

Under this equivalence, the unit η corresponds to the map

$$f_{n-1}^*: (n-1)\text{Pr}_A \rightarrow (n-1)\text{Pr}_B,$$

which indeed has a right adjoint $f_{n-1,*}$ in $n\text{Pr}_A$.

For ε , note that there is an equivalence

$$\begin{aligned} \text{End}_{(n+1)\text{Pr}_A}(n\text{Pr}_B) &= \text{End}_{\text{Mod}_{n\text{Pr}_A}}(\text{Mod}_{(n-1)\text{Pr}_B}(n\text{Pr}_A), \text{Mod}_{(n-1)\text{Pr}_B}(n\text{Pr}_A)) \\ &\simeq \text{Mod}_{(n-1)\text{Pr}_B \otimes_{(n-1)\text{Pr}_A} (n-1)\text{Pr}_B}(n\text{Pr}_A). \end{aligned}$$

Under this equivalence, the counit ε corresponds to the ‘multiplication map’

$$(n-1)\text{Pr}_B \otimes_{(n-1)\text{Pr}_A} (n-1)\text{Pr}_B \rightarrow (n-1)\text{Pr}_B.$$

The left-hand side is equivalent to $(n-1)\text{Pr}_{B \otimes_A B}$, and under this equivalence the map agrees with ∇_{n-1}^* induced by the codiagonal map $\nabla: B \otimes_A B \rightarrow B$. The map $B \rightarrow B \otimes_A B$ is obtained from $A \rightarrow B$ by base change, and ∇ is a retraction of it. Thus ∇ is countably presented by Lemma 2.12, so ∇_{n-1}^* admits a $(n-1)\text{Pr}_{B \otimes_A B}$ -linear right adjoint.

For the claim about base change, we apply the same result inside the 3-category $\text{Ar}((n+1)\text{Pr}_A)$, the arrow category of $(n+1)\text{Pr}_A$. Since adjoints in this arrow category are precisely the adjointable squares, this gives the claim. Finally, the projection formula for $f_{n,\sharp}$ follows from the identification $f_{n,\sharp} \simeq f_{n,*}$ and the projection formula for $f_{n,*}$ proved in Remark 2.7(d).

Finally, for the duality data on $n\text{Pr}_B$, we are using the following two maps:

$$\begin{aligned} n\text{Pr}_A &\xrightarrow{f_n^*} n\text{Pr}_B \xrightarrow{\nabla_{n,*}} n\text{Pr}_B \otimes_{n\text{Pr}_A} n\text{Pr}_B \\ n\text{Pr}_B \otimes_{n\text{Pr}_A} n\text{Pr}_B &\xrightarrow{\nabla_n^*} n\text{Pr}_B \xrightarrow{f_{n,*}} n\text{Pr}_A. \end{aligned}$$

The triangle identities are an immediate consequence of base change. □

3 Descent for quasicoherent sheaves

Talk given by Melvin.

We use the following output of the previous talk:

- For $A \in \mathbf{CAlg}(\mathbf{Sp})$, we defined categories $n\mathbf{Pr}_A$ by

$$0\mathbf{Pr}_A := \mathbf{D}(A) := \mathbf{Mod}_A(\mathbf{Sp}), \quad (n+1)\mathbf{Pr}_A := \mathbf{Mod}_{n\mathbf{Pr}_A}((n+1)\mathbf{Pr}) \simeq \mathbf{Mod}_{n\mathbf{Pr}_A}(1\mathbf{Pr}).$$

- The assignment $\mathbf{Spec}(A) \mapsto n\mathbf{Pr}_A$ defines a six-functor formalism on $\mathbf{Aff}_{\mathbb{S}}$ valued in $1\mathbf{Pr}$.
- For $n \geq 1$, we have ambidexterity: if $f: \mathbf{Spec}(A) \rightarrow \mathbf{Spec}(B)$ is countably presented, then the right adjoint $f_{n,*}: n\mathbf{Pr}_A \rightarrow n\mathbf{Pr}_B$ is also a left adjoint to f_n^* in a canonical way.

3.1 Hyperdescent

The goal of this chapter is the following hyperdescent statement:

Theorem 3.1 (cf. [Mat16, Corollary 3.33, Proposition 3.45]). *The functor $n\mathbf{Pr}_{(-)}^!: \mathbf{Aff}_{\mathbb{S}}^{\mathrm{op}} \rightarrow 1\mathbf{Pr}$ (defined on maps by the right adjoints $f^!$) is a hypersheaf with respect to the fpqc topology for each $n \geq 0$.*

Remark 3.2. For $n \geq 1$, the maps $f^!$ agree with f^* by ambidexterity.

To make sense of this statement, we recall the following two definitions:

Definition 3.3. Let C be a site with finite limits. A *hypercover* of $A \in C$ is a simplicial object $A_{\bullet} \in \mathbf{Fun}(\Delta^{\mathrm{op}}, C)$ together with a map $A_0 \rightarrow A$ which is a cover in C , such that if we right Kan extend $A_{\bullet}$ to $A(-) \in \mathbf{Fun}(\mathbf{PSh}_{\mathrm{fin}}(\Delta)^{\mathrm{op}}, C)$, the map $A(\Delta^n) \rightarrow A(\partial\Delta^n)$ is a cover in C .

Definition 3.4. Recall the following basic definitions:

- For a static ring $B_0 \in \mathbf{CAlg}(\mathbf{Ab})$ and a static B_0 -module M_0 , we say that M_0 is *flat* if the functor $M_0 \otimes_{B_0} -: \mathbf{Mod}_{B_0}(\mathbf{Ab}) \rightarrow \mathbf{Ab}$ is exact. It is called *faithfully flat* if, in addition, the functor $M_0 \otimes_{B_0} -$ is conservative, i.e. detects isomorphisms. (Equivalently it detects exact sequences.)
- A morphism $B_0 \rightarrow A_0$ in $\mathbf{CAlg}(\mathbf{Ab})$ is called *(faithfully) flat* if A_0 is (faithfully) flat as a B_0 -module.

- A morphism $B \rightarrow A$ in $\text{CAlg}(\text{Sp})$ is *(faithfully) flat* if the map $\pi_0 B \rightarrow \pi_0 A$ in $\text{CAlg}(\text{Ab})$ is (faithfully) flat and for each $i \in \mathbb{Z}$ the multiplication map $\pi_i B \otimes_{\pi_0 B} \pi_0 A \rightarrow \pi_i A$ is an isomorphism.
- A morphism $\text{Spec}(A) \rightarrow \text{Spec}(B)$ in $\text{Aff}_{\mathbb{S}}$ is called (faithfully) flat if its associated map $B \rightarrow A$ in $\text{CAlg}(\text{Sp})$ is.

Definition 3.5. The *fpqc topology* on $\text{Aff}_{\mathbb{S}}$ is the Grothendieck topology in which a sieve over $\text{Spec}(B)$ is declared to be a covering sieve if it contains a finite collection of morphisms $\text{Spec}(A_i) \rightarrow \text{Spec}(B)$ such that the induced map $\text{Spec}(\prod_{i=1}^n A_i) \rightarrow \text{Spec}(B)$ is faithfully flat.

Observe that a functor $F: \text{Aff}_{\mathbb{S}}^{\text{op}} \rightarrow C$ is a hypersheaf with respect to the fpqc topology if and only if the following two properties are satisfied:

- The functor F preserves finite products.
- For an fpqc-hypercover $\text{Spec}(A_{\bullet}) \rightarrow \text{Spec}(A)$, the induced map

$$F(\text{Spec}(A)) \rightarrow \lim_{[n] \in \Delta} F(\text{Spec}(A_n))$$

is an isomorphism.

The proof has three parts; the first two treat countably presented fpqc-hypercovers:

- We first prove the theorem for $n = 0$.
- We then prove the general case by induction on n .
- Finally, we reduce arbitrary fpqc-hypercovers to the countably presented case.

3.2 The base case $n = 0$

We will start by showing that the assignment $\text{Spec}(A) \mapsto D(A)$ satisfies fpqc-hyperdescent. Since we have $D(\prod_i A_i) \xrightarrow{\sim} \prod_i D(A_i)$, it remains to show that for an fpqc-hypercover $\text{Spec}(A_{\bullet}) \rightarrow \text{Spec}(A)$ in $\text{Aff}_{\mathbb{S}}$ the functor

$$D(A) \longrightarrow \lim_{[n] \in \Delta}^! D(A_{\bullet})$$

is an equivalence. Here the limit on the right is taken along the functors $f^!$.

Until Section 3.4, we assume that all terms A_m are countably presented over A . The general case will then follow by approximation.

3.2.1 Step 1: The functor is fully faithful

We start by showing that this comparison functor is fully faithful.

The comparison functor $D(A) \rightarrow \lim_{[n] \in \Delta}^! D(A_{\bullet})$ admits a left adjoint, given by sending an object $(M_n)_{[n] \in \Delta}$ to the colimit $\text{colim}_{[n] \in \Delta}^{\text{op}} (f_n)_* M_n$, where we denote by $f_n: \text{Spec}(A_n) \rightarrow \text{Spec}(A)$ the maps constituting the hypercover. So we must show that for any object $M \in D(A)$, the counit map

$$\text{colim}_{[n] \in \Delta}^{\text{op}} (f_n)_* f_n^! M \rightarrow M$$

is an isomorphism of A -modules. Note that we may rewrite the left-hand side as

$$\operatorname{colim}_{[n] \in \Delta^{\text{op}}} (f_n)_* f_n^! M = \operatorname{colim}_{[n] \in \Delta^{\text{op}}} (f_n)_* \underline{\operatorname{Hom}}_{A_n}(A_n, f_n^! M) \simeq \operatorname{colim}_{[n] \in \Delta^{\text{op}}} (f_n)_* \underline{\operatorname{Hom}}_A(A_n, M).$$

We may now filter Δ by the subcategories $\Delta_{\leq n}$ to rewrite this colimit in turn as

$$\operatorname{colim}_{m \geq 0} \operatorname{colim}_{[n] \in (\Delta_{\leq m})^{\text{op}}} \underline{\operatorname{Hom}}_A(A_n, M) \simeq \operatorname{colim}_{m \geq 0} \underline{\operatorname{Hom}}_A(\lim_{[n] \in \Delta_{\leq m}} A_{\bullet}, M),$$

where we use that a colimit over $\Delta_{\leq m}$ is a finite colimit and is thus preserved by the exact functor $\underline{\operatorname{Hom}}_A(-, M) : \mathcal{D}(A) \rightarrow \mathcal{D}(A)$. If we similarly write $M \simeq \underline{\operatorname{Hom}}_A(A, M) \simeq \operatorname{colim}_{m \geq 0} \underline{\operatorname{Hom}}_A(A, M)$, then the comparison map is obtained by applying $\underline{\operatorname{Hom}}_A(-, M)$ to the canonical map of pro-systems $A \rightarrow \{\lim_{\Delta_{\leq m}} A_{\bullet}\}_{m \geq 0}$ in $\mathcal{D}(A)$. So we need to show that this map is a pro-isomorphism in $\mathcal{D}(A)$.

We first justify the reduction to the connective case.

Lemma 3.6. *Let $A \rightarrow A_{\bullet}$ be an fpqc-hypercover of commutative ring spectra. Put $A^+ := \tau_{\geq 0} A$ and $A_n^+ := \tau_{\geq 0} A_n$. Then $A^+ \rightarrow A_{\bullet}^+$ is an fpqc-hypercover. Moreover, for every $n \geq 0$ we have a canonical isomorphism*

$$A_n \simeq A \otimes_{A^+} A_n^+,$$

and for every $m \geq 0$ the canonical map

$$A \otimes_{A^+} \lim_{[n] \in \Delta_{\leq m}} A_n^+ \longrightarrow \lim_{[n] \in \Delta_{\leq m}} A_n$$

is an isomorphism in $\mathcal{D}(A)$. Consequently, to show that

$$A \rightarrow \{\lim_{[n] \in \Delta_{\leq m}} A_n\}_{m \geq 0}$$

is a pro-isomorphism in $\mathcal{D}(A)$, it suffices to prove the analogous statement for $A^+ \rightarrow A_{\bullet}^+$.

Proof. We first record the basic flatness input. If $R \rightarrow S$ is flat, then $R^+ := \tau_{\geq 0} R \rightarrow S^+ := \tau_{\geq 0} S$ is flat and the canonical map

$$R \otimes_{R^+} S^+ \rightarrow S$$

is an isomorphism. More generally, for every R -algebra T there is a canonical isomorphism

$$T^+ \otimes_{R^+} S^+ \simeq \tau_{\geq 0}(T \otimes_R S).$$

Indeed, both statements follow by comparing homotopy groups, using the defining condition for flatness:

$$\pi_i S \simeq \pi_i R \otimes_{\pi_0 R} \pi_0 S.$$

Let K be a finite simplicial set, and write A_K for the commutative ring spectrum corresponding to the value of the right Kan extension of $\operatorname{Spec}(A_{\bullet})$ on K ; similarly, write A_K^+ for the corresponding object associated with $\operatorname{Spec}(A_{\bullet}^+)$. We claim, by induction over the cells of K , that

$$A_K^+ \simeq \tau_{\geq 0} A_K,$$

and that A_K is flat over A .

The claim is clear for $K = \emptyset$, where $A_K = A$. Suppose that L is obtained from K by attaching an r -simplex,

$$L = K \cup_{\partial \Delta^r} \Delta^r.$$

Then

$$A_L \simeq A_K \otimes_{A_{\partial\Delta^r}} A_r.$$

The matching map $A_{\partial\Delta^r} \rightarrow A_r$ is faithfully flat by the hypercover condition. Hence $A_K \rightarrow A_L$ is flat, and so A_L is flat over A . Moreover, by the flatness input above and the induction hypothesis, we get

$$\tau_{\geq 0} A_L \simeq \tau_{\geq 0} A_K \otimes_{\tau_{\geq 0} A_{\partial\Delta^r}} \tau_{\geq 0} A_r \simeq A_K^+ \otimes_{A_{\partial\Delta^r}^+} A_r^+ \simeq A_L^+.$$

This proves the induction.

Applying this to $K = \partial\Delta^r$, we see that the r -th matching map for $A^+ \rightarrow A_\bullet^+$ is the connective cover of the faithfully flat map

$$A_{\partial\Delta^r} \rightarrow A_r.$$

It is therefore faithfully flat, so $A^+ \rightarrow A_\bullet^+$ is again an fpqc-hypercover.

Since A_n is flat over A , the flatness input also gives $A_n \simeq A \otimes_{A^+} A_n^+$ for every n . Finally, the last displayed isomorphism follows because the functor

$$A \otimes_{A^+} -: D(A^+) \rightarrow D(A)$$

is exact and $\Delta_{\leq m}$ is finite:

$$A \otimes_{A^+} \lim_{[n] \in \Delta_{\leq m}} A_n^+ \simeq \lim_{[n] \in \Delta_{\leq m}} (A \otimes_{A^+} A_n^+) \simeq \lim_{[n] \in \Delta_{\leq m}} A_n.$$

The final assertion follows by applying this exact base-change functor to the pro-isomorphism for the connective hypercover. $\square$

By Lemma 3.6, we may assume from now on that both A and all of the A_i are connective.

Now, to show that the map $A \rightarrow \{\lim_{\Delta_{\leq m}} A_\bullet\}_{m \geq 0}$ is an isomorphism in $\text{Pro}(D(A))$, we may equivalently show its cofiber is zero. The cofiber is given by the pro-system $\{C_m\}_{m \geq 0}$, where we set

$$C_m := \text{cofib}(A \rightarrow \lim_{\Delta_{\leq m}} A_\bullet).$$

It will thus suffice to show that the map $C_{m+2} \rightarrow C_m$ is the zero map in $D(A)$ for each $m \geq 0$. For this, recall the following definition:

Definition 3.7. A morphism $X \rightarrow Y$ in $D(A)$ is called a *phantom map* if for every perfect A -module Z and every map $Z \rightarrow X$, the composite $Z \rightarrow X \rightarrow Y$ is nullhomotopic. In particular, taking $Z = A[i]$ shows that a phantom map induces the zero map $\pi_i(X) \rightarrow \pi_i(Y)$ for each $i \in \mathbb{Z}$.

Lemma 3.8 ([HPS97, Theorem 4.2.5]). *The composite of two phantom maps is nullhomotopic.*

Proof. The homotopy category of $D(A)$ is a Brown category, generated by the perfect A -modules, so the cited theorem applies. $\square$

In light of the lemma, it remains to show that each map $C_{m+1} \rightarrow C_m$ is a phantom map. We now make the following claim:

Claim 3.9. For each $m \geq 0$, there is a countably presented flat connective A -module Q_m such that

$$C_m := \text{cofib}(A \rightarrow \lim_{\Delta_{\leq m}} A_\bullet) \simeq Q_m[-m].$$

We will prove this claim below, but let us first explain why it implies that $C_{m+1} \rightarrow C_m$ is phantom. For this, recall the following spectral analogue of Lazard's theorem:

Theorem 3.10 (Lazard, [Lur17, Theorem 7.2.2.15]). *Let A be a connective ring spectrum. Then every connective countably presented flat A -module M is a sequential colimit of finite rank free A -modules:*

$$M \simeq \operatorname{colim}_{i \geq 0} A^{\oplus n_i}.$$

Combining this with Claim 3.9, we may write $C_{m+1} \simeq \operatorname{colim}_{i \geq 0} A^{\oplus n_i}[-m-1]$. If Z is any perfect A -module and $Z \rightarrow C_{m+1}$ is any map, then it factors through some finite stage $Z \rightarrow A^{\oplus n_i}[-m-1]$. It therefore suffices to show that every map $A[-m-1] \rightarrow C_m$ is nullhomotopic. Such maps are classified by $\pi_{-m-1}(C_m)$, which vanishes because $C_m \simeq Q_m[-m]$ with Q_m connective and flat. Thus $C_{m+1} \rightarrow C_m$ is phantom.

We now prove Claim 3.9. Since each A_r is flat over the connective ring spectrum A , there is a canonical equivalence

$$A_r \simeq A \otimes_{\pi_0 A} \pi_0 A_r.$$

The exact functor $A \otimes_{\pi_0 A} -$ commutes with the finite limits defining the partial totalizations. It therefore suffices to compute the cofiber after replacing $A \rightarrow A_\bullet$ by the fpqc-hypercover of static rings $\pi_0 A \rightarrow \pi_0 A_\bullet$.

Let $N^\bullet$ be the normalized cochain complex of the augmented cosimplicial A -module $A \rightarrow A_\bullet$:

$$A \rightarrow N^0 \rightarrow N^1 \rightarrow \dots$$

After the faithfully flat base change $A \rightarrow A_0$, the augmented hypercover is split, so this complex becomes contractible. Faithful flatness therefore implies that it is exact. The same argument applies after every base change in A .

The Dold–Kan description of partial totalizations now gives

$$C_m \simeq \operatorname{coker}(N^{m-1} \rightarrow N^m)[-m],$$

where $N^{-1} := A$. Each N^m is a direct summand of the countably presented flat A -module A_m , hence is countably presented and flat. Exactness after arbitrary base change identifies the displayed cokernel with a submodule of N^{m+1} whose inclusion remains injective after arbitrary base change. Since N^{m+1} is flat, the equational criterion for flatness shows that this cokernel is flat. It is countably presented because it is a cokernel of a map between countably presented modules. Base-changing this description from $\pi_0 A$ back to A proves Claim 3.9.

All in all, we have now shown that the functor $D(A) \rightarrow \lim_{[n] \in \Delta}^! D(A_n)$ is fully faithful.

3.2.2 Step 2: Essential surjectivity

We now show the functor $D(A) \rightarrow \lim_{[n] \in \Delta}^! D(A_n)$ is an equivalence.

Step 2a: We first reduce to the case where $A_\bullet$ is a Čech cover. To this end, consider the following commutative square:

$$\begin{array}{ccc} A & \longrightarrow & A_i \\ \downarrow & & \downarrow \\ A_0 & \longrightarrow & A_0 \otimes_A A_i. \end{array}$$

After base change along $A \rightarrow A_0$, the hypercover is split. Hence

$$D(A_0) \xrightarrow{\sim} \lim_{[i] \in \Delta}^! D(A_0 \otimes_A A_i),$$

and, more generally, $D(A_0^{\otimes_A k}) \xrightarrow{\sim} \lim_{[i] \in \Delta}^! D(A_0^{\otimes_A k} \otimes_A A_i)$ for all $k \geq 1$. So we are done if we can show that

$$D(A) \xrightarrow{\sim} \lim_{[n] \in \Delta}^! D(A_0^{\otimes_A (n+1)}) \quad \text{and} \quad D(A_i) \xrightarrow{\sim} \lim_{[n] \in \Delta}^! D(A_0^{\otimes_A (n+1)} \otimes_A A_i),$$

which is the special case of Čech descent. Applying these two Čech-descent equivalences levelwise and then interchanging the two resulting limits identifies descent for $A \rightarrow A_\bullet$ with descent for the Čech nerve of $A \rightarrow A_0$.

Step 2b: Assume now that $A_n \cong A_0^{\otimes_A (n+1)}$ is the Čech cover of $A \rightarrow A_0$. To show that the map $D(A) \rightarrow \lim_{[n] \in \Delta}^! D(A_n)$ is an isomorphism in Pr^R , we may pass to left adjoints and show that

$$\text{colim}_{[n] \in \Delta}^{\text{op}} D(A_n) \rightarrow D(A)$$

is an isomorphism in Pr^L . Here the colimit is taken along the functors $(f_n)_*: D(A_n) \rightarrow D(A)$. Recall that each of these functors lives in $\text{Pr}_{D(A)} = \text{Mod}_{D(A)}(\text{Pr})$, and hence the colimit may be computed in $\text{Mod}_{D(A)}(\text{Pr})$. Also note that if we apply the functor $D(A_0) \otimes_{D(A)} -: \text{Mod}_{D(A)}(\text{Pr}) \rightarrow \text{Mod}_{D(A_0)}(\text{Pr})$, the augmented simplicial diagram admits a splitting, since $D(A_0) \otimes_{D(A)} D(A_n) \cong D(A_{n+1})$. It follows that the map $\text{colim}_{[n] \in \Delta}^{\text{op}} D(A_n) \rightarrow D(A)$ becomes an isomorphism after applying $D(A_i) \otimes_{D(A)} -$ for each i , and thus by passing to colimits also after applying $(\text{colim}_i D(A_i)) \otimes_{D(A)} -$.

Set $L := \text{colim}_* D(A_\bullet)$ and let $u: L \rightarrow D(A)$ be the augmentation. By Step 1, the right adjoint $D(A) \rightarrow L$ is fully faithful, so u is a localization and admits a $D(A)$ -linear section. Thus $D(A)$ is a retract of L in $\text{Pr}_{D(A)}^L$. On the other hand, the splitting after base change to A_0 shows that u becomes an isomorphism after tensoring with every $D(A_i)$ and hence after tensoring with L . Therefore $u = u \otimes_{D(A)} D(A)$ is a retract of the isomorphism $u \otimes_{D(A)} L$, and is itself an isomorphism. Equivalently, the composite

$$D(A) \rightarrow \text{colim}_* D(A_\bullet) \rightarrow D(A)$$

is the identity in $\text{Pr}_{D(A)}$.

This finishes the proof in the countably presented case, for $n = 0$.

Lemma 3.11 ([Sch26, Lemma 4.11]). *Let $\mathcal{V} \in \text{CAlg}(1\text{Pr})$ and let I be a countable category. Suppose that $C: I \rightarrow \text{Mod}_{\mathcal{V}}(1\text{Pr})$ takes values in dualizable $\mathcal{V}$ -linear categories and that every transition functor is a left adjoint internal to $\text{Mod}_{\mathcal{V}}(1\text{Pr})$. Then $\text{colim}_I C$ is dualizable.*

3.3 The induction step for $n > 0$

We now assume that $n > 0$, and that we have already proved the claim for $n - 1$. Observe that we again have an adjunction

$$F_{n,\#}: \lim n\text{Pr}_{A_\bullet} \rightleftarrows n\text{Pr}_A: F_n^*.$$

Also, in this case each of the individual adjunctions $n\text{Pr}_A \rightleftarrows n\text{Pr}_{A_\bullet}$ is an ambidextrous adjunction.

We first show that F_n^* is fully faithful, or equivalently that the counit map

$$F_{n,\sharp} F_n^* \rightarrow \text{id}$$

is an isomorphism. Note that this adjunction is $n\text{Pr}_A$ -linear, so it suffices to show this claim on the monoidal unit: we need to show that the map $F_{n,\sharp} F_n^*(n-1)\text{Pr}_A \rightarrow (n-1)\text{Pr}_A$ is an isomorphism.

For $n = 1$, this map is $F_{1,\sharp} F_1^* D(A) \xrightarrow{\sim} D(A)$, which is the base case we just showed.

For $n \geq 2$, the object $F_{n,\sharp} F_n^*(\mathbb{1})$ is the countable colimit of the objects $(f_i)_{n,\sharp}(\mathbb{1})$ along internal left adjoints. The affine dualizability statement Proposition 2.19, together with Lemma 3.11, shows that it is dualizable. We may therefore dualize the counit

$$F_{n,\sharp} F_n^*(\mathbb{1}) \longrightarrow \mathbb{1}.$$

Under affine ambidexterity, its dual is precisely the comparison map one categorical level lower,

$$(n-1)\text{Pr}_A \longrightarrow \lim_{[i] \in \Delta} (n-1)\text{Pr}_{A_i},$$

which is an isomorphism by the induction hypothesis. Hence the counit is an isomorphism and F_n^* is fully faithful.

For essential surjectivity, the reduction of Step 2a again reduces us to the Čech nerve of $A \rightarrow A_0$. We must show that the unit $\text{id} \rightarrow F_n^* F_{n,\sharp}$ is an isomorphism. After base change along $A \rightarrow A_0$, the augmented Čech nerve is split, so the unit becomes an isomorphism. The pullback $n\text{Pr}_A \rightarrow n\text{Pr}_{A_0}$ is conservative: this follows from the full faithfulness just proved, applied to the same Čech nerve. Thus the unit was already an isomorphism over A .

This finishes the proof.

3.4 Reduction to the countably presented case

Finally, we show that the claim for $n = 0$ can be reduced to the countably presented case. We use the following lemma:

Lemma 3.12. *Every faithfully flat map $A \rightarrow B$ of commutative ring spectra is an $\aleph_1$ -filtered colimit of countably presented such maps.*

Proof. First, if A is static, then so is B . This case is due to Waterhouse;¹ let us quickly recall the argument. By Lazard's theorem, the flat A -modules are the Ind-category of the finite projective A -modules, so we can write the underlying module of B as an $\aleph_1$ -filtered colimit of countably presented flat modules M_i . Since each M_i is countably presented, it is $\aleph_1$ -compact as an A -module, so $\text{Hom}_A(M_i, -)$ commutes with $\aleph_1$ -filtered colimits. We will use two consequences of this: any map from a countably presented A -module into $B = \text{colim}_i M_i$ factors through some M_i , and any two such maps that agree after composing to B already agree at some stage M_i .

The tensor product $M_i \otimes_A M_i$ is again countably presented, so the composite $M_i \otimes_A M_i \rightarrow B \otimes_A B \rightarrow B$ of the canonical map with the multiplication of B factors through some M_j , $j = j(i)$, compatibly with the maps to B ; similarly the unit $A \rightarrow B$ factors through some M_{i_0} . Choosing inductively a

¹W. C. Waterhouse, *Basically bounded functors and flat sheaves*, Pacific J. Math. **57** (1975), no. 2, 597–610.

sequence $i_0, i_1, \dots$ with $i_{n+1} \geq j(i_n)$ yields compatible maps $M_{i_n} \otimes_A M_{i_n} \rightarrow M_{i_{n+1}}$, and the colimit $M := \operatorname{colim}(M_{i_0} \rightarrow M_{i_1} \rightarrow \dots)$ acquires a multiplication

$$M \otimes_A M = \operatorname{colim}_n (M_{i_n} \otimes_A M_{i_n}) \rightarrow \operatorname{colim}_n M_{i_{n+1}} = M$$

and a unit $A \rightarrow M_{i_0} \rightarrow M$, both compatible with $M \rightarrow B$.

The commutativity, associativity, and unitality of this multiplication each amount to an equality of two maps out of a countably presented module ($M_{i_n}^{\otimes 2}$, resp. $M_{i_n}^{\otimes 3}$, resp. M_{i_n}) that holds after composing into the commutative A -algebra B . By the second consequence above, each such identity holds after passing to a later stage; as there are only countably many of them, they can all be arranged to hold on a single cofinal subsystem. On such a subsystem M is a countably presented flat A -algebra equipped with a map to B , and these algebras form an $\aleph_1$ -filtered system with colimit B . This writes B in the desired form.

In general, write the map $A \rightarrow B$ as the $\aleph_1$ -filtered colimit of all countably presented A -algebras B_i mapping to B . We want to see that the set of i for which $A \rightarrow B_i$ is faithfully flat is cofinal. We get a corresponding presentation of $\pi_0 B$ as the colimit of the $\pi_0 B_i$, and by the static case, for any i there is some j so that $\pi_0 B_i \rightarrow \pi_0 B_j$ factors through a countably presented faithfully flat $\pi_0 A$ -algebra mapping to $\pi_0 B$. Taking a sequence $i_0, i_1, \dots$ which successively incorporates these factorizations, the colimit $B_j = \operatorname{colim}(B_{i_0} \rightarrow B_{i_1} \rightarrow \dots)$ is still countably presented and has the property that $\pi_0 A \rightarrow \pi_0 B_j$ is faithfully flat. For any fixed $n \in \mathbb{Z}$, the same approximation argument arranges that

$$\pi_0 B_j \otimes_{\pi_0 A} \pi_n A \longrightarrow \pi_n B_j$$

is an isomorphism. Since there are only countably many n , one further diagonal sequence arranges these isomorphisms simultaneously for all n . $\square$

As a consequence, every fpqc-hypercover is an $\aleph_1$ -filtered colimit of countably presented fpqc-hypercovers. One constructs the approximation inductively in the simplicial degree, using that each matching object is a finite limit and applying Lemma 3.12 to the matching map; a diagonal countable refinement makes the construction compatible in all degrees.

Now, given any hypercover $\operatorname{Spec}(A_\bullet) \rightarrow \operatorname{Spec}(A)$, we have

$$\lim_{\Delta}^! n\operatorname{Pr}_{A_\bullet} \simeq \lim_{\Delta}^! \operatorname{colim}_r n\operatorname{Pr}_{A_{r,\bullet}} \simeq \operatorname{colim}_r \lim_{\Delta}^! n\operatorname{Pr}_{A_{r,\bullet}} \simeq n\operatorname{Pr}_A,$$

where we use that $A \mapsto n\operatorname{Pr}_A$ preserves $\aleph_1$ -filtered colimits and that, in $1\operatorname{Pr}$, $\aleph_1$ -filtered colimits commute with countable limits. This finishes the proof of the general form of the theorem.

4 Quasicoherent sheaves of higher categories II

Talk given by Bastiaan.

We now pass from affine spectral schemes ($\mathrm{Aff}_{\mathbb{S}} := \mathrm{CAlg}(\mathrm{Sp})^{\mathrm{op}}$) to fpqc-stacks.

On affines, Talk 2 introduced the functors

$$\mathrm{Aff}_{\mathbb{S}}^{\mathrm{op}} \rightarrow \mathrm{CAlg}((n+1)\mathrm{Pr}) = \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/}, \quad \mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$$

for all $n \geq 0$, and showed they canonically extend to six-functor formalisms in which the countably presented maps are taken as the $!$ -able maps, with $f_! = f_*$. Denote by

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}}) \subseteq \mathrm{PSh}(\mathrm{Aff}_{\mathbb{S}})$$

the subcategory of *small* fpqc-sheaves. Here *small* means that the sheaf is a small colimit of representable objects. (In fact, we will mostly restrict to sheaves which are countable colimits of affines.) By the main result of Talk 3, we obtain a unique limit-preserving extension

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\mathrm{op}} \rightarrow \mathrm{CAlg}(1\mathrm{Pr})_{n\mathrm{Pr}/}, \quad X \mapsto n\mathrm{Pr}_X.$$

Since the endomorphism functor $\mathrm{End}: (n+1)\mathrm{Pr} \rightarrow n\mathrm{Pr}$ commutes with limits, this means that every fpqc-stack X defines a Stefanich ring

$$\mathcal{O}(X) := (\Omega^{\infty}\Gamma(X), \mathrm{D}(X), 1\mathrm{Pr}_X, 2\mathrm{Pr}_X, \dots),$$

with n -th component $\mathcal{O}(X)_n := (n-1)\mathrm{Pr}_X$ (using the low-degree convention $0\mathrm{Pr}_X := \mathrm{D}(X)$). Here $\Gamma(X) := \mathrm{End}_{\mathrm{D}(X)}(\mathbb{1}_{\mathrm{D}(X)}) \in \mathrm{CAlg}(\mathrm{Sp})$ denotes the spectrum of global functions on X ; more generally, for a map $f: Y \rightarrow X$ we write

$$\mathcal{O}(Y/X)_0 := f_* \mathbb{1}_{\mathrm{D}(Y)} \in \mathrm{D}(X)$$

for the level-zero relative structure object.¹ This extends the affine assignment $\mathrm{Spec}(A) \mapsto \mathcal{O}(\mathrm{Spec}(A))$ to a functor

$$\mathcal{O}: \mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\mathrm{op}} \rightarrow \mathrm{StRing}.$$

For $n = 0$, the value $0\mathrm{Pr}_X = \mathrm{D}(X)$ is the usual derived ∞ -category of quasicoherent sheaves on X , at least when X is a countable colimit of affines.

¹This notation is chosen to be compatible with the later notation $(\mathcal{O}(Y)/\mathcal{O}(X))_m$ for the relative objects of a map of Gestalten.

Remark 4.1. In general, there is a difference between taking the limit defining $n\mathrm{Pr}_X$ inside $1\mathrm{Pr} = \mathrm{Pr}_{\mathbb{N}_1}^L$ or inside the larger category Pr^L of all presentable categories. Usually, $\mathrm{QCoh}(X)$ is defined using limits in Pr^L . For countable colimits of affines the two limits agree.

The rest of the chapter has four goals:

- For $n \geq 1$, we extend the assignment $X \mapsto n\mathrm{Pr}_X$ to a six-functor formalism, with $!$ -able maps given by so-called ‘bi-countably presented maps’;
- We show that “every bi-countably presented map is étale”, and for $n \geq 2$ that in fact “every bi-countably presented map is proper”;
- We deduce a Künneth formula for sheaves of $(n-1)$ -presentable categories on a fiber product;
- We conclude that the Stefanich ring functor preserves countable colimits and finite limits.

4.1 Bi-countably presented maps

We will now extend the assignment $X \mapsto n\mathrm{Pr}_X$ to a six-functor formalism. We may think of this as a ‘higher’ version of the six-functor formalism on $X \mapsto \mathrm{D}(X)$. A first question is: what are the $!$ -able maps? It turns out that at the higher categorical level, the class of $!$ -able maps contains essentially *all* maps. The relevant class of maps is the following:

Definition 4.2. Let $A \in \mathrm{CAlg}(\mathrm{Sp})$. A map $Z \rightarrow \mathrm{Spec}(A)$ of fpqc-sheaves is *bi-countably presented over $\mathrm{Spec}(A)$* if Z is a countable colimit, in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})_{/\mathrm{Spec}(A)}$, of affines $\mathrm{Spec}(B_i)$ with $A \rightarrow B_i$ countably presented.

A map $f: Y \rightarrow X$ of fpqc-stacks is *bi-countably presented* if for every affine $\mathrm{Spec}(A) \rightarrow X$, the fiber product $Y \times_X \mathrm{Spec}(A)$ is bi-countably presented over $\mathrm{Spec}(A)$. Let E denote the class of bi-countably presented maps.

Lemma 4.3. Let $S = \mathrm{Spec}(A)$, and let $p: Z \rightarrow S$ be bi-countably presented. Then every section $s: S \rightarrow Z$ of p is bi-countably presented as a map to Z .

Proof. We first record two elementary fpqc-local reductions. Let $S' = \mathrm{Spec}(A') \rightarrow S = \mathrm{Spec}(A)$ be faithfully flat and countably presented. A map $W \rightarrow S$ is bi-countably presented if and only if its base change $W_{S'} \rightarrow S'$ is bi-countably presented. The forward implication is base change. Conversely, write $S'_\bullet$ for the Čech nerve of $S' \rightarrow S$. Since $A \rightarrow A'$ is countably presented, each $S'_m \rightarrow S$ is countably presented by Lemma 2.12. If $W_{S'}$ is a countable colimit of affines countably presented over S' , then each $W_{S'_m}$ is a countable colimit of affines countably presented over S . By fpqc-descent,

$$W \simeq |W_{S'_\bullet}|,$$

and the indexing category Δ^{op} is countable. Hence W is again a countable colimit of affines countably presented over S .

Second, let

$$Z \simeq \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i)$$

be a countable affine presentation over S , and let $T = \operatorname{Spec}(C) \rightarrow S$ be affine. Any finite collection of sections $T \rightarrow Z_T := Z \times_S T$ factors, after a faithfully flat countably presented base change $T' \rightarrow T$, through stages of the induced presentation

$$Z_T \simeq \operatorname{colim}_{i \in I} \operatorname{Spec}(B_i \otimes_A C).$$

Indeed, by the definition of sheafification, this factorization exists after some fpqc-cover of T . By Lemma 3.12, and because the stages are countably presented over T , this cover can be refined by a faithfully flat countably presented affine cover. For finitely many sections we take a common refinement.

We now prove the lemma. Let $T = \operatorname{Spec}(C) \rightarrow Z$ be an affine test object. We must show that

$$T \times_Z S \rightarrow T$$

is bi-countably presented. After base change along $T \rightarrow S$, the map $Z_T \rightarrow T$ has two sections: the section t induced by the given map $T \rightarrow Z$, and the section s_T obtained by pulling back s . By the fpqc-local reductions above, we may pass to a faithfully flat countably presented cover of T and assume that both sections factor through stages of a countable presentation of Z_T .

Let $\operatorname{Aff}_T^{\operatorname{cp}}$ be the full subcategory of affine T -schemes which are countably presented over T . It has finite limits by Lemma 2.12. Let $\tilde{Z}$ be the corresponding countable colimit of representables in $\operatorname{PSh}(\operatorname{Aff}_T^{\operatorname{cp}})$. The two sections $s_T, t: T \rightarrow Z_T$ now come from maps

$$* \rightrightarrows \tilde{Z}$$

in this presheaf category, where $*$ denotes the terminal representable. Consider the presheaf fiber product

$$P := * \times_{\tilde{Z}} *.$$

The objects $*$ and $\tilde{Z}$ are $\aleph_1$ -compact in $\operatorname{PSh}(\operatorname{Aff}_T^{\operatorname{cp}})$, and Proposition A.10 shows that P is again $\aleph_1$ -compact. Since presheaf categories are generated by representables, P is a countable colimit of representables from $\operatorname{Aff}_T^{\operatorname{cp}}$. Sheafification preserves finite limits, so the sheafification of P is precisely

$$T \times_{Z_T} T \simeq T \times_Z S.$$

Thus, fpqc-locally on T , the map $T \times_Z S \rightarrow T$ is bi-countably presented. The first reduction descends this conclusion to T itself. $\square$

Proposition 4.4. *The class E is closed under composition, base change, sections, and formation of diagonals.*

Proof. Closure under base change is immediate from the definition. For closure under composition, let $Z \rightarrow Y \rightarrow X$ be two maps in E and let $\operatorname{Spec}(A) \rightarrow X$ be affine. Write

$$Y_A := Y \times_X \operatorname{Spec}(A) \simeq \operatorname{colim}_{i \in I} \operatorname{Spec}(B_i),$$

where I is countable and each $A \rightarrow B_i$ is countably presented. By base change, $Z_A \rightarrow Y_A$ is bi-countably presented. Applying the definition of bi-countably presented to the affine test objects $\operatorname{Spec}(B_i) \rightarrow Y_A$, each

$$Z_A \times_{Y_A} \operatorname{Spec}(B_i)$$

is a countable colimit of affines $\mathrm{Spec}(C_{i,k})$ with $B_i \rightarrow C_{i,k}$ countably presented. Since countably presented maps are closed under composition, each $A \rightarrow C_{i,k}$ is countably presented. Universality of colimits in sheaves gives

$$Z_A \simeq \mathrm{colim}_{i \in I} (Z_A \times_{Y_A} \mathrm{Spec}(B_i)) \simeq \mathrm{colim}_{(i,k)} \mathrm{Spec}(C_{i,k}),$$

and the final indexing category is still countable. Hence $Z \rightarrow X$ lies in E .

It remains to discuss sections and diagonals. The diagonal $\Delta_f: Y \rightarrow Y \times_X Y$ is a section of the first projection

$$Y \times_X Y \rightarrow Y.$$

This projection is a base change of f , hence lies in E . Since E is already closed under base change, it is therefore enough to know that E is closed under sections.

Let $p: Y \rightarrow X$ lie in E , let $s: X \rightarrow Y$ be a section, and test s against an affine map $a: \mathrm{Spec}(A) \rightarrow Y$. Base change p along the composite $\mathrm{Spec}(A) \xrightarrow{a} Y \xrightarrow{p} X$. Since $p \in E$, the resulting map $Y_A := Y \times_X \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(A)$ is bi-countably presented over $\mathrm{Spec}(A)$. The map a gives one section of $Y_A \rightarrow \mathrm{Spec}(A)$, and $s \circ p \circ a$ gives another. Moreover

$$X \times_Y \mathrm{Spec}(A) \simeq \mathrm{Spec}(A) \times_{Y_A} \mathrm{Spec}(A)$$

with respect to these two sections. By Lemma 4.3, the first section $\mathrm{Spec}(A) \rightarrow Y_A$ is bi-countably presented as a map to Y_A . Pulling it back along the second section shows that

$$\mathrm{Spec}(A) \times_{Y_A} \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(A)$$

is bi-countably presented over $\mathrm{Spec}(A)$. Thus $s \in E$, so E is closed under sections and therefore under diagonals. $\square$

4.2 A six-functor formalism for $n\mathrm{Pr}$ at $n \geq 1$

We fix $n \geq 1$. Given a map $f: Y \rightarrow X$ of fpqc-stacks, we denote the resulting symmetric monoidal pullback functor by

$$f_n^*: n\mathrm{Pr}_X \rightarrow n\mathrm{Pr}_Y,$$

and its right adjoint by $f_{n,*}$. We denote by $\otimes_{n\mathrm{Pr}_X}$ the tensor product of $n\mathrm{Pr}_X$.

Remark 4.5. Throughout this talk, we will frequently abuse notation and regard $n\mathrm{Pr}_Y$ as an object of $(n+1)\mathrm{Pr}_X$ by setting

$$n\mathrm{Pr}_Y := f_{n+1,*}(\mathbb{1}_Y) \in (n+1)\mathrm{Pr}_X.$$

For $p_X: X \rightarrow \mathrm{pt}$, we get a ‘forgetful functor’ $p_{X,n+1,*}: (n+1)\mathrm{Pr}_X \rightarrow (n+1)\mathrm{Pr}$, and we claim that the image of $f_{n+1,*}(\mathbb{1}_Y)$ under this functor is $n\mathrm{Pr}_Y$. Indeed, if we write $Y \simeq \mathrm{colim}_{\mathrm{Spec}(A) \rightarrow Y} \mathrm{Spec}(A)$, then we may write $p_{X,n+1,*}f_{n+1,*}(\mathbb{1}) = p_{Y,n+1,*}(\mathbb{1})$ as a limit over $p_{A,n+1,*}(\mathbb{1}) = n\mathrm{Pr}_A$, which is $n\mathrm{Pr}_Y$ by definition.

Theorem 4.6. *Let $f: Y \rightarrow X$ be a bi-countably presented map of fpqc-stacks, and let $n \geq 1$.*

(i) The functor

$$f_n^*: n\mathrm{Pr}_X \rightarrow n\mathrm{Pr}_Y$$

admits a left adjoint $f_{n,\#}$ in $(n+1)\mathrm{Pr}_X$. In particular, $f_{n,\#}$ commutes with all small colimits and with tensoring.

(ii) The object $n\mathrm{Pr}_Y \in (n+1)\mathrm{Pr}_X$ is $\aleph_1$ -compact.

(iii) The formation of $f_{n,\#}$ commutes with base change: for every cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{v} & Y \\ f' \downarrow & & \downarrow f \\ X' & \xrightarrow{u} & X \end{array}$$

the Beck–Chevalley transformation

$$u_n^* f_{n,\#} \rightarrow f'_{n,\#} v_n^*$$

is an isomorphism.

(iv) For $n \geq 2$, the object $f_{n,\#}(\mathbb{1}) \in n\mathrm{Pr}_X$ is dualizable.

(v) For $n \geq 2$, the left adjoint $f_{n,\#}$ is also right adjoint to f_n^* , i.e. $f_{n,\#} \simeq f_{n,*}$. In particular, $f_{n,*}$ also commutes with small colimits, tensoring, and base change.

In other words: at level $n \geq 1$, “all maps are étale”, and at level $n \geq 2$, “all maps are proper”.

Proof. Step 1: The affine case. We first assume that $X = \mathrm{Spec}(A)$ is affine and prove (i)–(iv). By assumption $Y = \mathrm{colim}_{i \in I} \mathrm{Spec}(B_i)$ is a countable colimit of affines $Y_i = \mathrm{Spec}(B_i)$, where each B_i is countably presented over A . Consider

$$f_n^*: n\mathrm{Pr}_X = n\mathrm{Pr}_A \rightarrow n\mathrm{Pr}_Y = \lim_{i \in I^{\mathrm{op}}} n\mathrm{Pr}_{B_i}.$$

Since $n \geq 1$ and all transition maps are countably presented, the result from Talk 2 shows that all the pullback functors involved admit linear left adjoints. Here countable presentation of a transition map follows from Lemma 2.12(iii), since both its source and target are countably presented over A . In particular, the limit defining $n\mathrm{Pr}_Y$ is along right adjoints, and hence agrees with the colimit along the corresponding left adjoints:

$$n\mathrm{Pr}_Y \simeq \mathrm{colim}_{i \in I, \#} n\mathrm{Pr}_{B_i}.$$

(We use countability of I here so that this colimit takes place inside the $\aleph_1$ -presentable setting $1\mathrm{Pr}$.) In this colimit description, the desired left adjoint is given as the colimit of the lower- $\#$ maps $n\mathrm{Pr}_{B_i} \rightarrow n\mathrm{Pr}_A$:

$$f_{n,\#}: n\mathrm{Pr}_Y \simeq \mathrm{colim}_{i \in I, \#} n\mathrm{Pr}_{B_i} \rightarrow n\mathrm{Pr}_A.$$

(See Horev and Yanovski [HY17, Corollary 1.3].) This is a map in $(n+1)\mathrm{Pr}_A = \mathrm{Mod}_{n\mathrm{Pr}_A}(1\mathrm{Pr})$, and in particular it commutes with all small colimits and with tensoring. This gives (i). We may explicitly write the resulting functor as

$$f_{n,\#}: \lim_{i \in I^{\mathrm{op}}} n\mathrm{Pr}_{B_i} \rightarrow n\mathrm{Pr}_A, \quad (Z_i) \mapsto \mathrm{colim}_{i \in I} (g_i)_{n,\#}(Z_i),$$

with $g_i: \operatorname{Spec}(B_i) \rightarrow \operatorname{Spec}(A)$ the affine structure map.

For (ii), since $\mathfrak{N}_1$ -compact objects in $(n+1)\operatorname{Pr}_A$ are closed under countable colimits, the $\mathfrak{N}_1$ -compactness of $n\operatorname{Pr}_Y$ follows from the $\mathfrak{N}_1$ -compactness of the $n\operatorname{Pr}_{B_i}$ established in Talk 2.

For (iii), let $A \rightarrow A'$ be a base change, and write $B'_i := B_i \otimes_A A'$ and $Y' := Y \times_{\operatorname{Spec}(A)} \operatorname{Spec}(A')$. Since colimits in sheaves are universal,

$$Y' \simeq \operatorname{colim}_{i \in I} \operatorname{Spec}(B'_i).$$

Hence the same construction applied over A' gives

$$f'_{n,\#}: n\operatorname{Pr}_{Y'} \rightarrow n\operatorname{Pr}_{A'}, \quad (Z'_i) \mapsto \operatorname{colim}_{i \in I} (g'_i)_{n,\#}(Z'_i),$$

where $g'_i: \operatorname{Spec}(B'_i) \rightarrow \operatorname{Spec}(A')$ is the base change of $g_i: \operatorname{Spec}(B_i) \rightarrow \operatorname{Spec}(A)$. The base-change property is known from Talk 2:

$$u_n^*(g_i)_{n,\#} \simeq (g'_i)_{n,\#}(v_i)_n^*.$$

Thus base change carries the diagram whose colimit defines $f_{n,\#}$ to the diagram whose colimit defines $f'_{n,\#}$. Since the indexing category I is countable, this colimit is taken inside $1\operatorname{Pr}$, and we obtain the desired Beck–Chevalley isomorphism

$$u_n^* f_{n,\#} \simeq f'_{n,\#} v_n^*.$$

For (iv), we evaluate at the unit. In the affine case, we have $(g_i)_{n,\#} \simeq (g_i)_{n,*}$ by ambidexterity from Talk 2, so

$$f_{n,\#}(\mathbb{1}) \simeq \operatorname{colim}_{i \in I} (g_i)_{n,*}(\mathbb{1}) \simeq \operatorname{colim}_{i \in I,*} (n-1)\operatorname{Pr}_{B_i},$$

where the colimit is along the lower-* functors. For $n \geq 2$ (so that $n-1 \geq 1$), the lower-* functors $(n-1)\operatorname{Pr}_{B_i} \rightarrow (n-1)\operatorname{Pr}_{B_j}$ are themselves linear left adjoints. Hence the colimit above is a countable colimit of dualizable objects along internal left adjoint functors, and is dualizable by Lemma 3.11.

Step 2: The general case. For general X , we may write X as a colimit of affine spectral schemes, providing an equivalence

$$(n+1)\operatorname{Pr}_X \simeq \lim_{\operatorname{Spec}(A) \rightarrow X} (n+1)\operatorname{Pr}_A.$$

This is a limit in $1\operatorname{Pr} = \operatorname{Pr}_{\mathfrak{N}_1}^L \simeq \operatorname{Rex}_{\mathfrak{N}_1}$, so on the level of underlying categories we have

$$((n+1)\operatorname{Pr}_X)^{\mathfrak{N}_1} = \lim_{\operatorname{Spec}(A) \rightarrow X} ((n+1)\operatorname{Pr}_A)^{\mathfrak{N}_1}.$$

Letting $Y_A := \operatorname{Spec}(A) \times_X Y$, we have already established that $n\operatorname{Pr}_{Y_A}$ is $\mathfrak{N}_1$ -compact in $(n+1)\operatorname{Pr}_A$, thus giving the $\mathfrak{N}_1$ -compactness of $n\operatorname{Pr}_Y$ in $(n+1)\operatorname{Pr}_X$. Moreover, to check that $f_n^*: n\operatorname{Pr}_X \rightarrow n\operatorname{Pr}_Y$ admits a left adjoint in $((n+1)\operatorname{Pr}_X)^{\mathfrak{N}_1}$, it will suffice to show that all maps $(f_A)_n^*: n\operatorname{Pr}_A \rightarrow n\operatorname{Pr}_{Y_A}$ admit left adjoints compatible with base change, which we established in Step 1. Similarly, the dualizability of $f_{n,\#}(\mathbb{1}) \in n\operatorname{Pr}_X$ may be checked after pullback to affines.

Step 3: Ambidexterity. For (v), we use the general ambidexterity result (Proposition 2.17) again: it suffices to check that the unit and counit of the adjunction $f_{n,\#} \dashv f_n^*$ admit right adjoints. For the counit $f_{n,\#} f_n^* \rightarrow \operatorname{id}$, we use that under the equivalence $\operatorname{End}_{(n+1)\operatorname{Pr}_X}(n\operatorname{Pr}_X, n\operatorname{Pr}_X) \simeq n\operatorname{Pr}_X$ this map corresponds to the map $f_{n,\#}(\mathbb{1}) \rightarrow \mathbb{1}$, so we need to show this admits a right adjoint in $n\operatorname{Pr}_X$. By (iv), the object $f_{n,\#}(\mathbb{1})$ is dualizable, so this is equivalent to its dual $\mathbb{1} \rightarrow f_{n,*}(\mathbb{1})$ admitting a left

adjoint in $n\text{Pr}_X$. But this dual map is the map $(n-1)\text{Pr}_X \rightarrow (n-1)\text{Pr}_Y$, which has a left adjoint in $n\text{Pr}_X$ by part (i) applied one level down (which uses $n-1 \geq 1$).

For the unit $\eta: \text{id} \rightarrow f_n^* f_{n,\#}$ in $\text{End}_{(n+1)\text{Pr}_X}(n\text{Pr}_Y, n\text{Pr}_Y)$ we use a base-change argument. Consider the cartesian square

$$\begin{array}{ccc} Y \times_X Y & \xrightarrow{p_2} & Y \\ p_1 \downarrow & & \downarrow f \\ Y & \xrightarrow{f} & X \end{array}$$

together with the diagonal $\delta: Y \rightarrow Y \times_X Y$. Since the class E is closed under formation of diagonals, δ is bi-countably presented, so the conclusions of parts (i)–(iv) are available for δ at level n . By base change (part (iii)), there is a canonical isomorphism

$$f_n^* f_{n,\#} \simeq p_{1,n,\#} p_{2,n}^*.$$

The level- n counit $\varepsilon_\delta: \delta_{n,\#} \delta_n^* \rightarrow \text{id}_{n\text{Pr}_{Y \times_X Y}}$ of $\delta_{n,\#} \dashv \delta_n^*$ then induces a 2-morphism

$$p_{1,n,\#} \varepsilon_\delta p_{2,n}^*: \text{id}_{n\text{Pr}_Y} = p_{1,n,\#} \delta_{n,\#} \delta_n^* p_{2,n}^* \rightarrow p_{1,n,\#} p_{2,n}^* \simeq f_n^* f_{n,\#}.$$

This 2-morphism agrees with the unit η_f , which one verifies by checking that together with ε_f it satisfies the triangle identities (a consequence of base change). But since we already showed that the counit ε_δ is a left adjoint and whiskering with the functors $p_{1,n,\#}$ and $p_{2,n}^*$ preserves adjunctions, this finishes the proof. $\square$

4.3 The Künneth formula

As a consequence of the theorem, we obtain the following Künneth formula:

Corollary 4.7. *Let $f: Y \rightarrow X$ and $f': Y' \rightarrow X$ be bi-countably presented maps, and let $n \geq 2$. Then the objects $(n-1)\text{Pr}_Y$ and $(n-1)\text{Pr}_{Y'}$ in $n\text{Pr}_X$ are dualizable, and the Künneth formula*

$$(n-1)\text{Pr}_Y \otimes_{n\text{Pr}_X} (n-1)\text{Pr}_{Y'} \xrightarrow{\sim} (n-1)\text{Pr}_{Y \times_X Y'}$$

holds, where the equivalence is implemented via the external tensor product.

Proof. By definition, $(n-1)\text{Pr}_Y$ as an object of $n\text{Pr}_X$ is $f_{n,*}(\mathbb{1}_{n\text{Pr}_Y})$. Since $n \geq 2$, Theorem 4.6 gives ambidexterity $f_{n,*} \simeq f_{n,\#}$, and hence

$$(n-1)\text{Pr}_Y \simeq f_{n,\#}(\mathbb{1}_{n\text{Pr}_Y}).$$

The same theorem then implies that this object is dualizable in $n\text{Pr}_X$; similarly for Y' .

For the Künneth formula, write $Z := Y \times_X Y'$, with projections

$$\begin{array}{ccc} Z & \xrightarrow{q} & Y' \\ p \downarrow & & \downarrow f' \\ Y & \xrightarrow{f} & X. \end{array}$$

The maps p , q , and $f \circ p = f' \circ q$ are again bi-countably presented by Proposition 4.4, so the relevant $\sharp$ -functors are available. Using the projection formula for $f_{n,\sharp}$, base change for the cartesian square above, and then the projection formula for $p_{n,\sharp}$, we then compute

$$\begin{aligned}
(n-1)\mathrm{Pr}_Y \otimes_{n\mathrm{Pr}_X} (n-1)\mathrm{Pr}_{Y'} &\simeq f_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Y}) \otimes_{n\mathrm{Pr}_X} f'_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_{Y'}}) \\
&\simeq f_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Y} \otimes_{n\mathrm{Pr}_Y} f_n^* f'_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_{Y'}})) \\
&\simeq f_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Y} \otimes_{n\mathrm{Pr}_Y} p_{n,\sharp} q_n^*(\mathbb{1}_{n\mathrm{Pr}_{Y'}})) \\
&\simeq f_{n,\sharp} p_{n,\sharp} (p_n^*(\mathbb{1}_{n\mathrm{Pr}_Y}) \otimes_{n\mathrm{Pr}_Z} q_n^*(\mathbb{1}_{n\mathrm{Pr}_{Y'}})) \\
&\simeq (f \circ p)_{n,\sharp}(\mathbb{1}_{n\mathrm{Pr}_Z}) \\
&\simeq (n-1)\mathrm{Pr}_Z.
\end{aligned}$$

In the penultimate line we used that pullback preserves monoidal units. Since $Z = Y \times_X Y'$, this is the desired equivalence. $\square$

In particular, for $n \geq 2$, sheaves of presentable $(n-1)$ -categories on a fiber product are determined by what they are on the factors and what sheaves of presentable n -categories are on the base. This is the first genuinely surprising statement about quasicohherent sheaves of higher categories: *both* parts of Corollary 4.7 fail at lower categorical levels:

- At level $n = 0$, it would say that $\mathcal{O}(Y/X)_0 \in \mathrm{D}(X)$ is dualizable (i.e. a perfect complex), or that $\mathcal{O}(Y/X)_0 \otimes_{\mathrm{D}(X)} \mathcal{O}(Y'/X)_0 \simeq \mathcal{O}((Y \times_X Y')/X)_0$, both of which are absurd without strong assumptions on Y and Y' .
- At level $n = 1$, it is often the case that $\mathrm{D}(Y) \in 1\mathrm{Pr}_X$ is dualizable and the Künneth formula holds (cf. work by Gaitsgory), but one still needs some assumption such as “ Y is a geometric stack over X ”.
- For $n \geq 2$, these results are essentially *always* true.

However, we cannot stay on a fixed categorical level (say $1\mathrm{Pr}$): the Künneth formula always uses the next categorical level on the base X .

4.4 Dropping the bi-countable-presentation hypothesis

We will now see that for some of the previous assertions we may drop the condition on the map being bi-countably presented if instead we require sufficient compactness conditions on X and Y .

Proposition 4.8. *Let $f : Y \rightarrow X$ be a map between $\aleph_1$ -compact objects of $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})$ (so both X and Y are countable colimits of affine spectral schemes by Lemma 4.10 below). For $n \geq 2$, the right adjoint*

$$f_{n,*} : n\mathrm{Pr}_Y \rightarrow n\mathrm{Pr}_X$$

of f_n^ admits a further right adjoint in $\mathrm{Mod}_{n\mathrm{Pr}_X}(\mathrm{Pr}^{\mathrm{L}})$, which is compatible with base change. Moreover, for any $f' : Y' \rightarrow X$ satisfying the same hypothesis, the Künneth formula*

$$(n-1)\mathrm{Pr}_Y \otimes_{n\mathrm{Pr}_X} (n-1)\mathrm{Pr}_{Y'} \simeq (n-1)\mathrm{Pr}_{Y \times_X Y'}$$

holds.

Remark 4.9. If X is affine, the conclusion says that $f_{n,*}$ is a map in $\text{Mod}_{n\text{Pr}_A}(\text{Pr}^L)$, but *not* in general in $\text{Mod}_{n\text{Pr}_A}(\text{Pr}_{\aleph_1}^L) = \text{Mod}_{n\text{Pr}_A}(1\text{Pr})$: it fails only to preserve $\aleph_1$ -compact objects.

The proof of the proposition requires various auxiliary lemmas, which are given afterwards.

Proof. We first reduce to the case where X is affine. Since X is $\aleph_1$ -compact, by Lemma 4.10 it is a countable colimit of affines,

$$X \simeq \text{colim}_j \text{Spec}(A_j).$$

By Lemma 4.10, $\aleph_1$ -compact objects of $\text{Shv}_{\text{fpqc}}(\text{Aff}_{\mathbb{S}})$ are closed under finite limits; hence each base change $Y_j := Y \times_X \text{Spec}(A_j)$ is again $\aleph_1$ -compact. For such countable colimits, the descent formula

$$n\text{Pr}_X \simeq \lim_j n\text{Pr}_{A_j}$$

holds not only in $1\text{Pr} = \text{Pr}_{\aleph_1}^L$, but also in Pr^L , hence on the level of underlying categories. Thus the desired assertions may be checked after pullback to the affine pieces $\text{Spec}(A_j) \rightarrow X$. We may therefore assume that $X = \text{Spec}(A)$.

By Lemma 4.11 below, we may choose a diagram $(Y_i)_{i \in I}$ over X , together with a compatible cone $Y \rightarrow Y_i$, such that each $f_i: Y_i \rightarrow X$ is bi-countably presented and

$$n\text{Pr}_Y \simeq \text{colim}_{i \in I} n\text{Pr}_{Y_i},$$

where the transition maps are given by pullback. The pullback f_n^* is a morphism in 1Pr , hence preserves $\aleph_1$ -compact objects. For each i , the map f_i is bi-countably presented. Hence, by Theorem 4.6, for $n \geq 2$ the functor

$$f_{i,n,*}: n\text{Pr}_{Y_i} \rightarrow n\text{Pr}_X$$

preserves colimits, is $n\text{Pr}_X$ -linear, and satisfies base change. By Lemma 4.12 below, if $\alpha_i: n\text{Pr}_{Y_i} \rightarrow n\text{Pr}_Y$ denotes the colimit structure functor and $p_{ij}: Y_j \rightarrow Y_i$ denotes the transition map associated to a morphism $i \rightarrow j$ in I , then

$$f_{n,*}(\alpha_i E) \simeq \text{colim}_{j \in I_{|i|}} f_{j,n,*}(p_{ij,n}^* E).$$

Thus $f_{n,*}$ is assembled from the functors $f_{j,n,*}$ by taking $\aleph_1$ -filtered colimits. Since filtered colimits of functors preserve colimits, $n\text{Pr}_X$ -linearity, and base change whenever the functors in the system do, it follows that $f_{n,*}$ also commutes with small colimits, tensors by $n\text{Pr}_X$, and base change.

With these properties established, the Künneth formula is proved formally just like in Corollary 4.7. $\square$

Lemma 4.10. *The inclusion*

$$\text{Shv}_{\text{fpqc}}(\text{Aff}_{\mathbb{S}}) \hookrightarrow \text{PSh}(\text{Aff}_{\mathbb{S}})$$

preserves $\aleph_1$ -filtered colimits. The same holds for the corresponding fpqc-sheaf categories on affines over any fixed affine base.

Moreover, these sheaf categories are $\aleph_1$ -compactly generated by affines, their $\aleph_1$ -compact objects are closed under finite limits, and sheafification sends $\aleph_1$ -compact presheaves to $\aleph_1$ -compact sheaves. In particular, for every affine $X = \text{Spec}(A)$, every affine object $\text{Spec}(B) \rightarrow X$ is $\aleph_1$ -compact in $\text{Shv}_{\text{fpqc}}(\text{Aff}_{\mathbb{S}})_{/X}$, and every $\aleph_1$ -compact object in this slice is a countable colimit of affine objects.

Proof. By definition of the fpqc topology on affines, covering sieves are tested on finite covering families. Since finite sets are $\aleph_1$ -small, Lemma A.12 shows that fpqc-sheaves are closed under $\aleph_1$ -filtered colimits in presheaves. The same argument applies on the fpqc site of affines over a fixed affine base.

Now apply Corollary A.13 with $\kappa = \aleph_1$, both on $\text{Aff}_{\mathbb{S}}$ and on the corresponding site over an affine base. This gives $\aleph_1$ -compact generation, closure of $\aleph_1$ -compact objects under finite limits, and preservation of $\aleph_1$ -compact objects under sheafification. Since representable presheaves are compact, affine sheaves are $\aleph_1$ -compact.

The sheaf category over an affine X is generated under colimits by affine objects over X . Hence its $\aleph_1$ -compact objects are retracts of countable colimits of affine objects; absorbing the retract by the usual countable telescope gives the stated countable affine presentation. $\square$

Lemma 4.11. *Let $X = \text{Spec}(A)$ be affine. Then every $\aleph_1$ -compact object $Y \in \text{Shv}_{\text{fpqc}}(\text{Aff}_{\mathbb{S}})_{/X}$ admits a compatible cone $Y \rightarrow Y_i$ over an $\aleph_1$ -filtered category I , where each $Y_i \rightarrow X$ is bi-countably presented, such that, for every $n \geq 1$, the induced functor*

$$\text{colim}_{i \in I} n\text{Pr}_{Y_i} \rightarrow n\text{Pr}_Y$$

along pullback is an equivalence in $1\text{Pr} = \text{Pr}_{\aleph_1}^{\text{L}}$.

Proof. By Lemma 4.10, the affine objects over X are $\aleph_1$ -compact. Since affine objects generate $\text{Shv}_{\text{fpqc}}(\text{Aff}_{\mathbb{S}})_{/X}$ under colimits, this sheaf category is $\aleph_1$ -compactly generated by affines. Thus an $\aleph_1$ -compact object over X admits a presentation by countably many affines: we have

$$Y \simeq \text{colim}_{m \in M} \text{Spec}(B_m)$$

for some countable diagram $B: M^{\text{op}} \rightarrow \text{CAlg}_A(\text{Sp})$. Since $\text{CAlg}_A(\text{Sp})$ is $\aleph_1$ -compactly generated, Lemma 1.5 shows that also $\text{Fun}(M^{\text{op}}, \text{CAlg}_A(\text{Sp}))$ is $\aleph_1$ -compactly generated, with $\aleph_1$ -compact objects given by the functors $M^{\text{op}} \rightarrow \text{CAlg}_A(\text{Sp})^{\aleph_1}$, i.e. the diagrams of countably presented A -algebras. Applying this to the diagram B , we obtain an $\aleph_1$ -filtered category I and diagrams $B_i: M^{\text{op}} \rightarrow \text{CAlg}_A(\text{Sp})$ such that each $B_i(m)$ is countably presented over A , and we have

$$B \simeq \text{colim}_{i \in I} B_i$$

in $\text{Fun}(M^{\text{op}}, \text{CAlg}_A(\text{Sp}))$. Define

$$Y_i := \text{colim}_{m \in M} \text{Spec}(B_i(m)).$$

Since M is countable and each $A \rightarrow B_i(m)$ is countably presented, each $Y_i \rightarrow X$ is bi-countably presented. The maps $B_i \rightarrow B$ induce a compatible cone $Y \rightarrow Y_i$ over X .

It remains to check the asserted formula after applying $n\text{Pr}_{(-)}$. First consider the affine case: if $B \simeq \text{colim}_i B_i$ in $\text{CAlg}_A(\text{Sp})$, then

$$n\text{Pr}_B \simeq \text{colim}_i n\text{Pr}_{B_i}.$$

The base case $n = 0$ follows from Theorem 1.24 applied to $C = \text{Sp}$: the functor $\text{CAlg}_A(\text{Sp}) \rightarrow \text{CAlg}(1\text{Pr})_{0\text{Pr}_A/}$, $B \mapsto 0\text{Pr}_B$, is a left adjoint and hence preserves colimits. For higher n the same argument applies inductively, applying the theorem with $C = n\text{Pr}$ and base algebra $(n-1)\text{Pr}_A$.

Now apply descent to the countable affine presentations of Y and the Y_i . Since $B \simeq \operatorname{colim}_i B_i$ in the functor category, we have $B_m \simeq \operatorname{colim}_i B_i(m)$ for every $m \in M$. Hence

$$\begin{aligned} n\operatorname{Pr}_Y &\simeq \lim_{m \in M^{\operatorname{op}}} n\operatorname{Pr}_{B_m} \\ &\simeq \lim_{m \in M^{\operatorname{op}}} \operatorname{colim}_i n\operatorname{Pr}_{B_i(m)}. \end{aligned}$$

Since M is countable and $1\operatorname{Pr}$ is $\aleph_1$ -compactly generated, $\aleph_1$ -filtered colimits commute with this countable limit in $1\operatorname{Pr}$. Thus

$$\begin{aligned} n\operatorname{Pr}_Y &\simeq \operatorname{colim}_i \lim_{m \in M^{\operatorname{op}}} n\operatorname{Pr}_{B_i(m)} \\ &\simeq \operatorname{colim}_i n\operatorname{Pr}_{Y_i}. \end{aligned} \quad \square$$

Lemma 4.12. *Let $X = \operatorname{Spec}(A)$, and let $(Y_i)_{i \in I}$ be a diagram over X , indexed by an $\aleph_1$ -filtered category I , equipped with a compatible cone $Y \rightarrow Y_i$ over X . Assume that*

$$n\operatorname{Pr}_Y \simeq \operatorname{colim}_{i \in I} n\operatorname{Pr}_{Y_i}$$

in $1\operatorname{Pr}$ along pullback. In particular, the structure functors $\alpha_i: n\operatorname{Pr}_{Y_i} \rightarrow n\operatorname{Pr}_Y$ preserve $\aleph_1$ -compact objects, and compact generators of $n\operatorname{Pr}_Y$ are obtained from compact generators of the stages under countable colimits. Write $f_i: Y_i \rightarrow X$ and $f: Y \rightarrow X$ for the structure maps. For a morphism $i \rightarrow j$ in I , write $p_{ij}: Y_j \rightarrow Y_i$ for the induced transition map. Assume that f_n^ preserves $\aleph_1$ -compact objects and that each right adjoint*

$$f_{i,n,*}: n\operatorname{Pr}_{Y_i} \rightarrow n\operatorname{Pr}_X$$

preserves small colimits. Then $f_{n,}$ preserves small colimits, and, if $\alpha_i: n\operatorname{Pr}_{Y_i} \rightarrow n\operatorname{Pr}_Y$ denotes the colimit structure functor, then*

$$f_{n,*}(\alpha_i E) \simeq \operatorname{colim}_{j \in I_{i/}} f_{j,n,*}(p_{ij,n}^* E)$$

for every $E \in n\operatorname{Pr}_{Y_i}$.

Proof. For fixed i , define

$$G_i(E) := \operatorname{colim}_{j \in I_{i/}} f_{j,n,*}(p_{ij,n}^* E), \quad E \in n\operatorname{Pr}_{Y_i}.$$

Since each $f_{j,n,*}$ preserves colimits, so does G_i . If $i \rightarrow k$ is a morphism in I , then the inclusion $I_{k/} \rightarrow I_{i/}$ is cofinal, and hence

$$G_k(p_{ik,n}^* E) \simeq G_i(E).$$

Thus the restrictions of the G_i to compact objects are compatible with the $1\operatorname{Pr}$ -colimit presentation of $n\operatorname{Pr}_Y$. Since compact generators of $n\operatorname{Pr}_Y$ are generated under countable colimits by the objects $\alpha_i E$ with $E \in (n\operatorname{Pr}_{Y_i})^{\aleph_1}$, these compatible restrictions determine a functor on $(n\operatorname{Pr}_Y)^{\aleph_1}$ preserving countable colimits. Extending it cocontinuously gives a colimit-preserving functor

$$G: n\operatorname{Pr}_Y \rightarrow n\operatorname{Pr}_X$$

with $G(\alpha_i E) \simeq G_i(E)$ for compact E . Since both sides of this formula preserve colimits in E , the same formula holds for all $E \in n\operatorname{Pr}_{Y_i}$.

We claim that G is right adjoint to f_n^* . It suffices to test this on $\aleph_1$ -compact generators $K \in (n\mathrm{Pr}_X)^{\aleph_1}$ and on $\aleph_1$ -compact generators of $n\mathrm{Pr}_Y$ represented by some stage $n\mathrm{Pr}_{Y_i}$. Here we use the $1\mathrm{Pr}$ -colimit presentation and the assumption that f_n^* preserves compact objects. Using compactness of K and the standard mapping-space formula for filtered colimits of categories, we compute

$$\begin{aligned} \mathrm{Hom}_{n\mathrm{Pr}_X}(K, G(\alpha_i E)) &\simeq \mathrm{colim}_{j \in I_i} \mathrm{Hom}_{n\mathrm{Pr}_X}(K, f_{j,n,*}(p_{ij,n}^* E)) \\ &\simeq \mathrm{colim}_{j \in I_i} \mathrm{Hom}_{n\mathrm{Pr}_{Y_j}}(f_{j,n}^* K, p_{ij,n}^* E) \\ &\simeq \mathrm{colim}_{j \in I_i} \mathrm{Hom}_{n\mathrm{Pr}_{Y_j}}(p_{ij,n}^* f_{i,n}^* K, p_{ij,n}^* E) \\ &\simeq \mathrm{Hom}_{n\mathrm{Pr}_Y}(f_n^* K, \alpha_i E). \end{aligned}$$

For fixed compact K , both sides preserve $\aleph_1$ -filtered colimits in E : on the left this uses that G preserves colimits and that K is $\aleph_1$ -compact, while on the right it uses that $f_n^* K$ is $\aleph_1$ -compact. Thus the equivalence extends from the compact generators represented by the stages to every $E \in n\mathrm{Pr}_Y$. It then extends from compact K to arbitrary K by writing K as a colimit of compact objects. Hence G is right adjoint to f_n^* , so $G \simeq f_{n,*}$. $\square$

4.5 The structure sheaf functor

We can summarize what we have shown in the language of Stefanich rings:

Corollary 4.13. *The functor*

$$\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})^{\aleph_1} \rightarrow (\mathrm{StRing}_{\mathbb{S}/})^{\mathrm{op}}, \quad X \mapsto \mathcal{O}(X)$$

commutes with countable colimits and finite limits.

If the target were itself an ∞ -topos, then this functor would have all the properties of a morphism of ∞ -topoi (except that we restrict to countable colimits rather than to all small colimits). In Chapter 7, we will make this idea precise using the concept of a *countably toposic category*.

Sketch of proof. Implicit in the statement is the claim that $\aleph_1$ -compact objects are closed under finite limits in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_{\mathbb{S}})$; see Corollary A.13 for a proof of this fact.

Commutation with countable colimits holds essentially by definition. To prove commutation with finite limits, it suffices to handle the terminal object (which works by definition of $\mathrm{StRing}_{\mathbb{S}/}$) and pullbacks. So consider a pullback diagram

$$\begin{array}{ccc} Y \times_X Y' & \longrightarrow & Y' \\ \downarrow & & \downarrow f' \\ Y & \xrightarrow{f} & X \end{array}$$

of $\aleph_1$ -compact fpqc-stacks. A priori, to compute the pushout of Stefanich rings, we would start by forming the infinite tuple

$$(\Gamma(Y) \otimes_{\Gamma(X)} \Gamma(Y'), \quad \mathrm{D}(Y) \otimes_{\mathrm{D}(X)} \mathrm{D}(Y'), \quad 1\mathrm{Pr}_Y \otimes_{1\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \quad \dots),$$

consisting of all the levelwise pushouts. These come with natural maps from each entry to endomorphisms of the unit of the next, but these are typically *not* isomorphisms; one needs to “spectrify”, i.e. apply the left adjoint to the inclusion of Stefanich rings into the larger category that allows for non-invertible comparison maps.

In the next talk, we will explain a convenient way of computing such tensor products. What we will see there is that we can make a ‘first approximation’ to the spectrification, given by:

$$(O(Y/X)_0 \otimes_{D(X)} O(Y'/X)_0, \quad D(Y) \otimes_{1\mathrm{Pr}_X} D(Y'), \quad 1\mathrm{Pr}_Y \otimes_{2\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}, \quad \dots).$$

Here on each level the tensor product is taken at the next-higher categorical level. (For example, in the zeroth slot we form the tensor product not over $\Gamma(X)$, i.e. in $\Gamma(X)$ -modules, but rather inside $D(X)$; this is finer, since there is a canonical functor $\mathrm{Mod}_{\Gamma(X)}(D(\mathbb{S})) \rightarrow D(X)$.) By Proposition 4.8, starting from the slot $1\mathrm{Pr}_Y \otimes_{2\mathrm{Pr}_X} 1\mathrm{Pr}_{Y'}$, this expression agrees with

$$(\Omega^\infty \Gamma(Y \times_X Y'), D(Y \times_X Y'), 1\mathrm{Pr}_{Y \times_X Y'}, \dots),$$

which is already a Stefanich ring, namely the one corresponding to $Y \times_X Y'$. As the first few terms do not affect the Stefanich ring structure, this gives the desired result. $\square$

Remark 4.14. A general slogan in this area is that “everything becomes affine under sufficient categorification”. The Künneth formulas above, which hold from level $1\mathrm{Pr}$ onwards, justify this slogan in some form. The slogan is a bit imprecise, however: stacks like $B^n \mathbb{G}_m$ are n -affine (in a sense to be made precise in the next talk) but not k -affine for $k < n$, so in particular $\bigsqcup_k B^k \mathbb{G}_m$ is not n -affine for any n . We will later see that even the infinite-dimensional affine space $\mathbb{A}^\infty = \mathrm{colim}_n \mathbb{A}^n$ is not n -affine for any n , although all its terms are 0-affine and the transition maps are cohomologically smooth (since complete intersections are cohomologically smooth in the quasicoherent six-functor formalism).

5 Stefanich rings

Talk given by Yiming.

Recall that we defined the $\mathfrak{N}_1$ -presentable category StRing of *Stefanich rings* as

$$\begin{aligned} \text{StRing} &:= \text{colim}_{1\text{Pr}}(\text{CAlg}(\text{An}) \xrightarrow{\text{Mod}_1^{\mathbb{P}}} \text{CAlg}(1\text{Pr}) \xrightarrow{\text{Mod}_2^{\mathbb{P}}} \text{CAlg}(2\text{Pr}) \xrightarrow{\text{Mod}_3^{\mathbb{P}}} \dots) \\ &\simeq \text{lim}_{\text{CAT}}(\text{CAlg}(\text{An}) \xleftarrow{\text{End}_1^{\mathbb{P}}} \text{CAlg}(1\text{Pr}) \xleftarrow{\text{End}_2^{\mathbb{P}}} \text{CAlg}(2\text{Pr}) \xleftarrow{\text{End}_3^{\mathbb{P}}} \dots), \end{aligned}$$

where we write $\text{Mod}_n^{\mathbb{P}}(R) := \text{Mod}_R((n-1)\text{Pr})$ and $\text{End}_n^{\mathbb{P}}(D) := \text{End}_D(\mathbb{1}_D)$.

We denote the objects of this category as $A = (A_0, A_1, A_2, \dots)$, where $A_n \in \text{CAlg}(n\text{Pr})$ and the structure isomorphisms $A_n \simeq \text{End}_{n+1}^{\mathbb{P}}(A_{n+1})$ are left implicit in the notation. Similarly, morphisms from A to B are collections of maps $f_{n-1}^*: A_n \rightarrow B_n$ compatible with the structural isomorphisms.

Remark 5.1. Given an fpqc-stack X , the associated Stefanich ring has $A_n = (n-1)\text{Pr}_X$. The indexing convention for $f_{n-1}^*: A_n \rightarrow B_n$ reflects the convention from the previous talk.

We may equip StRing with the cocartesian monoidal structure; we will have a look at the resulting “tensor product” of Stefanich rings in more detail below. Since the functors $\text{End}_n^{\mathbb{P}}$ preserve the monoidal unit, we see that the monoidal unit in StRing is the initial Stefanich ring $\mathbb{P} := (*, \text{An}, 1\text{Pr}, 2\text{Pr}, \dots)$, with n -th component $\mathbb{P}_n := (n-1)\text{Pr}$.

Since $\mathbb{P}$ is initial, every Stefanich ring A receives a unique map $a: \mathbb{P} \rightarrow A$. The individual maps $a_{n-1}^*: (n-1)\text{Pr} \rightarrow A_n$ in $\text{CAlg}(\text{Pr})$ are obtained using the identification $\text{CAlg}(n\text{Pr}) \simeq \text{CAlg}(\text{Pr})_{(n-1)\text{Pr}/}$. By passing to right adjoints, we obtain ‘forgetful functors’

$$a_{n-1,*}: A_n \rightarrow (n-1)\text{Pr}, \quad B_{n-1} \mapsto \underline{\text{Hom}}_{A_n}(\mathbb{1}_{A_n}, B_{n-1}).$$

Given $B_{n-1} \in A_n$, we refer to $a_{n-1,*}(B_{n-1})$ as its *underlying* $(n-1)$ -category.

Notation 5.2. We will denote the monoidal unit of A_n by

$$(A/A)_{n-1} := \mathbb{1}_{A_n} \in \text{CAlg}(A_n).$$

Observe that we have $a_{n-1,*}((A/A)_{n-1}) = \text{End}_{A_n}(\mathbb{1}_{A_n}) = A_{n-1}$.

5.1 Internal Stefanich rings

We will now provide a limit description of the slice category $\text{StRing}_{A/}$ under any Stefanich ring A , exhibiting A -algebras in Stefanich rings as some form of internal Stefanich rings in A .

Construction 5.3. Let $f: A \rightarrow B$ be a morphism of Stefanich rings. For any $n \geq 1$, the symmetric monoidal functor $f_{n-1}^*: A_n \rightarrow B_n$ admits a lax symmetric monoidal right adjoint $f_{n-1,*}: B_n \rightarrow A_n$, which therefore induces a functor of the form

$$f_{n-1,*}: \text{CAlg}(B_n) \rightarrow \text{CAlg}(A_n).$$

Observe that the composite $\text{CAlg}(B_n) \xrightarrow{f_{n-1,*}} \text{CAlg}(A_n) \xrightarrow{\text{fgt}} \text{CAlg}((n-1)\text{Pr})$ is the forgetful functor $\text{CAlg}(B_n) \rightarrow \text{CAlg}((n-1)\text{Pr})$.

We then define

$$(B/A)_{n-1} := f_{n-1,*}(\mathbb{1}_{B_n}) \in \text{CAlg}(A_n).$$

Note that the underlying $(n-1)$ -presentable category of $(B/A)_{n-1}$ is B_{n-1} . So we may think of $(B/A)_{n-1}$ as a lift of B_{n-1} to a commutative algebra in A_n . For $f = \text{id}_A$ this recovers Notation 5.2.

Construction 5.4. We now construct an internal form of the module-endomorphism adjunction: For a Stefanich ring A and $m \geq 1$, we will produce an adjunction

$$\text{CAlg}(A_m) \begin{matrix} \xrightarrow{\text{Mod}_m^A} \\ \xleftarrow{\text{End}_m^A} \end{matrix} \text{CAlg}(A_{m+1}).$$

Recall from Notation 1.25 that, more generally, for $n \geq 1$ and $C \in \text{CAlg}(n\text{Pr})$ we have an adjunction

$$\text{CAlg}(C) \begin{matrix} \xrightarrow{\text{Mod}_{(-)}(C)} \\ \xleftarrow{\text{End}_{(-)}(\mathbb{1})} \end{matrix} \text{CAlg}(\text{Mod}_C(n\text{Pr})),$$

where the left adjoint sends R to $\text{Mod}_R(C)$ and the right adjoint sends D to $\text{End}_D(\mathbb{1}_D)$. Here we are using that C is an algebra over $(n-1)\text{Pr}$ in 1Pr , so that

$$\text{Mod}_C(1\text{Pr}) \simeq \text{Mod}_C(\text{Mod}_{(n-1)\text{Pr}}(1\text{Pr})) \simeq \text{Mod}_C(n\text{Pr}),$$

giving the $(n-1)\text{Pr}$ -linear structure for free.

Now let A be a Stefanich ring. Applying this adjunction with $C = m\text{Pr}$, we see that the structural equivalence

$$A_m \simeq \text{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}})$$

in $\text{CAlg}(m\text{Pr})$ corresponds to a symmetric monoidal functor

$$L_m^A: \text{Mod}_{A_m}(m\text{Pr}) \rightarrow A_{m+1}$$

in $\text{CAlg}((m+1)\text{Pr})$. It admits a right adjoint, which we denote by

$$U_m^A: A_{m+1} \rightarrow \text{Mod}_{A_m}(m\text{Pr}), \quad X \mapsto \underline{\text{Hom}}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}, X).$$

We use the same notation for the induced functors on commutative algebra objects. Finally, we write

$$\text{Mod}_m^A := L_m^A \circ \text{Mod}_{(-)}(A_m): \text{CAlg}(A_m) \rightarrow \text{CAlg}(A_{m+1})$$

for the internal module functor, and

$$\text{End}_m^A := \text{End}_{(-)}(\mathbb{1}) \circ U_m^A: \text{CAlg}(A_{m+1}) \rightarrow \text{CAlg}(A_m)$$

for its right adjoint. Explicitly, we have

$$\text{Mod}_m^A(R) = L_m^A(\text{Mod}_R(A_m)) \quad \text{and} \quad \text{End}_m^A(D) = \text{End}_{U_m^A(D)}(\mathbb{1}).$$

Observation 5.5. The functor $U_m^A: \text{CAlg}(A_{m+1}) \rightarrow \text{Mod}_{A_m}(m\text{Pr})$ preserves the monoidal unit: the unit of the adjunction $L_m^A \dashv U_m^A$ takes the form

$$A_m \rightarrow U_m^A(\mathbb{1}_{A_{m+1}}) = \text{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}),$$

which is the structure isomorphism of the Stefanich ring A .

Proposition 5.6. *Let $A \in \text{StRing}$ and $m \geq 1$. The internal module functor*

$$\text{Mod}_m^A: \text{CAlg}(A_m) \rightarrow \text{CAlg}(A_{m+1})$$

is fully faithful.

Proof. Since Mod_m^A is a morphism in 1Pr , by Lemma 1.10 it suffices to check that the unit of the adjunction $\text{Mod}_m^A \dashv \text{End}_m^A$ is an equivalence on $\aleph_1$ -compact objects, i.e. on countably presented objects of $\text{CAlg}(A_m)$. Let $R \in \text{CAlg}(A_m)$ be countably presented. Unwinding definitions, we get

$$\text{End}_m^A(\text{Mod}_m^A(R)) \simeq \text{End}_{U_m^A L_m^A(\text{Mod}_R(A_m))}(\mathbb{1}) = \text{End}_{T(\text{Mod}_R(A_m))}(\mathbb{1}),$$

where we use the abbreviated notation

$$T := U_m^A L_m^A: \text{Mod}_{A_m}(m\text{Pr}) \rightarrow \text{Mod}_{A_m}(m\text{Pr}).$$

To analyze this endomorphism category, note:

- The functors U_m^A and L_m^A are both maps in $(m+1)\text{Pr}$, hence so is T . Since $m \geq 1$, this means that T is (at least) a 2-functor, so it preserves adjunctions.
- By Observation 5.5, the unit map $A_m \rightarrow T(A_m)$ is an isomorphism.

We now consider the free R -module functor

$$F_R: A_m \rightarrow \text{Mod}_R(A_m), \quad X \mapsto R \otimes X.$$

It admits an A_m -linear right adjoint $G_R: \text{Mod}_R(A_m) \rightarrow A_m$, given by forgetting the R -module structure. Since R is countably presented, the functor G_R preserves $\aleph_1$ -compact objects, and so this adjunction lives inside $\text{Mod}_{A_m}(m\text{Pr})$. So applying the 2-functor T to the adjunction $F_R \dashv G_R$, we obtain an adjunction

$$T(F_R): T(A_m) \rightleftarrows T(\text{Mod}_R(A_m)) : T(G_R)$$

in $\text{Mod}_{A_m}(m\text{Pr})$. Since T is lax monoidal, it induces a functor on commutative algebras, and hence sends the symmetric monoidal functor F_R to a symmetric monoidal functor $T(F_R)$; consequently the unit object of $T(\text{Mod}_R(A_m))$ is $T(F_R)(\mathbb{1}_{T(A_m)})$. Combining all of this, we may now compute:

$$\begin{aligned} \text{End}_m^A(\text{Mod}_m^A(R)) &= \text{End}_{T(\text{Mod}_R(A_m))}(\mathbb{1}) \\ &\simeq \text{End}_{T(\text{Mod}_R(A_m))}(T(F_R)(\mathbb{1}_{T(A_m)})) \\ &\simeq \underline{\text{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(G_R)T(F_R)(\mathbb{1}_{T(A_m)})) \\ &\simeq \underline{\text{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(G_R \circ F_R)(\mathbb{1}_{T(A_m)})) \\ &\simeq \underline{\text{Hom}}_{T(A_m)}(\mathbb{1}_{T(A_m)}, T(R \otimes -)(\mathbb{1}_{T(A_m)})) \\ &\simeq \underline{\text{Hom}}_{A_m}(\mathbb{1}_{A_m}, (R \otimes -)(\mathbb{1}_{A_m})) \\ &\simeq R, \end{aligned}$$

where in the second-to-last equivalence we use that under the equivalence $A_m \xrightarrow{\sim} T(A_m)$, the endofunctor $T(R \otimes -)$ corresponds to $R \otimes -: A_m \rightarrow A_m$. Under this identification, the unit map $R \rightarrow \text{End}_m^A(\text{Mod}_m^A(R))$ is the map induced by the action of R on the free rank-one R -module $F_R(\mathbb{1}_{A_m})$. Under the adjunction $T(F_R) \dashv T(G_R)$, this action corresponds to the identity of $G_R F_R(\mathbb{1}_{A_m}) \simeq R$. Thus the unit is an equivalence on countably presented R , and Mod_m^A is fully faithful. $\square$

Definition 5.7. Let $A \in \text{StRing}$. The category of *internal Stefanich rings in A* is the colimit

$$\text{StRing}(A) := \text{colim}_{\text{IPr}}(\text{CAlg}(A_1) \xrightarrow{\text{Mod}_1^A} \text{CAlg}(A_2) \xrightarrow{\text{Mod}_2^A} \dots)$$

formed along the internal module functors. Equivalently, since each Mod_m^A is a left adjoint with right adjoint End_m^A , it may be computed as the limit

$$\text{StRing}(A) \simeq \lim_{\text{CAT}}(\text{CAlg}(A_1) \xleftarrow{\text{End}_1^A} \text{CAlg}(A_2) \xleftarrow{\text{End}_2^A} \dots).$$

We now show that A -algebras in Stefanich rings are the same as Stefanich rings internal to A .

Proposition 5.8. *Let $A \in \text{StRing}$. The assignments $B \mapsto (B/A)_n$, for $n \geq 0$, determine an equivalence*

$$\text{StRing}_{A/} \xrightarrow{\sim} \text{StRing}(A), \quad B \mapsto ((B/A)_0, (B/A)_1, (B/A)_2, \dots).$$

Proof. By passing to slices, we obtain

$$\text{StRing}_{A/} = \lim_{\text{CAT}}(\dots \xrightarrow{\text{End}_3^{\mathbb{P}}} \text{CAlg}(2\text{Pr})_{A_2/} \xrightarrow{\text{End}_2^{\mathbb{P}}} \text{CAlg}(1\text{Pr})_{A_1/} \xrightarrow{\text{End}_1^{\mathbb{P}}} \text{CAlg}(\text{An})_{A_0/}).$$

The entry $\text{CAlg}(\text{An})_{A_0/}$ is redundant for the limit, and we will ignore it. Under the equivalences $\text{CAlg}(m\text{Pr})_{A_m/} \simeq \text{CAlg}(\text{Mod}_{A_m}(m\text{Pr}))$, the transition maps $\text{CAlg}((m+1)\text{Pr})_{A_{m+1}/} \rightarrow \text{CAlg}(m\text{Pr})_{A_m/}$ admit the following simple description. An object D of the source is an A_{m+1} -linear monoidal category; the transition sends it to $\text{End}_D(\mathbb{1}_D)$, regarded over $\text{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}) \simeq A_m$. Thus the transition is the composite of

$$\text{CAlg}(\text{Mod}_{A_{m+1}}((m+1)\text{Pr})) \xrightarrow{\text{End}_{(-)}(\mathbb{1})} \text{CAlg}(A_{m+1})$$

and

$$\text{CAlg}(A_{m+1}) \xrightarrow{U_m^A} \text{CAlg}(\text{Mod}_{A_m}(m\text{Pr})),$$

where we use the notation from Construction 5.4. Hence we may alternatively compute $\text{StRing}_{A/}$ as the limit of the chain

$$\dots \rightarrow \text{CAlg}(\text{Mod}_{A_{m+1}}((m+1)\text{Pr})) \xrightarrow{\text{End}_{(-)}(\mathbb{1})} \text{CAlg}(A_{m+1}) \xrightarrow{U_m^A} \text{CAlg}(\text{Mod}_{A_m}(m\text{Pr})) \xrightarrow{\text{End}_{(-)}(\mathbb{1})} \text{CAlg}(A_m) \rightarrow \dots$$

Removing the intermediate $\text{CAlg}(\text{Mod}_{A_m}(m\text{Pr}))$ -vertices gives the limit over the composites

$$\text{End}_m^A = \text{End}_{(-)}(\mathbb{1}) \circ U_m^A: \text{CAlg}(A_{m+1}) \rightarrow \text{CAlg}(A_m),$$

hence the identification

$$\text{StRing}_{A/} \simeq \lim_{\text{CAT}}(\text{CAlg}(A_1) \xleftarrow{\text{End}_1^A} \text{CAlg}(A_2) \xleftarrow{\text{End}_2^A} \dots) = \text{StRing}(A),$$

where the last equality is the limit description of Definition 5.7. Under this equivalence, an object $f : A \rightarrow B$ is sent in degree $n \geq 0$ to

$$f_{n,*}(\mathbb{1}_{B_{n+1}}) = (B/A)_n \in \text{CAlg}(A_{n+1}),$$

that is, to the tuple $((B/A)_0, (B/A)_1, (B/A)_2, \dots)$, as claimed. $\square$

Corollary 5.9. *The functor $\text{StRing}_{A/} \rightarrow \text{CAlg}(A_{n+1}), B \mapsto (B/A)_n$ admits a fully faithful left adjoint*

$$\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}_{A/}.$$

Proof. By Proposition 5.6, the internal module functors Mod_m^A are fully faithful left adjoints. Hence Lemma 2.4 shows that, for every $n \geq 0$, the canonical functor

$$\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}(A)$$

is fully faithful and left adjoint to the projection $\text{StRing}(A) \rightarrow \text{CAlg}(A_{n+1})$. The claim now follows from the equivalence $\text{StRing}_{A/} \simeq \text{StRing}(A)$ of Proposition 5.8, under which this projection corresponds to the functor $B \mapsto (B/A)_n$. $\square$

In Proposition 5.6 we showed that the composite

$$\text{Mod}_m^A : \text{CAlg}(A_m) \xrightarrow{\text{Mod}_{(-)}(A_m)} \text{CAlg}(\text{Mod}_{A_m}(m\text{Pr})) \xrightarrow{L_m^A} \text{CAlg}(A_{m+1})$$

is fully faithful. While the functor $\text{Mod}_{(-)}(A_m)$ is also always fully faithful, the functor L_m^A is usually not. Nevertheless, we still have the following statement:

Lemma 5.10. *Let $A \in \text{StRing}$ and $m \geq 1$. The functor*

$$L_m^A : \text{Mod}_{A_m}(m\text{Pr}) \rightarrow A_{m+1}$$

is fully faithful on dualizable objects. (In particular, it is fully faithful on $\aleph_1$ -filtered colimits of dualizable objects, cf. Lemma 1.10.)

Proof. Set $C := \text{Mod}_{A_m}(m\text{Pr})$ and $D := A_{m+1}$, and let U_m^A denote the right adjoint of L_m^A . Let $X, Y \in C$ be dualizable. Since L_m^A is symmetric monoidal, it preserves duals, and we have a commutative diagram

$$\begin{array}{ccc} \text{Hom}_C(X \otimes Y^\vee, \mathbb{1}_C) & \xrightarrow{\sim} & \text{Hom}_C(X, Y) \\ \downarrow & & \downarrow \\ \text{Hom}_D(L_m^A(X \otimes Y^\vee), \mathbb{1}_D) & \xrightarrow{\sim} & \text{Hom}_D(L_m^A(X), L_m^A(Y)). \end{array}$$

The horizontal maps are the usual identifications coming from duality, given by tensoring with Y and precomposing with the coevaluation $\mathbb{1}_C \rightarrow Y^\vee \otimes Y$. Thus it suffices to show that the left vertical map is an equivalence. Composing this map with the adjunction equivalence

$$\text{Hom}_D(L_m^A(X \otimes Y^\vee), \mathbb{1}_D) \simeq \text{Hom}_C(X \otimes Y^\vee, U_m^A(\mathbb{1}_D))$$

we see that it suffices to show that the unit map $\mathbb{1}_C \rightarrow U_m^A L_m^A \mathbb{1}_C = U_m^A \mathbb{1}_D$ is an isomorphism. But this map is the canonical comparison map $A_m \xrightarrow{\sim} \text{End}_{A_{m+1}}(\mathbb{1}_{A_{m+1}})$, which is an isomorphism as A is a Stefanich ring.

Finally, L_m^A preserves $\aleph_1$ -compact objects, so its right adjoint U_m^A preserves $\aleph_1$ -filtered colimits. The standard compact-generation argument of Lemma 1.10 therefore extends the preceding calculation from dualizable objects to their $\aleph_1$ -filtered colimits. $\square$

5.2 Shifting

We will now define the *shift functor* on StRing .

Construction 5.11. The forgetful functors $\text{CAlg}((n+1)\text{Pr}) \rightarrow \text{CAlg}(n\text{Pr})$ induce a functor

$$\begin{array}{ccc} \text{StRing} & \xrightarrow{\cong} & \lim(\cdots \rightarrow \text{CAlg}(3\text{Pr}) \rightarrow \text{CAlg}(2\text{Pr}) \rightarrow \text{CAlg}(1\text{Pr})) \\ \text{shift} \downarrow & & \downarrow \\ \text{StRing} & \xrightarrow{=} & \lim(\cdots \rightarrow \text{CAlg}(2\text{Pr}) \rightarrow \text{CAlg}(1\text{Pr}) \rightarrow \text{CAlg}(\text{An})) \end{array}$$

given on objects by $A = (A_1, A_2, \dots) \mapsto (A_2, A_3, \dots)$.

Lemma 5.12. *The shift functor induces an equivalence*

$$\text{shift}: \text{StRing} \xrightarrow{\sim} \text{StRing}_{(1\text{Pr}, 2\text{Pr}, \dots)}/.$$

In particular, for any $n \geq 0$, iterated shifting induces an equivalence

$$\text{shift}^n: \text{StRing} \xrightarrow{\sim} \text{StRing}_{(n\text{Pr}, (n+1)\text{Pr}, \dots)}/.$$

Proof. For each $n \geq 1$, we have the identification

$$\text{CAlg}((n+1)\text{Pr}) = \text{CAlg}(\text{Mod}_{n\text{Pr}}(1\text{Pr})) = \text{CAlg}(1\text{Pr})_{n\text{Pr}/} = (\text{CAlg}(1\text{Pr})_{(n-1)\text{Pr}/})_{n\text{Pr}/} = \text{CAlg}(n\text{Pr})_{n\text{Pr}/}.$$

These identifications are compatible with the transition functors. Under them, the forgetful functor

$$\text{CAlg}((n+1)\text{Pr}) \rightarrow \text{CAlg}(n\text{Pr})$$

is the projection $\text{CAlg}(n\text{Pr})_{n\text{Pr}/} \rightarrow \text{CAlg}(n\text{Pr})$, while the same identification remembers the map from $n\text{Pr}$. Since each omitted lower-degree term in the limit presentation of StRing is forced by the next one, we may start the limit at $\text{CAlg}(2\text{Pr})$ and obtain

$$\begin{aligned} \text{StRing} &\simeq \lim_{\text{CAT}}(\cdots \rightarrow \text{CAlg}(4\text{Pr}) \rightarrow \text{CAlg}(3\text{Pr}) \rightarrow \text{CAlg}(2\text{Pr})) \\ &\simeq \lim_{\text{CAT}}(\cdots \rightarrow \text{CAlg}(3\text{Pr})_{3\text{Pr}/} \rightarrow \text{CAlg}(2\text{Pr})_{2\text{Pr}/} \rightarrow \text{CAlg}(1\text{Pr})_{1\text{Pr}/}) \\ &\simeq (\lim_{\text{CAT}}(\cdots \rightarrow \text{CAlg}(3\text{Pr}) \rightarrow \text{CAlg}(2\text{Pr}) \rightarrow \text{CAlg}(1\text{Pr})))_{(1\text{Pr}, 2\text{Pr}, \dots)}/ \\ &\simeq \text{StRing}_{(1\text{Pr}, 2\text{Pr}, \dots)}/. \end{aligned}$$

Here we use that slices commute with limits in CAT . The resulting functor is precisely shift by construction. The iterated statement follows by applying the same argument repeatedly. $\square$

5.3 Spectrification and colimits of Stefanich rings

Throughout this section we fix a Stefanich ring A and study colimits in the category $\text{StRing}(A)$ of internal Stefanich rings in A . Limits and $\aleph_1$ -filtered colimits in $\text{StRing}(A)$ are computed pointwise, since they are preserved by the transition functors End_n^A in the limit description of Definition 5.7. Arbitrary colimits are harder to describe directly, since they are not (necessarily) preserved by the functors End_n^A .

Definition 5.13. Let

$$\mathcal{P}_A := \prod_{n \geq 0} \text{CAlg}(A_{n+1}),$$

and define an endofunctor

$$\text{End}_\bullet^A : \mathcal{P}_A \rightarrow \mathcal{P}_A, \quad (B_n)_{n \geq 0} \mapsto (\text{End}_{n+1}^A(B_{n+1}))_{n \geq 0}.$$

The category of *internal preStefanich rings in A* is the lax equalizer

$$\text{PreStRing}(A) := \mathcal{P}_A \times_{\mathcal{P}_A \times \mathcal{P}_A} \text{Ar}(\mathcal{P}_A),$$

where $\mathcal{P}_A \rightarrow \mathcal{P}_A \times \mathcal{P}_A$ is $(\text{id}, \text{End}_\bullet^A)$ and $\text{Ar}(\mathcal{P}_A) \rightarrow \mathcal{P}_A \times \mathcal{P}_A$ is the source-target functor. Thus an internal preStefanich ring is a tuple $B = (B_0, B_1, B_2, \dots)$ with $B_n \in \text{CAlg}(A_{n+1})$, together with structure maps

$$\alpha_n : B_n \rightarrow \text{End}_{n+1}^A(B_{n+1})$$

that are not assumed to be equivalences. By Definition 5.7, the category $\text{StRing}(A)$ identifies with the full subcategory of $\text{PreStRing}(A)$ spanned by those objects for which all α_n are equivalences.

Lemma 5.14. *The category $\text{PreStRing}(A)$ admits small colimits, and the projection $\text{PreStRing}(A) \rightarrow \mathcal{P}_A$ creates them. In particular, colimits in $\text{PreStRing}(A)$ are computed pointwise.*

Proof. Let $I \rightarrow \text{PreStRing}(A)$ be a diagram, written as (B^i, α^i) for $i \in I$. Let $B := \text{colim}_i B^i$ in $\mathcal{P}_A$, so that $B_n \simeq \text{colim}_i B_n^i$ in $\text{CAlg}(A_{n+1})$. The maps $\alpha^i : B^i \rightarrow \text{End}_\bullet^A(B^i)$ assemble to a morphism of diagrams $B^\bullet \rightarrow \text{End}_\bullet^A(B^\bullet)$. Hence we obtain a structure map

$$B \simeq \text{colim}_i B^i \rightarrow \text{colim}_i \text{End}_\bullet^A(B^i) \rightarrow \text{End}_\bullet^A(\text{colim}_i B^i) = \text{End}_\bullet^A(B),$$

where the second arrow is the canonical comparison map induced by the cocone $B^i \rightarrow B$. This defines an object (B, α) of $\text{PreStRing}(A)$.

For any $(Y, \beta) \in \text{PreStRing}(A)$, the hom anima $\text{Hom}_{\text{PreStRing}(A)}((B, \alpha), (Y, \beta))$ is the homotopy equalizer of the two maps

$$\text{Hom}_{\mathcal{P}_A}(B, Y) \rightrightarrows \text{Hom}_{\mathcal{P}_A}(B, \text{End}_\bullet^A Y)$$

defined by $\beta \circ (-)$ and $\text{End}_\bullet^A(-) \circ \alpha$. Since B is the colimit of the B^i in $\mathcal{P}_A$, this homotopy equalizer identifies with

$$\lim_i \text{Hom}_{\text{PreStRing}(A)}((B^i, \alpha^i), (Y, \beta)).$$

Thus (B, α) has the universal property of the colimit. $\square$

What we will do is to embed $\text{StRing}(A)$ in $\text{PreStRing}(A)$ and then reflect back.

Lemma 5.15. *The inclusion $\text{StRing}(A) \hookrightarrow \text{PreStRing}(A)$ admits a left adjoint.*

We refer to this left adjoint as *spectrification*.

Proof. Define a functor

$$T: \text{PreStRing}(A) \rightarrow \text{PreStRing}(A), \quad (B_n)_{n \geq 0} \mapsto (B'_n)_{n \geq 0},$$

by setting $B'_n = \text{End}_{n+1}^A(B_{n+1})$, with structure maps obtained by applying the functors End_{n+1}^A to the structure maps of B . This admits a natural transformation $\eta: \text{id} \rightarrow T$, given by the structure maps of an internal preStefanich ring. For $X \in \text{PreStRing}(A)$, define $X^{(0)} := X$, $X^{(\alpha+1)} := T(X^{(\alpha)})$, and

$$X^{(\lambda)} := \text{colim}_{\alpha < \lambda} X^{(\alpha)}$$

for limit ordinals λ .

We first show that $X^{(\omega_1)}$ lies in $\text{StRing}(A)$. Each End_n^A is the right adjoint of the morphism Mod_n^A of IPr from Construction 5.4, hence preserves $\aleph_1$ -filtered colimits; therefore the functor T preserves $\aleph_1$ -filtered colimits. Hence

$$T(X^{(\omega_1)}) \simeq T(\text{colim}_{\alpha < \omega_1} X^{(\alpha)}) \simeq \text{colim}_{\alpha < \omega_1} T(X^{(\alpha)}) = \text{colim}_{\alpha < \omega_1} X^{(\alpha+1)} \simeq X^{(\omega_1)}.$$

The last isomorphism uses that the successor ordinals are cofinal in ω_1 . Under these identifications, the structure map $X^{(\omega_1)} \rightarrow T(X^{(\omega_1)})$ is the canonical map

$$\text{colim}_{\alpha < \omega_1} X^{(\alpha)} \rightarrow \text{colim}_{\alpha < \omega_1} X^{(\alpha+1)},$$

hence is an isomorphism. Thus $X^{(\omega_1)}$ is an internal Stefanich ring in A .

It remains to prove the universal property. Let $Y \in \text{StRing}(A)$. We claim by transfinite induction that the natural map

$$\text{Hom}_{\text{PreStRing}(A)}(X^{(\alpha)}, Y) \rightarrow \text{Hom}_{\text{PreStRing}(A)}(X, Y)$$

is an equivalence for every ordinal $\alpha \leq \omega_1$. The claim is clear for $\alpha = 0$. For a successor ordinal, it suffices to show that

$$\eta_{X^{(\alpha)}}^*: \text{Hom}_{\text{PreStRing}(A)}(TX^{(\alpha)}, Y) \rightarrow \text{Hom}_{\text{PreStRing}(A)}(X^{(\alpha)}, Y)$$

is an equivalence. Since Y lies in $\text{StRing}(A)$, the map $\eta_Y: Y \rightarrow TY$ is an equivalence. An inverse to $\eta_{X^{(\alpha)}}^*$ is therefore given by applying T to a map $X^{(\alpha)} \rightarrow Y$ and then using η_Y^{-1} to identify TY with Y . The two composites are the identity by naturality of η and the identity $T\eta = \eta T$, which follows directly from the definition of η by the structure maps.

For a limit ordinal λ , the object $X^{(\lambda)}$ is the colimit of the $X^{(\alpha)}$ for $\alpha < \lambda$, so

$$\text{Hom}_{\text{PreStRing}(A)}(X^{(\lambda)}, Y) \simeq \lim_{\alpha < \lambda} \text{Hom}_{\text{PreStRing}(A)}(X^{(\alpha)}, Y) \simeq \text{Hom}_{\text{PreStRing}(A)}(X, Y).$$

Taking $\alpha = \omega_1$ gives

$$\text{Hom}_{\text{StRing}(A)}(X^{(\omega_1)}, Y) \simeq \text{Hom}_{\text{PreStRing}(A)}(X, Y),$$

naturally in X and Y . Thus $X \mapsto X^{(\omega_1)}$ is left adjoint to the inclusion. $\square$

Corollary 5.16. *We may compute colimits in $\text{StRing}(A)$ by computing them in $\text{PreStRing}(A)$ (where they are computed pointwise) and then reflecting them back into $\text{StRing}(A)$. $\square$*

Remark 5.17. The Stefanich ring $\mathbb{P}$ is initial in StRing , so the slice $\text{StRing}_{\mathbb{P}/}$ is all of StRing . Hence Proposition 5.8 yields an equivalence $\text{StRing} \simeq \text{StRing}(\mathbb{P})$, and we abbreviate $\text{PreStRing} := \text{PreStRing}(\mathbb{P})$. Specializing the above to $A = \mathbb{P}$, colimits in StRing are computed pointwise in PreStRing and then spectrified.

Observation 5.18. Let $B = (B_0, B_1, B_2, \dots)$ be an internal preStefanich ring in A . Assume that there is some $n \geq 0$ such that for each $m \geq n$ the canonical map $B_m \rightarrow \text{End}_{m+1}^A(B_{m+1})$ is an isomorphism in A_{m+1} . Then the spectrification of B agrees with B in degrees $m \geq n$, and is in lower degrees given by repeatedly applying $\text{End}_\bullet^A$.

5.4 Tensor products of Stefanich rings

Consider a diagram of the form $B \leftarrow A \rightarrow C$. We wish to compute the pushout $B \otimes_A C$ in StRing . By Remark 5.17 (the case $A = \mathbb{P}$ of Corollary 5.16), we can do this by computing the pushout in PreStRing , where colimits are pointwise by Lemma 5.14. This gives the preStefanich ring

$$(B_0 \otimes_{A_0} C_0, B_1 \otimes_{A_1} C_1, \dots),$$

and the actual tensor product is obtained by spectrifying this object.

In certain situations, it is useful to do a relative version of the same construction which uses the slice description from Proposition 5.8.

Construction 5.19. Fix a Stefanich ring $A \in \text{StRing}$. By Proposition 5.8 we have $\text{StRing}_{A/} \simeq \text{StRing}(A)$, and by Corollary 5.16 colimits in $\text{StRing}(A)$ are computed by passing to $\text{PreStRing}(A)$, where they are pointwise, and then spectrifying.

Now let $B, C \in \text{StRing}_{A/}$. Under the equivalence of Proposition 5.8, they correspond to the objects of $\text{PreStRing}(A)$ given by the sequences $((B/A)_m)_{m \geq 0}$ and $((C/A)_m)_{m \geq 0}$, where $(B/A)_m \in \text{CAlg}(A_{m+1})$. Their coproduct in $\text{PreStRing}(A)$ is therefore the pointwise tensor product

$$((B/A)_0 \otimes (C/A)_0, (B/A)_1 \otimes (C/A)_1, \dots),$$

where the m -th tensor product is formed in $\text{CAlg}(A_{m+1})$. The coproduct in $\text{StRing}_{A/}$ is obtained by spectrifying this internal preStefanich ring.

Note that this method of computing the tensor product was already used at the end of the previous talk, in the proof of Corollary 4.13.

Construction 5.20. In a yet more refined way, let $g : A \rightarrow C$ be a morphism of Stefanich rings. The cobase-change functor

$$g^* : \text{StRing}_{A/} \rightarrow \text{StRing}_{C/}, \quad B \mapsto B \otimes_A C$$

is computed by passing from $\text{PreStRing}(A)$ to $\text{PreStRing}(C)$. Namely, first apply levelwise extension of scalars

$$g_m^* : \text{CAlg}(A_{m+1}) \rightarrow \text{CAlg}(C_{m+1})$$

to the sequence attached to B . This produces the internal preStefanich ring in C

$$(g_0^*(B/A)_0, g_1^*(B/A)_1, g_2^*(B/A)_2, \dots).$$

The structure maps are obtained by applying g_m^* to the structure maps of B and then using the natural comparison

$$g_m^*(\text{End}_{m+1}^A(R)) \rightarrow \text{End}_{m+1}^C(g_{m+1}^*(R)), \quad R \in \text{CAlg}(A_{m+2}).$$

In other words, for $R_m = (B/A)_m$, the structure map is the composite

$$g_m^* R_m \rightarrow g_m^* \text{End}_{m+1}^A(R_{m+1}) \rightarrow \text{End}_{m+1}^C(g_{m+1}^* R_{m+1}).$$

Spectrifying this pre-object in $\text{PreStRing}(C)$ gives $B \otimes_A C$.

During the seminar, this talk also covered the theory of affine maps of Stefanich rings. For the sake of exposition, that material has been collected in Section 6.2 of the following chapter.

6 Affine, proper and étale maps of Stefanich rings

Talk given by Tashi.

In this talk, we will introduce a variety of properties of maps of Stefanich rings: affine maps, prim maps, proper maps, suave maps, and étale maps.

6.1 Recollections and notations

Before we start, let us recall the notation for the transition maps attached to a morphism of Stefanich rings. First recall the external construction. Let

$$\mathbb{P} = (*, \text{An}, 1\text{Pr}, 2\text{Pr}, \dots)$$

denote the initial Stefanich ring, with $\mathbb{P}_n = (n-1)\text{Pr}$. The absolute transition adjunctions are the adjunctions

$$\text{Mod}_n^{\mathbb{P}} : \text{CAlg}((n-1)\text{Pr}) \rightleftarrows \text{CAlg}(n\text{Pr}) : \text{End}_n^{\mathbb{P}}.$$

Thus a Stefanich ring A consists of objects $A_n \in \text{CAlg}(n\text{Pr})$ together with structure equivalences

$$A_n \xrightarrow{\sim} \text{End}_{n+1}^{\mathbb{P}}(A_{n+1}) = \text{End}_{A_{n+1}}(\mathbb{1}_{A_{n+1}}).$$

By adjunction, these are equivalently encoded by symmetric monoidal left adjoints

$$L_n^A : \text{Mod}_{A_n}(n\text{Pr}) \rightarrow A_{n+1}.$$

In the previous talk, we repeated this construction internally to a fixed Stefanich ring A . Namely, for $m \geq 1$ there are internal module-endomorphism adjunctions

$$\text{Mod}_m^A : \text{CAlg}(A_m) \rightleftarrows \text{CAlg}(A_{m+1}) : \text{End}_m^A.$$

These adjunctions give the category $\text{StRing}(A)$ of Stefanich rings internal to A , and Proposition 5.8 identifies it with the slice category $\text{StRing}_{A/}$. Under this equivalence, a map $f : A \rightarrow B$ is sent to the internal Stefanich ring B/A whose degree n term is

$$(B/A)_n = f_{n,*}(\mathbb{1}_{B_{n+1}}) \in \text{CAlg}(A_{n+1}),$$

as in Construction 5.3. In particular, B/A comes equipped with internal structure equivalences

$$(B/A)_n \xrightarrow{\sim} \text{End}_{n+1}^A((B/A)_{n+1}).$$

Adjoining these equivalences gives maps in $\text{CAlg}(A_{n+2})$

$$L_n^{B/A} : \text{Mod}_{n+1}^A((B/A)_n) = L_{n+1}^A(\text{Mod}_{(B/A)_n}(A_{n+1})) \rightarrow (B/A)_{n+1}.$$

For $B = A$, this gives the corresponding maps $L_n^{A/A}$ for the unit internal Stefanich ring A/A .

Now regard f as a morphism $A/A \rightarrow B/A$ in $\text{StRing}(A)$. In degree n , this gives a map

$$(f/A)_{n-1}^* : (A/A)_n \rightarrow (B/A)_n$$

in $\text{CAlg}(A_{n+1})$; the shifted subscript follows the convention used for pullback functors. The fact that these maps define a morphism of internal Stefanich rings means that they are compatible with the internal structure equivalences. After adjunction, this compatibility says that the square

$$\begin{array}{ccc} \text{Mod}_{n+1}^A((A/A)_n) & \xrightarrow{\text{Mod}_{n+1}^A((f/A)_{n-1}^*)} & \text{Mod}_{n+1}^A((B/A)_n) \\ L_n^{A/A} \downarrow & & \downarrow L_n^{B/A} \\ (A/A)_{n+1} & \xrightarrow{(f/A)_n^*} & (B/A)_{n+1} \end{array}$$

commutes. Since Mod_{n+1}^A preserves monoidal units, $L_n^{A/A}$ is an equivalence, and thus we see that the map $(f/A)_n^*$ factors as follows:

$$(A/A)_{n+1} \rightarrow \text{Mod}_{n+1}^A((B/A)_n) \xrightarrow{L_n^{B/A}} (B/A)_{n+1}.$$

For later use, we also fix notation for internal Hom objects relative to A . Let $m \geq 1$ and let $D \in \text{CAlg}(A_{m+1})$. For $X, Y \in D$, we write

$$\underline{\text{Hom}}_D^A(X, Y) := \underline{\text{Hom}}_{U_m^A(D)}(X, Y) \in A_m,$$

where $U_m^A : A_{m+1} \rightarrow \text{Mod}_{A_m}(m\text{Pr})$ is the right adjoint from Construction 5.4. In particular,

$$\text{End}_m^A(D) = \underline{\text{Hom}}_D^A(\mathbb{1}_D, \mathbb{1}_D).$$

6.2 Affine maps

During the seminar, this material was presented as part of the previous talk; it has been collected here for the sake of exposition.

Definition 6.1. Consider $n \geq 0$, and let $A \in \text{StRing}$ be a Stefanich ring. Then $B \in \text{StRing}_{A/}$ is called *n-affine* if it lies in the essential image of the functor $\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}_{A/}$.

Thus an n -affine object over A is obtained from a single algebra

$$R \in \text{CAlg}(A_{n+1})$$

by inserting it as the relative algebra in degree n and then applying the left adjoint transition functors in all higher degrees. In other words, if B is the n -affine object determined by R , then

$$(B/A)_n \simeq R, \quad (B/A)_{m+1} \simeq \text{Mod}_{m+1}^A((B/A)_m) \quad (m \geq n),$$

while the lower relative algebras are forced by the right adjoint transitions End_m^A . Equivalently, for an arbitrary $B \in \text{StRing}_{A/}$, the object B is n -affine precisely when the relative structure maps

$$L_m^{B/A} : \text{Mod}_{m+1}^A((B/A)_m) \rightarrow (B/A)_{m+1}$$

are equivalences for all $m \geq n$.

The fully faithful embedding $\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}_{A/}$ already says that the recursive construction of an n -affine object does not lose information. We will also need a slightly different recovery statement: in high degrees, the relative algebra $(B/A)_{m+1}$ is obtained by applying L_{m+1}^A to the A_{m+1} -linear category $\text{Mod}_{(B/A)_m}(A_{m+1})$, and we want to recover this linear category from its image in A_{m+2} . This is where Lemma 5.10 enters. If $R \in \text{CAlg}(A_{m+1})$ is countably presented, then $\text{Mod}_R(A_{m+1})$ is dualizable in $\text{Mod}_{A_{m+1}}((m+1)\text{Pr})$. More generally, after writing R as an $\aleph_1$ -filtered colimit of countably presented algebras, and using that $R \mapsto \text{Mod}_R(A_{m+1})$ preserves such colimits, $\text{Mod}_R(A_{m+1})$ is an $\aleph_1$ -filtered colimit of dualizable objects. On this class, the functor

$$L_{m+1}^A : \text{Mod}_{A_{m+1}}((m+1)\text{Pr}) \rightarrow A_{m+2}$$

is fully faithful.

Remark 6.2. We have a factorization $\text{CAlg}(A_{n+1}) \hookrightarrow \text{CAlg}(A_{n+2}) \hookrightarrow \text{StRing}_{A/}$. In particular, any n -affine Stefanich ring under A is also $(n+1)$ -affine.

Here are some basic properties of affineness:

Proposition 6.3. *Let $n \geq 0$ and $A \in \text{StRing}$.*

- (1) *The class of n -affines over A is closed under small colimits.*
- (2) *If $B \in \text{StRing}_{A/}$ is n -affine, then the comparison map $\text{Mod}_{(B/A)_m}(A_{m+1}) \rightarrow B_{m+1}$ is an equivalence for $m \geq n$.*
- (3) *Given a map $g : A \rightarrow C$ in StRing and if $B \in \text{StRing}_{A/}$ is n -affine, then also $g^*B = B \otimes_A C \in \text{StRing}_{C/}$ is n -affine. Moreover, we have*

$$g_m^*(B/A)_m \simeq (B \otimes_A C/C)_m$$

for $m \geq n$.

- (4) *Given maps $f : A \rightarrow B$ and $h : B \rightarrow C$ in StRing with composite $g : A \rightarrow C$, if f is n -affine, then g is n -affine if and only if h is.*
- (5) *If $B \in \text{StRing}_{A/}$ is n -affine, then the codiagonal $\nabla : B \otimes_A B \rightarrow B$ is n -affine; consequently all iterated codiagonals of $A \rightarrow B$ are n -affine.*

Proof. (1) is clear, since $\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}_{A/}$ is a fully faithful left adjoint.

(2) Let $m \geq n$, and let U_{m+1}^A denote the right adjoint of

$$L_{m+1}^A : \text{Mod}_{A_{m+1}}((m+1)\text{Pr}) \rightarrow A_{m+2}.$$

The comparison map in question is the unit map

$$\mathrm{Mod}_{(B/A)_m}(A_{m+1}) \rightarrow U_{m+1}^A L_{m+1}^A(\mathrm{Mod}_{(B/A)_m}(A_{m+1})),$$

followed by the identification of the target with B_{m+1} . Indeed, by the recursive description above, since B is n -affine over A , we have

$$\mathrm{Mod}_{m+1}^A((B/A)_m) \simeq (B/A)_{m+1}.$$

Moreover, by the discussion preceding the proposition, after writing $(B/A)_m$ as an $\aleph_1$ -filtered colimit of countably presented algebras, the object $\mathrm{Mod}_{(B/A)_m}(A_{m+1})$ is an $\aleph_1$ -filtered colimit of dualizable objects. Hence Lemma 5.10 implies that the above unit map is an equivalence. Applying the displayed identification, the target becomes

$$U_{m+1}^A((B/A)_{m+1}) = \mathrm{fgt}((B/A)_{m+1}) \simeq B_{m+1},$$

as desired.

(3) We use the description of cobase change from Construction 5.20. The pre-object whose spectrification computes g^*B is represented in $\mathrm{PreStRing}(C)$ by the sequence

$$(g_0^*(B/A)_0, g_1^*(B/A)_1, g_2^*(B/A)_2, \dots).$$

For $m \geq n$, the recursive description of n -affineness gives

$$(B/A)_{m+1} \simeq \mathrm{Mod}_{m+1}^A((B/A)_m).$$

Extension of scalars is compatible with these transition functors, in the sense that for $R \in \mathrm{CAlg}(A_{m+1})$ we have

$$g_{m+1}^* \mathrm{Mod}_{m+1}^A(R) \simeq \mathrm{Mod}_{m+1}^C(g_m^* R).$$

Thus the transition maps of the above preStefanich ring are already equivalences in all degrees $m \geq n$. Spectrification therefore leaves those degrees unchanged, and the result is the n -affine Stefanich ring over C determined by $g_n^*(B/A)_n$. In particular,

$$g_m^*(B/A)_m \simeq (B \otimes_A C/C)_m$$

for $m \geq n$.

(4) Assume B is n -affine over A , so that it is the image of $(B/A)_n \in \mathrm{CAlg}(A_{n+1})$. Slicing the fully faithful functor

$$\mathrm{CAlg}(A_{n+1}) \hookrightarrow \mathrm{StRing}_{A/}$$

at this object gives a fully faithful functor

$$\mathrm{CAlg}(A_{n+1})_{(B/A)_n/} \hookrightarrow \mathrm{StRing}_{B/},$$

whose essential image consists precisely of those objects $C \in \mathrm{StRing}_{B/}$ for which the composite $A \rightarrow C$ is n -affine. On the other hand, by part (2),

$$B_{n+1} \simeq \mathrm{Mod}_{(B/A)_n}(A_{n+1}),$$

and therefore

$$\mathrm{CAlg}(B_{n+1}) \simeq \mathrm{CAlg}(A_{n+1})_{(B/A)_n/}.$$

Under this equivalence, the functor $\mathrm{CAlg}(B_{n+1}) \hookrightarrow \mathrm{StRing}_{B/}$ has the same essential image as the sliced functor above. Thus C is n -affine over B if and only if it is n -affine over A .

(5) Let $\iota: B \rightarrow B \otimes_A B$ be a coprojection, i.e. the cobase change of the structure map $f: A \rightarrow B$ along itself; it is n -affine by (3). Hence (4), applied to $A \xrightarrow{f} B \xrightarrow{\iota} B \otimes_A B$, shows that the structure map $\eta: A \rightarrow B \otimes_A B$ is n -affine. As $\nabla \circ \eta = f$, a second application of (4), now to $A \xrightarrow{\eta} B \otimes_A B \xrightarrow{\nabla} B$, shows that ∇ is n -affine. The statement for iterated codiagonals follows by induction, since each is the codiagonal of the n -affine map obtained at the previous step. $\square$

Remark 6.4. The converse to part (2) is *not* necessarily true. Let k be a field, let

$$f: * \rightarrow B^2\mathbb{G}_m$$

be the base point, and let $A \rightarrow B$ be the corresponding map of Stefanich rings. One can show that f is 1-affine, so the comparison in part (2) is an equivalence for $m \geq 1$. It is also an equivalence for $m = 0$: here $A_1 = B_1 = D(k)$ and $(B/A)_0 = k$, so the comparison is just

$$\mathrm{Mod}_k(D(k)) \simeq D(k).$$

Nevertheless f is not 0-affine. Indeed, if it were 0-affine, then B would be determined over A by the algebra $(B/A)_0 \in \mathrm{CAlg}(A_1)$. But $(B/A)_0 = k$ is the unit algebra, and the 0-affine object corresponding to the unit is just the identity object $A \in \mathrm{StRing}_{A/}$. This would force $A \simeq B$, i.e. $* \simeq B^2\mathbb{G}_m$, which is false.

6.3 Prim maps

Definition 6.5. The map $f: A \rightarrow B$ is called n -prim if for all $m \geq n$ the map

$$(f/A)_m^*: (A/A)_{m+1} \rightarrow (B/A)_{m+1}$$

admits a right adjoint $(f/A)_{m,*}: (B/A)_{m+1} \rightarrow (A/A)_{m+1}$ in A_{m+2} .

Observation 6.6. If f is n -prim, the composite $(f/A)_{m,*}(f/A)_m^*: (A/A)_{m+1} \rightarrow (A/A)_{m+1}$ defines an endomorphism of the unit in A_{m+2} . We claim that it is sent to $(B/A)_m$ under the equivalence $\mathrm{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$ that evaluates an endomorphism at the $\mathbb{1} \in A_{m+1}$. Indeed, since the forgetful 2-functor $A_{m+2} \rightarrow (m+2)\mathrm{Pr}$ preserves adjunctions, the internal adjunction $(f/A)_m^* \dashv (f/A)_{m,*}$ in A_{m+2} refines the external adjunction $f_m^* \dashv f_{m,*}$, and we compute

$$(f/A)_{m,*}(f/A)_m^*(\mathbb{1}_{A_{m+1}}) \simeq f_{m,*}f_m^*(\mathbb{1}_{A_{m+1}}) = (B/A)_m.$$

Lemma 6.7. Let $f: A \rightarrow B$ be an n -prim map of Stefanich rings. Then, for every $m \geq n$, the relative algebra

$$(B/A)_m \in \mathrm{CAlg}(A_{m+1})$$

is countably presented. Consequently, the m -affine truncation of B over A is countably presented. In particular, if f is also m -affine, then f is countably presented and m -proper.

Proof. By n -primness, the relative pullback

$$(f/A)_m^*: \mathbb{1} \rightarrow (B/A)_{m+1}$$

admits an internal right adjoint $(f/A)_{m,*}$ in A_{m+2} . The underlying functor of this internal right adjoint is a morphism in the relevant $\aleph_1$ -presentable category and therefore preserves $\aleph_1$ -compact objects. The monoidal unit of $(B/A)_{m+1}$ is $\aleph_1$ -compact, and Observation 6.6 identifies

$$(f/A)_{m,*}(\mathbb{1}) \simeq (B/A)_m$$

as an object of A_{m+1} . Thus the underlying A_{m+1} -module of $(B/A)_m$ is $\aleph_1$ -compact. By Lemma A.1, applied with $\kappa = \aleph_1$, the algebra $(B/A)_m$ is countably presented.

The fully faithful functor $\text{CAlg}(A_{m+1}) \hookrightarrow \text{StRing}_{A/}$ defining m -affine Stefanich algebras is a morphism in 1Pr , so it carries $(B/A)_m$ to a countably presented m -affine truncation. If f itself is m -affine, it agrees with this truncation and is therefore countably presented; it is then m -proper by Proposition 6.15(1). $\square$

Remark 6.8. The lemma does not assert that an arbitrary n -prim map is countably presented as an object of $\text{StRing}_{A/}$. It asserts that all of its relative algebras in the prim range, and hence all affine truncations in that range, are countably presented. Countable presentation of the map itself follows when the map is affine in one of these degrees.

Proposition 6.9. *Let $A \in \text{StRing}$.*

- (1) *If B/A is n -affine and countably presented, then B/A is n -prim.*
- (2) *The n -prim Stefanich A -algebras are closed under all countable colimits. Moreover, for every $m \geq n$, the formation of these colimits commutes with the functor*

$$\text{StRing}_{A/} \rightarrow \text{CAlg}(A_{m+1}), \quad B \mapsto (B/A)_m.$$

- (3) *Consider a pushout square of Stefanich rings*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \lrcorner & \downarrow g' \\ C & \xrightarrow{f'} & C \otimes_A B. \end{array}$$

If f is n -prim, then so is f' , and for every $m \geq n$ the canonical map

$$g_m^*(B/A)_m \rightarrow (B \otimes_A C/C)_m$$

is an isomorphism (i.e. we do not need to “spectrify”). Moreover, the internal Beck–Chevalley map

$$(g/A)_m^*(f/A)_{m,*} \rightarrow (f'/A)_{m,*}(g'/A)_m^*$$

is an isomorphism of morphisms $(B/A)_{m+1} \rightarrow (C/A)_{m+1}$ in A_{m+2} .

- (4) *The n -prim maps are closed under composition.*

Proof. (1) Since B/A is n -affine and countably presented, $(B/A)_n$ is countably presented in $\text{CAlg}(A_{n+1})$. The functor $R \mapsto \text{Mod}_R(A_{m+1})$ preserves countably presented objects by Lemma 2.8, and L_{m+1}^A preserves them because it is a morphism in $(m+2)\text{Pr}$. Induction along the n -affine presentation therefore shows that $(B/A)_m$ is countably presented for every $m \geq n$.

The map $(f/A)_m^*: (A/A)_{m+1} \rightarrow (B/A)_{m+1}$ factors as

$$(A/A)_{m+1} \rightarrow \text{Mod}_{A_{m+1}}^A((B/A)_m) \xrightarrow{L_m^{B/A}} (B/A)_{m+1}.$$

If B/A is n -affine, then the second map is an equivalence for $m \geq n$, and so this composite has a right adjoint in A_{m+2} if and only if the first map admits a right adjoint. By definition, the first map is the image under the 2-functor $L_{m+1}^A: \text{Mod}_{A_{m+1}}((m+1)\text{Pr}) \rightarrow A_{m+2}$ of the free module functor

$$A_{m+1} \rightarrow \text{Mod}_{(B/A)_m}(A_{m+1}).$$

If $(B/A)_m \in \text{CAlg}(A_{m+1})$ is countably presented, then this free module functor admits a right adjoint in $\text{Mod}_{A_{m+1}}((m+1)\text{Pr})$ by part (f) of Remark 2.7. Applying the 2-functor L_{m+1}^A preserves this adjunction and supplies the required right adjoint in A_{m+2} .

(4) Let $g: B \rightarrow C$ be a further n -prim map, with composite $h = g \circ f: A \rightarrow C$. Fix $m \geq n$. Since g is n -prim, the relative map $(g/B)_m^*: \mathbb{1} \rightarrow (C/B)_{m+1}$ admits a right adjoint in B_{m+2} . The forgetful functor $B_{m+2} \rightarrow A_{m+2}$ is a 2-functor of the underlying 2-categories, and any 2-functor preserves adjunctions; applying it to the previous adjunction exhibits a right adjoint of

$$(B/A)_{m+1} \xrightarrow{(g/A)_m^*} (C/A)_{m+1}$$

in A_{m+2} . Since f is n -prim, the relative map $(f/A)_m^*: \mathbb{1} \rightarrow (B/A)_{m+1}$ also admits a right adjoint in A_{m+2} . Composing the two right adjoints exhibits a right adjoint of

$$(h/A)_m^*: \mathbb{1} \xrightarrow{(f/A)_m^*} (B/A)_{m+1} \xrightarrow{(g/A)_m^*} (C/A)_{m+1}$$

in A_{m+2} . As $m \geq n$ was arbitrary, h is n -prim.

(3) Recall from Construction 5.20 that the relative tensor product $C \otimes_A B \in \text{StRing}_{C/}$ may be computed as the spectrification of the internal preStefanich ring

$$(g_0^*(B/A)_0, g_1^*(B/A)_1, g_2^*(B/A)_2, \dots)$$

in C . We show that in degrees $m \geq n$ the levels $g_m^*(B/A)_m$ already assemble into a Stefanich ring, i.e. that the canonical maps $g_m^*(B/A)_m \rightarrow \text{End}_{m+1}^C(g_{m+1}^*(B/A)_{m+1})$ are isomorphisms; it then follows from Observation 5.18 that $g^*(B/A)_\bullet$ already agrees with $(C \otimes_A B/C)_\bullet$ in these degrees.

Fix $m \geq n$. Since f is n -prim, we have the internal adjunction $(f/A)_m^* \dashv (f/A)_{m,*}$ in A_{m+2} , and by the observation preceding the proposition the algebra $(B/A)_m \in \text{CAlg}(A_{m+1})$ is the value of the composite

$$(A/A)_{m+1} \xrightarrow{(f/A)_m^*} (B/A)_{m+1} \xrightarrow{(f/A)_{m,*}} (A/A)_{m+1}$$

under the equivalence $\text{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$. Applying the 2-functor $g_{m+1}^*: A_{m+2} \rightarrow C_{m+2}$ to this adjunction yields an internal adjunction

$$(C/C)_{m+1} \xrightarrow{g_{m+1}^*((f/A)_m^*)} g_{m+1}^*((B/A)_{m+1}) \xrightarrow{g_{m+1}^*((f/A)_{m,*})} (C/C)_{m+1}$$

in C_{m+2} . The functor $g_{m+1}^*((f/A)_m^*)$ is symmetric monoidal and so preserves the monoidal unit. We may now compute the C_{m+1} -linear endomorphisms of the unit in $g_{m+1}^*((B/A)_{m+1}) \in \text{CAlg}(C_{m+2})$ as

$$\begin{aligned}
\text{End}_{m+1}^C(g_{m+1}^*((B/A)_{m+1})) &= \underline{\text{Hom}}_{g_{m+1}^*((B/A)_{m+1})}(\mathbb{1}, \mathbb{1}) \\
&= \underline{\text{Hom}}_{g_{m+1}^*((B/A)_{m+1})}(g_{m+1}^*((f/A)_m^*) \mathbb{1}, g_{m+1}^*((f/A)_m^*) \mathbb{1}) \\
&\simeq \underline{\text{Hom}}_{(C/C)_{m+1}}(\mathbb{1}, g_{m+1}^*((f/A)_{m,*}(f/A)_m^*) \mathbb{1}) \\
&\simeq g_{m+1}^*((f/A)_{m,*}(f/A)_m^*) \mathbb{1} \\
&\stackrel{(1)}{\simeq} g_m^*((f/A)_{m,*}(f/A)_m^*) \mathbb{1} \\
&\stackrel{(2)}{\simeq} g_m^*((B/A)_m),
\end{aligned}$$

where the equivalence (1) uses the fact that the map of Stefanich rings $g: A \rightarrow C$ comes with a commutative diagram of the form

$$\begin{array}{ccc}
\text{End}_{A_{m+2}}(\mathbb{1}) & \xrightarrow{g_{m+1}^*} & \text{End}_{C_{m+2}}(\mathbb{1}) \\
\downarrow \simeq \text{ev}_{\mathbb{1}} & & \downarrow \simeq \text{ev}_{\mathbb{1}} \\
A_{m+1} & \xrightarrow{g_m^*} & C_{m+1},
\end{array}$$

while (2) holds by Observation 6.6. We conclude that $g^*(B/A)_m = (C \otimes_A B/C)_m$ for $m \geq n$. Moreover the map $(f'/C)_m^*: (C/C)_{m+1} \rightarrow g^*(B/A)_{m+1} = (C \otimes_A B/C)_{m+1}$ admits the right adjoint $g_{m+1}^*((f/A)_{m,*})$ in C_{m+2} , so f' is n -prim.

It remains to prove the internal Beck–Chevalley statement: we need to show that the square

$$\begin{array}{ccc}
(A/A)_{m+1} & \xrightarrow{(g/A)_m^*} & (C/A)_{m+1} \\
(f/A)_m^* \downarrow & & \downarrow (f'/A)_m^* \\
(B/A)_{m+1} & \xrightarrow{(g'/A)_m^*} & (B \otimes_A C/A)_{m+1}
\end{array}$$

in A_{m+2} is vertically right adjointable. By the preceding proof, this square is isomorphic to the naturality square

$$\begin{array}{ccc}
(A/A)_{m+1} & \xrightarrow{\eta} & g_{m+1,*}g_{m+1}^*(A/A)_{m+1} \\
(f/A)_m^* \downarrow & & \downarrow g_{m+1,*}g_{m+1}^*(f/A)_m^* \\
(B/A)_{m+1} & \xrightarrow{\eta} & g_{m+1,*}g_{m+1}^*(B/A)_{m+1}
\end{array}$$

of the unit $\eta: \text{id} \rightarrow g_{m+1,*}g_{m+1}^*$ of the 2-categorical adjunction $g_{m+1}^*: A_{m+2} \rightleftarrows C_{m+2}: g_{m+1,*}$. The vertical adjointability of this square is formal.

(2) Every countable colimit can be assembled from finite coproducts and sifted countable colimits. Finite coproducts are covered by (3) and (4): the coproduct of B and C in $\text{StRing}_{A/}$ is $B \otimes_A C$, and the map $A \rightarrow B \otimes_A C$ factors as an n -prim map followed by a cobase change of an n -prim map. The levelwise assertion follows at the same time from the base-change isomorphism in (3).

It remains to consider a countable sifted category I and a diagram $i \mapsto B(i)$ of n -prim Stefanich A -algebras. For $m \geq n$, set

$$R_m := \text{colim}_{i \in I} (B(i)/A)_m \in \text{CAlg}(A_{m+1}).$$

Since I is sifted, this colimit is also the colimit of the underlying objects of A_{m+1} . Apply the lemma below to the countable category $I^\triangleleft$, with value $\mathbb{1}$ at its initial object and value $(B(i)/A)_{m+1}$ at $i \in I$. It follows that the unit map $u_m: \mathbb{1} \rightarrow R_{m+1}$ admits a right adjoint and that the endomorphism $u_m^R u_m$ is the colimit of the corresponding endomorphisms for the $B(i)$. By Observation 6.6, evaluation at the unit identifies this colimit with R_m . Consequently

$$R_m \xrightarrow{\sim} \text{End}_{m+1}^A(R_{m+1})$$

for every $m \geq n$. Thus these levels already satisfy the Stefanich condition, and spectrification does not change them. They are therefore the levels of $\text{colim}_{i \in I} B(i)$, the displayed adjunctions show that this colimit is n -prim, and the formula for R_m proves the asserted compatibility with the functors $B \mapsto (B/A)_m$. $\square$

Lemma 6.10. *Let I be a countable category with an initial object 0 , let $\mathbb{C} \in 2\text{Pr}$, and consider a functor $X: I \rightarrow \mathbb{C}$. Assume that each $F_i: X_0 \rightarrow X_i$ has a right adjoint F_i^R . Then the functor $F: X_0 \rightarrow \text{colim}_I X$ admits a right adjoint $F^R: \text{colim}_{i \in I} X_i \rightarrow X_0$, and we have*

$$F^R F = \text{colim}_{i \in I} F_i^R F_i \in \mathbb{C}(X_0, X_0).$$

The unit $\text{id}_{X_0} \rightarrow F^R F$ is induced by the initial term $F_0^R F_0 = \text{id}_{X_0}$.

Proof. We argue by the (enriched) Yoneda lemma. For each object $Y \in \mathbb{C}$, the represented functor

$$\mathbb{C}(-, Y): \mathbb{C}^{\text{op}} \rightarrow 1\text{Pr}$$

sends a 1-morphism that is a left adjoint in $\mathbb{C}$ to a right adjoint functor of categories. Conversely, a 1-morphism G of $\mathbb{C}$ is a left adjoint as soon as $\mathbb{C}(-, Y)(G)$ admits a right adjoint for every Y , and these right adjoints are compatible with the maps $\mathbb{C}(-, Y) \rightarrow \mathbb{C}(-, Y')$ induced by morphisms $Y \rightarrow Y'$ (the Beck–Chevalley condition); as usual, it suffices to test this on $Y = X_0$ and $Y = \text{colim } X$. We may therefore assume $\mathbb{C} = 1\text{Pr}$.

In 1Pr this is the standard comparison between the colimit presentation $\text{colim}_{i \in I} X_i$ in Pr^{L} and the limit presentation $\lim_{i \in I^{\text{op}}} X_i$ formed along the right adjoints. Writing $\iota_i: X_i \rightarrow \text{colim}_I X$ for the structure maps, each ι_i admits a right adjoint ι_i^R ; since $0 \in I$ is initial we have $F = \iota_0$, and hence a right adjoint $F^R := \iota_0^R$. Under the limit presentation F^R is computed by the partial right adjoints, and the Beck–Chevalley condition required above holds by construction. Tracing through this description, the composite $F^R F$ is computed as the colimit

$$F^R F = \text{colim}_{i \in I} F_i^R F_i \in \mathbb{C}(X_0, X_0),$$

and the unit $\text{id} \rightarrow F^R F$ is the inclusion of the initial index $i = 0$, for which $F_0 = \text{id}_{X_0}$. Alternatively, one writes down the unit and counit of the adjunction explicitly and checks the triangle identities directly. $\square$

Lemma 6.11 (Prim projection formula). *Let $f: A \rightarrow B$ be an n -prim map of Stefanich rings. Then for all $m \geq n$, $X \in A_{m+1}$ and $Y \in B_{m+1}$, the canonical map*

$$X \otimes_{A_{m+1}} f_{m,*}(Y) \rightarrow f_{m,*}(f_m^*(X) \otimes_{B_{m+1}} Y)$$

is an isomorphism.

Proof. Fix $m \geq n$. By the definition of n -primness, the relative pullback

$$(f/A)_m^* : (A/A)_{m+1} \rightarrow (B/A)_{m+1}$$

admits a right adjoint $(f/A)_{m,*}$ as a 1-morphism in A_{m+2} . Apply the 2-functor underlying

$$U_{m+1}^A : A_{m+2} \rightarrow \text{Mod}_{A_{m+1}}((m+1)\text{Pr}).$$

Since 2-functors preserve adjunctions, this gives an adjunction of A_{m+1} -module objects whose underlying adjunction is

$$f_m^* : A_{m+1} \rightleftarrows B_{m+1} : f_{m,*}.$$

In particular, the right adjoint $f_{m,*}$ is A_{m+1} -linear, where the action of A_{m+1} on B_{m+1} is induced by the symmetric monoidal functor f_m^* . Its linearity isomorphism is therefore

$$X \otimes_{A_{m+1}} f_{m,*}(Y) \xrightarrow{\sim} f_{m,*}(f_m^*(X) \otimes_{B_{m+1}} Y).$$

This isomorphism is adjoint to the composite obtained from the counit $f_m^* f_{m,*}(Y) \rightarrow Y$; hence it is precisely the canonical projection map in the statement. $\square$

The relative objects of n -prim maps satisfy a Künneth formula for tensor products over A :

Lemma 6.12 (Künneth for prim maps). *Let $f : A \rightarrow B$ and $g : A \rightarrow C$ be n -prim maps of Stefanich rings. Then*

$$(B \otimes_A C/A)_n \simeq (B/A)_n \otimes_{A_{n+1}} (C/A)_n.$$

Proof. Let $\text{pr} : C \rightarrow B \otimes_A C$ be the cobase change of f along g , so that $\text{pr} \circ g : A \rightarrow B \otimes_A C$ is the structure map. By functoriality of the relative $*$ -pushforward,

$$(B \otimes_A C/A)_n = (\text{pr} \circ g)_{n,*}(\mathbb{1}) = g_{n,*}(\text{pr}_{n,*}(\mathbb{1})) = g_{n,*}((B \otimes_A C/C)_n).$$

Since pr is the cobase change of the n -prim map f along g , Proposition 6.9(3) gives $(B \otimes_A C/C)_n \simeq g_n^*((B/A)_n)$, and therefore

$$(B \otimes_A C/A)_n \simeq g_{n,*}g_n^*((B/A)_n).$$

As g is n -prim, the right adjoint $g_{n,*}$ satisfies the projection formula, so

$$(B \otimes_A C/A)_n \simeq g_{n,*}g_n^*((B/A)_n) \simeq (B/A)_n \otimes_{A_{n+1}} g_{n,*}(\mathbb{1}) = (B/A)_n \otimes_{A_{n+1}} (C/A)_n. \quad \square$$

6.4 Proper maps

Note that one missing feature of the previous proposition is that n -prim maps are not necessarily closed under diagonals.

Example 6.13. Let k be a field of characteristic 0 (for simplicity), and let X be a variety over k . The map of Stefanich rings corresponding to the de Rham stack $X_{\text{dR}} \rightarrow \text{Spec}(k)$ is 0-prim over k , but its diagonal is not. Thus n -primness genuinely fails to be stable under passage to diagonals.

This motivates the following definition.

Definition 6.14. A map $f: A \rightarrow B$ of Stefanich rings is called *n-proper* if f and all its iterated codiagonals

$$\nabla_f: B \otimes_A B \rightarrow B, \quad \nabla_{\nabla_f}: B \otimes_{B \otimes_A B} B \rightarrow B, \quad \dots$$

are all *n*-prim.

Proposition 6.15. *Let $A \in \text{StRing}$.*

- (1) *If $B \in \text{StRing}_{A|}$ is *n*-affine and countably presented, then B is *n*-proper.*
- (2) *The full subcategory of $\text{StRing}_{A|}$ spanned by the *n*-proper A -algebras is closed under countable colimits.*
- (3) *The subcategory in (2) is also closed under cobase change.*
- (4) *Let $f: A \rightarrow B$ be *n*-proper, and let $g: B \rightarrow C$ be a map. Then g is *n*-prim if and only if gf is *n*-prim. Moreover, g is *n*-proper if and only if gf is *n*-proper.*

Proof. All four statements follow from Proposition 6.9, together with the stability of *n*-affine maps under passing to iterated codiagonals (Proposition 6.3(5)).

(1) If B is *n*-affine and countably presented over A , then so is every iterated codiagonal of $f: A \rightarrow B$, by the stability just mentioned and the fact that countable presentability is preserved under cobase change. Hence each iterated codiagonal is *n*-prim by Proposition 6.9(1), so f is *n*-proper.

(2) It suffices to treat sifted colimits and finite coproducts separately. For a sifted colimit, every iterated codiagonal of the colimit map is itself a sifted colimit of cobase changes of the corresponding iterated codiagonals of the diagram, so Proposition 6.9(2) and (3) apply degreewise. Finite coproducts reduce to stability under cobase change and composition, which are parts (3) and (4) below.

(3) The formation of iterated codiagonals commutes with cobase change. Hence, if all codiagonals of f are *n*-prim, then Proposition 6.9(3) shows that all codiagonals of the cobase change f' are *n*-prim as well, i.e. f' is *n*-proper.

(4) An iterated codiagonal of a composite can be written as a composite of cobase changes of the iterated codiagonals of the two individual maps. Together with Proposition 6.9(3) and (4), this shows that *n*-prim and *n*-proper maps are closed under composition; in particular, if g is *n*-prim (resp. *n*-proper), then so is gf . For the converse, assume f is *n*-proper and gf is *n*-prim (resp. *n*-proper). Factor g as

$$B \rightarrow B \otimes_A C \rightarrow C,$$

where the first map is the cobase change of $gf: A \rightarrow C$ along f , hence *n*-prim (resp. *n*-proper), and the second map is a cobase change of the codiagonal $\nabla_f: B \otimes_A B \rightarrow B$, hence *n*-proper, and in particular *n*-prim. By the composition statement just proved, g is *n*-prim (resp. *n*-proper). $\square$

Being 0-proper is closely related to being *rigid* in the sense of Gaitsgory–Rozenblyum [GR17].

Proposition 6.16. *Let $A \in \text{StRing}$, and let B be a countably presented 1-affine Stefanich A -algebra. Then B is 0-proper over A if and only if $(B/A)_1 \in \text{CAlg}(A_2)$ is rigid in A_2 , i.e. the following two conditions hold.*

(1) The unit map $\mathbb{1} \rightarrow (B/A)_1$ admits a right adjoint in A_2 .

(2) The multiplication map $(B/A)_1 \otimes_{A_2} (B/A)_1 \rightarrow (B/A)_1$ admits a right adjoint in

$$\text{Mod}_{(B/A)_1 \otimes_{A_2} (B/A)_1}(A_2).$$

Proof. Since B is 1-affine and countably presented, it is automatically 1-proper by Proposition 6.15(1). Thus being 0-proper amounts to having, in addition, the relevant right adjoints at the lowest level.

If B is 0-proper, then (1) and (2) hold by applying the 0-prim condition to $f: A \rightarrow B$ and to its codiagonal $\nabla_f: B \otimes_A B \rightarrow B$, using that by Proposition 6.3(2) we have $(B \otimes_A B)_2 \simeq \text{Mod}_{(B/A)_1 \otimes_{A_2} (B/A)_1}(A_2)$.

Conversely, it suffices to show that (1) and (2) imply that the codiagonal $\nabla_f: B \otimes_A B \rightarrow B$ is 0-affine. Indeed, the codiagonal is then certainly 0-proper, and together with $A \rightarrow B$ being 0-prim this shows that f is 0-proper. Now the codiagonal is 1-affine and 0-prim, and it admits a section, the coprojection $B \rightarrow B \otimes_A B$, which is 1-affine as a cobase change of f . Hence it is 0-affine by Lemma 6.17. $\square$

The following lemma is often useful to reduce the level of affineness.

Lemma 6.17. *Let $f: A \rightarrow B$ be a map of Stefanich rings. Assume that f is $(n+1)$ -affine and n -prim, and that there is an $(n+1)$ -affine map $g: B \rightarrow A$ with $fg = \text{id}_B$. Then f is n -affine.*

Proof. By shifting we may assume $n = 0$. Replacing A by the image of $(B/A)_0 \in \text{CAlg}(A_1)$ in StRing_{A_1} , we may assume that $(B/A)_0 = \mathbb{1} \in \text{CAlg}(A_1)$; we must then show that $A \rightarrow B$ is an equivalence. By 1-affineness it suffices to show that $\mathbb{1} \rightarrow (B/A)_1$ is an equivalence.

Since f is 0-prim, the map $(f/A)_0^*: \mathbb{1} \rightarrow (B/A)_1$ admits a right adjoint $(f/A)_{0,*}$ in A_2 . Moreover, the unit $\text{id} \rightarrow (f/A)_{0,*}(f/A)_0^*$ in $A_1 \simeq \text{End}_{A_2}(\mathbb{1})$ is an equivalence, since it is the map $\mathbb{1} \rightarrow (B/A)_0 = \mathbb{1}$. We would like to conclude that $\mathbb{1} \rightarrow (B/A)_1$ is an equivalence because g splits it; this is not quite true, but the following weaker statement is. After applying $g_{1,*}$, the map $\mathbb{1} \rightarrow (B/A)_1$ becomes the map $(A/B)_1 \rightarrow \mathbb{1}$, which is split by $g_0^*: \mathbb{1} \rightarrow (A/B)_1$. Since $g_{1,*}$ preserves the fully faithful adjunction, $\mathbb{1} \rightarrow (B/A)_1$ becomes an equivalence after applying $g_{1,*}$. But g is 1-affine, so $g_{1,*}: A_2 \simeq \text{Mod}_{(A/B)_1}(B_2) \rightarrow B_2$ is conservative, which finishes the proof. $\square$

6.5 Suave maps

Definition 6.18. A map $f: A \rightarrow B$ of Stefanich rings is called n -suave if the following two conditions are satisfied.

(1) For each $m \geq n$, the map $(f/A)_m^*: \mathbb{1} = (A/A)_{m+1} \rightarrow (B/A)_{m+1}$ admits a left adjoint

$$(f/A)_{m,\sharp}: (B/A)_{m+1} \rightarrow (A/A)_{m+1}$$

in A_{m+2} .

This in particular results in a left adjoint $f_{m,\sharp} : B_{m+1} \rightarrow A_{m+1}$ to f_m^* . We define the m -suave dualizing object of B over A as $(B/A)_m^! := f_{m,\sharp}(\mathbb{1}_{B_{m+1}}) \in A_{m+1}$. The counit of the adjunction defines a map

$$(f/A)_{m-1,!} : (B/A)_m^! \rightarrow (A/A)_m$$

in A_{m+1} .

(2) For each $m \geq n+1$, the map $(f/A)_{m-1,!}$ admits a right adjoint

$$(f/A)_{m-1}^! : (A/A)_m \rightarrow (B/A)_m^!$$

in A_{m+1} .

Since the functor $f_{m,\sharp}$ is oplax symmetric monoidal, the object $(B/A)_m^!$ is canonically a commutative coalgebra in A_{m+1} . Moreover, if we write $(-)^{\vee} := \underline{\text{Hom}}_{A_{m+1}}(-, \mathbb{1}) : A_{m+1}^{\text{op}} \rightarrow A_{m+1}$ for the weak duality functor, then we get by adjunction that

$$((B/A)_m^!)^{\vee} = \underline{\text{Hom}}_{A_{m+1}}(f_{m,\sharp}(\mathbb{1}_{B_{m+1}}), \mathbb{1}_{A_{m+1}}) \simeq \underline{\text{Hom}}_{A_{m+1}}(\mathbb{1}_{A_{m+1}}, f_{m,*}(\mathbb{1}_{B_{m+1}})) \simeq (B/A)_m.$$

Under this equivalence, the counit map $(f/A)_{m-1,!} : (B/A)_m^! \rightarrow (A/A)_m$ dualizes to the unit map $(f/A)_{m-1}^* : (A/A)_m \rightarrow (B/A)_m$. Since the duality functor $(-)^{\vee} : (A/A)_{m+1} \rightarrow (A/A)_{m+1}$ preserves adjunctions, we obtain an isomorphism

$$(f/A)_{m-1,\sharp} \simeq ((f/A)_{m-1}^!)^{\vee}$$

of maps $(B/A)_m \rightarrow (A/A)_m$ in A_{m+1} . In particular, condition (2) is a strengthening of the existence of the left adjoint for $(A/A)_m \rightarrow (B/A)_m$. One should think of the existence of the right adjoint $(B/A)_m^! \rightarrow \mathbb{1}$ as the “correct” condition to put, but its mere formulation requires the existence of the left adjoint one level higher.

Lemma 6.19. *Let $f : A \rightarrow B$ be n -suave. For each $m \geq n$, there is a preferred equivalence*

$$f_{m,!} f_m^! \mathbb{1}_{A_{m+1}} \simeq (B/A)_m^!.$$

Proof. Note that the duality functor $(-)^{\vee} : A_{m+2}^{\text{op}} \rightarrow A_{m+2}$ restricts to the identity on $\text{End}_{A_{m+2}}(\mathbb{1})$. In particular, the composite $(f/A)_{m,!} (f/A)_m^! : (A/A)_{m+1} \rightarrow (A/A)_{m+1}$ is identified with the composite $((f/A)_{m,!})^{\vee} ((f/A)_m^!)^{\vee}$. By the previous observation, this is identified with the composite

$$(A/A)_{m+1} \xrightarrow{(f/A)_m^*} (B/A)_{m+1} \xrightarrow{(f/A)_{m,\sharp}} (A/A)_{m+1}.$$

All in all, we conclude that

$$f_{m,!} f_m^! \mathbb{1}_{A_{m+1}} \simeq f_{m,\sharp} f_m^* \mathbb{1}_{A_{m+1}} = (B/A)_m^!.$$

□

Proposition 6.20. (1) *The n -suave maps are closed under cobase change. More precisely, consider a pushout square of Stefanich rings*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & \lrcorner & \downarrow g' \\ C & \xrightarrow{f'} & B \otimes_A C. \end{array}$$

If f is n -suave, then so is f' , and for $m \geq n$ the canonical map

$$g_m^*(B/A)_m^! \rightarrow (B \otimes_A C/C)_m^!$$

is an isomorphism. Moreover, the internal Beck–Chevalley map

$$(f'/A)_{m,\#}(g'/A)_m^* \rightarrow (g/A)_m^*(f/A)_{m,\#}$$

is an isomorphism of morphisms $(B/A)_{m+1} \rightarrow (C/A)_{m+1}$ in A_{m+2} .

(2) The n -suave maps are closed under composition.

Proof. (1) By Construction 5.20, the relative tensor product $B \otimes_A C \in \text{StRing}_{C/}$ is the spectrification of the internal preStefanich ring $(g_m^*(B/A)_m)_{m \geq 0}$ in C . Following Lemma 5.15, the first spectrification step T replaces the m -th level by $\text{End}_{C_{m+1}}^C(g_{m+1}^*(B/A)_{m+1})$. We compute this endomorphism object, show that the result already forms a Stefanich C -algebra in degrees $m \geq n$, and read off both the suave structure of f' and the base change of the dualizing object. Throughout we write $(-)^{\vee} := \underline{\text{Hom}}_{C_{m+1}}(-, \mathbb{1})$ for the weak duality functor.

The endomorphism object. Fix $m \geq n$. The unit map of the algebra $g_{m+1}^*(B/A)_{m+1} \in \text{CAlg}(C_{m+2})$ is the symmetric monoidal functor

$$u := g_{m+1}^*((f/A)_m^*) : \mathbb{1} \rightarrow g_{m+1}^*(B/A)_{m+1}.$$

Since f is n -suave, $(f/A)_m^*$ admits a left adjoint $(f/A)_{m,\#}$ in A_{m+2} . As the 2-functor $g_{m+1}^* : A_{m+2} \rightarrow C_{m+2}$ preserves adjunctions, it follows that u admits the left adjoint $u_{\#} := g_{m+1}^*((f/A)_{m,\#})$. The C_{m+1} -linear adjunction $u_{\#} \dashv u$ then yields

$$\text{End}_{C_{m+1}}^C(g_{m+1}^*(B/A)_{m+1}) = \underline{\text{Hom}}_{g_{m+1}^*(B/A)_{m+1}}(u \mathbb{1}, u \mathbb{1}) \simeq \underline{\text{Hom}}_{(C/C)_{m+1}}(u_{\#} u \mathbb{1}, \mathbb{1}) = (u_{\#} u \mathbb{1})^{\vee}.$$

This is dual to the computation in the prim case Proposition 6.9(3): there one moves the right adjoint $(f/A)_{m,*}$ into the second variable of $\underline{\text{Hom}}$, here one moves the left adjoint $(f/A)_{m,\#}$ out of the first. Exactly as in Observation 6.6, the endomorphism $(f/A)_{m,\#}(f/A)_m^*$ of $\mathbb{1}$ in A_{m+2} evaluates at the unit to $f_{m,\#} f_m^* \mathbb{1} = (B/A)_m^!$; applying g^* and using the compatibility of g^* with evaluation at the unit, we obtain $u_{\#} u \mathbb{1} \simeq g_m^*((B/A)_m^!)$ and therefore

$$\text{End}_{C_{m+1}}^C(g_{m+1}^*(B/A)_{m+1}) \simeq (g_m^*(B/A)_m^!)^{\vee} \quad (m \geq n).$$

This already forms a Stefanich C -algebra. Set $P_m := (g_m^*(B/A)_m^!)^{\vee}$; we verify the Stefanich condition $\text{End}_{P_m}(\mathbb{1}) \simeq P_{m-1}$ for $m \geq n+1$. The second suave condition for f gives an adjunction

$$\mathbb{1} \xrightarrow{(f/A)_{m-1}^!} (B/A)_m^! \xrightarrow{(f/A)_{m-1,!}} \mathbb{1} \quad \in \quad A_{m+1},$$

with composite $(B/A)_{m-1}^! \in A_m = \text{End}_{A_{m+1}}(\mathbb{1})$. Applying $g_m^* : A_{m+1} \rightarrow C_{m+1}$ yields

$$\mathbb{1} \xrightarrow{g_m^*(f/A)_{m-1}^!} g_m^*(B/A)_m^! \xrightarrow{g_m^*(f/A)_{m-1,!}} \mathbb{1} \quad \in \quad C_{m+1},$$

with composite $g_{m-1}^*(B/A)_{m-1}^!$, and dualizing via $(-)^{\vee}$ turns this into an adjunction

$$\mathbb{1} \rightarrow P_m \rightarrow \mathbb{1} \quad \in \quad C_{m+1},$$

again with composite $g_{m-1}^*(B/A)_{m-1}^!$, in which P_m is a commutative algebra with the first map as unit and the second as its left adjoint. By the predual computation above, the composite is a predual of $\text{End}_{P_m}(\mathbb{1})$, so

$$\text{End}_{P_m}(\mathbb{1}) = (g_{m-1}^*(B/A)_{m-1}^!)^\vee = P_{m-1},$$

as required. We conclude that a single application of $\text{End}_{(-)}(\mathbb{1})$ already produces a Stefanich ring in degrees $m \geq n$, so by Lemma 5.15 the spectrification agrees with it there:

$$(B \otimes_A C/C)_m \simeq (g_m^*(B/A)_m^!)^\vee \quad (m \geq n).$$

Warning: Unlike the prim case, the right-hand side cannot in general be replaced by $g_m^*(B/A)_m$: the comparison map $g_m^*(B/A)_m \rightarrow (g_m^*(B/A)_m^!)^\vee$ is an isomorphism only when g_m^* commutes with the weak dual of $(B/A)_m^!$. By Lemma 6.26 below this holds for example when f is in addition n -prim, so that $(B/A)_m^!$ is dualizable. But in general, the relative algebra genuinely requires spectrifying and it is only the dualizing object that base-changes, as we now record.

Suaveness of f' . The previous computation shows that for $m \geq n$, the map $(f'/C)_m^*: \mathbb{1} \rightarrow (B \otimes_A C)_{m+1}$ is the weak dual of the map

$$g_{m+1}^*((f/A)_{m,!}): g_{m+1}^*(B/A)_{m+1}^! \rightarrow g_{m+1}^*(A/A) = \mathbb{1}.$$

Since $(f/A)_{m,!}$ admits a right adjoint $(f/A)_m^!$, so does $g_{m+1}^*((f/A)_{m,!})$, and hence $(f'/C)_m^*$ admits a left adjoint $(f'/C)_{m,\#}$. We may then compute

$$\begin{aligned} (B \otimes_A C/C)_m^! &= (f'/C)_{m,\#}(f'/C)_m^* \mathbb{1} \\ &= g_{m+1}^*((f/A)_{m,!}(f/A)_m^!) \mathbb{1} \\ &\stackrel{(1)}{\simeq} g_m^*((f/A)_{m,!}(f/A)_m^! \mathbb{1}) \\ &\stackrel{(2)}{\simeq} g_m^*((B/A)_m^!), \end{aligned}$$

where (1) holds since g_{m+1}^* restricts to g_m^* on endomorphisms of the unit and (2) is Lemma 6.19. Under this equivalence, the counit map $(f'/C)_{m-1,!}: (B \otimes_A C/C)_m^! \rightarrow \mathbb{1}$ is the image under g_m^* of the counit map $(f/A)_{m-1,!}: (B/A)_m^! \rightarrow \mathbb{1}$, and hence it admits a right adjoint. This shows that f' is n -suave, and that we have

$$g_m^*(B/A)_m^! \xrightarrow{\sim} (B \otimes_A C/C)_m^! \quad (m \geq n).$$

Internal Beck–Chevalley. We now deduce the internal Beck–Chevalley isomorphism by dualizing the argument given in the prim case in Proposition 6.9(3). We need to show that the commutative square

$$\begin{array}{ccc} (A/A)_{m+1} & \xrightarrow{(g/A)_m^*} & (C/A)_{m+1} \\ (f/A)_m^* \downarrow & & \downarrow (f'/A)_m^* \\ (B/A)_{m+1} & \xrightarrow{(g'/A)_m^*} & (B \otimes_A C/C)_{m+1} \end{array}$$

in A_{m+2} is vertically left adjointable. By the computations we have done before, this square may be identified with the naturality square

$$\begin{array}{ccc} (A/A)_{m+1} & \xrightarrow{\eta} & g_{m+1,*}g_{m+1}^*(A/A)_{m+1} \\ ((f/A)_m^!)^\vee \downarrow & & \downarrow g_{m+1,*}g_{m+1}^*((f/A)_m^!)^\vee \\ ((B/A)_{m+1}^!)^\vee & \xrightarrow{\eta} & g_{m+1,*}g_{m+1}^*((B/A)_{m+1}^!)^\vee \end{array}$$

of the unit $\eta: \text{id} \rightarrow g_{m+1,*}g_{m+1}^*$ of the adjunction $g_{m+1}^*: A_{m+2} \rightleftarrows C_{m+2} : g_{m+1,*}$. In other words, the square in question is the image of the functor $((f/A)_m^!)^\vee$ under the 2-functor $A_{m+2} \rightarrow \text{Fun}([1], C_{m+2})$ corresponding to η . Since this 2-functor preserves adjunctions, it follows that the given commutative square is vertically left adjointable.

(2) Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be n -suave, with composite $h: A \rightarrow C$. For $m \geq n$ the functor $f_{m+1,*}: B_{m+2} \rightarrow A_{m+2}$ preserves adjunctions, so the left adjoint of $(g/B)_m^*$ in B_{m+2} results in a left adjoint of the induced map $(g/A)_m^*: (B/A)_{m+1} \rightarrow (C/A)_{m+1}$ in A_{m+2} . Since adjoints compose, this gives a left adjoint for the composite

$$(h/A)_m^*: (A/A)_{m+1} \xrightarrow{(f/A)_m^*} (B/A)_{m+1} \xrightarrow{(g/A)_m^*} (C/A)_{m+1},$$

which is the first suave condition for h . For the second condition, the resulting maps

$$(C/A)_m^! \rightarrow (B/A)_m^! \rightarrow (A/A)_m$$

each admit a right adjoint for $m \geq n+1$ (the first since g is n -suave, the second since f is), hence so does their composite. Thus h is n -suave. $\square$

Lemma 6.21 (Suave projection formula). *Let $f: A \rightarrow B$ be an n -suave map of Stefanich rings. Then for all $m \geq n$, $X \in A_{m+1}$ and $Y \in B_{m+1}$, the canonical map*

$$f_{m,\#}(Y \otimes_{B_{m+1}} f_m^*(X)) \rightarrow f_{m,\#}(Y) \otimes_{A_{m+1}} X$$

is an isomorphism.

Proof. Fix $m \geq n$. By the definition of n -suaveness, the relative pullback

$$(f/A)_m^*: (A/A)_{m+1} \rightarrow (B/A)_{m+1}$$

admits a left adjoint $(f/A)_{m,\#}$ as a 1-morphism in A_{m+2} . Apply the 2-functor underlying

$$U_{m+1}^A: A_{m+2} \rightarrow \text{Mod}_{A_{m+1}}((m+1)\text{Pr}).$$

Since 2-functors preserve adjunctions, this gives an adjunction of A_{m+1} -module objects whose underlying adjunction is

$$f_{m,\#}: B_{m+1} \rightleftarrows A_{m+1} : f_m^*.$$

In particular, the left adjoint $f_{m,\#}$ is A_{m+1} -linear, where the action of A_{m+1} on B_{m+1} is induced by the symmetric monoidal functor f_m^* . Its linearity isomorphism is therefore

$$f_{m,\#}(Y \otimes_{B_{m+1}} f_m^*(X)) \xrightarrow{\sim} f_{m,\#}(Y) \otimes_{A_{m+1}} X.$$

This isomorphism is adjoint to the composite obtained from the unit $Y \rightarrow f_m^* f_{m,\#}(Y)$; hence it is precisely the canonical projection map in the statement. $\square$

The dualizing objects of n -suave maps satisfy a Künneth formula, dual to Lemma 6.12.

Lemma 6.22 (Künneth for suave maps). *Let $f: A \rightarrow B$ and $g: A \rightarrow C$ be n -suave maps of Stefanich rings. Then*

$$(B \otimes_A C/A)_n^! \longrightarrow (B/A)_n^! \otimes_{A_{n+1}} (C/A)_n^!$$

is an isomorphism, where the displayed map is the canonical Künneth map obtained by base change and the projection formula.

Proof. Let $\text{pr}: C \rightarrow B \otimes_A C$ be the cobase change of f along g , so that $\text{pr} \circ g: A \rightarrow B \otimes_A C$ is the structure map. By functoriality of the relative $\sharp$ -pushforward,

$$(B \otimes_A C/A)_n^! = (\text{pr} \circ g)_{n,\sharp}(\mathbb{1}) \xrightarrow{\sim} g_{n,\sharp}(\text{pr}_{n,\sharp}(\mathbb{1})) = g_{n,\sharp}((B \otimes_A C/C)_n^!).$$

Since pr is the cobase change of the n -suave map f along g , Proposition 6.20(1) gives $(B \otimes_A C/C)_n^! \xrightarrow{\sim} g_n^*((B/A)_n^!)$, and therefore

$$(B \otimes_A C/A)_n^! \xrightarrow{\sim} g_{n,\sharp} g_n^*((B/A)_n^!) \xrightarrow{\sim} (B/A)_n^! \otimes_{A_{n+1}} g_{n,\sharp}(\mathbb{1}) = (B/A)_n^! \otimes_{A_{n+1}} (C/A)_n^!,$$

using the projection formula for the left adjoint $g_{n,\sharp}$. By the definition of the base-change and projection maps as mates, this composite is the canonical Künneth map in the statement. $\square$

6.6 Étale maps

Definition 6.23. We say that $f: A \rightarrow B$ is n -étale if f and all its codiagonals are n -suave.

Proposition 6.24. Let $A \in \text{StRing}$ and $n \geq 0$.

- (1) If $f: A \rightarrow B$ is n -étale and $g: A \rightarrow C$ is any map of Stefanich rings, then the cobase change $f': C \rightarrow B \otimes_A C$ is n -étale.
- (2) Let $f: A \rightarrow B$ be n -étale, and let $g: B \rightarrow C$ be a map. Then g is n -suave if and only if gf is n -suave, and g is n -étale if and only if gf is n -étale.

Proof. Since f is n -étale precisely when f and all of its iterated codiagonals are n -suave, both statements follow from Proposition 6.20 by exactly the argument used to deduce Proposition 6.15 from Proposition 6.9. Concretely, the iterated codiagonals of a cobase change (resp. of a composite) are cobase changes (resp. composites of cobase changes) of the iterated codiagonals of the original map(s), and n -suave maps are closed under cobase change and composition by Proposition 6.20. $\square$

6.7 Ambidexterity

In the previous talks we saw that, for $n \geq 1$, all bi-countably presented fpqc stacks give rise to ambidextrous adjunctions, so that the corresponding maps of Stefanich rings are 1-étale and 2-proper. Abstracting that argument, with the general ambidexterity result Proposition 2.17 in place of the geometric input, yields the following.

Proposition 6.25. Let $f: A \rightarrow B$ be a map of Stefanich rings.

- (1) If f and $\nabla_f: B \otimes_A B \rightarrow B$ are both n -prim, then f is $(n+1)$ -suave.
- (2) If f is n -proper, then f is $(n+1)$ -étale.
- (3) If f and ∇_f are n -suave, then f is $(n+1)$ -prim.
- (4) If f is n -étale, then f is $(n+1)$ -proper.

Moreover, in any of these cases, we have an identification $f_{m,\sharp} \simeq f_{m,*}$ for each $m \geq n + 1$.

Proof. By shifting we may assume $n = 0$.

(1) Suppose f and ∇_f are 0-prim. Fix $m \geq 1$ and consider the map

$$(f/A)_m^*: \mathbb{1} = (A/A)_{m+1} \rightarrow (B/A)_{m+1}$$

in the $(\infty, 3)$ -category A_{m+2} (which lies in $(m+2)\text{Pr}$, hence in 3Pr , since $m \geq 1$). As f is 0-prim, $(f/A)_m^*$ admits a right adjoint $(f/A)_{m,*}$ in A_{m+2} ; we verify that the unit η and counit ε of this adjunction admit right adjoints, so that Proposition 2.17 applies.

For η , evaluation at the unit gives an equivalence $\text{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$, under which, as in the observation following Definition 6.5, the unit $\eta: \text{id} \rightarrow (f/A)_{m,*}(f/A)_m^*$ corresponds to the map $(f/A)_{m-1}^*: \mathbb{1} \rightarrow (B/A)_m$. Since f is 0-prim and $m-1 \geq 0$, this map admits a right adjoint in A_{m+1} .

For ε , the counit lies in $\text{End}_{A_{m+2}}((B/A)_{m+1})$. As in the proof of the ambidexterity property of the functors $f_{n,*}$ in Section 2.4, there is an equivalence

$$\text{End}_{A_{m+2}}((B/A)_{m+1}) \simeq \text{Mod}_{(B \otimes_A B/A)_m}(A_{m+2}),$$

under which ε corresponds to the relative map of the codiagonal $\nabla_f: B \otimes_A B \rightarrow B$ at level m . Since ∇_f is 0-prim, this map admits a right adjoint.

Hence Proposition 2.17, applied in A_{m+2} , shows that $(f/A)_{m,*}$ is also a left adjoint $(f/A)_{m,\sharp}$ of $(f/A)_m^*$, and provides a canonical identification $(f/A)_{m,\sharp} \simeq (f/A)_{m,*}$ for $m \geq 1$. By Lemma 6.26 this common functor preserves dualizable objects; in particular $(B/A)_m^! = (f/A)_{m,\sharp}(\mathbb{1})$ is dualizable for $m \geq 1$. As recorded after Definition 6.18, the second suave condition is automatic once $(B/A)_m^!$ is dualizable. Thus f is 1-suave.

(2) If f is 0-proper, then every iterated codiagonal ξ of f has the property that ξ and ∇_ξ are again 0-prim, since the iterated codiagonals of ξ are among those of f . By part (1), each such ξ is 1-suave, so f is 1-étale.

(3) Suppose f and ∇_f are 0-suave. For $m \geq 1$, the map $(f/A)_m^*$ admits a left adjoint $(f/A)_{m,\sharp}$ in A_{m+2} , as f is 0-suave. We apply the dual version of Proposition 2.17 (cf. the remark following it), and so must check that the unit and counit of $(f/A)_{m,\sharp} \dashv (f/A)_m^*$ admit right adjoints. The counit corresponds, via the equivalence $\text{End}_{A_{m+2}}(\mathbb{1}) \xrightarrow{\sim} A_{m+1}$, to the map $(f/A)_{m-1,!}$, which admits a right adjoint by the second condition in the definition of 0-suave for f . The unit corresponds, under the relation-object identification used in part (1), to the analogous datum for the codiagonal ∇_f , which admits a right adjoint since ∇_f is 0-suave. Hence the dual ambidexterity result shows that $(f/A)_{m,\sharp}$ is also a right adjoint of $(f/A)_m^*$ for $m \geq 1$, so f is 1-prim.

(4) If f is 0-étale, then every iterated codiagonal ξ of f has ξ and ∇_ξ both 0-suave, so ξ is 1-prim by part (3); hence f is 1-proper.

Finally, the argument above produces the ambidextrous identification for $(f/A)_m^*$ with $m \geq 1$ in the shifted ($n = 0$) picture; unshifting, the canonical identification of Proposition 2.17 reads $(f/A)_{m,\sharp} \simeq (f/A)_{m,*}$ for $m \geq n + 1$ in cases (1) and (3), and the same identification holds in cases (2) and (4), applied to f itself. By Lemma 6.26 this common functor preserves dualizable objects. $\square$

The following lemma was used above; in particular, being both n -prim and n -suave forces $f_{m,*}$ and $f_{m,\#}$ to preserve dualizable objects.

Lemma 6.26. *Let $\varphi: C \rightarrow D$ be a morphism in $\mathbf{CAlg}(\mathbf{Pr}^{\mathbf{L}})$, and suppose φ admits both a C -linear left adjoint $\varphi_{\#}$ and a C -linear right adjoint φ_* . Then both $\varphi_{\#}$ and φ_* preserve dualizable objects.*

Proof. Let $N \in D$ be dualizable, with dual $N^{\vee}$. Since φ is symmetric monoidal, it preserves dualizable objects, and C -linearity of the adjoints yields the projection formulas

$$\varphi_{\#}(Y \otimes \varphi(M)) \simeq \varphi_{\#}(Y) \otimes M, \quad \varphi_*(Y \otimes \varphi(M)) \simeq \varphi_*(Y) \otimes M$$

for $Y \in D$ and $M \in C$. Using the adjunction $\varphi_{\#} \dashv \varphi \dashv \varphi_*$ together with these formulas, we compute, naturally in $M \in C$,

$$\mathrm{Hom}_C(\varphi_{\#}N, M) \simeq \varphi_* \mathrm{Hom}_D(N, \varphi(M)) \simeq \varphi_*(N^{\vee} \otimes \varphi(M)) \simeq \varphi_*(N^{\vee}) \otimes M.$$

Setting $M = \mathbb{1}$ gives $\mathrm{Hom}_C(\varphi_{\#}N, \mathbb{1}) \simeq \varphi_*(N^{\vee})$, so the displayed natural equivalence becomes $\mathrm{Hom}_C(\varphi_{\#}N, M) \simeq \mathrm{Hom}_C(\varphi_{\#}N, \mathbb{1}) \otimes M$. Hence $\varphi_{\#}N$ is dualizable, with dual $\varphi_*(N^{\vee})$. Applying this conclusion to $N^{\vee}$ shows that $\varphi_{\#}(N^{\vee})$ is dualizable with dual $\varphi_*(N)$, so $\varphi_*(N)$ is dualizable as well. $\square$

Corollary 6.27. *Let $n \geq 0$ and assume that $f: A \rightarrow B$ is a map of Stefanich rings which is both n -suave and n -prim. Then the suave dualizing object $(B/A)_m^!$ is dualizable in A_{m+1} for all $m \geq n$, with dual $(B/A)_m$.*

Proof. We have $(B/A)_m^! = f_{m,\#}(\mathbb{1}_{A_{m+1}})$, which is dualizable by Lemma 6.26. Its dual is $f_{m,*}(\mathbb{1}_{A_{m+1}}) = (B/A)_m$. $\square$

7 Gestalten

Talk given by Giovanni.

In the previous talk we introduced suave and étale maps of Stefanich rings. The goal of this talk is to assemble the n -étale maps over a fixed Stefanich ring A into a category and to show that its opposite is a *countably toposic category*, in the sense made precise below. Passing to opposites throughout, this exhibits the category of *Gestalten*, the opposite of StRing , as a setting in which the étale maps play the role of the open immersions in a topos.

Throughout we freely use the relative objects $(B/A)_m$ and $(B/A)_m^!$ attached to a map $f: A \rightarrow B$ of Stefanich rings, together with the relative maps $(f/A)_m^*$, $(f/A)_{m,\#}$, $(f/A)_{m,!}$, etc., from Chapter 6.

7.1 Countable limits of étale algebras

Recall from Definitions 6.18 and 6.23 that, for $n \geq 0$, a map $f: A \rightarrow B$ is n -suave if for each $m \geq n$ the relative map $(f/A)_m^*: \mathbb{1} = (A/A)_{m+1} \rightarrow (B/A)_{m+1}$ admits a left adjoint $(f/A)_{m,\#}$ in A_{m+2} , whose value $(B/A)_m^! := (f/A)_{m,\#}(\mathbb{1})$ we call the relative *dualizing object*, and if moreover, for each $m \geq n+1$, the resulting counit $(f/A)_{m-1,!}: (B/A)_m^! \rightarrow (A/A)_m$ admits a right adjoint; and that f is n -étale if f and all its iterated codiagonals are n -suave. By Propositions 6.20 and 6.24, the n -suave and n -étale maps are stable under cobase change and composition, the dualizing objects satisfy the base-change formula

$$g_m^*(B/A)_m^! \simeq (B \otimes_A C/C)_m^! \quad (m \geq n)$$

for any $g: A \rightarrow C$, and n -étale maps enjoy right cancellation: if $f: A \rightarrow B$ is n -étale and $g: B \rightarrow C$ is arbitrary, then g is n -suave (resp. n -étale) if and only if gf is.

We now turn to the behaviour of suave algebras under *limits*. The classical theory offers a cautionary example.

Example 7.1. Static étale algebras are not closed under countable limits. For instance, if k is a field, then every finite product k^m is étale over k , but the limit $\lim(\cdots \rightarrow k^3 \rightarrow k^2 \rightarrow k) \simeq \prod_{\mathbb{N}} k$ is not étale over k , since it is not of finite presentation (it is not even noetherian).

Étale Stefanich rings do not suffer from this defect, essentially because of ambidexterity. The key adjointability input is the following.

Lemma 7.2. *Let $\mathbb{C} \in 2\text{Pr}$ be a 2-presentable category and let I be a countable category.*

- (1) *Let $X_\bullet: I \rightarrow \mathbb{C}$ be a diagram and let $F: Y \rightarrow \lim_{i \in I} X_i$ be a morphism. Write $F_i := \text{pr}_i \circ F: Y \rightarrow X_i$ for its composite with the projection to X_i . If each F_i admits a left adjoint*

$F_{i,\#}: X_i \rightarrow Y$, then F admits a left adjoint $F_{\#}$, and there is a canonical equivalence

$$F_{\#}F \simeq \operatorname{colim}_{i \in I^{\operatorname{op}}} F_{i,\#}F_i \in \operatorname{End}_{\mathbb{C}}(Y).$$

- (2) Let $X_{\bullet}: I \rightarrow \mathbb{C}$ be a diagram and let $G: \operatorname{colim}_{i \in I} X_i \rightarrow Y$ be a morphism. Write $G_i := G \circ \iota_i: X_i \rightarrow Y$ for its composite with the colimit structure map. If each G_i admits a right adjoint $G_{i,*}: Y \rightarrow X_i$, then G admits a right adjoint G_* , and there is a canonical equivalence

$$GG_* \simeq \operatorname{colim}_{i \in I} G_i G_{i,*} \in \operatorname{End}_{\mathbb{C}}(Y).$$

Proof. The two statements are dual, so we treat (1). As in the proof of Lemma 6.10, the represented functors $\mathbb{C}(-, Z): \mathbb{C}^{\operatorname{op}} \rightarrow \mathbf{1Pr}$ jointly detect and reflect adjunctions of 1-morphisms (and the displayed formula may be checked after applying them), so we may assume $\mathbb{C} = \mathbf{1Pr} = \operatorname{Pr}_{\aleph_1}^{\mathbf{L}}$. There a countable limit $\lim_{i \in I} X_i$ is computed as the corresponding limit in $\mathbf{Cat}$, equivalently as the limit in $\operatorname{Pr}^{\mathbf{R}}$ formed along the right adjoints of the projections. A morphism F into such a limit admits a left adjoint as soon as each component F_i does, by assembling the partial left adjoints; this is where countability of I enters, ensuring the relevant Beck–Chevalley colimits exist and are computed in $\operatorname{Pr}^{\mathbf{L}}$. Tracing through the limit presentation, the composite $F_{\#}F$ is computed as the colimit $\operatorname{colim}_{i \in I^{\operatorname{op}}} F_{i,\#}F_i$, the transition maps being the mates of the identifications $X_{\alpha}F_i \simeq F_j$ for $\alpha: i \rightarrow j$ in I . $\square$

Proposition 7.3. *Let $A \in \operatorname{StRing}$ and $n \geq 0$.*

- (1) *The n -suave A -algebras are closed under countable limits in $\operatorname{StRing}_{A/}$. Moreover, for $m \geq n$ the formation of the dualizing object sends countable limits of n -suave algebras to colimits: if $B = \lim_{i \in I} B_i$ with each B_i n -suave over A , then*

$$(B/A)_m^! \simeq \operatorname{colim}_{i \in I^{\operatorname{op}}} (B_i/A)_m^!.$$

- (2) *For any map $g: A \rightarrow C$, the base-change functor $- \otimes_A C: \operatorname{StRing}_{A/} \rightarrow \operatorname{StRing}_{C/}$ preserves countable limits of n -suave algebras.*

Proof. By shifting (Lemma 5.12) we may assume $n = 0$. Let I be countable and let $B_{\bullet}: I \rightarrow \operatorname{StRing}_{A/}$ be a diagram of 0-suave algebras, with limit $B := \lim_{i \in I} B_i$, which exists since $\operatorname{StRing}_{A/}$ is presentable. Since limits of internal Stefanich rings are computed levelwise (Proposition 5.8), we have $(B/A)_m \simeq \lim_{i \in I} (B_i/A)_m$ in $\operatorname{CAlg}(A_{m+1})$ for every m .

Part (1). Fix $m \geq 0$. To produce the dualizing object we must show that the map

$$\mathbb{1} = (A/A)_{m+1} \rightarrow (B/A)_{m+1} \simeq \lim_{i \in I} (B_i/A)_{m+1}$$

admits a left adjoint in A_{m+2} . For each i the map $\mathbb{1} \rightarrow (B_i/A)_{m+1}$ admits a left adjoint, since B_i is 0-suave over A . Applying Lemma 7.2(1) in A_{m+2} produces the desired left adjoint, together with the identification

$$(B/A)_m^! \simeq \operatorname{colim}_{i \in I^{\operatorname{op}}} (B_i/A)_m^!$$

in A_{m+1} . For the second suave condition, under this identification the counit $(B/A)_m^! \rightarrow \mathbb{1}$ is the colimit over I^{op} of the counits $(B_i/A)_m^! \rightarrow \mathbb{1}$, each of which admits a right adjoint because B_i is

0-suave. As $(B/A)_m^!$ is the countable colimit of the $(B_i/A)_m^!$, Lemma 7.2(2) shows that the counit $(B/A)_m^! \rightarrow 1$ admits a right adjoint. Hence B is 0-suave, proving (1).

Part (2). Fix $g : A \rightarrow C$. As 0-suave maps are stable under base change (Proposition 6.20(1)), the C -algebras $B \otimes_A C$ and $B_i \otimes_A C$ are 0-suave, and we must show that the canonical map

$$\left(\lim_{i \in I} B_i \right) \otimes_A C \rightarrow \lim_{i \in I} (B_i \otimes_A C)$$

is an equivalence. By the base-change formula of Proposition 6.20(1) and the fact that g_m^* preserves colimits, we have for every $m \geq 0$

$$(B \otimes_A C/C)_m^! \simeq g_m^*(B/A)_m^! \simeq g_m^* \operatorname{colim}_{i \in I^{\text{op}}} (B_i/A)_m^! \simeq \operatorname{colim}_{i \in I^{\text{op}}} (B_i \otimes_A C/C)_m^!.$$

On the other hand, applying part (1) to the diagram $i \mapsto B_i \otimes_A C$ of 0-suave C -algebras gives

$$\left(\left(\lim_{i \in I} (B_i \otimes_A C) \right) / C \right)_m^! \simeq \operatorname{colim}_{i \in I^{\text{op}}} (B_i \otimes_A C/C)_m^!.$$

Thus both C -algebras have the same dualizing objects in every degree. Since for suave algebras $((B/A)_m^!)^\vee \simeq (B/A)_m$ (see the discussion after Definition 6.18), they also have the same relative levels, showing that the canonical map is an equivalence. $\square$

7.2 Countably toposic categories

Recall the definition of a topos:

Definition 7.4. A *topos* is an accessible left exact localization of a presheaf category $\mathbf{PSh}(C)$ on a small category C ; that is, a category $\mathcal{X}$ for which there exist a small category C and an adjunction $\mathbf{PSh}(C) \rightleftarrows \mathcal{X}$ whose left adjoint preserves finite limits and whose right adjoint is fully faithful and accessible.

In particular, every topos is presentable. In ordinary category theory, Giraud's theorem characterizes topoi by intrinsic exactness properties; the same holds in the ∞ -categorical setting, the third axiom below (effectivity of groupoids) being the one that distinguishes ∞ -topoi from 1-topoi.

Definition 7.5. A presentable category $\mathcal{X}$ *satisfies the Giraud axioms* if:

- (G1) *Colimits are universal:* for every map $f : X \rightarrow Y$ in $\mathcal{X}$, the pullback functor $f^* : \mathcal{X}_Y \rightarrow \mathcal{X}_X$ preserves all colimits;
- (G2) *Coproducts are disjoint:* for all $X, Y \in \mathcal{X}$, the pullback of the span $X \rightarrow X \sqcup Y \leftarrow Y$ is initial;
- (G3) *Groupoid objects are effective:* for every groupoid object $X_\bullet : \Delta^{\text{op}} \rightarrow \mathcal{X}$ with geometric realization $|X_\bullet|$, the canonical map from $X_\bullet$ to the Čech nerve of $X_0 \rightarrow |X_\bullet|$ is an equivalence; equivalently, $X_n \simeq X_0 \times_{|X_\bullet|} \cdots \times_{|X_\bullet|} X_0$ for all $n \geq 0$.

Theorem 7.6 (Lurie [Lur09, Theorem 6.1.0.6]). *Let $\mathcal{X}$ be a presentable category. The following are equivalent:*

- (1) $\mathcal{X}$ is a topos.

- (2) For some (equivalently, any) regular cardinal κ such that $\mathcal{X}$ is κ -compactly generated and $\mathcal{X}^\kappa$ is closed under finite limits, the colimit-preserving extension $\mathrm{PSh}(\mathcal{X}^\kappa) \rightarrow \mathcal{X}$ of the inclusion preserves finite limits.
- (3) $\mathcal{X}$ satisfies the Giraud axioms.

We are interested in the following ‘countable analogue’ of the definition of a topos:

Definition 7.7. A small category $\mathcal{Y}$ with finite limits and countable colimits is a *countably toposic category* if the presentable category $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ is a topos.

Remark 7.8. Scholze uses “countably topos”, and Giovanni used “countable topos” during his talk; we avoid both, the former being grammatically awkward and the latter misleading, since the category in question is not countable and not a topos.

Since $\mathcal{Y}$ has countable colimits, the unit $\mathcal{Y} \rightarrow \mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ identifies $\mathcal{Y}$ with the full subcategory $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})^{\aleph_1}$ of $\aleph_1$ -compact objects. We deduce the following characterization.

Proposition 7.9. Let $\mathcal{Y}$ be a small category with finite limits and countable colimits. The following are equivalent.

- (1) $\mathcal{Y}$ is a countably toposic category.
- (2) The countable-colimit-preserving extension $\mathrm{PSh}(\mathcal{Y})^{\aleph_1} \rightarrow \mathcal{Y}$ of the identity preserves finite limits.
- (3) There exist a small category C with finite limits and countable colimits and a finite-limit-preserving, countable-colimit-preserving localization $\mathrm{PSh}(C)^{\aleph_1} \rightarrow \mathcal{Y}$.
- (4) $\mathcal{Y}$ satisfies the countable Giraud axioms: countable colimits are universal, coproducts are disjoint, and groupoid objects are effective.

Proof. (1) $\Rightarrow$ (2): If $\mathcal{Y}$ is countably toposic then $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ is a topos. Using $\mathcal{Y} \simeq \mathrm{Ind}_{\aleph_1}(\mathcal{Y})^{\aleph_1}$ together with the finite limits of $\mathcal{Y}$, we may apply Theorem 7.6(2) with $\kappa = \aleph_1$: the colimit-preserving functor $\mathrm{PSh}(\mathcal{Y}) \rightarrow \mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ preserves finite limits. Restricting to $\aleph_1$ -compact objects, which are closed under finite limits in both source and target by Proposition A.10 and Theorem 7.6(2), yields (2).

(2) $\Rightarrow$ (3): Take $C = \mathcal{Y}$. The functor $\mathrm{PSh}(\mathcal{Y})^{\aleph_1} \rightarrow \mathcal{Y}$ preserves countable colimits and finite limits, and it is a localization: its fully faithful right adjoint is induced by the Yoneda embedding $\mathcal{Y} \hookrightarrow \mathrm{PSh}(\mathcal{Y})^{\aleph_1}$.

(3) $\Rightarrow$ (4): Let $L: \mathrm{PSh}(C)^{\aleph_1} \rightarrow \mathcal{Y}$ be a left exact localization as in (3). First, $\mathrm{PSh}(C)^{\aleph_1}$ satisfies the countable Giraud axioms: indeed $\mathrm{PSh}(C) \simeq \mathrm{Ind}_{\aleph_1}(\mathrm{PSh}(C)^{\aleph_1})$ is a presheaf topos, hence satisfies the Giraud axioms, and these restrict to the $\aleph_1$ -compact objects. Here countable colimits of $\aleph_1$ -compact objects remain $\aleph_1$ -compact, and the latter are closed under finite limits by Proposition A.10. It then suffices to know that the countable Giraud axioms are stable under left exact localizations, which is Lemma 7.10 below.

(4) $\Rightarrow$ (1): Since $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ is presentable, by Theorem 7.6 it is enough to verify the Giraud axioms for $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})$. Every small colimit in $\mathrm{Ind}_{\aleph_1}(\mathcal{Y})$ is an $\aleph_1$ -filtered colimit of countable colimits of

objects of $\mathcal{Y}$, while finite limits commute with $\aleph_1$ -filtered colimits. Thus universality of colimits and disjointness of coproducts propagate from $\mathcal{Y}$ to $\text{Ind}_{\aleph_1}(\mathcal{Y})$. For effectivity, since Δ^{op} is countable, the canonical functor

$$\text{Ind}_{\aleph_1}(\text{Fun}(\Delta^{\text{op}}, \mathcal{Y})) \longrightarrow \text{Fun}(\Delta^{\text{op}}, \text{Ind}_{\aleph_1}(\mathcal{Y}))$$

is an equivalence. The full subcategory of groupoid objects is defined by finite-limit conditions, so it is likewise generated under $\aleph_1$ -filtered colimits by the groupoid objects in $\mathcal{Y}$. Geometric realization commutes with these filtered colimits, as do the finite limits forming the Čech nerve. Effectivity in $\mathcal{Y}$ therefore implies effectivity in $\text{Ind}_{\aleph_1}(\mathcal{Y})$. $\square$

Lemma 7.10. *The countable Giraud axioms are stable under left exact localizations.*

Proof. Let $L: \mathcal{X} \rightleftarrows \mathcal{Y} : \iota$ be a left exact localization, with $\mathcal{X}$ satisfying the countable Giraud axioms. Finite limits in $\mathcal{Y}$ are computed in $\mathcal{X}$ through the fully faithful right adjoint ι , while colimits in $\mathcal{Y}$ are obtained by applying L to the corresponding colimits in $\mathcal{X}$. Universality of countable colimits and disjointness of coproducts then transport from $\mathcal{X}$ to $\mathcal{Y}$ exactly as in the localization step of the proof of Theorem 7.6, using that L is left exact and colimit-preserving and that ι is left exact and fully faithful. Effectivity of groupoid objects is preserved because L carries groupoid objects to groupoid objects and commutes with the formation of Čech nerves and geometric realizations, while ι reflects equivalences. $\square$

7.3 The category of étale algebras as a topos

We now prove that $(\text{StRing}_{A/}^{n\text{-ét}})^{\text{op}}$ is a countably toposic category, where $\text{StRing}_{A/}^{n\text{-ét}} \subseteq \text{StRing}_{A/}$ denotes the full subcategory on the n -étale algebras. We first record that this is a legitimate small category with finite limits.

Remark 7.11. Let $A \in \text{StRing}$ and $n \geq 0$.

- (1) $\text{StRing}_{A/}^{n\text{-ét}}$ has finite colimits: the identity $A = A$ is n -étale, giving an initial object, and n -étale maps are stable under cobase change (Proposition 6.24(1)), giving pushouts.
- (2) $\text{StRing}_{A/}^{n\text{-ét}}$ is essentially small. Indeed, if $A \rightarrow B$ is n -étale then it is $(n+1)$ -proper (Proposition 6.25), and for $m \geq n+1$ ambidexterity identifies $(B/A)_m \simeq (B/A)_m^!$ as a dualizable object of A_{m+1} (Proposition 6.25 and Lemma 6.26). Since the unit of A_{m+1} is $\aleph_1$ -compact, the dualizable objects are $\aleph_1$ -compact and hence form an essentially small subcategory (cf. Appendix A); as a Stefanich algebra under A is determined by the sequence $((B/A)_m)_{m \geq n+1}$ via the identification $\text{StRing}_{A/} \simeq \lim(\text{CAlg}(A_1) \leftarrow \text{CAlg}(A_2) \leftarrow \cdots)$, the n -étale A -algebras form an essentially small category.

Consequently $C := (\text{StRing}_{A/}^{n\text{-ét}})^{\text{op}}$ is a small category with finite limits.

The key technical input is the following. Let C be a small category with finite limits, and suppose given an $\aleph_1$ -accessible left exact localization $L: \text{PSh}(C) \rightleftarrows \text{Shv}(C) : \iota$. (The notation is suggestive, but we do *not* assume that $\text{Shv}(C)$ is the category of sheaves for a Grothendieck topology.) Assume

that C is *subcanonical*, meaning that the composite $C \xrightarrow{y} \mathrm{PSh}(C) \xrightarrow{L} \mathrm{Shv}(C)$ is fully faithful. Fix $A \in \mathrm{StRing}$ and let

$$\mathcal{O}: C^{\mathrm{op}} \rightarrow \mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}}$$

be a finite-colimit-preserving functor which is a *sheaf*, in the sense that its right Kan extension to $\mathrm{PSh}(C)$ inverts L -equivalences. Equivalently, the right Kan extension of $\mathcal{O}$ along y^{op} factors through a limit-preserving functor

$$\mathcal{O}: \mathrm{Shv}(C)^{\mathrm{op}} \rightarrow \mathrm{StRing}_{A/}, \quad \mathcal{O}(F) \simeq \lim_{(y(X) \rightarrow F) \in C/F} \mathcal{O}(X),$$

which exists since $\mathrm{StRing}_{A/}$ is presentable.

Theorem 7.12. *In the situation above, the functor $\mathcal{O}: (\mathrm{Shv}(C)^{\aleph_1})^{\mathrm{op}} \rightarrow \mathrm{StRing}_{A/}$ factors through $\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}}$, and it preserves finite colimits.*

The proof occupies the rest of this section; we first deduce the main result.

Theorem 7.13. *Let $A \in \mathrm{StRing}$ and $n \geq 0$. Then $(\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}})^{\mathrm{op}}$ is a countably toposic category.*

Proof. By Remark 7.11, $C := (\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}})^{\mathrm{op}}$ is a small category with finite limits. Equip it with the trivial topology, so that $\mathrm{PSh}(C) = \mathrm{Shv}(C)$ and C is trivially subcanonical, and apply Theorem 7.12 to the identity sheaf $\mathcal{O} = \mathrm{id}: C^{\mathrm{op}} = \mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}} \rightarrow \mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}}$, which preserves finite colimits. We obtain a finite-colimit-preserving functor

$$(\mathrm{PSh}(C)^{\aleph_1})^{\mathrm{op}} \rightarrow \mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}}.$$

Passing to opposite categories gives a functor

$$\mathrm{PSh}(C)^{\aleph_1} \rightarrow (\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}})^{\mathrm{op}} = C$$

which preserves finite limits and countable colimits and is a localization with fully faithful right adjoint the Yoneda embedding. By the equivalence (2) $\Leftrightarrow$ (1) of Proposition 7.9, C is a countably toposic category. $\square$

Corollary 7.14. *The full subcategory $\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}} \subseteq \mathrm{StRing}_{A/}$ is closed under countable limits.*

Proof. By Theorem 7.13 the category $(\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}})^{\mathrm{op}}$ is countably toposic, hence in particular has countable colimits; equivalently, $\mathrm{StRing}_{A/}^{n\text{-}\acute{\mathrm{e}}\mathrm{t}}$ has countable limits. That these agree with the limits computed in $\mathrm{StRing}_{A/}$ follows from Proposition 7.3(1): a countable limit of n -suave algebras formed in $\mathrm{StRing}_{A/}$ is again n -suave. The iterated codiagonals of such a limit are computed as countable limits of the corresponding codiagonals, using that base change preserves countable limits of suave algebras (Proposition 7.3(2)), and these are again n -suave, so the limit is n -étale. $\square$

Proof of Theorem 7.12. Since the original functor sends the terminal object of C to A and fiber products in C to tensor products, the finite-colimit assertion for its extension will follow from the Künneth formula proved below. We must also show that for every $\aleph_1$ -compact $X \in \mathrm{Shv}(C)^{\aleph_1}$ the algebra $\mathcal{O}(X)$ is n -étale over A . Our strategy is to show:

- for every map $f: Y \rightarrow X$ in $\mathrm{Shv}(C)^{\aleph_1}$ the map $\mathcal{O}(f): \mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ is n -suave (Claim 1);

- $\mathcal{O}$ satisfies a Künneth formula (Claim 2).

Granting both, the iterated codiagonals of $\mathcal{O}(f)$ are again of the form $\mathcal{O}(f')$: by Künneth they correspond to the diagonals $Y \rightarrow Y \times_X Y, Y \rightarrow Y \times_{(Y \times_X Y)} Y, \dots$, hence are n -suave by Claim 1. Thus $\mathcal{O}(f)$ is n -étale; taking X to be the terminal object of $\mathrm{Shv}(C)^{\aleph_1}$, for which $\mathcal{O}(X) = \mathcal{O}(*) = A$, shows that $\mathcal{O}(Y)$ is n -étale over A .

Claim 1. *For every map $f: Y \rightarrow X$ in $\mathrm{Shv}(C)^{\aleph_1}$, the map $\mathcal{O}(f): \mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ is n -suave.*

Suppose first that $X \in C$ is representable. Since Y is $\aleph_1$ -compact, in the slice $\mathrm{Shv}(C)_{/X}^{\aleph_1}$ it may be written as a countable colimit $Y \simeq \mathrm{colim}_{i \in I} Y_i$ of representables $Y_i \in C_{/X}$. As $\mathcal{O}$ sends countable colimits in $\mathrm{Shv}(C)$ to countable limits in $\mathrm{StRing}_{A/}$,

$$\mathcal{O}(Y) \simeq \lim_{i \in I} \mathcal{O}(Y_i) \in \mathrm{StRing}_{\mathcal{O}(X)/}.$$

Both $A \rightarrow \mathcal{O}(X)$ and $A \rightarrow \mathcal{O}(Y_i)$ are n -étale, so the second-out-of-three property of Proposition 6.24(2) shows that each relative map $\mathcal{O}(X) \rightarrow \mathcal{O}(Y_i)$ is n -étale, hence n -suave. By Proposition 7.3(1) the countable limit $\mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ is again n -suave, with

$$(\mathcal{O}(Y)/\mathcal{O}(X))_m^! \simeq \mathrm{colim}_{i \in I^{\mathrm{op}}} (\mathcal{O}(Y_i)/\mathcal{O}(X))_m^! \quad (m \geq n).$$

Moreover, Proposition 7.3(2) shows that for a map $g: X' \rightarrow X$ of representables base change is compatible:

$$\mathcal{O}(X') \otimes_{\mathcal{O}(X)} \mathcal{O}(Y) \simeq \mathcal{O}(X' \times_X Y), \quad \mathcal{O}(X') \otimes_{\mathcal{O}(X)} (\mathcal{O}(Y)/\mathcal{O}(X))_m^! \simeq (\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^!.$$

Now let $X \in \mathrm{Shv}(C)^{\aleph_1}$ be arbitrary. Here we cannot apply Proposition 7.3 directly, since the base need not be representable; we argue by descent. For each map $g: X' \rightarrow X$ with $X' \in C$ representable, form the pullback $X' \times_X Y$, which is again $\aleph_1$ -compact, since $\aleph_1$ -compact sheaves are closed under finite limits (Corollary A.13). By the representable case the map $\mathcal{O}(X') \rightarrow \mathcal{O}(X' \times_X Y)$ is n -suave, with dualizing objects $(\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^! \in \mathcal{O}(X')_{m+1}$ for $m \geq n$. For a further representable $X'' \rightarrow X' \rightarrow X$, the base-change compatibility above gives

$$\mathcal{O}(X'') \otimes_{\mathcal{O}(X')} (\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^! \simeq (\mathcal{O}(X'' \times_X Y)/\mathcal{O}(X''))_m^!,$$

so the assignment $X' \mapsto (\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^!$ is a descent datum for the sheaf $\mathcal{O}(-)_{m+1}$. As $\mathcal{O}$ is a sheaf, it descends to a cocommutative coalgebra object $B_m^! \in \mathcal{O}(X)_{m+1}$ with $g^* B_m^! \simeq (\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^!$ for every representable $g: X' \rightarrow X$. The trick we use is to relate properties of $B_m^!$ to properties of the map $\mathcal{O}(X) \rightarrow \mathcal{O}(Y)$.

For $m \geq n+1$, the counit $(\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^! \rightarrow \mathbb{1}$ admits a right adjoint, since $\mathcal{O}(X') \rightarrow \mathcal{O}(X' \times_X Y)$ is n -suave. These right adjoints are compatible with base change, so the counit $h_{m-1,!}: B_m^! \rightarrow \mathbb{1}$ admits a right adjoint in $\mathcal{O}(X)_{m+1}$. It follows that the induced map $\mathbb{1} \rightarrow (B_m^!)^\vee$ on duals admits a *left* adjoint. This dual may be computed as follows:

$$\begin{aligned} (B_m^!)^\vee &\simeq \lim_{g: X' \rightarrow X} g_{m,*}((\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m^!)^\vee \\ &\simeq \lim_{g: X' \rightarrow X} g_{m,*}(\mathcal{O}(X' \times_X Y)/\mathcal{O}(X'))_m \\ &\simeq (\mathcal{O}(Y)/\mathcal{O}(X))_m, \end{aligned}$$

where we used $((B/A)_m^!)^\vee \simeq (B/A)_m$ for suave algebras and, in the last step, that $\mathcal{O}$ is a sheaf (so that $\mathcal{O}(Y) \simeq \lim_{g: X' \rightarrow X} \mathcal{O}(X' \times_X Y)$). This in particular shows that the map $(f/\mathcal{O}(X))_{m-1}^*: \mathbb{1} \rightarrow (B_m^!)^\vee = (\mathcal{O}(Y)/\mathcal{O}(X))_m$ admits a left adjoint

$$(f/\mathcal{O}(X))_{m-1,\sharp}: (\mathcal{O}(Y)/\mathcal{O}(X))_m \rightarrow (\mathcal{O}(X)/\mathcal{O}(X))_m.$$

We claim that the value at the unit is $B_{m-1}^!$, showing that $B_{m-1}^! \simeq (\mathcal{O}(Y)/\mathcal{O}(X))_{m-1}^!$. As $m \geq n+1$, this identifies the dualizing objects in every degree at least n . Together with the right adjoints to the counits $B_m^! \rightarrow \mathbb{1}$ for $m \geq n+1$, this proves both conditions for n -suaveness.

To see this, recall from Lemma 6.19 that we may alternatively compute it as the image of the unit under the composite

$$\mathbb{1} \xrightarrow{h_{m-1}^!} B_m^! \xrightarrow{h_{m-1,!}} \mathbb{1}.$$

By descent, this is

$$h_{m-1,!} h_{m-1}^! \mathbb{1} \simeq \lim_{g: X' \rightarrow X} g_{m-1,!} g_{m-1}^! \mathbb{1} \simeq B_{m-1}^!.$$

We conclude that $\mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ is n -suave.

Claim 2 (Künneth formula). *For every pullback square*

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

in $\mathrm{Shv}(C)^{\mathbb{S}^1}$, the natural map $\mathcal{O}(X') \otimes_{\mathcal{O}(X)} \mathcal{O}(Y) \rightarrow \mathcal{O}(Y')$ is an equivalence in $\mathrm{StRing}_{A/}$.

Set $Y' = X' \times_X Y$. Suppose first that $X' = S \in C$ is representable, with structure map $g: S \rightarrow X$. By Claim 1 and the construction of $B_m^!$, the dualizing objects of $\mathcal{O}(X) \rightarrow \mathcal{O}(Y)$ pull back along g to $\mathcal{O}(S) \otimes_{\mathcal{O}(X)} (\mathcal{O}(Y)/\mathcal{O}(X))_m^! \simeq (\mathcal{O}(S \times_X Y)/\mathcal{O}(S))_m^!$. Since base change of an n -suave map is computed on dualizing objects (Proposition 6.20(1)), it follows that $\mathcal{O}(S) \otimes_{\mathcal{O}(X)} \mathcal{O}(Y) \simeq \mathcal{O}(S \times_X Y)$.

For general $X' \rightarrow X$, consider $\varphi: \mathcal{O}(X') \otimes_{\mathcal{O}(X)} \mathcal{O}(Y) \rightarrow \mathcal{O}(Y')$. We check that φ is an equivalence after base change along every representable $S \rightarrow X'$. The source becomes

$$\mathcal{O}(S) \otimes_{\mathcal{O}(X')} (\mathcal{O}(X') \otimes_{\mathcal{O}(X)} \mathcal{O}(Y)) \simeq \mathcal{O}(S) \otimes_{\mathcal{O}(X)} \mathcal{O}(Y) \simeq \mathcal{O}(S \times_X Y),$$

while the target becomes

$$\mathcal{O}(S) \otimes_{\mathcal{O}(X')} \mathcal{O}(Y') \simeq \mathcal{O}(S \times_{X'} Y') \simeq \mathcal{O}(S \times_X Y),$$

both by the representable case (the second applied to $S \rightarrow X'$ and $Y' \rightarrow X'$); under these identifications φ becomes the identity of $\mathcal{O}(S \times_X Y)$. Since $\mathcal{O}$ is a sheaf, equivalences of $\mathcal{O}(X')$ -algebras may be detected after base change to representables over X' , so φ is an equivalence. This proves the Künneth formula, and with it the theorem. $\square$

7.4 Gestalten

Definition 7.15. The category of *Gestalten* is the opposite category $\mathrm{Gest} := \mathrm{StRing}^{\mathrm{op}}$. For $A \in \mathrm{StRing}$ we write $\mathrm{Gest}(A) \in \mathrm{Gest}$ for the corresponding Gestalt.

Dualizing the properties of maps of Stefanich rings introduced in Chapter 6, we obtain notions of n -affine, n -prim, n -proper, n -suave, and n -étale map of Gestalten. In this language, Theorem 7.13 asserts that for every $A \in \text{StRing}$ and $n \geq 0$ the category of n -étale Gestalten over $\text{Gest}(A)$ is countably toposic.

Corollary 7.16. *Let $f: X \rightarrow Y$ be a map in Gest , let $Y' \rightarrow Y$ be an effective epimorphism, and let $f': X' = X \times_Y Y' \rightarrow Y'$ be the base change. Then f has property $\mathcal{P}_n$ if and only if f' does, where $\mathcal{P}_n$ is any of n -affine, n -prim, n -proper, n -suave, n -étale.*

Proof. The “only if” direction is stability under base change. Conversely, write $Y'_\bullet$ for the Čech nerve of $Y' \rightarrow Y$. Since $Y' \rightarrow Y$ is an effective epimorphism, for every m we have

$$O(Y)_m \simeq \lim_{[k] \in \Delta} O(Y'_k)_m,$$

and the analogous descent descriptions hold for the relative objects of f . If f' is n -affine, its relative commutative algebras in all degrees at least n descend through these limits, as do the identifications with their internal module categories, so f is n -affine. If f' is n -prim, the objects $f'_{m-1,*}(\mathbb{1})$ and the units $\mathbb{1} \rightarrow f'_{m-1,*}(\mathbb{1})$ form descent data for $m \geq n+1$; their internal right adjoints descend as well, proving that f is n -prim. The same argument with $f'_{m-1,\#}(\mathbb{1})$ and the adjoints to the counits proves the n -suave case. Finally, n -properness and n -étaleness reduce to the preceding cases by applying the same argument to all iterated diagonals. Hence each listed property descends. $\square$

8 Injections and surjections of Gestalten

Talk given by Shai.

The goal of this chapter is to understand injective and surjective maps of Gestalten, and to relate surjectivity to Mathew's notion of *descendability* [Mat16; Mat18].

Throughout, for $X \in \text{Gest}$ we write $\mathcal{O}(X)_n \in \text{CAlg}(n\text{Pr})$ for the n -th level of the corresponding Stefanich ring, and for a map $f: Y \rightarrow X$ of Gestalten we write $(\mathcal{O}(Y)/\mathcal{O}(X))_m$ and $(\mathcal{O}(Y)/\mathcal{O}(X))_m^!$, with transition maps $f_m^*, f_{m,*}, f_{m,\#}, f_{m,!}$, for the relative objects and maps of Chapter 6, dualized along $\text{Gest} = \text{StRing}^{\text{op}}$. Dualizing the fully faithful embedding $\text{CAlg}(A_{n+1}) \hookrightarrow \text{StRing}_{A/}$ of Definition 6.1, whose essential image is the n -affine objects, we write $A \mapsto \text{Gest}_X(A)$ for the resulting equivalence

$$\text{CAlg}(\mathcal{O}(X)_{n+1}) \simeq (\text{Gest}_X^{n\text{-affine}})^{\text{op}},$$

so that $\text{Gest}_X(A) \rightarrow X$ is the n -affine Gestalt over X corresponding to $A \in \text{CAlg}(\mathcal{O}(X)_{n+1})$.

8.1 Injections of Gestalten

Definition 8.1. A map $f: Y \rightarrow X$ of Gestalten is *injective* if its diagonal $\Delta_f: Y \rightarrow Y \times_X Y$ is an isomorphism. Equivalently, the Čech nerve of f is constant.

8.1.1 Closed immersions and idempotent algebras

We first relate closed immersions to idempotent *algebras*, via the relative algebras $(B/A)_m$ of an n -prim map (Definitions 6.5 and 6.14).

Proposition 8.2. *Let $f: Y \rightarrow X$ be an injective map of Gestalten. Then f is n -prim if and only if it is n -proper. In this case f is n -affine and the algebra $(\mathcal{O}(Y)/\mathcal{O}(X))_n$ is idempotent in $\mathcal{O}(X)_{n+1}$. This assignment defines an equivalence of posets*

$$\{n\text{-proper injections } Z \hookrightarrow X\} \longleftrightarrow \{\text{idempotent algebras in } \mathcal{O}(X)_{n+1}\}.$$

Proof. As the diagonal of f is an isomorphism, so are all its iterated diagonals, and isomorphisms are automatically n -prim; hence n -prim is equivalent to n -proper for injective maps. By Lemma 6.12 and injectivity, the diagonal of f induces an isomorphism

$$(\mathcal{O}(Y)/\mathcal{O}(X))_n \otimes_{\mathcal{O}(X)_{n+1}} (\mathcal{O}(Y)/\mathcal{O}(X))_n \simeq (\mathcal{O}(Y \times_X Y)/\mathcal{O}(X))_n \xrightarrow{\sim} (\mathcal{O}(Y)/\mathcal{O}(X))_n,$$

exhibiting $(\mathcal{O}(Y)/\mathcal{O}(X))_n$ as an idempotent algebra. In particular, we see that the functor

$$(\mathcal{O}(-)/\mathcal{O}(X))_n: \text{Gest}_X^{\text{op}} \rightarrow \text{CAlg}(\mathcal{O}(X)_{n+1})$$

carries n -proper injections to idempotent algebras. In the other direction, recall from Definition 6.1 that this functor admits a fully faithful, symmetric monoidal left adjoint, which we will denote by

$$\text{Gest}_X(-): \text{CAlg}(\mathcal{O}(X)_{n+1}) \rightleftarrows \text{Gest}_X^{\text{op}}.$$

Given an idempotent algebra A in $\mathcal{O}(X)_{n+1}$, we show that the map $\text{Gest}_X(A) \rightarrow X$ is an n -proper injection. Since $\text{Gest}_X(-)$ is symmetric monoidal, it sends the codiagonal $A \otimes_{\mathcal{O}(X)_{n+1}} A \xrightarrow{\sim} A$ of A to the diagonal $\text{Gest}_X(A) \rightarrow \text{Gest}_X(A) \times_X \text{Gest}_X(A)$, and so since A is idempotent this is an isomorphism. This shows that $\text{Gest}_X(A) \rightarrow X$ is an injection. Writing $A_m := (\mathcal{O}(\text{Gest}_X(A))/\mathcal{O}(X))_m$ for $m \geq n$, each A_m is idempotent (as it is the image of A under the inclusion $\text{CAlg}(\mathcal{O}(X)_{n+1}) \hookrightarrow \text{CAlg}(\mathcal{O}(X)_{m+1})$) and, by affineness, $\mathcal{O}(\text{Gest}_X(A))_{m+1} = \text{Mod}_{A_m}(\mathcal{O}(X)_{m+1})$. This gives an internal adjunction in $\text{Mod}_{\mathcal{O}(X)_{m+1}}(1\text{Pr})$

$$A_m \otimes -: \mathcal{O}(X)_{m+1} \rightarrow \text{Mod}_{A_m}(\mathcal{O}(X)_{m+1}),$$

whose right adjoint is fully faithful since A_m is idempotent. Applying $\text{Mod}_{\mathcal{O}(X)_{m+1}}(1\text{Pr}) \rightarrow \mathcal{O}(X)_{m+2}$, we obtain an internal adjunction in $\mathcal{O}(X)_{m+2}$ of the form

$$\mathbb{1} \rightleftarrows (\mathcal{O}(\text{Gest}_X(A))/\mathcal{O}(X))_{m+1}.$$

As $m \geq n$ was arbitrary, $\text{Gest}_X(A) \rightarrow X$ is an n -prim injection.

We conclude that there is an adjunction of posets

$$(\mathcal{O}(-)/\mathcal{O}(X))_n: \{n\text{-proper injections } Z \hookrightarrow X\} \rightleftarrows \{\text{idempotent algebras in } \mathcal{O}(X)_{n+1}\} : \text{Gest}_X(-).$$

Since $\text{Gest}_X(-)$ is fully faithful, the unit $A \xrightarrow{\sim} (\mathcal{O}(\text{Gest}_X(A))/\mathcal{O}(X))_n$ is an isomorphism. For the counit, let $Z \hookrightarrow X$ be an n -proper injection; we must show its n -affinization map $Z \rightarrow \text{Gest}_X((\mathcal{O}(Z)/\mathcal{O}(X))_n)$ is an isomorphism. By cancellation properties, this map is again an n -proper injection, now with trivial relative level- n algebra. It thus suffices to show that any n -proper injection $Z \rightarrow X$ satisfying $\mathbb{1} \xrightarrow{\sim} (\mathcal{O}(Z)/\mathcal{O}(X))_n \simeq \mathbb{1}$ is an isomorphism.

So let $i: Z \hookrightarrow X$ be an n -proper injection with $\mathbb{1} \xrightarrow{\sim} (\mathcal{O}(Z)/\mathcal{O}(X))_n$. We prove by induction on $m \geq n$ that $\mathbb{1} \xrightarrow{\sim} (\mathcal{O}(Z)/\mathcal{O}(X))_m$; the case $m = n$ is the hypothesis. Suppose $m > n$ and assume the claim for $m - 1$. By assumption on i , the map $i_{m-1}^*: \mathbb{1}_X \rightarrow (\mathcal{O}(Z)/\mathcal{O}(X))_m$ has an internal right adjoint $i_{m-1,*}$ in $\mathcal{O}(X)_{m+1}$, and by the inductive hypothesis the unit of this adjunction, $\mathbb{1} \rightarrow (\mathcal{O}(Z)/\mathcal{O}(X))_{m-1}$, is an isomorphism in $\text{End}_{\mathcal{O}(X)_{m+1}}(\mathbb{1}) = \mathcal{O}(X)_m$. Hence i_{m-1}^* is fully faithful. On the other hand, idempotence of the target algebra implies that $i_{m-1,*}$ is fully faithful. Thus the adjunction is an equivalence, and $\mathbb{1} \rightarrow (\mathcal{O}(Z)/\mathcal{O}(X))_m$ is an isomorphism. This completes the induction and the proof. $\square$

Definition 8.3. A map of Gestalten is a *closed immersion* if it is a 0-prim injection. By Proposition 8.2, closed immersions $Z \hookrightarrow X$ are classified by idempotent algebras in $\mathcal{O}(X)_1$.

Example 8.4. The idempotent algebra $\mathbb{Z}[\frac{1}{p}] \in \text{D}(\mathbb{Z})$ corresponds to a *closed immersion* $\text{Gest}(\mathbb{Z}[\frac{1}{p}]) \hookrightarrow \text{Gest}(\mathbb{Z})$.

8.1.2 Open immersions and idempotent coalgebras

We now dualize the previous subsection, relating open immersions to idempotent *coalgebras*. Recall from Definitions 6.18 and 6.23 that an n -suave map $f: A \rightarrow B$ comes with dualizing objects $(B/A)_m^! := f_{m,\#}(1)$, built from the left adjoints $f_{m,\#}$ of its structure maps, and that f is n -étale if all its codiagonals are n -suave. Since $f_{m,\#}$ is left adjoint to a symmetric monoidal functor f_m^* , it is automatically oplax symmetric monoidal. Consequently, $(B/A)_m^!$ is a cocommutative coalgebra in A_{m+1} , whose (weak) dual is the algebra $(B/A)_m$; the counit $f_{m-1,!}: (B/A)_m^! \rightarrow 1$ similarly dualizes to the unit f_{m-1}^* of $(B/A)_m$.

Lemma 8.5. *Let $\mathcal{V} \in \text{CAlg}(1\text{Pr})$, and let D be an idempotent cocommutative coalgebra in $\mathcal{V}$. Then the inclusion*

$$\iota_D: \text{coMod}_D(\mathcal{V}) \hookrightarrow \mathcal{V}$$

admits the symmetric monoidal right adjoint $- \otimes D$, and exhibits $\text{coMod}_D(\mathcal{V})$ as a smashing colocalization of $\mathcal{V}$. Moreover, for idempotent coalgebras D, D' in $\mathcal{V}$ there is a canonical equivalence

$$\text{coMod}_D(\mathcal{V}) \otimes_{\mathcal{V}} \text{coMod}_{D'}(\mathcal{V}) \simeq \text{coMod}_{D \otimes_{\mathcal{V}} D'}(\mathcal{V}).$$

Finally, regarded as an object of $\text{Mod}_{\mathcal{V}}(1\text{Pr})$, the category $W_D := \text{coMod}_D(\mathcal{V})$ is simultaneously an idempotent algebra and an idempotent coalgebra, with multiplication and comultiplication mutually inverse. Consequently, if $\mathcal{W} \in \text{CAlg}(1\text{Pr})$ and $F: \text{Mod}_{\mathcal{V}}(1\text{Pr}) \rightarrow \mathcal{W}$ is any symmetric monoidal functor, then

$$\text{Mod}_{F(W_D)}(\mathcal{W}) \simeq \text{coMod}_{F(W_D)}(\mathcal{W}),$$

both sides identifying with the full subcategory of $F(W_D)$ -local objects.

Proof. Since $D \rightarrow D \otimes_{\mathcal{V}} D$ is an equivalence, the comonad $T_D := - \otimes D$ on $\mathcal{V}$ is idempotent. Thus its comodules are precisely the T_D -colocal objects, and the inclusion of this full subcategory admits T_D as right adjoint. The localized tensor product has unit D , and the idempotence of D makes T_D symmetric monoidal as a functor $\mathcal{V} \rightarrow \text{coMod}_D(\mathcal{V})$.

The tensor formula follows from the same description: both sides are smashing colocalizations of $\mathcal{V}$, and the right adjoints of their inclusions into $\mathcal{V}$ are canonically identified with $- \otimes D \otimes_{\mathcal{V}} D'$.

Under this equivalence with $D' = D$, the multiplication of W_D is induced by the inverse of the comultiplication $D \xrightarrow{\sim} D \otimes_{\mathcal{V}} D$. The colocalization adjunction also regards W_D as an idempotent coalgebra in $\text{Mod}_{\mathcal{V}}(1\text{Pr})$, with counit ι_D . These algebra and coalgebra structures have inverse multiplication and comultiplication. This property is preserved by any symmetric monoidal functor F . For an object whose multiplication and comultiplication are inverse equivalences, module structures and comodule structures are the same datum: both say that the canonical map to tensoring with that object is an equivalence, with the action and coaction inverse to each other. $\square$

Proposition 8.6. *Let $f: Y \rightarrow X$ be an injective map of Gestalten. Then f is n -suave if and only if it is n -étale. In this case $(\mathcal{O}(Y)/\mathcal{O}(X))_n^!$ is an idempotent coalgebra in $\mathcal{O}(X)_{n+1}$, and this assignment defines an equivalence of posets*

$$\{n\text{-étale injections } U \hookrightarrow X\} \longleftrightarrow \{\text{idempotent coalgebras in } \mathcal{O}(X)_{n+1}\},$$

whose inverse sends C to $\text{Gest}_X(\text{coMod}_C(\mathcal{O}(X)_{n+1}))$.

Proof. As Δ_f is an isomorphism, all iterated diagonals of f are isomorphisms, and isomorphisms are n -suave; so n -suave is equivalent to n -étale for injective maps.

Assume f is n -suave. By injectivity, we have $Y \xrightarrow{\sim} Y \times_X Y$, and so by the Künneth formula for dualizing objects from Lemma 6.22 we get an isomorphism

$$(O(Y)/O(X))_n^! \otimes_{O(X)_{n+1}} (O(Y)/O(X))_n^! \simeq (O(Y \times_X Y)/O(X))_n^! \xrightarrow{\sim} (O(Y)/O(X))_n^!,$$

exhibiting $(O(Y)/O(X))_n^!$ as an idempotent coalgebra.

Conversely, let C be an idempotent coalgebra in $O(X)_{n+1}$. Its comodules form a full subcategory

$$\iota: \text{coMod}_C(O(X)_{n+1}) \hookrightarrow O(X)_{n+1},$$

on which the cofree functor $-\otimes C$ is a symmetric monoidal right adjoint to ι , exhibiting $\text{coMod}_C(O(X)_{n+1})$ as a smashing colocalization of $O(X)_{n+1}$. Through the structure map $-\otimes C$ this is a commutative algebra in $O(X)_{n+2}$, and so determines an $(n+1)$ -affine Gestalt

$$U := \text{Gest}_X(\text{coMod}_C(O(X)_{n+1})).$$

We claim $U \rightarrow X$ is the desired n -étale injection, with $(O(U)/O(X))_n^! = C$.

First, $U \rightarrow X$ is an injection. Under the symmetric monoidal equivalence $\text{CAlg}(O(X)_{n+2}) \simeq (\text{Gest}_X^{(n+1)\text{-affine}})^{\text{op}}$, its diagonal corresponds to the codiagonal of the algebra $\text{coMod}_C(O(X)_{n+1})$ in $O(X)_{n+2}$. Since

$$\text{coMod}_C(O(X)_{n+1}) \otimes_{O(X)_{n+1}} \text{coMod}_C(O(X)_{n+1}) \simeq \text{coMod}_{C \otimes_{O(X)_{n+1}} C}(O(X)_{n+1}),$$

this codiagonal is the functor induced by the inverse of the comultiplication $C \xrightarrow{\sim} C \otimes_{O(X)_{n+1}} C$; hence the codiagonal is an isomorphism and $U \rightarrow X$ is an injection.

It remains to show that $U \rightarrow X$ is n -suave; being an injection, it is then n -étale. The idea is to realize each level of U as comodules over a canonical idempotent coalgebra, bootstrapping up from C .

We shall use the smashing-idempotent package from Lemma 8.5. In particular, the conclusions are preserved after applying the symmetric monoidal transition functors $\text{Mod}_{O(X)_{m+1}}(1\text{Pr}) \rightarrow O(X)_{m+2}$.

Now define idempotent coalgebras recursively by $C_n := C$ and

$$C_{m+1} := \text{coMod}_{C_m}(O(X)_{m+1}) \in O(X)_{m+2},$$

each one an idempotent coalgebra by Lemma 8.5, taken with $\mathcal{V} = O(X)_{m+1}$ and $D = C_m$. We show by induction that

$$O(U)_{m+1} = \text{coMod}_{C_m}(O(X)_{m+1}) \quad (m \geq n).$$

The base case $m = n$ is the construction of U . For the induction step, $(n+1)$ -affineness gives $O(U)_{m+2} = \text{Mod}_{(O(U)/O(X))_{m+1}}(O(X)_{m+2})$, and by the inductive hypothesis the relative algebra $(O(U)/O(X))_{m+1}$ is $\text{coMod}_{C_m}(O(X)_{m+1}) = C_{m+1}$; Lemma 8.5 identifies modules and comodules over C_{m+1} in $O(X)_{m+2}$, so

$$O(U)_{m+2} = \text{Mod}_{C_{m+1}}(O(X)_{m+2}) = \text{coMod}_{C_{m+1}}(O(X)_{m+2}),$$

completing the induction.

At each level $m \geq n$ the structure map $f_m^*: \mathbb{1} \rightarrow (O(U)/O(X))_{m+1} = \text{coMod}_{C_m}(O(X)_{m+1})$ is the cofree functor $- \otimes C_m$, whose left adjoint in $O(X)_{m+2}$ is the inclusion ι_m ; this is the required left adjoint $f_{m,\#}$, and $(O(U)/O(X))_m^! = \iota_m(\mathbb{1}) = C_m$. The same smashing-idempotent structure provides the right adjoint to each counit $f_{m-1,!}$, so both conditions of Definition 6.18 hold and $U \rightarrow X$ is n -suave. In particular $(O(U)/O(X))_n^! = C_n = C$, as claimed.

It remains to recover an n -étale injection $U \hookrightarrow X$ from its dualizing coalgebra $C := (O(U)/O(X))_n^!$. The injection $U \rightarrow X$ factors through its $(n+1)$ -affinization $U \rightarrow \text{Gest}_X(\text{coMod}_C(O(X)_{n+1}))$, again an n -étale injection, now with trivial dualizing coalgebra; replacing X by this Gestalt, it suffices to show that an n -étale injection $U \rightarrow X$ with $(O(U)/O(X))_n^! \simeq \mathbb{1}_X$ is an isomorphism. We show by induction that $(O(U)/O(X))_m^! \simeq \mathbb{1}_X$ for all $m \geq n$. Assume the claim for m . As $j: U \rightarrow X$ is n -suave, the map $j_{m,!}: (O(U)/O(X))_{m+1}^! \rightarrow \mathbb{1}_X$ admits a right adjoint $j_m^!$ in $O(X)_{m+2}$, and the composite

$$\mathbb{1} \xrightarrow{j_m^!} (O(U)/O(X))_{m+1}^! \xrightarrow{j_{m,!}} \mathbb{1}_X$$

is, as an object of $\text{End}_{O(X)_{m+2}}(\mathbb{1}) = O(X)_{m+1}$, given by $(O(U)/O(X))_m^!$, while the counit of the adjunction is the isomorphism $j_{m-1,!}: (O(U)/O(X))_m^! \xrightarrow{\sim} \mathbb{1}_X$ furnished by the inductive hypothesis. Hence the right adjoint $j_m^!$ is fully faithful. Coidempotence of $(O(U)/O(X))_{m+1}^!$ implies that the counit map $j_{m,!}$ is fully faithful as well, by the dual of the idempotent-algebra argument above. Thus the adjunction $j_{m,!} \dashv j_m^!$ is an equivalence, so $j_{m,!}$ is an isomorphism, completing the induction. $\square$

Definition 8.7. A map of Gestalten is an *open immersion* if it is a 0-étale injection. By Proposition 8.6, open immersions $U \hookrightarrow X$ are classified by idempotent coalgebras in $O(X)_1$.

8.2 Complements

Definition 8.8. Let $f: Y \rightarrow X$ be a map of Gestalten. Its *complement* is the subpresheaf $X \setminus Y$ of $\text{Hom}_{\text{Gest}}(-, X)$ given by

$$(X \setminus Y)(S) = \{S \rightarrow X \mid S \times_X Y = \emptyset\}.$$

If this presheaf is representable by some $Z \rightarrow X$, we call Z the complement of Y in X .

Remark 8.9. In general $X \setminus Y$ need not be representable. If f lies in $\text{Gest}^{n\text{-ét}}$ for some n , then it might follow from abstract topos-theoretic arguments that $X \setminus Y$ is representable.

Proposition 8.10. Let $j: U \hookrightarrow X$ be an n -étale injection, corresponding to the idempotent coalgebra $C := (O(U)/O(X))_n^!$ in $O(X)_{n+1}$, and let

$$A := \text{cofib}(C \rightarrow \mathbb{1}) \in O(X)_{n+1}.$$

Assume that $n \geq 1$, or that $O(X)_1$ is pointed. Then A is an idempotent algebra, and the n -proper injection $\text{Gest}_X(A) \hookrightarrow X$ is the complement of U .

Proof. Since the tensor product preserves cofibers in each variable, idempotency of C gives

$$A \otimes C = \text{cofib}(C \otimes C \rightarrow C) = 0, \quad \text{whence} \quad A \otimes A = \text{cofib}(A \otimes C \rightarrow A) = \text{cofib}(0 \rightarrow A) = A,$$

so A is an idempotent algebra, complementary to C : the sequence $C \rightarrow \mathbb{1} \rightarrow A$ is a cofiber sequence with $A \otimes C = 0$.

Let $g: S \rightarrow X$ be a map. On one side, $S \times_X U \hookrightarrow S$ is the n -étale injection over S with dualizing coalgebra $(\mathcal{O}(S \times_X U)/\mathcal{O}(S))_n^\dagger = g_n^*(C)$, so $S \times_X U$ is empty if and only if $g_n^*(C) = 0$. On the other, $S \times_X \text{Gest}_X(A) \hookrightarrow S$ is the n -proper injection over S with idempotent algebra $(\mathcal{O}(S \times_X \text{Gest}_X(A))/\mathcal{O}(S))_n = g_n^*(A)$, so g factors through $\text{Gest}_X(A) \hookrightarrow X$ if and only if $g_n^*(A) \simeq \mathbb{1}$.

Applying g_n^* to the cofiber sequence $C \rightarrow \mathbb{1} \rightarrow A$ yields a complementary pair $g_n^*(C), g_n^*(A)$ in $\mathcal{O}(S)_{n+1}$, again with $g_n^*(A) = \text{cofib}(g_n^*(C) \rightarrow \mathbb{1})$ and $g_n^*(C) \otimes g_n^*(A) = 0$. For such a pair one member vanishes precisely when the other is the unit: if $g_n^*(C) = 0$ then $g_n^*(A) = \text{cofib}(0 \rightarrow \mathbb{1}) = \mathbb{1}$, and if $g_n^*(A) = \mathbb{1}$ then $g_n^*(C) = g_n^*(C) \otimes g_n^*(A) = g_n^*(C \otimes A) = 0$. Hence g factors through $\text{Gest}_X(A)$ if and only if $S \times_X U = \emptyset$; that is, $\text{Gest}_X(A)$ represents the complement $X \setminus U$. $\square$

Remark 8.11. If instead one is given an n -proper injection $i: Z \hookrightarrow X$, one regards i as $(n+1)$ -étale by ambidexterity (Proposition 6.25) and applies Proposition 8.10 one level higher to obtain its complement.

Warning 8.12. The passage $C \mapsto A = \text{cofib}(C \rightarrow \mathbb{1})$ is not reversible when $\mathcal{O}(X)_{n+1}$ is not stable¹: it may happen that $A = 0$ even though $C \rightarrow \mathbb{1}$ is not an isomorphism. So an open immersion can have empty closed complement without being everything.

Corollary 8.13. *If $\mathcal{O}(X)_1$ is pointed, then the complement of an open immersion into X is a closed immersion. In particular, this holds for Gestalten over $\text{Gest}(\text{An}_*)$. The converse holds for Gestalten over $\text{Spec}(\mathbb{S})$.* $\square$

Corollary 8.14. *Consider a 1-presentable stable category $E \in 1\text{Pr}_{\text{st}}$. Then there are (order-preserving or order-reversing) bijections between the following four posets:*

- (1) Closed immersions $Z \hookrightarrow \text{Gest}_{\mathbb{S}}(E)$;
- (2) Idempotent algebras A in E ;
- (3) Open immersions $U \hookrightarrow \text{Gest}_{\mathbb{S}}(E)$;
- (4) Idempotent coalgebras C in E .

$\square$

Example 8.15. The map $\text{Gest}(\mathbb{Z}[\frac{1}{p}]) \rightarrow \text{Gest}(\mathbb{Z})$ of Example 8.4 is a closed immersion, given by the idempotent algebra $\mathbb{Z}[\frac{1}{p}] \in \text{D}(\mathbb{Z})$. As $\text{D}(\mathbb{Z})$ is stable, its complement is the open immersion given by the complementary idempotent coalgebra $C := \text{fib}(\mathbb{Z} \rightarrow \mathbb{Z}[\frac{1}{p}]) \simeq (\mathbb{Z}[\frac{1}{p}]/\mathbb{Z})[-1]$, the Gestalt corresponding to

$$\text{coMod}_C(\text{D}(\mathbb{Z})) = \{M \mid M \otimes \mathbb{Z}[\frac{1}{p}] = 0\} \xrightarrow[\sim]{M \mapsto M_p^\wedge} \text{D}(\mathbb{Z})_p^\wedge.$$

The two immersions assemble into the arithmetic fracture recollement of $\text{D}(\mathbb{Z})$:

$$\begin{array}{ccccc} & \xleftarrow{(-)[1/p]} & & \xleftarrow{\quad} & \\ \text{D}(\mathbb{Z}[\frac{1}{p}]) & \xrightarrow{\quad} & \text{D}(\mathbb{Z}) & \xrightarrow{C \otimes -} & \text{coMod}_C(\text{D}(\mathbb{Z})) \\ & \xleftarrow{\quad} & & \xleftarrow{M \mapsto M_p^\wedge} & \end{array}$$

¹Note that stability can only happen for $n = 0$.

Tensoring the cofiber sequence $C \rightarrow \mathbb{1} \rightarrow \mathbb{Z}[\frac{1}{p}]$ with M recovers the fracture fiber sequence

$$C \otimes M \rightarrow M \rightarrow M[\frac{1}{p}].$$

8.3 Surjections of Gestalten

We will now introduce the dual notion of a *surjection* of Gestalten. Even though the category Gest as a whole is not a topos, the suave surjections of Gestalten will admit a behavior similar to that of effective epimorphisms in a topos.

Definition 8.16. A map $g: Y \rightarrow X$ of Gestalten is *surjective*, or a *cover*, if the map

$$\text{im}(g) := \text{colim}_{[m] \in \Delta^{\text{op}}} Y^{\times_X(m+1)} \rightarrow X$$

is an isomorphism.

Remark 8.17. Assume that g is n -étale, so that it is a morphism in the countably toposic category $\text{Gest}_{/X}^{n\text{-ét}}$ of Theorem 7.13. Then g is surjective precisely when it is a cover in the sense of topos theory.

Lemma 8.18. *Let $f: Y \rightarrow X$ be an n -suave surjection of Gestalten. Then the pullback functor $f^*: \text{Gest}_{/X} \rightarrow \text{Gest}_{/Y}$ is conservative.*

Proof. Let $\varphi: A \rightarrow B$ be a morphism over X such that $f^*\varphi: A \times_X Y \rightarrow B \times_X Y$ is an isomorphism over Y ; we must show φ is an isomorphism. Since f is surjective, $\text{colim}_{[m] \in \Delta^{\text{op}}} Y^{\times_X(m+1)} \simeq X$. Each map $Y^{\times_X(m+1)} \rightarrow X$ is n -suave by iterated base change, so Proposition 7.3(2) gives, for any $T \rightarrow X$,

$$T \times_X \text{colim}_{[m]} Y^{\times_X(m+1)} \simeq \text{colim}_{[m]} (T \times_X Y)^{\times_T(m+1)}.$$

Applying this with $T = A$ and using surjectivity of f :

$$A \simeq A \times_X \text{colim}_{[m]} Y^{\times_X(m+1)} \simeq \text{colim}_{[m]} (A \times_X Y)^{\times_A(m+1)},$$

and likewise $B \simeq \text{colim}_{[m]} (B \times_X Y)^{\times_B(m+1)}$. In simplicial degree m the induced map between these Čech nerves identifies with the base change of $f^*\varphi$ along $Y^{\times_X(m+1)} \rightarrow Y$, and is therefore an isomorphism. Taking colimits, φ induces an isomorphism $A \xrightarrow{\sim} B$ over X . $\square$

Lemma 8.19. *Let $f: Y \rightarrow X$ be an n -suave map of Gestalten. Then the image $U := \text{colim}_{[m] \in \Delta^{\text{op}}} Y^{\times_X(m+1)}$ is an n -suave injection $U \hookrightarrow X$.*

Proof. By base change and stability under colimits, each $Y^{\times_X(m+1)} \rightarrow X$ is n -suave, hence so is $U \rightarrow X$. It remains to show $U \rightarrow X$ is a monomorphism, i.e. that the diagonal $U \rightarrow U \times_X U$ is an isomorphism, or equivalently that the first projection $U \times_X U \rightarrow U$ is an isomorphism.

It will suffice to show that the projection $Y \times_X U \rightarrow Y$ is an isomorphism. By Proposition 7.3(2), base change along $Y \rightarrow X$ commutes with the colimit of the n -suave diagram $[m] \mapsto Y^{\times_X(m+1)}$, giving

$$Y \times_X U \simeq \text{colim}_{[m]} Y^{\times_X(m+2)}.$$

The simplicial object $[m] \mapsto Y^{\times_X(m+2)}$ carries extra degeneracies $s_{-1}: Y^{\times_X(m+2)} \rightarrow Y^{\times_X(m+3)}$ given by inserting a copy of Y at position 0 via the diagonal $\Delta: Y \rightarrow Y \times_X Y$. These make it a split augmented simplicial object with augmentation Y (via the first projection), so $Y \times_X U \simeq \operatorname{colim}_{[m]} Y^{\times_X(m+2)} \xrightarrow{\sim} Y$.

Base change along $U \rightarrow X$ also commutes with the simplicial colimit, and therefore

$$U \times_X U \simeq \operatorname{colim}_{[m]} U \times_X Y^{\times_X(m+1)} \simeq \operatorname{colim}_{[m]} Y^{\times_X(m+1)} = U.$$

Thus $U \rightarrow X$ is an injection. It follows that the Čech nerves of $Y \rightarrow U$ and $Y \rightarrow X$ agree, so $Y \rightarrow U$ is surjective. Moreover, $Y \rightarrow U$ is the base change of the n -suave map $Y \rightarrow X$, hence is n -suave. $\square$

Proposition 8.20. *Every n -suave map $f: Y \rightarrow X$ of Gestalten admits a (necessarily unique) factorization*

$$Y \twoheadrightarrow U \hookrightarrow X$$

into an n -suave surjection followed by an n -suave injection.

Proof. The factorization is provided by Lemma 8.19: set $U := \operatorname{colim}_{[m] \in \Delta^{\text{op}}} Y^{\times_X(m+1)}$, so that $U \hookrightarrow X$ is an n -suave injection by that lemma, and $Y \rightarrow U$ is an n -suave surjection by definition. For uniqueness, any factorization $Y \twoheadrightarrow V \hookrightarrow X$ with $V \rightarrow X$ a monomorphism gives $Y^{\times_V(m+1)} \simeq Y^{\times_X(m+1)}$ for all m , hence $V \simeq \operatorname{colim}_{[m]} Y^{\times_V(m+1)} \simeq U$. $\square$

Corollary 8.21. *An n -suave map of Gestalten is an isomorphism if and only if it is both surjective and injective.* $\square$

Proposition 8.22. *Let $f: Y \rightarrow X$ be an n -suave map of Gestalten. The following are equivalent.*

- (1) *f is surjective.*
- (2) *The map $\operatorname{colim}_{[m]} (\mathcal{O}(Y^{\times_X(m+1)}) / \mathcal{O}(X))_n^\dagger \rightarrow \mathbb{1}$ is an isomorphism.*
- (3) *The map $\operatorname{colim}_{[m]} ((\mathcal{O}(Y) / \mathcal{O}(X))_n^\dagger)^{\otimes(m+1)} \rightarrow \mathbb{1}$ is an isomorphism.*
- (4) *The pullback functor $f_n^*: \mathcal{O}(X)_{n+1} \rightarrow \mathcal{O}(Y)_{n+1}$ is conservative.*

Proof. (1) $\Leftrightarrow$ (2): Consider the epi-mono factorization $Y \twoheadrightarrow U \hookrightarrow X$ of f from Proposition 8.20. The dualizing coalgebra of $U \hookrightarrow X$ is

$$(\mathcal{O}(U) / \mathcal{O}(X))_n^\dagger \simeq \operatorname{colim}_{[m]} (\mathcal{O}(Y^{\times_X(m+1)}) / \mathcal{O}(X))_n^\dagger.$$

By Proposition 8.6, the injection $U \rightarrow X$ is an isomorphism precisely when this coalgebra is the unit. This is condition (2), while $U = X$ is equivalent to surjectivity of f .

(2) $\Leftrightarrow$ (3) is Lemma 6.22.

(1) $\Rightarrow$ (4): Surjectivity and descent give

$$\mathcal{O}(X)_{n+1} \simeq \lim_{[m] \in \Delta} \mathcal{O}(Y^{\times_X(m+1)})_{n+1}.$$

A map in this limit is an isomorphism if its pullback to Y is an isomorphism, since all of its further pullbacks to the terms of the Čech nerve are then isomorphisms. Thus f_n^* is conservative.

(4) $\Rightarrow$ (3): By conservativity it suffices to check the map of (3) after base change to $\mathcal{O}(Y)_{n+1}$. After base change it is the corresponding augmented bar construction for the projection $Y \times_X Y \rightarrow Y$. This projection is split by the diagonal, so the augmented simplicial object has extra degeneracies and its colimit is the unit. $\square$

For $n = 0$ these conditions are easy to check directly. For higher n one instead wants a criterion phrased in terms of *generation*: being a cover is closely related to the functor

$$f_{n-1,!} : \mathrm{Hom}_{\mathcal{O}(X)_{n+1}}(\mathbb{1}, (\mathcal{O}(Y)/\mathcal{O}(X))_n^!) \rightarrow \mathcal{O}(X)_n$$

generating the target quickly. If f is in addition $(n-1)$ -proper, the source identifies with $\mathcal{O}(Y)_n$ and the displayed functor with $f_{n-1,*} : \mathcal{O}(Y)_n \rightarrow \mathcal{O}(X)_n$.

We record the following 2-categorical lemma, a special case of the condensation formalism of Gaiotto and Johnson-Freyd [GJ19].

Lemma 8.23 (Gaiotto–Johnson–Freyd). *Let $\mathbb{C} \in 2\mathrm{Pr}$, let $X, Y \in \mathbb{C}$, and suppose there are maps $F : Y \rightarrow X$ and $G : X \rightarrow Y$ such that id_Y is a retract of GF in $\mathrm{End}_{\mathbb{C}}(Y)$. Then Y lies in the 1Pr-linear subcategory of $\mathbb{C}$ generated by X .*

Proof. We may replace $\mathbb{C}$ by the 1Pr-linear subcategory atomically generated by X and Y , so that X is atomic in $\mathbb{C}$. Let $i : \mathbb{C}_X \subset \mathbb{C}$ be the subcategory generated by X ; it admits a linear right adjoint $i^R : \mathbb{C} \rightarrow \mathbb{C}_X$, and we set $Y' := i^R(Y) \in \mathbb{C}_X$, regarded also as an object of $\mathbb{C}$ via i . There is a comparison map $H : Y' \rightarrow Y$, which we must show is an equivalence. Applying i^R to F and G , the object Y' inherits maps $F' : Y' \rightarrow X$ and $G' : X \rightarrow Y'$ such that $\mathrm{id}_{Y'}$ is a retract of $G'F'$.

By the Yoneda embedding we may assume $\mathbb{C} = 1\mathrm{Pr}$, so that X, Y, Y' are presentable categories and $H : Y' \rightarrow Y$ is a functor we must show is an equivalence. Every object $y \in Y$ is a retract

$$y \rightarrow GF(y) = H(G'F(y)) \rightarrow y,$$

so every object of Y is a retract of an object in the image of H , and it suffices to show H is fully faithful. We first check this on the image of G' . For $x_1, x_2 \in X$, the composite

$$\mathrm{Hom}_Y(Gx_1, Gx_2) \xrightarrow{F} \mathrm{Hom}_X(FGx_1, FGx_2) \xrightarrow{G} \mathrm{Hom}_Y(GFGx_1, GFGx_2) \rightarrow \mathrm{Hom}_Y(Gx_1, Gx_2),$$

where the last map uses the two structure maps of the retraction $\mathrm{id}_Y \Rightarrow GF \Rightarrow \mathrm{id}_Y$, is the identity. Thus $\mathrm{Hom}_Y(Gx_1, Gx_2)$ is the image of the resulting idempotent on $\mathrm{Hom}_X(FGx_1, FGx_2)$. The identical construction for Y' exhibits $\mathrm{Hom}_{Y'}(G'x_1, G'x_2)$ as the image of the same idempotent, because $F'G' = FG$ and the retraction data on Y' is obtained by applying i^R . Hence H induces an isomorphism

$$\mathrm{Hom}_{Y'}(G'x_1, G'x_2) \xrightarrow{\sim} \mathrm{Hom}_Y(Gx_1, Gx_2).$$

As every object of Y' and every object of Y is a retract of an object in the image of G' and G , respectively, this proves that H is fully faithful. It is therefore an equivalence. $\square$

Proposition 8.24. *Let $f : Y \rightarrow X$ be n -suave, $n \geq 1$. Consider the functor*

$$\mathrm{colim}_{[m] \in \Delta^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{O}(X)_{n+1}}(\mathbb{1}, (\mathcal{O}(Y^{\times_X(m+1)})/\mathcal{O}(X))_n^!) \rightarrow \mathcal{O}(X)_n \quad \in \quad 1\mathrm{Pr}.$$

If $\mathbb{1}$ is a retract of an object in the image of this functor, then f is surjective.

Proof. Let $C := \operatorname{colim}_{[m] \in \Delta^{\text{op}}} (O(Y^{\times_X(m+1)})/O(X))_n^!$, the idempotent coalgebra corresponding to the image of f (Proposition 8.22), with counit $F: C \rightarrow \mathbb{1}$. The hypothesis exhibits $\mathbb{1}$ as a retract of an object in the image of

$$F: \operatorname{Hom}_{O(X)_{n+1}}(\mathbb{1}, C) \rightarrow \operatorname{Hom}_{O(X)_{n+1}}(\mathbb{1}, \mathbb{1}) = O(X)_n,$$

so there is $G: \mathbb{1} \rightarrow C$ with id a retract of FG . By Lemma 8.23 applied in $O(X)_{n+1}$ with $Y = \mathbb{1}$, $X = C$, the unit $\mathbb{1}$ lies in the 1Pr-linear subcategory of $O(X)_{n+1}$ generated by C . The category of C -comodules is a 1Pr-linear subcategory containing C , hence contains this generated subcategory; so $\mathbb{1}$ is a C -comodule, $\operatorname{coMod}_C(O(X)_{n+1}) = O(X)_{n+1}$, and $C = \mathbb{1}$. By Proposition 8.22(2), f is surjective. $\square$

We also record a converse, assuming X is n -prim.

Proposition 8.25. *Let $n \geq 1$ and let X be an n -prim Gestalt (or, more generally, assume the functor $n\text{Pr} \rightarrow O(X)_{n+1}$ has a colimit-preserving right adjoint). Let $f: Y \rightarrow X$ be an n -suave surjection. Then*

$$\operatorname{colim}_{[m] \in \Delta^{\text{op}}} \operatorname{Hom}_{O(X)_{n+1}}(\mathbb{1}, (O(Y^{\times_X(m+1)})/O(X))_n^!) \xrightarrow{\sim} O(X)_n,$$

and the image of $f_{n-1,!}$ generates $O(X)_n$ under simplicial colimits.

Proof. The colimit-preserving right adjoint is $\operatorname{Hom}_{O(X)_{n+1}}(\mathbb{1}, -)$, so the displayed isomorphism follows from Proposition 8.22(2). Any $M \in O(X)_n$ is the geometric realization of the simplicial object obtained by composing

$$O(X)_n \rightarrow \operatorname{Hom}_{O(X)_{n+1}}(\mathbb{1}, (O(Y^{\times_X(\bullet+1)})/O(X))_n^!) \rightarrow O(X)_n,$$

where the first map is the right adjoint, on underlying categories, of the second. This can be checked by conservativity after base change along f , where this bar resolution is split. The colimit takes values in the image of $f_{n-1,!}$. $\square$

Definition 8.26. A map of Gestalten is called *eventually suave* if it is m -suave for some $m \geq 0$.

Corollary 8.27. *Let $f: Y \rightarrow X$ be an n -prim, eventually suave map of Gestalten. Then the map*

$$Y \rightarrow \operatorname{Gest}_X((O(Y)/O(X))_n)$$

is surjective. In particular, f is surjective if and only if $\operatorname{Gest}_X((O(Y)/O(X))_n) \rightarrow X$ is surjective.

Proof. Replacing X by $\operatorname{Gest}_X((O(Y)/O(X))_n)$, we may assume $(O(Y)/O(X))_n = \mathbb{1}$. Assume f is m -suave, $m \geq n + 2$ without loss of generality; we induct on m .

Base case $m = n + 2$. Shifting, we may assume $n = 0$, so $f_{0,*}(\mathbb{1}) = \mathbb{1}$. As f is both 2-prim (being 0-prim) and 2-suave, $(O(Y)/O(X))_2^! \in O(X)_3$ is dualizable with dual $(O(Y)/O(X))_2$. As f is 1-prim, the unit map $f_1^*: \mathbb{1} \rightarrow (O(Y)/O(X))_2$ has a right adjoint $f_{1,*}$ in $O(X)_3$, so dually the counit $f_{1,!}: (O(Y)/O(X))_2^! \rightarrow \mathbb{1}$ has a left adjoint $f_1^!$ in $O(X)_3$. Dualizing identifies the composite

$$\mathbb{1} \xrightarrow{f_1^!} (O(Y)/O(X))_2^! \xrightarrow{f_{1,!}} \mathbb{1} \quad \text{with} \quad \mathbb{1} \xrightarrow{f_1^*} (O(Y)/O(X))_2 \xrightarrow{f_{1,*}} \mathbb{1},$$

which, as an object of $O(X)_2 = \operatorname{End}_{O(X)_3}(\mathbb{1})$, is given by tensoring with $f_{1,*}(\mathbb{1})$. Now $f_{1,*}(\mathbb{1}) \in O(X)_2$ has $\mathbb{1}$ as a retract: the unit $\mathbb{1} \rightarrow f_{1,*}(\mathbb{1})$ has a right adjoint (as f is 0-prim), and the composite

$\mathbb{1} \rightarrow f_{1,*}(\mathbb{1}) \rightarrow \mathbb{1}$ is tensoring with $f_{0,*}(\mathbb{1}) = \mathbb{1}$. So $\mathbb{1} \in \mathcal{O}(X)_2$ is a retract of an object in the image of

$$f_{1,!}: \operatorname{Hom}_{\mathcal{O}(X)_3}(\mathbb{1}, (\mathcal{O}(Y)/\mathcal{O}(X))_2^!) \rightarrow \mathcal{O}(X)_2,$$

and Proposition 8.24 (with $n = 2$) shows that f is surjective.

Induction step. Assume the claim for $(m - 1)$ -suave maps, and let f be n -prim and m -suave. Let

$$Y' := \operatorname{Gest}_X((\mathcal{O}(Y)/\mathcal{O}(X))_{m-2})$$

be the $(m - 2)$ -affine truncation of Y . By the base case, $Y \rightarrow Y'$ is surjective. Since f is n -prim, Lemma 6.7 shows that $(\mathcal{O}(Y)/\mathcal{O}(X))_{m-2}$, and hence the $(m - 2)$ -affine map $g: Y' \rightarrow X$, is countably presented. The map g is still n -prim: in degrees $k \leq m - 2$ its relative pullback maps agree with those of f , while in degrees $k > m - 2$ the required right adjoints follow from Proposition 6.9(1), applied to the countably presented $(m - 2)$ -affine map g . By Proposition 6.15(1), the map g is $(m - 2)$ -proper, and hence $(m - 1)$ -suave by ambidexterity (Proposition 6.25). The induction hypothesis shows that g is surjective. As $Y \rightarrow Y'$ is surjective as well, so is the composite $Y \rightarrow X$. $\square$

8.4 Descendability

We now relate surjectivity to Mathew's notion of descendability [Mat16, §§3–4], see also Mathew [Mat18].

Definition 8.28. Let $\mathbb{C}$ be a stably symmetric monoidal category and $R \in \operatorname{Alg}(\mathbb{C})$.

- (1) An object $M \in \mathbb{C}$ is *R -nilpotent* if it lies in the thick $\otimes$ -ideal $\operatorname{Thick}^\otimes(R)$ generated by R .
- (2) R is *descendable* if the unit $\mathbb{1} \in \mathbb{C}$ is R -nilpotent.

Proposition 8.29 (Mathew [Mat18, Proposition 2.32]). *Let $R \in \operatorname{Alg}(\mathbb{C})$ with $I := \operatorname{fib}(\mathbb{1} \rightarrow R)$. The following are equivalent.*

- (1) R is descendable.
- (2) The map $I \rightarrow \mathbb{1}$ is $\otimes$ -nilpotent: $I^{\otimes N} \rightarrow \mathbb{1}$ is null for some N .
- (3) For some finite m , the unit map $\mathbb{1} \rightarrow \lim_{\Delta_{\leq m}} R^{\otimes \bullet+1}$, into the m -th partial totalization of the Amitsur (cobar) complex of R , admits a retract.

Corollary 8.30. *Let $f: Y \rightarrow X$ be a 0-prim, eventually suave map of Gestalten over $\operatorname{Spec}(\mathbb{S})$. If $f_{0,*}(\mathbb{1}) \in \mathcal{O}(X)_1$ is descendable, then f is surjective.*

Proof. By Corollary 8.27 we may assume Y is 0-affine over X . Then f is 0-proper, so by ambidexterity $(\mathcal{O}(Y)/\mathcal{O}(X))_1^! = (\mathcal{O}(Y)/\mathcal{O}(X))_1$, and likewise for the finite fiber powers of Y over X . For each finite m , the functor

$$\operatorname{colim}_{\Delta_{\leq m}^{\operatorname{op}}} \operatorname{Hom}_{\mathcal{O}(X)_2}(\mathbb{1}, (\mathcal{O}(Y^{\times_X(\bullet+1)})/\mathcal{O}(X))_1^!) = \operatorname{colim}_{\Delta_{\leq m}^{\operatorname{op}}} (\mathcal{O}(Y^{\times_X(\bullet+1)})/\mathcal{O}(X))_1 \rightarrow \mathcal{O}(X)_1$$

has a left adjoint, whose unit is the finite limit of the units at the individual levels: the indexing category is finite, and finite limits agree with finite colimits in the stable category $\mathcal{O}(X)_1$, so this finite limit of left adjoints is again computed levelwise. Evaluated at $\mathbb{1} \in \mathcal{O}(X)_1$, this unit is the map

$$\mathbb{1} \rightarrow \lim_{\Delta_{\leq m}} f_{0,*}(\mathbb{1})^{\otimes \bullet+1}$$

into the m -th partial totalization of the Amitsur complex of $f_{0,*}(\mathbb{1})$. By descendability of $f_{0,*}(\mathbb{1})$ and Proposition 8.29(3), some finite m makes this map admit a retract. Then $\mathbb{1}$ is a retract of an object in the image of the displayed functor, and Proposition 8.24 shows that f is surjective. $\square$

Corollary 8.31. *Let X be a 0-prim Gestalt over $\mathrm{Spec}(\mathbb{S})$, and let $f: Y \rightarrow X$ be a 0-prim, eventually suave map. Then f is surjective if and only if $f_{0,*}(\mathbb{1})$ is descendable in $\mathcal{O}(X)_1$.*

Proof. The implication $\Leftarrow$ is Corollary 8.30. Conversely, suppose f is surjective; by Corollary 8.27 we may assume f is 0-affine. The image of $f_{0,*}: \mathcal{O}(Y)_1 \rightarrow \mathcal{O}(X)_1$ lies in $f_{0,*}(\mathbb{1})$ -modules, so by Proposition 8.25 every object of $\mathcal{O}(X)_1$, in particular $\mathbb{1}$, is a geometric realization of $f_{0,*}(\mathbb{1})$ -modules. As X is 0-prim, the unit map $\mathbb{1} \rightarrow \mathcal{O}(X)_1$ admits the internal right adjoint $\mathrm{Hom}_{\mathcal{O}(X)_1}(\mathbb{1}, -)$, so $\mathbb{1} \in \mathcal{O}(X)_1$ is compact; it is therefore already a retract of a finite stage of this realization, i.e. a retract of an object of the thick subcategory generated by $f_{0,*}(\mathbb{1})$ -modules. Thus $\mathbb{1} \in \mathrm{Thick}^{\otimes}(f_{0,*}(\mathbb{1}))$, which is descendability. $\square$

9a Affine and coaffine maps of Gestalten

During the seminar, the contents of Chapters 9a and 9b were discussed in the same talk, which came before Chapter 8. The material has been expanded and reorganized into two separate chapters for the sake of clarity of exposition.

The goal of this chapter is to isolate the affine-coaffine factorization system and use it to study how affineness interacts with diagonals.

Throughout the chapter, we keep using the notations $\text{Gest}_X(-): (\text{StRing}_{\mathcal{O}(X)/})^{\text{op}} \rightarrow \text{Gest}_{/X}$ and $\mathcal{O}(-)/\mathcal{O}(X): (\text{Gest}_{/X})^{\text{op}} \rightarrow \text{StRing}_{\mathcal{O}(X)/}$ for the inverse equivalence between Gestalten over a fixed Gestalt X and Stefanich rings over the global sections $\mathcal{O}(X)$ of X .

9.1 The affine-coaffine factorization system

Definition 9.1. Recall from Corollary 5.9 that for a Stefanich ring A , the functor $\text{StRing}_{A/} \rightarrow \text{CAlg}(A_{n+1}), B \mapsto (B/A)_n$ admits a fully faithful left adjoint $\text{CAlg}(A_{n+1}) \rightarrow \text{StRing}_{A/}$. This defines a functor

$$(-)^{n\text{-affine}}: \text{StRing}_{A/} \rightarrow \text{StRing}_{A/}, B \mapsto (B/A)^{n\text{-affine}}$$

equipped with a natural transformation $(B/A)^{n\text{-affine}} \rightarrow B$ of Stefanich A -algebras. We refer to this as the *n-affinization of B over A* .

By dualizing, we similarly obtain for every map $Y \rightarrow X$ of Gestalten a factorization

$$Y \rightarrow (Y/X)^{n\text{-affine}} \rightarrow X,$$

with $(Y/X)^{n\text{-affine}}$ the *n-affinization of Y over X* .

Definition 9.2. A map $f: Y \rightarrow X$ of Gestalten is called *n-coaffine* if the unit map

$$\mathbb{1}_{\mathcal{O}(X)_{n+1}} \rightarrow (\mathcal{O}(Y)/\mathcal{O}(X))_n$$

is an isomorphism in $\text{CAlg}(\mathcal{O}(X)_{n+1})$.

Observation 9.3. If $n < m$, then every n -affine map is m -affine, and every m -coaffine map is n -coaffine.

Proof. The assertion for affine maps follows, with $A = \mathcal{O}(X)$, from the factorization $\text{CAlg}(A_{n+1}) \hookrightarrow \text{CAlg}(A_{m+1}) \hookrightarrow \text{StRing}_{A/}$. For coaffine maps, if $(\mathcal{O}(Y)/\mathcal{O}(X))_m$ is the unit, then applying the endomorphism transition functors $\text{End}_m^{\mathcal{O}(X)}, \dots, \text{End}_{n+1}^{\mathcal{O}(X)}$ identifies each lower relative algebra with the corresponding unit. $\square$

Corollary 9.4. *If $f: Y \rightarrow X$ is both n -affine and n -coaffine, then f is an isomorphism.*

Proof. An n -affine map is classified by its relative algebra $(O(Y)/O(X))_n \in \text{CAlg}(O(X)_{n+1})$. If f is also n -coaffine, this algebra is the unit. The n -affine Gestalt over X classified by the unit algebra is the identity of X . $\square$

Lemma 9.5. *Let X be a Gestalt, let $m, n \geq 0$, and let $\varphi: Z \rightarrow W$ be an m -affine map between Gestalten over X . Then its n -affinization over X ,*

$$\varphi^{n\text{-affine}}: (Z/X)^{n\text{-affine}} \rightarrow (W/X)^{n\text{-affine}},$$

is again m -affine.

Proof. Write $Z' := (Z/X)^{n\text{-affine}}$ and $W' := (W/X)^{n\text{-affine}}$, both n -affine over X by construction. Thus $\varphi^{n\text{-affine}}: Z' \rightarrow W'$ is a map between n -affine Gestalten over X , and is therefore itself n -affine by the cancellation property of Proposition 6.3(4).

If $m \geq n$, this already gives the claim, since an n -affine map is m -affine by Observation 9.3.

If $m \leq n$, it remains to upgrade n -affineness to m -affineness. By the recursive criterion for affineness following Definition 6.1, a map that is n -affine is m -affine as soon as its relative structure maps

$$L_i^{O(Z')/O(W')}: \text{Mod}_{i+1}^{O(W')}((O(Z')/O(W'))_i) \rightarrow (O(Z')/O(W'))_{i+1}$$

are equivalences in the remaining degrees $m \leq i < n$. In the inverse-limit description of Stefanich algebras from Corollary 5.9, n -affinization leaves the relative algebra data and their transition maps through degree n unchanged. Consequently, in the displayed range the relative structure maps for $\varphi^{n\text{-affine}}$ identify with those for φ . They are equivalences because φ is m -affine. This verifies the criterion, so $\varphi^{n\text{-affine}}$ is m -affine. $\square$

We record the following observation.

Proposition 9.6. *The n -coaffine maps and the n -affine maps form a factorization system on Gest. The factorization of a map $Y \rightarrow X$ is*

$$Y \rightarrow (Y/X)^{n\text{-affine}} \rightarrow X.$$

Proof. Write $Y' := (Y/X)^{n\text{-affine}}$, and let $g: Y' \rightarrow X$ be the structure map. By construction, g is n -affine and the map $Y \rightarrow Y'$ identifies the relative algebra of Y' over X in degree n with that of Y over X . By transitivity of relative right adjoints, applying $g_{n,*}$ to the unit map

$$\mathbb{1}_{O(Y')_{n+1}} \rightarrow (O(Y)/O(Y'))_n$$

gives precisely this identification over X . Since g is n -affine, $g_{n,*}$ is the forgetful functor from a module category and is conservative. The displayed unit is therefore an isomorphism, i.e. $Y \rightarrow Y'$ is n -coaffine.

It remains to check orthogonality. Let $i: Y \rightarrow Z$ be n -coaffine, let $p: W \rightarrow X$ be n -affine, and consider a commutative square

$$\begin{array}{ccc} Y & \longrightarrow & W \\ i \downarrow & & \downarrow p \\ Z & \longrightarrow & X. \end{array}$$

We regard Y and Z as objects of $\text{Gest}/_X$ through the bottom map. Since p is n -affine, with classifying algebra $R := (O(W)/O(X))_n \in \text{CAlg}(O(X)_{n+1})$, maps into W over X are computed by maps out of R :

$$\text{Hom}_{\text{Gest}/_X}(T, W) \simeq \text{Hom}_{\text{CAlg}(O(X)_{n+1})}(R, (O(T)/O(X))_n)$$

naturally in $T \in \text{Gest}/_X$. Applying this to $T = Z$ and $T = Y$, the map

$$\text{Hom}_{\text{Gest}/_X}(Z, W) \rightarrow \text{Hom}_{\text{Gest}/_X}(Y, W)$$

is an equivalence, because i being n -coaffine implies that

$$(O(Z)/O(X))_n \rightarrow (O(Y)/O(X))_n$$

is an isomorphism: if $A = O(X)$, $C = O(Z)$, and $B = O(Y)$, then the relative algebra for $A \rightarrow B$ is obtained by applying the right adjoint for $A \rightarrow C$ to the relative algebra for $C \rightarrow B$, which is the unit by n -coaffineness. This shows that the square we started with has a contractible anima of fillers.

The usual retract argument then identifies the two classes with the corresponding left and right orthogonal classes. $\square$

9.2 Affineness and diagonals

The main result of this section, Proposition 9.9 below, shows that, under suitable hypotheses, passing to the diagonal drops the level of affineness by precisely one. Part (2) of this proposition generalizes results of Gaitsgory on 1-affineness [Gai15], and related results for the general case appeared in Stefanich's thesis [Ste21].

Lemma 9.7. *Let $f: Y \rightarrow X$ be a map of Gestalten. If*

$$f_n^*: O(X)_{n+1} \rightarrow O(Y)_{n+1}$$

is an equivalence, then f is n -coaffine.

Proof. Let $f_{n,*}$ denote the right adjoint of f_n^* . Since f_n^* is an equivalence, the adjunction unit identifies the unit of $O(X)_{n+1}$ with

$$f_{n,*} \mathbb{1}_{O(X)_{n+1}} = (O(Y)/O(X))_n.$$

This is exactly the condition that f is n -coaffine. $\square$

Lemma 9.8. *Let $h: Y \rightarrow X$ be an $(n+1)$ -prim and $(n+1)$ -coaffine map of Gestalten. Then the diagonal $\Delta_h: Y \rightarrow Y \times_X Y$ is n -coaffine. In particular, if Δ_h is n -affine, h is injective.*

Proof. By $(n+1)$ -coaffineness, the unit map

$$\mathbb{1}_{O(X)_{n+2}} \rightarrow (O(Y)/O(X))_{n+1}$$

is an isomorphism. The Künneth formula for prim maps (Lemma 6.12), applied to the two copies of h , therefore gives

$$(O(Y \times_X Y)/O(X))_{n+1} \simeq (O(Y)/O(X))_{n+1} \otimes_{O(X)_{n+2}} (O(Y)/O(X))_{n+1} \simeq \mathbb{1}_{O(X)_{n+2}}.$$

Under this identification, the map induced by Δ_h on relative algebras over X is the multiplication map $\mathbb{1} \otimes \mathbb{1} \rightarrow \mathbb{1}$, hence is an isomorphism. Passing to underlying categories gives an equivalence

$$(\Delta_h)_n^*: \mathcal{O}(Y \times_X Y)_{n+1} \rightarrow \mathcal{O}(Y)_{n+1}.$$

Thus Lemma 9.7 shows that Δ_h is n -coaffine.

If Δ_h is also n -affine, then Corollary 9.4 shows that Δ_h is an isomorphism. Thus h is injective. $\square$

Proposition 9.9. *Let $n \geq 0$, and let $f: Y \rightarrow X$ be a map of Gestalten.*

- (1) *If f is $(n+1)$ -affine and Δ_f is n -prim, then Δ_f is n -affine.*
- (2) *If Δ_f is n -affine and f is $(n+1)$ -prim, then f is $(n+1)$ -affine.*

In particular, if f is n -proper, then f is $(n+1)$ -affine if and only if Δ_f is n -affine.

Proof. Part (1) is an instance of Lemma 6.17, applied to the codiagonal

$$\Delta_f^*: A = \mathcal{O}(Y \times_X Y) \rightarrow B = \mathcal{O}(Y),$$

which is what the diagonal Δ_f becomes under the equivalence $\text{Gest}/X \simeq (\text{StRing}_{\mathcal{O}(X)})^{\text{op}}$. We verify the three hypotheses of that lemma. First, since n -affine maps are stable under passing to codiagonals (Proposition 6.3(5)), the codiagonal Δ_f^* of the $(n+1)$ -affine map f is again $(n+1)$ -affine. Second, Δ_f^* is n -prim by assumption. Third, a projection $\text{pr}: Y \times_X Y \rightarrow Y$ yields a section $\text{pr}^*: \mathcal{O}(Y) \rightarrow \mathcal{O}(Y \times_X Y)$ of Δ_f^* , and pr^* is $(n+1)$ -affine because pr is a base change of f (Proposition 6.3(3)). Thus Lemma 6.17 applies and shows that Δ_f^* , and therefore Δ_f , is n -affine.

For part (2), consider the unique factorization of f as an $(n+1)$ -coaffine map followed by an $(n+1)$ -affine map

$$Y \xrightarrow{h} Y' := (Y/X)^{(n+1)\text{-affine}} \xrightarrow{g} X.$$

We first verify that Corollary 8.27 applies to f . Set

$$A := \mathcal{O}(X), \quad R := (\mathcal{O}(Y)/A)_{n+1} \in \text{CAlg}(A_{n+2}).$$

By Lemma 6.7, the algebra R is countably presented. The Künneth formula for the $(n+1)$ -prim map f gives

$$(\mathcal{O}(Y \times_X Y)/A)_{n+1} \simeq R \otimes_{A_{n+2}} R.$$

The relative algebra of Δ_f in degree $n+1$ is R , regarded as an algebra over $R \otimes_{A_{n+2}} R$ through multiplication. Both R and $R \otimes_{A_{n+2}} R$ are countably presented as A_{n+2} -algebras, so Lemma A.4(4), with $\kappa = \aleph_1$, shows that R is countably presented as an $R \otimes_{A_{n+2}} R$ -algebra. Since Δ_f is n -affine and hence $(n+1)$ -affine, it follows that Δ_f is a countably presented $(n+1)$ -affine map. Thus Δ_f is $(n+1)$ -prim by Proposition 6.9(1). Ambidexterity (Proposition 6.25(1)) now shows that f is $(n+2)$ -suave.

Consequently Corollary 8.27, applied with $n+1$ in place of n , shows that h is surjective. Moreover, the algebra R is countably presented, so the $(n+1)$ -affine map g is countably presented and hence $(n+1)$ -proper by Proposition 6.15(1). Applying the cancellation statement Proposition 6.15(4) to $f = g \circ h$, the $(n+1)$ -primness of f implies that h is $(n+1)$ -prim.

By construction, h is $(n + 1)$ -coaffine. It follows from Lemma 9.8 that Δ_h is n -coaffine, and so it remains to show that Δ_h is also n -affine.

Since Δ_f is assumed n -affine and n -affine maps are closed under left cancellation and base change, it suffices to show that Δ_g is n -affine:

$$\begin{array}{ccc}
 Y & & \\
 \Delta_h \downarrow & \searrow \Delta_f & \\
 Y \times_{Y'} Y & \longrightarrow & Y \times_X Y \\
 \downarrow & \lrcorner & \downarrow \\
 Y' & \xrightarrow{\Delta_g} & Y' \times_X Y'.
 \end{array}$$

By assumption, the map $\Delta_f: Y \rightarrow Y \times_X Y$ is n -affine. Since $(n + 1)$ -affinization preserves n -affine maps by Lemma 9.5, it thus remains to show that Δ_g is the image of Δ_f under the functor

$$(-)^{(n+1)\text{-affine}}: \text{Gest}_{/X} \rightarrow \text{Gest}_{/X}^{(n+1)\text{-affine}}.$$

But indeed, since the map $f: Y \rightarrow X$ is $(n + 1)$ -prim, the Künneth formula for prim maps (Lemma 6.12) implies that we have

$$(\mathcal{O}(Y \times_X Y)/\mathcal{O}(X))_{n+1} \xrightarrow{\sim} (\mathcal{O}(Y)/\mathcal{O}(X))_{n+1} \otimes_{\mathcal{O}(X)_{n+2}} (\mathcal{O}(Y)/\mathcal{O}(X))_{n+1}.$$

Under the equivalence

$$\text{CAlg}(\mathcal{O}(X)_{n+2}) \simeq (\text{Gest}_X^{(n+1)\text{-affine}})^{\text{op}},$$

the object Y' corresponds to $(\mathcal{O}(Y)/\mathcal{O}(X))_{n+1}$, and fiber products of $(n + 1)$ -affine Gestalten over X correspond to tensor products of the corresponding algebras in $\text{CAlg}(\mathcal{O}(X)_{n+2})$. This identifies the $(n + 1)$ -affinization of $Y \times_X Y$ over X with $Y' \times_X Y'$, and identifies the $(n + 1)$ -affinization of Δ_f with Δ_g , as claimed. It follows that Δ_h is both n -affine and n -coaffine, hence an isomorphism by Corollary 9.4. Thus h is a monomorphism. Since it is also surjective, it is an isomorphism. This concludes the proof. $\square$

9b Examples of Gestalten

Talk given by Sil.

The goal of this chapter is to discuss examples of Gestalten.

Convention 9.10. Throughout this chapter, we will fix n large enough, and work inside $\text{Gest}_{/X}^{n\text{-ét}}$ for some Gestalt X , mostly $X = *$ or $X = \text{Gest}(\text{D}(R))$ for some $R \in \text{CAlg}(\text{Sp})$. Since the inclusions $\text{Gest}_{/X}^{n\text{-ét}} \hookrightarrow \text{Gest}_{/X}^{(n+1)\text{-ét}}$ are fully faithful, it doesn't matter which precise choice of n we pick as long as all objects that are involved are n -étale.

9.3 Basic examples

We start with some elementary examples.

Example 9.11 (The point). The *terminal Gestalt* $*$ $\in \text{Gest}_{/X}$ is the identity map $\text{id}_X: X \rightarrow X$. This is 0-affine, 0-étale, and 0-proper.

Example 9.12 (The empty set). The *initial Gestalt* $\emptyset \in \text{Gest}_{/X}$ is the Gestalt associated to the zero ring:

$$\emptyset = \text{Gest}(*, *, *, \dots).$$

We claim that this is always 1-affine. For each $n \geq 0$, we have $(\mathcal{O}(\emptyset)/\mathcal{O}(X))_n = *$. But for $n \geq 2$, the category $\mathcal{O}(X)_n$ is a module over the pointed category 1Pr , hence $\mathcal{O}(X)_n$ itself is pointed, so that $\mathcal{O}(\emptyset)_n = \text{Mod}_{(\mathcal{O}(\emptyset)/\mathcal{O}(X))_{n-1}}(\mathcal{O}(X)_n) = \text{Mod}_*(\mathcal{O}(X)_n) = *$ as desired.

Since $\emptyset$ is 1-affine, it is in particular 1-proper. Moreover, it is 0-étale, as it is a colimit (an empty colimit) of 0-étale objects over X . It is 0-affine if and only if $\mathcal{O}(X)_1$ is pointed, or equivalently whenever X is a Gestalt over $\text{Gest}(\text{An}_*)$.

Example 9.13 (Finite sets). Let I be a finite set. Then we have the corresponding Gestalt

$$I := \bigsqcup_{x \in I} * \in \text{Gest}_{/X}.$$

If $X = \text{Gest}(A)$, we may express this as $\text{Gest}(\prod_I A_0, \prod_I A_1, \prod_I A_2, \dots)$, since colimits of Gestalten correspond to limits of Stefanich rings.

Note that I is 0-étale over X , as a colimit of 0-étale Gestalten. We claim it is also 1-affine: we have $(\mathcal{O}(I)/\mathcal{O}(X))_{n-1} = \prod_I \mathbb{1} \in \text{CAlg}(A_n)$, and so

$$\mathcal{O}(I)_n = \text{Mod}_{(\mathcal{O}(I)/\mathcal{O}(X))_{n-1}}(\mathcal{O}(X)_n) = \text{Mod}_{\prod_I \mathbb{1}}(A_n) \simeq \prod_I \text{Mod}_{\mathbb{1}}(A_n) \simeq \prod_I A_n$$

for all $n \geq 2$; here we use that A_n is semiadditive, since it is a module over 1Pr , which is semiadditive. If A_1 is semiadditive, then I is even 0-affine.

Example 9.14 (Countable sets). Similarly as in the previous example, for any countable set I we may define

$$I = \coprod_I * \in \text{Gest}/X.$$

Once again this is 0-étale and 1-affine; this uses the fact that 1Pr is not just semiadditive but even countably semiadditive, in that countable products agree with countable coproducts.

Example 9.15 (Countable anima). Generalizing the previous two examples, there exists a unique countable-colimit-preserving functor

$$\text{An}^{\text{ctbl}} \rightarrow \text{Gest}_{/X}^{0\text{-ét}}$$

given on the point by $* \mapsto *$. Every countable anima is 0-étale, hence 1-proper. However, it may not be 1-affine.

As a counterexample, we will show by induction on n that the Gestalt $B^{n+1}(\mathbb{Z}/2)$ is not n -affine over $\text{Gest}(\text{D}(\mathbb{Q}))$.

We first show, by induction on n , that it is 0-proper. It is 0-étale, hence 1-proper. Its diagonal is the projection

$$B^n(\mathbb{Z}/2) \times B^{n+1}(\mathbb{Z}/2) \longrightarrow B^{n+1}(\mathbb{Z}/2),$$

which is 0-proper by the induction hypothesis and base change. The initial case uses that the finite anima $\mathbb{Z}/2$ is 0-proper over $\text{Gest}(\text{D}(\mathbb{Q}))$. It therefore remains only to show that $B^{n+1}(\mathbb{Z}/2)$ is 0-prim. In other words, we need to show that the functor

$$\text{D}(\mathbb{Q}) \rightarrow \text{D}(\mathbb{Q})^{B^{n+1}(\mathbb{Z}/2)} \simeq \text{Mod}_{\mathbb{Q}[\Omega(B^{n+1}(\mathbb{Z}/2))]}(\text{D}(\mathbb{Q})) \simeq \text{Mod}_{\mathbb{Q}[B^n(\mathbb{Z}/2)]}(\text{D}(\mathbb{Q}))$$

has a $\text{D}(\mathbb{Q})$ -linear right adjoint. For $n \geq 1$, we have $H_*(B^n(\mathbb{Z}/2); \mathbb{Q}) \cong \mathbb{Q}$, so that $\mathbb{Q}[B^n(\mathbb{Z}/2)]$ is the monoidal unit in $\text{D}(\mathbb{Q})$ and this functor is even an isomorphism. In the remaining case $n = 0$, the functor is $\text{D}(\mathbb{Q}) \rightarrow \text{Fun}(B(\mathbb{Z}/2), \text{D}(\mathbb{Q}))$, whose right adjoint is given by taking $\mathbb{Z}/2$ -fixed points; we must show that this right adjoint preserves colimits. But since $\text{D}(\mathbb{Q})$ is rational, the norm map $(-)_h \rightarrow (-)^h$ is an isomorphism, so the right adjoint is also a left adjoint and hence preserves colimits.

We may now use Proposition 9.9(1) to inductively show that $B^{n+1}(\mathbb{Z}/2)$ is not n -affine. First, $B(\mathbb{Z}/2)$ is not 0-affine: if it were, then the category $\text{D}(\mathbb{Q})^{B(\mathbb{Z}/2)}$ of rational $\mathbb{Z}/2$ -representations would be equivalent to

$$\text{Mod}_{\mathbb{Q}^{B(\mathbb{Z}/2)}}(\text{D}(\mathbb{Q})) \simeq \text{D}(\mathbb{Q}),$$

because $B(\mathbb{Z}/2)$ has trivial rational cohomology. This contradicts the existence of nontrivial rational $\mathbb{Z}/2$ -representations.

Now let $n \geq 1$ and assume for contradiction that $B^{n+1}(\mathbb{Z}/2)$ is n -affine. Since its diagonal is $(n-1)$ -prim, Proposition 9.9(1), applied with $n-1$, shows that the diagonal is $(n-1)$ -affine. Hence the pullback square

$$\begin{array}{ccc} B^n(\mathbb{Z}/2) & \xrightarrow{\quad} & B^{n+1}(\mathbb{Z}/2) \\ \downarrow & \lrcorner & \downarrow \Delta \\ * & \xrightarrow{\quad} & B^{n+1}(\mathbb{Z}/2) \times B^{n+1}(\mathbb{Z}/2) \end{array}$$

would imply that $B^n(\mathbb{Z}/2)$ is $(n-1)$ -affine, contradicting the induction hypothesis.

9.4 Examples from algebraic geometry

Recall that for a Gestalt of the form $X = \text{Gest}(\text{D}(R))$ for some ring spectrum $R \in \text{CAlg}(\text{Sp})$, we defined a functor

$$\text{Gest}(-) : \text{Shv}_{\text{fpqc}}(\text{Aff}_R)^{\mathbb{N}_1} \rightarrow \text{Gest}_X, \quad Y \mapsto \text{Gest}(\Omega^\infty \Gamma(Y), \text{D}(Y), 1\text{Pr}_Y, 2\text{Pr}_Y, \dots).$$

In fact, we can define things more generally.

Example 9.16 (Affine line). There is a functor

$$\mathcal{O}(-)_0 : (\text{Gest}_X)^{\text{op}} \rightarrow \text{An}, \quad Y \mapsto \mathcal{O}(Y)_0.$$

Writing $A := \mathcal{O}(X) \in \text{StRing}$, this is represented by the *affine line*

$$\mathbb{A}_X^1 := \text{Gest}_X(A_0\{x\}),$$

where $A_0\{x\}$ is the free E_∞ -algebra on one generator x over A_0 :

$$\text{Hom}_{\text{Gest}_X}(Y, \mathbb{A}_X^1) \simeq \text{Hom}_{\text{CAlg}(A_1)}(A_0\{x\}, (\mathcal{O}(Y)/\mathcal{O}(X))_0) \simeq \mathcal{O}(Y)_0,$$

where the first equivalence is the adjunction $\text{Gest}_X(-) \dashv (\mathcal{O}(-)/\mathcal{O}(X))_0$ of Definition 6.1, using $\mathbb{A}_X^1 = \text{Gest}_X(A_0\{x\})$. Note that $\mathbb{A}_X^1$ is 0-affine, hence 0-proper and 1-étale.

Over the absolute base, this is $\text{Gest}(*\{x\})$, where

$$*\{x\} = \bigsqcup_{n \geq 0} */\Sigma_n$$

is the free E_∞ -monoid (in *anima*) on one generator.

Example 9.17 (Flat affine line). Over $\mathbb{S}$, or even over $\mathbb{Z}$, the free E_∞ -algebra $\mathbb{Z}\{x\} = \bigoplus_{n \geq 0} \mathbb{Z}[*/\Sigma_n]$ is *not* isomorphic to the polynomial algebra $\mathbb{Z}[x]$; rather, it has contributions coming from the group homology of symmetric groups. After base change to $\mathbb{Q}$, the two versions of the polynomial algebra agree.

Over a general base $X = \text{Gest}(A)$, we will denote this other version of the affine line by

$$\mathbb{G}_a := \text{Gest}_X(A_0[x]),$$

where $A_0[x] = \bigsqcup_{n \geq 0} A_0 \cdot x^n$ is the polynomial algebra over A_0 . This is 0-affine, and hence 0-proper and 1-étale.

Over the absolute base, we have $\mathbb{G}_a = \text{Gest}(\mathbb{N})$, the natural numbers.

Example 9.18 (Multiplicative group). Consider the subfunctor

$$\text{GL}_1 \subseteq \mathbb{A}_X^1$$

sending a Gestalt $Y \in \text{Gest}_X$ to the invertible elements $\mathcal{O}(Y)_0^\times$ of the commutative ring $\mathcal{O}(Y)_0$. This is representable by the Gestalt

$$\text{GL}_1 := \text{Gest}_X(A_0\{x^\pm\}),$$

the free E_∞ -algebra over A_0 on an invertible element x .

There is again another version:

$$\mathbb{G}_m := \mathrm{GL}_1 \times_{\mathbb{A}^1} \mathbb{G}_a = \mathrm{Gest}_X(A_0[x^{\pm 1}]).$$

Over the absolute base, this is $\mathrm{Gest}(\mathbb{Z})$, where we regard $\mathbb{Z}$ as a “multiplicative group”: it represents the functor taking Y to $\mathrm{Hom}_{\mathrm{CAlg}(\mathrm{An})}(\mathbb{Z}, \mathcal{O}(Y)_0)$.

Over $D(\mathbb{Q})$ (or just $D_{\geq 0}(\mathbb{Q})$), GL_1 and $\mathbb{G}_m$ are isomorphic.

Example 9.19 (Projective line). We define the *projective line* as the pushout

$$\mathbb{P}_X^1 := \mathbb{A}_X^1 \sqcup_{\mathrm{GL}_1} \mathbb{A}_X^1$$

in $\mathrm{Gest}/_X$. Here the two maps $\mathrm{GL}_1 \rightarrow \mathbb{A}^1$ are the maps represented by the algebra maps $A_0\{x\} \rightarrow A_0\{x^{\pm 1}\}$ given by $x \mapsto x$ and $x \mapsto x^{-1}$.

As a pushout of 1-étale Gestalten, this is 1-étale. It is not 0-affine, but its diagonal is 0-affine: The product of two copies of $\mathbb{A}_X^1 \sqcup_{\mathrm{GL}_1} \mathbb{A}_X^1$ is covered by four copies of $\mathbb{A}^1 \times \mathbb{A}^1$ and the preimage of the diagonal is either $\mathbb{A}^1$ embedded diagonally, or GL_1 embedded via $x \mapsto (x, x^{-1})$, and both maps are 0-affine as maps between 0-affines.

While we may already define $\mathbb{P}_X^1$ over the absolute base, it turns out that before passing to the stable setting of $D(\mathbb{S})$, this is not well-behaved, and is different from the object one might expect. For concreteness, we explain the situation over a field k .

Proposition 9.20. *Let k be a field. Then the map*

$$\mathrm{Gest}_X(k[T^{\pm 1}]) \sqcup \mathrm{Gest}_X(k[T, \frac{1}{1-T}]) \rightarrow \mathrm{Gest}_X(k[T])$$

over $X = \mathrm{Gest}(D_{\geq 0}(k))$ is not surjective. After base change to $\mathrm{Gest}(D(k))$, it becomes surjective.

Proof. The fact that it is surjective after base change to $D(k)$ follows from fpqc descent for the affine functor $\mathrm{Spec}(A) \mapsto \mathcal{O}(\mathrm{Spec}(A))$, assembled from Theorem 3.1, together with Theorem 7.12. Indeed, after base change the displayed map is the image of the usual Zariski cover $D(T) \cup D(1-T) = \mathrm{Spec}(k[T])$, whose Čech nerve realizes to $\mathrm{Spec}(k[T])$ in $\mathrm{Shv}_{\mathrm{fpqc}}(\mathrm{Aff}_k)^{\mathbf{N}^1}$.

We now show that the original map is not surjective. If the displayed map were surjective, then Proposition 8.25, applied with $n = 1$, would imply that the image of

$$D_{\geq 0}(k[T^{\pm 1}]) \times D_{\geq 0}(k[T, \frac{1}{1-T}]) \rightarrow D_{\geq 0}(k[T])$$

generates $D_{\geq 0}(k[T])$ under small colimits.

However, the mapping anima from any object in this image to $k[T] \in D_{\geq 0}(k[T])$ is contractible. Indeed, because $k[T]$ is 0-truncated, every $k[T]$ -linear map $M \rightarrow k[T]$ with M connective is detected on $\pi_0(M)$, and the mapping anima is discrete. If $M \in D_{\geq 0}(k[T^{\pm 1}])$, the image of such a map in $k[T]$ is divisible by all powers of T , hence lies in $\bigcap_{r \geq 0} T^r k[T] = 0$. Similarly, maps from objects of $D_{\geq 0}(k[T, \frac{1}{1-T}])$ have image in $\bigcap_{r \geq 0} (1-T)^r k[T] = 0$. The full subcategory of objects whose mapping anima to $k[T]$ is contractible is closed under colimits, so $k[T]$ cannot be generated by the displayed image. $\square$

Remark 9.21. After base change to $\text{Gest}(\mathbb{D}(\mathbb{S}))$, the descent argument in the previous proof, together with Theorems 3.1 and 7.12, shows that

$$\mathbb{A}^1 \sqcup_{\text{GL}_1} \mathbb{A}^1$$

comes from the obvious fpqc stack, namely the derived scheme obtained by gluing two affine lines along the invertible locus. In particular, it is 0-proper. If one instead forms the version

$$\mathbb{G}_a \sqcup_{\mathbb{G}_m} \mathbb{G}_a$$

using ordinary polynomial algebras rather than free E_∞ -algebras, then the resulting Gestalt is not only 0-proper but even 0-suave, which is apparently not the case for the free E_∞ -version above. Over $\mathbb{D}(\mathbb{Q})$, the two choices of polynomial algebra agree, so the distinction between these two projective lines disappears.

For our last example we work over $\mathbb{D}(\mathbb{Q})$.

Example 9.22. Consider the functor

$$\text{pic}(-): (\text{Gest}/\mathbb{D}(\mathbb{Q}))^{\text{op}} \rightarrow \text{An}, X \mapsto \mathcal{O}(X)_1^\times.$$

This is represented by a 2-étale Gestalt

$$\text{Pic} \in \text{Gest}/\mathbb{D}(\mathbb{Q}).$$

Construction 9.23. Over the base $X_0 := \text{Gest}(\mathbb{D}(\mathbb{Q}))$, let

$$\mu_2 := \text{Gest}_{X_0}(\mathbb{Q}[x]/(x^2 - 1))$$

denote the group of second roots of unity. Since 2 is invertible in $\mathbb{Q}$, this is isomorphic to the constant sheaf $\underline{\mathbb{Z}/2}$.

We construct a map

$$\text{Pic} \rightarrow \mu_2$$

of sheaves of spectra on $\text{Gest}/\mathbb{D}(\mathbb{Q})$, where we identify $\text{CGrp}(\text{An})$ -valued sheaves with $\text{Sp}_{\geq 0}$ -valued sheaves. Given an invertible object $L \in \mathcal{O}(X)_1^\times = \text{pic}(X)$, the swap map $c: L \otimes L \cong L \otimes L$ provides a map

$$1 \cong L^{-\otimes 2} \otimes (L \otimes L) \xrightarrow{\sim} L^{-\otimes 2} \otimes (L \otimes L) \cong 1,$$

which provides an element of μ_2 .

Proposition 9.24. *There is a fiber sequence of sheaves of spectra on $\text{Gest}/\mathbb{D}(\mathbb{Q})$ of the form*

$$*/\mathbb{G}_m \rightarrow \text{Pic} \rightarrow \mu_2.$$

Proof. The map $\text{Pic} \rightarrow \mu_2$ is surjective, since the $\mathbb{Q}$ -module $\mathbb{Q}[1]$ has -1 as a swap. In particular, we get a fiber sequence

$$\text{fib}(\text{Pic} \rightarrow \mu_2) \rightarrow \text{Pic} \rightarrow \mu_2.$$

We will now show that the map $* \rightarrow \text{fib}(\text{Pic} \rightarrow \mu_2)$ is surjective, i.e. that any $L \in \mathcal{O}(X)_1^\times$ such that the swap map $c: L \otimes L \rightarrow L \otimes L$ is trivial is already locally trivial. Write $A = \mathcal{O}(X)$. The action of Σ_n on $L^{\otimes n}$ is uniquely trivial for all $n \geq 1$. Indeed, this coherent action is given by a map

$$B\Sigma_n \rightarrow B\text{Aut}(L^{\otimes n}) \simeq B\text{Aut}(\mathbb{1}) = BA_0^\times.$$

Since X lies over $D(\mathbb{Q})$, the commutative algebra A_0 refines to an object of $\text{CAlg}(D_{\geq 0}(\mathbb{Q}))$. Thus $A_0^\times$ is a connective E_∞ -group whose higher homotopy groups are $\mathbb{Q}$ -vector spaces. The induced homomorphism $\Sigma_n \rightarrow \pi_0(A_0^\times)$ is trivial because transpositions act trivially by assumption, and transpositions generate Σ_n . All remaining obstruction and ambiguity groups are positive-degree group cohomology groups of the finite group Σ_n with coefficients in rational vector spaces, hence vanish. Therefore the displayed map is uniquely nullhomotopic.

We may therefore form the commutative algebra

$$B := \bigoplus_{n \in \mathbb{Z}} L^{\otimes n} \in \text{CAlg}(A_1),$$

with multiplication induced by the identifications $L^{\otimes m} \otimes L^{\otimes n} \simeq L^{\otimes(m+n)}$. Tensoring with L shifts the grading, so

$$L \otimes B \simeq B$$

in $\text{Mod}_B(A_1)$. Hence L becomes trivial after base change along the 0-affine map $\text{Gest}_X(B) \rightarrow X$. Moreover, the inclusion $\mathbb{1} \rightarrow B$ of the degree-zero summand splits in A_1 by projection onto degree zero, so $B \otimes -$ is conservative. By the conservative criterion of Proposition 8.22, the map $\text{Gest}_X(B) \rightarrow X$ is a cover. This proves that $* \rightarrow \text{fib}(\text{Pic} \rightarrow \mu_2)$ is surjective.

Finally, $* \times_{\text{Pic}} *$ classifies automorphisms of the trivial line bundle, which is precisely $\mathbb{G}_m$. Since $* \rightarrow \text{fib}(\text{Pic} \rightarrow \mu_2)$ is surjective, its Čech nerve presents the fiber as the quotient $*/\mathbb{G}_m$, giving the desired identification $\text{fib}(\text{Pic} \rightarrow \mu_2) \simeq */\mathbb{G}_m$. $\square$

Remark 9.25. We have $\pi_0 \text{Pic}(D(\mathbb{Q})) = \mathbb{Z}$, given by the shifts $\mathbb{Q}[n]$ for $n \in \mathbb{Z}$. However, the proof shows that the classes with n even are locally trivial in Gestalten. To trivialize them, one has to pass to the periodic E_∞ -rings

$$\bigoplus_{m \in \mathbb{Z}} \mathbb{Q}[nm].$$

Only for even n does this admit a commutative multiplication.

We will later study higher versions of this story, comparing $B^n \mathbb{G}_m$ with the moduli space of invertible objects of A_n . In the terminology used in the literature, this studies “invertible topological quantum field theories”; these are expected to be classified by a spectrum essentially given by an Anderson dual of the sphere. From this viewpoint, the μ_2 above should be seen as dual to the first stable homotopy group of spheres, $\pi_1 \mathbb{S} = \mathbb{Z}/2$. We will in fact see that the moduli space of invertible objects in the A_n defines a sheaf of spectra on Gestalten which is an internal Anderson dual of the sphere, proving a precise version of this expectation.

9.5 Examples from condensed and analytic mathematics

So far our examples have started from commutative algebras in simple symmetric monoidal base categories, such as $D(\mathbb{S})$ or $D(k)$, producing 0-affine Gestalten, from which more general ones were

obtained by gluing, i.e. by taking colimits. We now turn to examples of a more analytic flavour, coming from condensed and analytic mathematics.

Construction 9.26 (Light condensed anima). Generalizing the finite and countable sets considered above, we work over $X = \text{Gest}(\mathcal{D}(\mathbb{S}))$ and send a light profinite set $S = \lim_i S_i$ to the Gestalt

$$\text{Gest}_{\text{Gest}(\mathcal{D}(\mathbb{S}))}(\text{Cont}(S, \mathbb{S})) = \lim_i (S_i \times \text{Gest}(\mathcal{D}(\mathbb{S}))),$$

where $\text{Cont}(S, \mathbb{S})$ is the Stefanich ring of continuous $\mathbb{S}$ -valued functions on S , and $S_i \times \text{Gest}(\mathcal{D}(\mathbb{S}))$ denotes the finite coproduct of copies of the base indexed by S_i . This could already be defined over the absolute base $\text{Gest}(\text{An})$; for Proposition 9.27 below it is important to work stably. By hyperdescent (Theorem 3.1), this assignment takes hypercovers to colimits in Gestalten, so the functor

$$\text{ProFin}^{\text{light}} \rightarrow \text{Gest}_{\text{Gest}(\mathcal{D}(\mathbb{S}))}^{0\text{-affine}}$$

extends to a countable-colimit- and finite-limit-preserving functor

$$\text{CondAn}^{\text{light}} \rightarrow \text{Gest}_{\text{Gest}(\mathcal{D}(\mathbb{S}))}^{1\text{-ét}}.$$

As an important example, we can identify the internal real number object.

Proposition 9.27 (Scholze [Sch26, Prop. 9.1]). *The internal real number object of Cauchy reals in $\text{Gest}_{\text{Gest}(\mathcal{D}(\mathbb{S}))}$ is given by the image of the real numbers $\mathbb{R} \in \text{CondAn}^{\text{light}}$ under Construction 9.26, and is 1-affine, equal to*

$$\text{Gest}(\text{Shv}(\mathbb{R}, \mathcal{D}(\mathbb{S}))).$$

Proof. The internal construction of light profinite sets yields the functor $\text{ProFin}^{\text{light}} \rightarrow \text{Gest}$ used in Construction 9.26. Under its extension to light condensed anima, the real numbers $\mathbb{R} \in \text{CondAn}^{\text{light}}$ are sent to $\text{Gest}(\text{Shv}(\mathbb{R}, \mathcal{D}(\mathbb{S})))$, using descendability of $f_{\mathbb{S}}$ for a locally finite cover $f: C \rightarrow \mathbb{R}$ of $\mathbb{R}$ by a disjoint union of Cantor sets.¹

Now let Cauchy denote the internal object of Cauchy sequences of rational numbers with fixed convergence, i.e. sequences $(x_1, x_2, \dots)$ of rational numbers with $|x_i - x_j| \leq 1/\min(i, j)$ for all i and j . This is closed inside $\prod_{\mathbb{N}} \mathbb{Q}$, and the latter is 1-affine as a countable limit of 1-affine Gestalten; in fact, Cauchy is precisely the Gestalt corresponding to the category of $\mathcal{D}(\mathbb{S})$ -valued sheaves on the topological space of Cauchy sequences. Using this, one constructs a map

$$\text{Cauchy} \rightarrow \text{Gest}(\text{Shv}(\mathbb{R}, \mathcal{D}(\mathbb{S}))).$$

This is surjective, as the cover $f: C \rightarrow \mathbb{R}$ above factors over Cauchy. Moreover, the preimage of $0 \in \mathbb{R}$ is precisely the internal object $\text{Null} \subseteq \text{Cauchy}$ of nullsequences (automatically with fixed convergence, $|x_i| \leq 1/i$). Thus the induced map

$$\text{Cauchy}/\text{Null} \rightarrow \text{Gest}(\text{Shv}(\mathbb{R}, \mathcal{D}(\mathbb{S})))$$

is an isomorphism, as desired. □

¹This can be checked most cleanly using a descendability criterion for sheaves due to Clausen.

Remark 9.28. The internal real numbers object can also be defined over the absolute base. The proof shows that it is then given by the image in Gestalten of

$$\text{Gest}(\text{Shv}(\text{Cauchy}, \text{An})) \rightarrow \text{Gest}(\text{Shv}(\mathbb{R}, \text{An})),$$

where Cauchy now denotes the topological space of Cauchy sequences with fixed convergence. It is not known whether this map is surjective before passing to $D(\mathbb{S})$. In the absence of surjectivity, the image is a priori only a 2-étale injection into $\text{Gest}(\text{Shv}(\mathbb{R}, \text{An}))$, and hence might only be 3-affine.

One can build more general examples by allowing more general presentable symmetric monoidal categories as coefficients. We mention two such families.

Example 9.29 (Almost rings and Tamarkin’s category). Almost rings yield examples of Gestalten, thereby incorporating “almost mathematics”. A related example is Tamarkin’s category W of completely and continuously $\mathbb{R}$ -filtered $\mathbb{S}$ -modules. The map

$$\text{Gest}(W) \rightarrow \text{Gest}(D(\mathbb{S}))$$

is an $\mathbb{R}_{>0}$ -torsor, where $\mathbb{R}_{>0} = \text{Gest}(\text{Shv}(\mathbb{R}_{>0}, D(\mathbb{S})))$ denotes the internal positive real number object, acting on W by multiplicatively rescaling the filtration. In other words, working over $\text{Gest}(D(\mathbb{S}))$ there is a nontrivial map

$$* \rightarrow */\mathbb{R}_{>0},$$

exhibiting highly nontrivial internal structure on the base point.

Example 9.30 (Analytic rings and analytic stacks). Analytic rings yield examples of Gestalten. Any analytic ring A , consisting of a light condensed ring together with a chosen full subcategory $D(A)$ of “complete” modules, defines a symmetric monoidal category $D(A)$ and hence a Stefanich ring; this assembles into a functor

$$\text{AnRing} \rightarrow \text{StRing}, \quad A \mapsto (\Omega^\infty \text{End}_{D(A)}(\mathbb{1}), D(A), 1\text{Pr}_{D(A)}, 2\text{Pr}_{D(A)}, \dots).$$

Analytic stacks are defined precisely so that this extends to a functor

$$\text{AnStck} \rightarrow \text{Gest}.$$

In general, the resulting Gestalten are only colimits of 1-affine Gestalten, hence only 2-étale. In practice, after fixing a base analytic ring A , the stacks under consideration are 1-étale over $\text{Gest}(D(A))$.

All of the examples above are colimits of 1-affine Gestalten. A natural source of interesting Gestalten that go *beyond* such colimits are six-functor formalisms, which we turn to in the next chapter.

10 Six-functor formalisms and Gestalten

Talk given by Ou. The material from Sections 10.3 to 10.5 was not treated during the seminar and was added afterwards for completeness.

Today we discuss how to assign a Stefanich ring A_D to a six-functor formalism D , resulting in a functor $[-]_D: C \rightarrow \text{Gest}$ called the *transmutation* associated to D .

Recall that a *six-functor formalism* on a category C is a lax symmetric monoidal functor

$$D: \text{Span}(C, E) \rightarrow \mathbf{1Pr},$$

assigning to each $X \in C$ a presentably symmetric monoidal category $D(X)$ of coefficients, to each map f an adjoint pair $f^* \dashv f_*$, and to each map f in a wide subcategory E of C closed under base change an adjunction $f_! \dashv f^!$, compatible via base change and the projection formula (see Appendix B for some recollections). As was observed in [Sch25a, Lecture X], it is an empirical fact that many six-functor formalisms on the category $C = \text{Sch}$ of schemes (separated of finite type, say) can be factored as a composite

$$\text{Span}(\text{Sch}) \xrightarrow{X \mapsto X^?} \text{Span}(\text{AnStck}) \xrightarrow{D_{\text{qc}}} \mathbf{1Pr},$$

where AnStck is the category of analytic stacks and D_{qc} is a six-functor formalism on AnStck given by quasi-coherent sheaves. The functor $X \mapsto X^?$ from schemes to analytic stacks is called a *transmutation*, following [Bha22, Remark 2.3.8]. Here are some examples:

- (1) For $D(\mathbb{Z})$ -valued sheaves we have $D(X, \mathbb{Z}) \simeq D_{\text{qc}}(X_{\text{Betti}})$, where X_{Betti} is the Betti stack of X .
- (2) For algebraic D -modules, we have $D\text{mod}(X) \simeq D_{\text{qc}}(X_{\text{dR}})$, where X_{dR} is the de Rham stack of X .
- (3) There is an assignment $X \mapsto X^{\Delta_A}$ encoding prismatic cohomology.
- (4) There is an assignment $X \mapsto X_{\text{dR}}^{\square}$ encoding the solid de Rham D -module formalism.

The transmutation perspective has the virtue of black-boxing all the category theory to the theory of analytic stacks, and reducing the construction of six-functor formalisms to a geometric construction $X \mapsto X^?$. (In fact, the situation is even better: the functor $X \mapsto X^?$ often commutes with finite limits and is compatible with Zariski gluing, which means that it is completely determined by the ring stack $(\mathbb{A}^1)^?$. See Section 11.4 for details.)

The goal of the talk is to develop an analogous story of transmutations that works for six-functor formalisms over an arbitrary geometric category C in place of schemes, where we replace the

category of analytic stacks with the category $\mathbf{Gest}$ of Gestalten. More precisely, given a presentable six-functor formalism $D: \mathbf{Span}(C, E) \rightarrow \mathbf{1Pr}$, we will define a finite-limit-preserving transmutation functor $[-]_D: C \rightarrow \mathbf{Gest}/[-]_D$ fitting in a triangle

$$\begin{array}{ccc} C^{\mathrm{op}} & \xrightarrow{\quad [-]_D \quad} & \mathbf{Gest}^{\mathrm{op}} \\ & \searrow D^{*, \otimes} & \downarrow \mathcal{O}(-)_! \\ & & \mathbf{CAlg}(\mathbf{1Pr}), \end{array}$$

where $D^{*, \otimes}$ is the $(f^*, \otimes)$ -part of D . Moreover, we will show that the Gestalt $[X]_D$ is 1-affine over $[*]_D$ for every $X \in C$ if and only if D satisfies the categorical Künneth formula, in the sense that the canonical functor

$$D(X) \otimes_{D(*)} D(Y) \xrightarrow{\cong} D(X \times Y)$$

is an equivalence for all $X, Y \in C$. We will also construct a six-functor formalism on $\mathbf{Gest}$ and discuss when also the shriek-functoriality of D can be recovered from $[-]_D$.

10.1 The category of kernels $\mathbb{K}_D$

Fix a category C with finite limits and a presentable six-functor formalism $D: \mathbf{Span}(C) \rightarrow \mathbf{1Pr}$. In this section, we introduce and study a presentable 2-category $\mathbb{K}_D$, which is the main ingredient for the construction of the transmutation functor $[-]_D$.

Definition 10.1 (Aoki [Aok26, Def. 2.4]). Let C be a category with finite limits. Regarding the six-functor formalism D as a commutative algebra in $\mathbf{Fun}(\mathbf{Span}(C), \mathbf{1Pr})$ with the Day convolution symmetric monoidal structure, the 2-category of kernels of D is the module category

$$\mathbb{K}_D := \mathbf{Mod}_D(\mathbf{Fun}(\mathbf{Span}(C), \mathbf{1Pr})).$$

This comes with a symmetric monoidal functor from $\mathbf{Span}(C)^{\mathrm{op}}$: the Yoneda embedding followed by the free D -module functor:

$$\mathbf{Span}(C)^{\mathrm{op}} \rightarrow \mathbf{Fun}(\mathbf{Span}(C), \mathbf{1Pr}) \rightarrow \mathbb{K}_D, \quad X \mapsto [X]_{\mathbb{K}_D}.$$

Since $[X]_{\mathbb{K}_D}$ is the free D -module on the representable at X , the free-forgetful adjunction and the Yoneda lemma give for each D -module $M \in \mathbb{K}_D$ an equivalence

$$\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) \xrightarrow{\sim} M(X).$$

In particular, as the unit of $\mathbb{K}_D$ is D itself, we see that

$$\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, \mathbb{1}) \xrightarrow{\sim} D(X).$$

Every object of $\mathbf{Span}(C)$ is canonically self-dual [HM24, p. 2.3.9], and the embedding is symmetric monoidal, so $[Y]_{\mathbb{K}_D} \simeq [Y]_{\mathbb{K}_D}^\vee$ and $[X]_{\mathbb{K}_D} \otimes [Y]_{\mathbb{K}_D} \simeq [X \times Y]_{\mathbb{K}_D}$; hence

$$\mathrm{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, [Y]_{\mathbb{K}_D}) \simeq \mathrm{Hom}_{\mathbb{K}_D}([X \times Y]_{\mathbb{K}_D}, \mathbb{1}) = D(X \times Y).$$

With more care, one may show that the composition functor in $\mathbb{K}_D$ between representables is given by the formula

$$D(Y \times Z) \times D(X \times Y) \rightarrow D(X \times Z), \quad (B, A) \mapsto \text{pr}_{XZ,!}(\text{pr}_{XY}^* A \otimes \text{pr}_{YZ}^* B).$$

Note that this definition of the 2-category of kernels is different from the one that usually appears in the literature, e.g. in [HM24]: Usually, one considers the canonical self-enrichment of $\text{Span}(C)$ via its closed monoidal structure, and then transfers the enrichment via the lax monoidal functor $D: \text{Span}(C) \rightarrow \text{Cat}$. (See Definition B.17.) The following result shows that $\mathbb{K}_D$ is the 1Pr-enriched presheaves on this 2-category.

Theorem 10.2 (Aoki [Aok26, Thm. 2.6]). *Let S be a small closed symmetric monoidal category and $\mathcal{V} \in \text{CAlg}(\text{Pr}^L)$, and let $D: S \rightarrow \mathcal{V}$ be a lax symmetric monoidal functor. Then there is a canonical equivalence*

$$\text{PSh}^{\mathcal{V}}(\tilde{S}^{\text{op}}) \simeq \text{Mod}_D(\text{Fun}(S, \mathcal{V})),$$

where $\tilde{S}$ is the $\mathcal{V}$ -enriched category obtained by base changing the self-enrichment of S along D .

Proof. The argument proceeds by monadicity.

Write $U: \text{An} \rightarrow \mathcal{V}$ for the unit functor. The *trivial* $\mathcal{V}$ -enrichment of S is the enriched category S^{triv} with the same objects as S and

$$\underline{\text{Hom}}_{S^{\text{triv}}}(a, b) := U(\text{Hom}_S(a, b)).$$

In this case, $\mathcal{V}$ -enriched presheaves are given by

$$\text{PSh}^{\mathcal{V}}((S^{\text{triv}})^{\text{op}}) \simeq \text{Fun}(S, \mathcal{V}).$$

On the other hand, the closed symmetric monoidal structure on S makes it enriched over itself, with enriched hom $\underline{\text{Hom}}_S(a, b) \in S$, and change of enrichment along D gives the $\mathcal{V}$ -enrichment

$$\underline{\text{Hom}}_{\tilde{S}}(a, b) := D(\underline{\text{Hom}}_S(a, b)).$$

Equip $\text{Fun}(S, \mathcal{V})$ with Day convolution. Its monoidal unit is the functor

$$\mathbb{1}_{\text{Day}}(s) = U(\text{Hom}_S(\mathbb{1}_S, s)).$$

The lax symmetric monoidal structure on D is equivalently a commutative algebra structure on $D \in \text{Fun}(S, \mathcal{V})$, and its unit $\mathbb{1}_{\text{Day}} \rightarrow D$ induces an identity-on-objects $\mathcal{V}$ -enriched functor

$$i: S^{\text{triv}} \rightarrow \tilde{S},$$

which on enriched homs is given by the map

$$U(\text{Hom}_S(a, b)) \simeq U(\text{Hom}_S(\mathbb{1}_S, \underline{\text{Hom}}_S(a, b))) \longrightarrow D(\underline{\text{Hom}}_S(a, b)).$$

Put

$$\mathcal{C} := \text{Fun}(S, \mathcal{V}), \quad \mathcal{P} := \text{PSh}^{\mathcal{V}}(\tilde{S}^{\text{op}}).$$

Restriction and enriched left Kan extension along i give an adjunction

$$i_!: \mathcal{C} \rightleftarrows \mathcal{P}: i^*.$$

Since i is the identity on objects, i^* is conservative. Colimits of enriched functors are computed objectwise, so i^* also preserves all colimits. The Barr–Beck theorem therefore identifies

$$\mathcal{P} \simeq \text{Mod}_T(C), \quad T := i^*i_!.$$

The category $\mathcal{P}$ carries the enriched Day convolution symmetric monoidal structure. Its unit is the enriched functor represented by $\mathbb{1}_S$, and hence

$$i^*(\mathbb{1}_{\mathcal{P}})(s) = \underline{\text{Hom}}_{\widetilde{S}}(\mathbb{1}_S, s) = D(\underline{\text{Hom}}_S(\mathbb{1}_S, s)) \simeq D(s).$$

In fact, this is an identification $i^*(\mathbb{1}_{\mathcal{P}}) \simeq D$ of commutative algebras in C : the algebra structure on the left is induced by the lax symmetric monoidal structure on i^* , which by construction is the lax symmetric monoidal structure on D .

It follows formally that i^* lifts to a functor

$$\widetilde{i}^*: \mathcal{P} \rightarrow \text{Mod}_D(C),$$

where for $Y \in \mathcal{P}$ the action map on i^*Y is given by the lax monoidal structure on i^* :

$$D \otimes i^*Y \simeq i^*(\mathbb{1}_{\mathcal{P}}) \otimes i^*Y \longrightarrow i^*(\mathbb{1}_{\mathcal{P}} \otimes Y) \simeq i^*Y.$$

Since $\widetilde{i}^*$ lies over i^* , it determines a canonical morphism of monads

$$\alpha: (-) \otimes D \longrightarrow T.$$

Concretely, its component at $P \in C$ is the projection map

$$\alpha_P: P \otimes D \simeq P \otimes i^*(\mathbb{1}_{\mathcal{P}}) \longrightarrow i^*(i_!P \otimes \mathbb{1}_{\mathcal{P}}) \simeq i^*i_!P.$$

The fact that α is a morphism of monads, including all of its higher coherences, is built into its construction from the lift $\widetilde{i}^*$.

It remains only to show that α is an equivalence. Both $T = i^*i_!$ and $(-) \otimes D$ preserve colimits and are $\mathcal{V}$ -linear. Moreover, $C = \text{PSh}^{\mathcal{V}}((S^{\text{triv}})^{\text{op}})$ is generated under colimits and the $\mathcal{V}$ -action by the enriched representables

$$h_a := \underline{\text{Hom}}_{S^{\text{triv}}}(a, -) = U(\text{Hom}_S(a, -)), \quad a \in S.$$

It therefore suffices to prove that α_{h_a} is an equivalence for every $a \in S$. Enriched left Kan extension takes the representable h_a to the corresponding representable of $\widetilde{S}$, and hence

$$T(h_a)(s) = i^*i_!(h_a)(s) \simeq \underline{\text{Hom}}_{\widetilde{S}}(a, s) = D(\underline{\text{Hom}}_S(a, s)).$$

On the other hand, the Day-convolution Yoneda formula, together with the adjunction $a \otimes - \dashv \underline{\text{Hom}}_S(a, -)$, gives a canonical equivalence

$$(h_a \otimes D)(s) \simeq D(\underline{\text{Hom}}_S(a, s)).$$

Under these identifications, α_{h_a} is the identity of the enriched representable $\underline{\text{Hom}}_{\widetilde{S}}(a, -)$, and is therefore an equivalence. We conclude that α is an equivalence of monads, and hence

$$\text{PSh}^{\mathcal{V}}(\widetilde{S}^{\text{op}}) = \mathcal{P} \simeq \text{Mod}_T(C) \simeq \text{Mod}_D(C) = \text{Mod}_D(\text{Fun}(S, \mathcal{V})),$$

as desired. $\square$

Corollary 10.3. *The category of kernels $\mathbb{K}_D$ from Definition 10.1 is the 1Pr-enriched presheaves on the 2-category of kernels K_D from Definition B.17.*

Proof. Take $\mathcal{V} = 1\text{Pr}$ and $S = \text{Span}(C)$. □

We will now show that the category of kernels is not very interesting if D satisfies the categorical Künneth formula: $\mathbb{K}_D$ is simply the module category $\text{Mod}_{D(*)}(1\text{Pr})$ in that case. We start by constructing a comparison functor.

Construction 10.4. The monoidal unit of $\mathbb{K}_D$ is $\mathbb{1} = [*]_{\mathbb{K}_D} = D$. Taking $X = *$ in the formula

$$\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X)$$

gives

$$\text{End}_{\mathbb{K}_D}(\mathbb{1}) = D(*).$$

The resulting action of $D(*)$ on $\mathbb{1}$ defines an induction functor

$$L_D: \text{Mod}_{D(*)}(1\text{Pr}) \rightarrow \mathbb{K}_D, \quad N \mapsto N \otimes_{D(*)} \mathbb{1}.$$

It is left adjoint to evaluation at the point,

$$G_D: \mathbb{K}_D \rightarrow \text{Mod}_{D(*)}(1\text{Pr}), \quad M \mapsto M(*) = \text{Hom}_{\mathbb{K}_D}(\mathbb{1}, M),$$

where the $D(*)$ -action on $M(*)$ is given by precomposition. Indeed, the adjunction $L_D \dashv G_D$ is the tensor-hom adjunction for the $D(*)$ -module object $\mathbb{1} \in \mathbb{K}_D$.

Proposition 10.5. *The induction functor $L_D: \text{Mod}_{D(*)}(1\text{Pr}) \rightarrow \mathbb{K}_D$ of Construction 10.4 is fully faithful, with essential image the restricted objects: $M \in \mathbb{K}_D$ lies in it if and only if $D(X) \otimes_{D(*)} M(*) \rightarrow M(X)$ is an equivalence for all X .*

Proof. Work with $\mathbb{K}_D = \text{Mod}_D(\text{Fun}(\text{Span}(C), 1\text{Pr}))$ and the formula $\text{Hom}_{\mathbb{K}_D}([X]_{\mathbb{K}_D}, M) = M(X)$ of Definition 10.1.

Evaluation is atomic. Recall that an object A of a 1Pr-module category C is 1Pr-atomic if the 1Pr-valued functor $\text{Hom}_C(A, -)$ preserves colimits and 1Pr-tensors. The formula above identifies the corepresentable of $[X]_{\mathbb{K}_D}$ with the evaluation $\text{ev}_X: M \mapsto M(X)$. Colimits in $\mathbb{K}_D$ are created by the forgetful functor to $\text{Fun}(\text{Span}(C), 1\text{Pr})$, where they are pointwise, so ev_X preserves colimits and is 1Pr-linear; thus each $[X]_{\mathbb{K}_D}$ is 1Pr-atomic. In particular $\mathbb{1} = [*]_{\mathbb{K}_D}$ is atomic, with $\text{End}_{\mathbb{K}_D}(\mathbb{1}) = \text{ev}_*(\mathbb{1}) = D(*)$.

Full faithfulness. By Construction 10.4, we have an adjunction $L_D \dashv G_D$. The functor L_D is fully faithful if and only if the unit $N \rightarrow G_D L_D(N)$ is an equivalence. As evaluation at $*$ is colimit-preserving and 1Pr-linear, it commutes with $N \otimes_{D(*)} (-)$, so

$$G_D L_D(N) = (N \otimes_{D(*)} \mathbb{1})(*) \simeq N \otimes_{D(*)} \mathbb{1}(*) = N \otimes_{D(*)} D(*) = N,$$

using $\mathbb{1}(*) = D(*)$. Hence $G_D L_D \simeq \text{id}$.

Essential image. As L_D is fully faithful with right adjoint G_D , an object M lies in its image if and only if the counit $L_D G_D(M) \rightarrow M$ is an equivalence. Evaluating $L_D G_D(M) = M(*) \otimes_{D(*)} \mathbb{1}$ at X (again using that ev_X commutes with $M(*) \otimes_{D(*)} (-)$, and $\mathbb{1}(X) = D(X)$) gives $D(X) \otimes_{D(*)} M(*)$, and the counit at X is the canonical map $D(X) \otimes_{D(*)} M(*) \rightarrow M(X)$. So M is restricted if and only if this is an equivalence for all X . □

Corollary 10.6. *The inclusion $L_D : \text{Mod}_{D(*)}(1\text{Pr}) \hookrightarrow \mathbb{K}_D$ is an equivalence if and only if D satisfies the categorical Künneth formula: for all objects $X, Y \in C$, we have*

$$D(X) \otimes_{D(*)} D(Y) \xrightarrow{\sim} D(X \times Y).$$

Proof. The objects in the image of L_D are closed under colimits and 1Pr -tensoring, and $\mathbb{K}_D$ is 1Pr -atomically generated by the $[Y]_{\mathbb{K}_D}$, so L_D is an equivalence iff every $[Y]_{\mathbb{K}_D}$ is restricted. By the $\text{Hom}_{\mathbb{K}_D}$ -formula, $[Y]_{\mathbb{K}_D}(*) = D(Y)$ and $[Y]_{\mathbb{K}_D}(X) = D(X \times Y)$, so this holds iff $D(X) \otimes_{D(*)} D(Y) \rightarrow D(X \times Y)$ is an equivalence for all X, Y , which is the categorical Künneth formula. $\square$

10.2 Transmutation

Our next goal is to use the category of kernels construction to define the transmutation functor associated to D . For each object $X \in C$, we may form the category of kernels associated to the restricted six-functor formalism

$$D_X : \text{Span}(C_{/X}) \rightarrow \text{Span}(C) \xrightarrow{D} 1\text{Pr},$$

giving rise to

$$\mathbb{K}_{D,X} := \text{Mod}_{D_X}(\text{Fun}(\text{Span}(C_{/X}), 1\text{Pr})) \in \text{CAlg}(2\text{Pr}).$$

For each map $f : Y \rightarrow X$ in C , left Kan extension along the symmetric monoidal pullback functor $\text{Span}(f^*) : \text{Span}(C_{/X}) \rightarrow \text{Span}(C_{/Y})$ induces a symmetric monoidal restriction functor

$$\text{Span}(f^*)_! : \text{Fun}(\text{Span}(C_{/X}), 1\text{Pr}) \rightarrow \text{Fun}(\text{Span}(C_{/Y}), 1\text{Pr}),$$

which by construction sends D_X to D_Y and hence induces a symmetric monoidal functor

$$f_1^* : \mathbb{K}_{D,X} \rightarrow \mathbb{K}_{D,Y}.$$

All these constructions are functorial in f , resulting in a functor

$$C^{\text{op}} \rightarrow \text{CAlg}(2\text{Pr}).$$

Proposition 10.7 (Scholze [Sch26, Prop. 9.4]). *For every morphism $f : Y \rightarrow X$, the functor $f_1^* : \mathbb{K}_{D,X} \rightarrow \mathbb{K}_{D,Y}$ admits canonically equivalent left and right adjoints, $f_{1,\#} \simeq f_{1,*}$. They satisfy the projection formula and their formation is compatible with base change.*

Proof. For the duration of the proof, write (and use the same notation for all other maps)

$$f_! : C_{/Y} \rightarrow C_{/X}$$

for postcomposition with f . It is left adjoint to the base-change functor $f^* : C_{/X} \rightarrow C_{/Y}$. The naturality squares of the unit and counit of $f_! \dashv f^*$ are cartesian: those for the counit are the usual base-change squares for the projections $W \times_X Y \rightarrow W$, while those for the unit are the corresponding squares for the graph maps $Z \rightarrow Z \times_X Y$. Applying both parts of [BH21, Corollary C.21] therefore gives a pair of adjunctions

$$\text{Span}(f_!) \dashv \text{Span}(f^*) \dashv \text{Span}(f_!).$$

Passing to 1Pr-enriched presheaves by precomposition reverses both adjunctions, giving

$$\mathrm{Span}(f^*)^* \dashv \mathrm{Span}(f_!)^* \dashv \mathrm{Span}(f^*)^*.$$

Here the middle functor goes from $\mathrm{Fun}(\mathrm{Span}(C/X), 1\mathrm{Pr})$ to $\mathrm{Fun}(\mathrm{Span}(C/Y), 1\mathrm{Pr})$. It is symmetric monoidal: the base-change functor f^* preserves finite products, so $\mathrm{Span}(f^*)$ is symmetric monoidal, and the adjunction $\mathrm{Span}(f^*) \dashv \mathrm{Span}(f_!)$ identifies $\mathrm{Span}(f_!)$ with left Kan extension along $\mathrm{Span}(f^*)$. To see that this adjunction passes further to module categories over D_X and D_Y , we need to verify the projection formula

$$\mathrm{Span}(f^*)^* N \otimes M \xrightarrow{\sim} \mathrm{Span}(f^*)^* (N \otimes \mathrm{Span}(f_!)^* M).$$

It is enough to check this when M and N are representable, since both sides preserve colimits and 1Pr-tensors in both variables. For objects $A \rightarrow X$ and $B \rightarrow Y$, the assertion reduces to the canonical isomorphism

$$f_!(B \times_Y f^* A) \simeq f_! B \times_X A,$$

which follows directly from the defining pullback square for $f^* A$. All in all, we obtain the desired pair of adjunctions

$$f_{1,\#} \dashv f_1^* \dashv f_{1,*},$$

where by construction we have an identification $f_{1,\#} = f_{1,*}$. The projection formula exhibits the adjoint $\mathrm{Span}(f_!)^*$ as a D_X -linear functor, so the two adjunctions pass to the corresponding module categories and give the projection formula for $f_{1,\#}$ and $f_{1,*}$.

It remains to show that these adjoints are compatible with base change. Consider a cartesian square

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ g' \downarrow & \lrcorner & \downarrow g \\ Y & \xrightarrow{f} & X. \end{array}$$

There is a canonical equivalence $f^* g_! \simeq g'_! f'^*$ between functors on slice categories. Passing successively to span categories, enriched presheaves, and modules gives

$$g_1^* f_{1,*} \simeq f'_{1,*} g_1'^*.$$

Under the canonical equivalences $f_{1,\#} \simeq f_{1,*}$ and $f'_{1,\#} \simeq f'_{1,*}$, this is also the base-change equivalence for the left adjoints. $\square$

The category of kernels already carries nontrivial geometry. The assignment

$$C \rightarrow \mathrm{Gest}, \quad X \mapsto \mathrm{Gest}(\mathbb{K}_{D,X}),$$

takes values in the 2-affine Gestalten, and Proposition 10.7 shows that for each map $f: X \rightarrow Y$ the induced map $\mathrm{Gest}(\mathbb{K}_{D,X}) \rightarrow \mathrm{Gest}(\mathbb{K}_{D,Y})$ is 2-proper and 1-prim. It is not 1-suave in general, though: the missing left adjoint $f_{2,\#}$ exists precisely when the assignment $X \mapsto \mathrm{Gest}(\mathbb{K}_{D,X})$ satisfies the Künneth formula, which it need not. To force the Künneth formula to hold (and thereby obtain maps that are 1-proper and 1-étale), we iterate the category of kernels construction.

Construction 10.8 (Iterated categories of kernels). Set $D^{(1)} := D$. By Proposition 10.7, the functors associated with a map $f: Y \rightarrow X$ have adjoints

$$f_{1,\#} \simeq f_{1,*}: \mathbb{K}_{D,Y} \longrightarrow \mathbb{K}_{D,X}$$

that are compatible with base change and projection formulas. Together with the lax symmetric monoidal structure of the kernel construction, these adjunctions assemble the assignment $X \mapsto \mathbb{K}_{D,X}$ into a presentable six-functor formalism

$$D^{(2)}: \text{Span}(C) \longrightarrow 2\text{Pr}, \quad D^{(2)}(X) := \mathbb{K}_{D,X}.$$

More generally, suppose that $D^{(n)}: \text{Span}(C) \rightarrow n\text{Pr}$ has been constructed. Restricting it to C/X and applying the kernel construction in $n\text{Pr}$ defines

$$D^{(n+1)}(X) := \mathbb{K}_{D^{(n)},X} := \text{Mod}_{D^{(n)}_X}(\text{Fun}(\text{Span}(C/X), n\text{Pr})) \in \text{CAlg}((n+1)\text{Pr}).$$

The higher-categorical form of Proposition 10.7 again provides base-change-compatible left and right adjoints, and hence makes $D^{(n+1)}$ a presentable six-functor formalism.

For $X \in C$ and $n \geq 1$, put $A_{D,X,n} := D^{(n)}(X)$, and set

$$A_{D,X,0} := \text{End}_{D(X)}(\mathbb{1}).$$

By the enriched-presheaf description of the kernel construction in Theorem 10.2, there are canonical equivalences

$$A_{D,X,n} \simeq \text{End}_{A_{D,X,n+1}}(\mathbb{1}) \quad (n \geq 0).$$

Consequently the sequence

$$A_{D,X} := (\text{End}_{D(X)}(\mathbb{1}), D(X), \mathbb{K}_{D,X}, \mathbb{K}_{D^{(2)},X}, \dots)$$

is a Stefanich ring, functorially in X .

Definition 10.9 (Transmutation). The structure map $X \rightarrow *$ induces a map of Stefanich rings $A_{D,*} \rightarrow A_{D,X}$. The *transmutation* of D is the functor

$$[-]_D: C \rightarrow \text{Gest}/\text{Gest}(A_{D,*}), \quad X \mapsto [X]_D := \text{Gest}(A_{D,X}).$$

Thus every $[X]_D$ is canonically a Gestalt over $[*]_D := \text{Gest}(A_{D,*})$.

Theorem 10.10. *For every presentable six-functor formalism D , the transmutation $[-]_D$ satisfies:*

- (1) *It commutes with finite limits;*
- (2) *For each $f: Y \rightarrow X$ in C , the map $[f]_D: [Y]_D \rightarrow [X]_D$ is 1-étale and 1-proper, and satisfies $f_{1,\#} \simeq f_{1,*}$;*
- (3) *The composite $O(-)_1 \circ [-]_D: C^{\text{op}} \rightarrow \text{CAlg}(1\text{Pr})$ recovers the $(f^*, \otimes)$ -functoriality of D .*

Proof. Adjointability. By construction the m -th level of $A_{D,X}$ has pullback the kernel pullback for $D^{(m-1)}$, so Proposition 10.7 applied to $D^{(m-1)}$ gives, for every $m \geq 2$, base-change-compatible left and right adjoints to $f_{m-1}^*: D^{(m)}(X) \rightarrow D^{(m)}(Y)$ with $f_{m-1,\#} \simeq f_{m-1,*}$. Unwinding the definitions of primness and suaveness for Stefanich rings, this is exactly the internal right-adjoint datum for 1-primness and the internal left-adjoint datum for 1-suaveness. The extra condition in n -suaveness (Definition 6.18) also holds: by the levelwise ambidexterity and Lemma 6.26, the adjoints preserve dualizable objects, so the dualizing object $(B/A)_m^! = f_{m,\#}(\mathbb{1})$ is dualizable, and for such objects the extra right-adjoint condition follows from the available left-adjoint one. Hence every $[f]_D$ is 1-prim and 1-suave.

Finite limits. The terminal object goes to $[*]_D = \text{Gest}(A_{D,*})$. For pullbacks, after passing to a slice it suffices to preserve binary products. Given $p_X: X \rightarrow S$ and $p_Y: Y \rightarrow S$, put $P = X \times_S Y$, with projections q_X, q_Y and structure map $p: P \rightarrow S$. The two projections induce a map of Stefanich rings over $A_{D,S}$ of the form

$$\Phi: A_{D,X} \otimes_{A_{D,S}} A_{D,Y} \rightarrow A_{D,P}.$$

We must show that this map is an isomorphism. Recall from Construction 5.19 the explicit description of the relative tensor product over the base $A_{D,S}$: we first form the levelwise tensor product in $\text{PreStRing}(A_{D,S})$, and then we spectrify. It will thus suffice to show that the levelwise tensor product in $\text{PreStRing}(A_{D,S})$ is isomorphic to $A_{D,P}$ (so that spectrifying is unnecessary). In level n , the comparison map in $\text{PreStRing}(A_{D,S})$ is given by the map

$$p_{X,n*} \mathbb{1}_X \otimes p_{Y,n*} \mathbb{1}_Y \rightarrow p_{n*} \mathbb{1}_P$$

in $\text{CAlg}((A_{D,S})_{n+1})$. But this follows from the projection formulas and base change established in Proposition 10.7:

$$p_{X,n*} \mathbb{1}_X \otimes p_{Y,n*} \mathbb{1}_Y \simeq p_{X,n*}(\mathbb{1}_X \otimes p_X^* p_{Y,n*} \mathbb{1}_Y) \simeq p_{X,n*} q_{X,n*} \mathbb{1}_P = p_{n*} \mathbb{1}_P,$$

using $p_X^* p_{Y,n*} \mathbb{1}_Y \simeq q_{X,n*} q_Y^* \mathbb{1}_Y = q_{X,n*} \mathbb{1}_P$. We conclude that $A_{D,X} \otimes_{A_{D,S}} A_{D,Y} \xrightarrow{\sim} A_{D,P}$, and so $[P]_D \xrightarrow{\sim} [X]_D \times_{[S]_D} [Y]_D$.

Properness/étaleness. In the Gestalt formalism 1-properness (resp. 1-étaleness) means 1-primness (resp. 1-suaveness) for the map and all its iterated diagonals. As $[-]_D$ preserves finite limits it preserves diagonals, $[\Delta_f]_D \simeq \Delta_{[f]_D}$, so applying the first part to f and to its iterated diagonals shows $[f]_D$ is 1-proper and 1-étale.

Recovering $D^{,\otimes}$.* By construction, the first level of $A_{D,X}$ is $O([X]_D)_1 = D(X)$, with pullback and tensor functoriality the original $(f^*, \otimes)$ of D ; hence $O(-)_1 \circ [-]_D$ recovers $D^{*,\otimes}$. $\square$

Remark 10.11. The proof of finite-limit preservation makes clear why in Construction 10.8 we need to iterate the kernel construction indefinitely: to compare the degree- n relative algebras, we work in the degree- $(n+1)$ algebra of the base object.

Corollary 10.12 (Künneth implies affineness). *Let $n \geq 1$, and suppose that $D^{(n)}$ satisfies the relative categorical Künneth formula: for every $S \in C$ and all $X, Y \in C/S$, the canonical map*

$$D^{(n)}(X) \otimes_{D^{(n)}(S)} D^{(n)}(Y) \longrightarrow D^{(n)}(X \times_S Y)$$

is an equivalence. Then $[S]_D$ is n -affine for every $S \in C$. Consequently every structure map $[S]_D \rightarrow []_D$ is n -affine.*

Proof. We show more generally that, for every $m \geq n$, the coefficient system $D^{(m)}$ satisfies the same relative Künneth formula, and that

$$D^{(m+1)}(S) \simeq \text{Mod}_{D^{(m)}(S)}(m\text{Pr})$$

for every $S \in C$.

Assume this has been proved in degree m . For a fixed $S \in C$, the restriction of $D^{(m)}$ to the slice C/S satisfies the categorical Künneth formula in the sense of Corollary 10.6. Applying the same corollary with 1Pr replaced by $m\text{Pr}$ identifies the category of kernels of this restricted formalism with

$$\text{Mod}_{D^{(m)}(S)}(m\text{Pr}).$$

By construction this category of kernels is exactly $D^{(m+1)}(S)$.

Now let $X, Y \in C/S$. Using the identifications just obtained and the tensor product formula for module categories, we get

$$\begin{aligned} D^{(m+1)}(X) \otimes_{D^{(m+1)}(S)} D^{(m+1)}(Y) &\simeq \text{Mod}_{D^{(m)}(X)}(m\text{Pr}) \otimes_{\text{Mod}_{D^{(m)}(S)}(m\text{Pr})} \text{Mod}_{D^{(m)}(Y)}(m\text{Pr}) \\ &\simeq \text{Mod}_{D^{(m)}(X) \otimes_{D^{(m)}(S)} D^{(m)}(Y)}(m\text{Pr}) \\ &\simeq \text{Mod}_{D^{(m)}(X \times_S Y)}(m\text{Pr}) \\ &\simeq D^{(m+1)}(X \times_S Y). \end{aligned}$$

Thus $D^{(m+1)}$ again satisfies relative Künneth, completing the induction.

For every $S \in C$, the Stefanich ring of $[S]_D$ has m -th level $D^{(m)}(S)$ for $m \geq 1$. The equivalences

$$D^{(m+1)}(S) \simeq \text{Mod}_{D^{(m)}(S)}(m\text{Pr}) \quad (m \geq n)$$

are precisely the affine comparison maps in degrees $m \geq n$. Hence $[S]_D$ is n -affine. Applying the cancellation property of Proposition 6.3(4) to the n -affine Gestalten $[S]_D$ and $[*]_D$ proves the final assertion. $\square$

10.3 Dictionary: six-functor notions vs. Gestalten

Consider a presentable six-functor formalism $D: \text{Span}(C) \rightarrow 1\text{Pr}$, and consider a map $f: Y \rightarrow X$ in C . Using the transmutation functor from the previous section, we now have two competing notions of cohomological properties (suaveness, primness, étaleness, properness):

- We may consider the map $[f]_D$ of Gestalten, and use the definitions from the framework of Gestalten established in Chapter 6;
- We also have the notions from the six-functor calculus in the 2-category of kernels, recalled in Section B.5.

We now show that these two notions agree with each other.

The comparison is concentrated entirely at the bottom layer. By Theorem 10.10 every $[f]_D$ is automatically 1-étale and 1-proper with $f_{1,\sharp} \simeq f_{1,*}$, so the layers above the bottom are forced by

ambidexterity and separate no maps. Writing $A := \mathcal{O}([X]_{\mathcal{D}})$ and $B := \mathcal{O}([Y]_{\mathcal{D}})$, the morphism $[f]_{\mathcal{D}}$ is attached to a map $A \rightarrow B$ of Stefanich rings, whose lowest two layers are the coefficient category and the module 2-category of kernels,

$$A_1 = \mathcal{D}(X), \quad A_2 = \mathbb{K}_{\mathcal{D},X}.$$

By Theorem 10.2 the module 2-category $\mathbb{K}_{\mathcal{D},X}$ is the $\mathcal{D}_X$ -enriched presheaf category on the kernel 2-category $K_{\mathcal{D},X}$ (Definition B.17), so the fully faithful 2-Yoneda embedding $K_{\mathcal{D},X} \hookrightarrow \mathbb{K}_{\mathcal{D},X}$ preserves and reflects adjunctions. By the construction of Section 10.2, the level-0 relative datum $(B/A)_1 \in \text{CAlg}(A_2)$ has unit map

$$(f/A)_0^*: \mathbb{1} \longrightarrow (B/A)_1,$$

and, under the embedding above, it is the unit kernel $\mathbb{1}_Y$ of f regarded as a 1-morphism $[X] \rightarrow [Y]$ in $K_{\mathcal{D},X}$. At every higher level $m \geq 1$ the relative pullback $(f/A)_m^*$ already admits adjoints on both sides (Theorem 10.10). Thus the only conditions of Definitions 6.5 and 6.18 not automatic for $[f]_{\mathcal{D}}$ sit at level $m = 0$, where they concern precisely the unit kernel $\mathbb{1}_Y$.

We will now spell this out in a bit more detail.

Lemma 10.13 (Suave and prim maps). *Let $f: Y \rightarrow X$ be a morphism in $\mathcal{C}$.*

- (1) *f is $\mathcal{D}$ -suave (Definition B.21) if and only if $[f]_{\mathcal{D}}$ is 0-suave (Definition 6.18).*
- (2) *f is $\mathcal{D}$ -prim (Definition B.21) if and only if $[f]_{\mathcal{D}}$ is 0-prim (Definition 6.5).*

Proof. (2) By Definition 6.5 and the reduction above, $[f]_{\mathcal{D}}$ is 0-prim if and only if its one non-automatic condition holds: the level-0 datum $\mathbb{1} \rightarrow (B/A)_1$ admits a right adjoint in $A_2 = \mathbb{K}_{\mathcal{D},X}$. Since 2-Yoneda reflects adjunctions, this says that the unit kernel $\mathbb{1}_Y$, regarded as the 1-morphism $[X] \rightarrow [Y]$ in $K_{\mathcal{D},X}$, is a *left* adjoint, i.e. $\mathbb{1}_Y \in \text{Prim}_f(Y)$ (Definition B.19), i.e. f is $\mathcal{D}$ -prim (Definition B.21).

(1) Dually, 0-suaveness of $[f]_{\mathcal{D}}$ asks that each $(f/A)_m^*$ admit a *left* adjoint, subject to the secondary adjointability condition of Definition 6.18. By the reduction above and Theorem 10.10 every requirement is automatic except the left adjoint at $m = 0$ and the secondary condition at $m = 1$. The latter holds because the level-1 dualizing object $(B/A)_1^! = f_{1,\#}(\mathbb{1})$ is dualizable: the ambidextrous level-1 adjoints preserve dualizable objects by Lemma 6.26. As explained immediately after Definition 6.18, dualizability identifies the required right adjoint with the dual of the already available left adjoint. As in (2), the left adjoint at $m = 0$ says that $\mathbb{1}_Y$ is a *right* adjoint 1-morphism $[X] \rightarrow [Y]$ in $K_{\mathcal{D},X}$. Via the self-duality $K_{\mathcal{D},X} \simeq K_{\mathcal{D},X}^{\text{op}}$, this is equivalent to saying that $\mathbb{1}_Y$ is a left adjoint $[Y] \rightarrow [X]$, i.e. that $\mathbb{1}_Y \in \text{Suave}_f(Y)$. We conclude that $[f]_{\mathcal{D}}$ is 0-suave if and only if f is $\mathcal{D}$ -suave. $\square$

Corollary 10.14 (Étale and proper maps). *Let $f: Y \rightarrow X$ be a truncated morphism in $\mathcal{C}$.*

- (1) *f is $\mathcal{D}$ -étale (Definition B.25) if and only if $[f]_{\mathcal{D}}$ is 0-étale (Definition 6.23).*
- (2) *f is $\mathcal{D}$ -proper (Definition B.25) if and only if $[f]_{\mathcal{D}}$ is 0-proper (Definition 6.14).*

Proof. Since $[-]_{\mathcal{D}}$ preserves finite limits, it carries the tower of iterated diagonals of f to the tower of iterated codiagonals of $[f]_{\mathcal{D}}$. Since both $\mathcal{D}$ -étaleness (resp. $\mathcal{D}$ -properness) of f and 0-étaleness

(resp. 0-properness) of $[f]_D$ ask for suaveness (resp. primness) for all iterated diagonals, the claim follows from Lemma 10.13 applied at each stage. (Truncatedness of f ensures the tower of iterated diagonals terminates at isomorphisms.) $\square$

f in C	$[f]_D$ in Gest
D-suave	0-suave
D-prim	0-prim
D-étale	truncated and 0-étale
D-proper	truncated and 0-proper

10.4 Extension to stacks

The transmutation extends from C to stacks for the Grothendieck topology intrinsically attached to D . Following Scholze [Sch25a, Definition 5.3], call a family $\{f_i: Y_i \rightarrow X\}_{i \in I}$ *universally D-!-descending* if, for every $T \rightarrow X$, the canonical functor

$$D(T) \longrightarrow \lim_{[p] \in \Delta}^! \prod_{(i_0, \dots, i_p) \in I^{p+1}} D(T \times_X Y_{i_0} \times_X \cdots \times_X Y_{i_p})$$

is an equivalence. Here the limit is formed using exceptional pullbacks. Let J_D be the Grothendieck topology generated by these families, and write $\mathrm{Shv}_D(C)$ for the resulting category of sheaves.

Proposition 10.15. *Let $\{f_i: Y_i \rightarrow X\}_{i \in I}$ be universally D-!-descending. Then*

$$\coprod_{i \in I} [Y_i]_D \longrightarrow [X]_D$$

is a cover of Gestalten.

Proof. Both the hypothesis and the conclusion are stable under base change. We may therefore replace C by $C_{/X}$ and assume that $X = *$.

First suppose that I is countable. By Theorem 10.10, each $[Y_i]_D \rightarrow [*]_D$ is 1-étale, and so is their coproduct. Under the kernel description, the relative degree-1 dualizing object associated to $[Z]_D \rightarrow [*]_D$ is the corepresentable $[Z]_{\mathbb{K}_D}$: for $p: Z \rightarrow *$, the identification $p_{1,\#} \simeq p_{1,*}$ gives

$$p_{1,\#}(\mathbb{1}_Z) \simeq p_{1,*}(\mathbb{1}_Z) \simeq [Z]_{\mathbb{K}_D}.$$

Thus the cover criterion of Proposition 8.22 reduces the claim to showing that the augmentation

$$\mathrm{colim}_{[p] \in \Delta^{\mathrm{op}}} \prod_{(i_0, \dots, i_p) \in I^{p+1}} [Y_{i_0} \times \cdots \times Y_{i_p}]_{\mathbb{K}_D} \longrightarrow \mathbb{1} \in \mathbb{K}_D = \mathcal{O}([*]_D)_2$$

is an equivalence. The category $\mathbb{K}_D$ is 1Pr-atomically generated by the corepresentables $[T]_{\mathbb{K}_D}$ for $T \in C$. Applying $\mathrm{Hom}_{\mathbb{K}_D}([T]_{\mathbb{K}_D}, -)$ to the displayed map gives

$$\mathrm{colim}_{[p] \in \Delta^{\mathrm{op}}} \prod_{(i_0, \dots, i_p) \in I^{p+1}} D(T \times Y_{i_0} \times \cdots \times Y_{i_p}) \longrightarrow D(T).$$

This is the left adjoint of the $!$ -descent functor for the base change $T \rightarrow *$, and is therefore an equivalence by hypothesis. Hence the augmentation in $\mathbb{K}_D$ is an equivalence, as required.

For a general indexing set I , let C_I denote the source of the analogous augmentation. The same atomic-generation argument, now using universal $!$ -descent for the entire family, shows that

$$C_I \xrightarrow{\sim} \mathbb{1}$$

is an equivalence. Moreover, C_I is the $\mathfrak{N}_1$ -filtered colimit of the analogous objects C_{I_0} as I_0 ranges over the countable subsets of I . Since $\mathbb{1} = [*]_{\mathbb{K}_D}$ is $\mathfrak{N}_1$ -compact, the inverse $\mathbb{1} \rightarrow C_I$ to the displayed equivalence factors through some C_{I_0} with I_0 countable. Consequently, the identity of $\mathbb{1}$ lies in the image of

$$\mathrm{Hom}_{\mathbb{K}_D}(\mathbb{1}, C_{I_0}) \longrightarrow \mathrm{End}_{\mathbb{K}_D}(\mathbb{1}).$$

Using the identification of C_{I_0} with the idempotent coalgebra of the countable family, Proposition 8.24 shows that $\coprod_{i \in I_0} [Y_i]_D \rightarrow [*]_D$ is a cover. Hence the same is true for the original family. $\square$

Corollary 10.16 (Scholze [Sch26, Corollary 9.8]). *The transmutation extends uniquely to a functor*

$$[-]_D : \mathrm{Shv}_D(C)^{\mathfrak{N}_1} \longrightarrow \mathrm{Gest}_{/[*]_D}^{1\text{-}\acute{\mathrm{e}}\mathrm{t}} : X \longmapsto [X]_D$$

which preserves finite limits and countable colimits.

Proof. The target is countably toposic by Theorem 7.13. By Proposition 10.15, the finite-limit-preserving functor $[-]_D : C \rightarrow \mathrm{Gest}_{/[*]_D}^{1\text{-}\acute{\mathrm{e}}\mathrm{t}}$ carries the generating J_D -covers to effective epimorphisms. The universal property of the $\mathfrak{N}_1$ -compact J_D -sheaf category therefore gives the asserted extension. Its construction as the left exact localization of the countable presheaf category shows that it preserves finite limits and countable colimits. $\square$

Thus the D -topology is detected by the canonical topology on Gestalten: universal $!$ -descent becomes effective descent after transmutation. The result is deliberately only a statement about the $\mathfrak{N}_1$ -compact part of the D -topos; this is the level on which the countable colimits used in the Gestalt formalism are available.

10.5 A six-functor formalism on Gestalten

Let $D : \mathrm{Span}(C) \rightarrow 1\mathrm{Pr}$ be a presentable six-functor formalism. We constructed a pullback-preserving functor

$$[-]_D : C \rightarrow \mathrm{Gest}$$

fitting in a commutative triangle as follows:

$$\begin{array}{ccc} C^{\mathrm{op}} & \xrightarrow{\quad [-]_D \quad} & \mathrm{Gest}^{\mathrm{op}} \\ & \searrow D^{*, \otimes} & \downarrow \mathcal{O}(-)_1 \\ & & \mathrm{CAlg}(1\mathrm{Pr}). \end{array}$$

A natural follow-up question arises: Can we also recover the shriek-functoriality of D ? Or, even better: does Gest admit a six-functor formalism whose restriction along $[-]_D$ agrees with D ?

The goal of this section is to give some partial answers to this question. Let's start by informally describing !-functors for a map of Gestalten. In the previous section we saw that for any map f in C , the induced map $[f]_D$ of Gestalten is 1-étale and 1-proper, and moreover comes with an identification $f_{1,\#} \cong f_{1,*}$. These conditions already yield !-functors: We get the counit map

$$f_{1,*}f_1^* \cong f_{1,\#}f_1^* \rightarrow \text{id}$$

which, when evaluated at the unit, yields a map

$$f_{0,!} : (O(Y)/O(X))_1 \rightarrow \mathbb{1}$$

in $O(X)_2$. Said differently, the identification $f_{1,\#} \cong f_{1,*}$ yields an identification $(O(Y)/O(X))_1^! \cong (O(Y)/O(X))_1$, along which one can transfer $f_{0,!}$. Taking $\text{Hom}_{O(X)_2}(\mathbb{1}, -)$, this in particular produces a map

$$f_{0,!} : O(Y)_1 \rightarrow O(X)_1.$$

This suggests that there exists a six-functor formalism on the category Gest with the following class of !-able maps:

Definition 10.17. A map $f : Y \rightarrow X$ of Gestalten is *quasi-!-able* if f is 1-étale and 1-proper. The map f is *!-able* if it is quasi-!-able and some iterated diagonal of f is 0-proper. The map f is *co-!-able* if it is quasi-!-able and some iterated diagonal of f is 0-étale.

Recall from Chapter 6 that these !-able maps automatically satisfy base change:

Proposition 10.18 (Prim and suave base change). *Consider a cartesian diagram*

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ g' \downarrow & & \downarrow g \\ Y & \xrightarrow{f} & X \end{array}$$

of Gestalten, and let $n \geq 0$.

(1) *If g is n -prim, then for every $m \geq n$ the Beck–Chevalley transformation*

$$f_m^* g_{m,*} \rightarrow g_{m,*}' f_m'^*$$

of functors $O(X')_{m+1} \rightarrow O(Y)_{m+1}$ is an isomorphism.

(2) *If f is n -suave, then for every $m \geq n$ the Beck–Chevalley transformation*

$$f_{m,\#}' g_m'^* \rightarrow g_m^* f_{m,\#}$$

of functors $O(Y)_{m+1} \rightarrow O(X')_{m+1}$ is an isomorphism.

In either case the Beck–Chevalley transformation refines to an isomorphism of 1-morphisms in the 2-category $O(X)_{m+2}$.

Proof. For (1), apply Proposition 6.9(3) to the pushout square of Stefanich rings obtained by applying $O(-)$, with the prim map $g^*: O(X) \rightarrow O(X')$ and base-change map $f^*: O(X) \rightarrow O(Y)$. For (2), apply Proposition 6.20(1) to the same pushout square, now with the suave map $f^*: O(X) \rightarrow O(Y)$ and base-change map $g^*: O(X) \rightarrow O(X')$. $\square$

Also recall that if functors admit both left and right adjoints, it is important to control their interaction in terms of a *double* Beck–Chevalley transformation.

Proposition 10.19. *Consider a cartesian diagram*

$$\begin{array}{ccc} Y' & \xrightarrow{f'} & X' \\ g' \downarrow & & \downarrow g \\ Y & \xrightarrow{f} & X \end{array}$$

of Gestalten, and assume that f is n -suave while g is n -prim. Then the double Beck–Chevalley map

$$f_{n,\#} g'_{n,*} \rightarrow g_{n,*} f'_{n,\#}$$

of maps from $O(Y')_{n+1}$ to $O(X)_{n+1}$ is an isomorphism. In fact, it is even an isomorphism of maps from $(O(Y')/O(X))_{n+1}$ to $\mathbb{1}$ in $O(X)_{n+2}$.

Proof. First, we claim that there is a natural transformation

$$f_{n+1,*} f_{n+1}^* \xrightarrow{f_{n,\#}} \text{id}$$

in $\text{End}(O(X)_{n+2})$. If f is n -étale, so that $f_{n+1,*}$ is linear, this is just obtained from $f_{n+1,*} \mathbb{1} \xrightarrow{f_{n,\#}} \mathbb{1}$ in $O(X)_{n+2}$ by letting $O(X)_{n+2}$ act on itself. In general, one first has a natural transformation

$$\text{id} \xrightarrow{f_{n,!}} f_{n+1,\#} f_{n+1}^*$$

in $\text{End}(O(X)_{n+2})$, by the similar argument, and then passes to right adjoints.

Write $h := g f' = f g': Y' \rightarrow X$ for the long diagonal of the square. Now consider the natural morphism $\mathbb{1} \rightarrow g_{n+1,*} \mathbb{1}$ in $O(X)_{n+2}$. Applying $f_{n+1,*} f_{n+1}^*$ to it, this becomes $f_{n+1,*} \mathbb{1} \rightarrow h_{n+1,*} \mathbb{1}$, and the existence of $f_{n,\#}$ amounts to the commutative square

$$\begin{array}{ccc} f_{n+1,*} \mathbb{1} & \longrightarrow & h_{n+1,*} \mathbb{1} \\ f_{n,\#} \downarrow & & \downarrow f'_{n,\#} \\ \mathbb{1} & \longrightarrow & g_{n+1,*} \mathbb{1} \end{array}$$

expressing the base change compatibility of $f_{n,\#}$.

On the other hand, we can also apply this reasoning to the right adjoint $g_{n,*}: g_{n+1,*} \mathbb{1} \rightarrow \mathbb{1}$ of $\mathbb{1} \rightarrow g_{n+1,*} \mathbb{1}$. Applying $f_{n+1,*} f_{n+1}^*$ to it, this becomes $g'_{n,*}: h_{n+1,*} \mathbb{1} \rightarrow f_{n+1,*} \mathbb{1}$, and then $f_{n,\#}: f_{n+1,*} f_{n+1}^* \rightarrow \text{id}$ gives the other composite from $h_{n+1,*} \mathbb{1}$ to $\mathbb{1}$. The displayed square identifies these two composites. Under the internal adjunction correspondence they are precisely the two sides of the double Beck–Chevalley transformation, proving that it is an isomorphism in $O(X)_{n+2}$ and hence also on the underlying functors. $\square$

In particular, we get the following corollary. (In principle one could state this with 1 replaced by any $n \geq 0$.)

Corollary 10.20. *Let $f: Y \rightarrow X$ be a map of Gestalten that is 1-étale and 1-proper. Then $f_{1,\#}, f_{1,*}: \mathcal{O}(Y)_2 \rightarrow \mathcal{O}(X)_2$ agree up to twist. More precisely, there are natural isomorphisms*

$$f_{1,\#}(p_{1,*}\Delta_{f,1,\#} \mathbb{1} \otimes -) \cong f_{1,*}, \quad f_{1,*}(p_{1,\#}\Delta_{f,1,*} \mathbb{1} \otimes -) \cong f_{1,\#}$$

and

$$p_{1,*}\Delta_{f,1,\#} \mathbb{1} \otimes p_{1,\#}\Delta_{f,1,*} \mathbb{1} \cong \mathbb{1} \in \mathcal{O}(Y)_2,$$

so both tensor factors are invertible. Here $Y \xrightarrow{\Delta_f} Y \times_X Y \xrightarrow{p} Y$ are the diagonal and first projection.

Proof. Indeed, using linearity of all intervening functors, as well as the double Beck–Chevalley isomorphism,

$$\begin{aligned} f_{1,\#}(p_{1,*}\Delta_{f,1,\#} \mathbb{1} \otimes M) &\cong f_{1,\#}p_{1,*}\Delta_{f,1,\#}\Delta_{f,1}^*p_1^*M \\ &\cong f_{1,\#}p_{1,*}\Delta_{f,1,\#}M \\ &\cong f_{1,*}q_{1,\#}\Delta_{f,1,\#}M \\ &\cong f_{1,*}M, \end{aligned}$$

where $q: Y \times_X Y \rightarrow Y$ is the second projection. The other isomorphism is analogous. Moreover,

$$\begin{aligned} p_{1,*}\Delta_{f,1,\#} \mathbb{1} \otimes p_{1,\#}\Delta_{f,1,*} \mathbb{1} &\cong p_{1,*}\Delta_{f,1,\#}\Delta_{f,1}^*p_1^*p_{1,\#}\Delta_{f,1,*} \mathbb{1} \\ &\cong p_{1,*}\Delta_{f,1,\#}p_{1,\#}\Delta_{f,1,*} \mathbb{1} \\ &\cong p_{1,*}\Delta_{f,1,*} \mathbb{1} \cong \mathbb{1}. \end{aligned} \quad \square$$

In particular, for any 1-étale and 1-proper $f: Y \rightarrow X$, we get a canonically associated invertible $\mathbb{D}_f \in \mathcal{O}(Y)_2$. If f is 0-étale or 0-proper, then one gets a trivialization $\mathbb{D}_f \cong \mathbb{1}$ by ambidexterity. If one relates f and $\Delta_f: Y \rightarrow Y \times_X Y$, one finds that $\mathbb{D}_f$ and $\mathbb{D}_{\Delta_f}$ are inverse. Thus, if f is !-able or co-!-able, one gets a trivialization $\mathbb{D}_f \cong \mathbb{1}$ by passing to some iterated diagonal.

Once one has such a trivialization, we get the counit map

$$f_{0,!}: f_{1,*} \mathbb{1} \cong f_{1,\#} \mathbb{1} \rightarrow \mathbb{1}$$

yielding a map $f_{0,!}: \mathcal{O}(Y)_1 \rightarrow \mathcal{O}(X)_1$ compatible with base change and composition. It is reasonable to expect that these define a six-functor formalism on !-able maps (or co-!-able maps), but at present it is not yet clear how to resolve all coherence issues.

Restricting to truncated maps, we can however get the desired six-functor formalism. Let

$$E_{\text{tr}} \subseteq \text{Gest}$$

denote the wide subcategory of truncated, 1-étale and 1-proper maps.

Proposition 10.21. *The assignment $X \mapsto \mathcal{O}(X)_1$ extends to a presentable six-functor formalism*

$$\mathcal{O}(-)_1: \text{Span}(\text{Gest}, E_{\text{tr}}) \longrightarrow \mathbf{1Pr}.$$

If $f: Y \rightarrow X$ lies in E_{tr} , the exceptional pushforward of this formalism is the map $f_{0,!}: \mathcal{O}(Y)_1 \rightarrow \mathcal{O}(X)_1$ constructed above.

Proof. First fix a Gestalt X , put $A := \mathcal{O}(X)$, and let

$$C_X := (E_{\text{tr}})_{/X}$$

be the category of truncated, 1-étale and 1-proper Gestalten over X . This category has finite limits, and every morphism in C_X again lies in E_{tr} : the relevant properties are stable under base change and composition, and the cancellation properties for the 1-étale and 1-proper conditions are part of Propositions 6.15 and 6.24. Working in a larger universe if necessary, we may therefore apply Theorem B.14 with all three classes equal to C_X .

The relative algebras define a symmetric monoidal functor to $\text{CAlg}(A_3)$; after forgetting the algebra structures, we regard this as a lax symmetric monoidal functor

$$\Phi_X: C_X^{\text{op}} \longrightarrow A_3, \quad (Y \rightarrow X) \longmapsto (\mathcal{O}(Y)/A)_2.$$

The symmetric monoidal structure is supplied by the Künneth formula Lemma 6.12, applied in degree 2. If $f: Y \rightarrow Z$ lies in C_X , then the induced map $(f/A)_1^*$ admits both internal adjoints, since f is 1-étale and 1-proper. The Beck–Chevalley, double Beck–Chevalley, and projection-formula requirements are exactly Propositions 10.18 and 10.19 and lemmas 6.11 and 6.21. Hence Theorem B.14 produces a lax symmetric monoidal 2-functor

$$\tilde{\Phi}_X: \text{Span}_2(C_X, C_X)_{C_X, C_X} \longrightarrow A_3$$

extending Φ_X .

The terminal object of C_X is id_X , and $\Phi_X(\text{id}_X) = (A/A)_2 \simeq \mathbb{1}_{A_3}$. Passing to endomorphisms of the unit as in Construction B.15 gives a lax symmetric monoidal functor

$$\text{Span}(C_X) \simeq \text{Hom}_{\text{Span}_2(C_X, C_X)_{C_X, C_X}}(\text{id}_X, \text{id}_X) \longrightarrow \text{End}_{A_3}(\mathbb{1}) \simeq A_2.$$

An object $Y \rightarrow X$ of C_X corresponds to the endomorphism span

$$\text{id}_X \xleftarrow{p_Y} Y \xrightarrow{p_Y} \text{id}_X.$$

Its image is the endomorphism $(p_Y/A)_{1,*}(p_Y/A)_1^*$ of the unit in A_3 . Under the equivalence $\text{End}_{A_3}(\mathbb{1}) \simeq A_2$, evaluation at the unit identifies this object with

$$(p_Y/A)_{1,*}(p_Y/A)_1^*(\mathbb{1}) \simeq (\mathcal{O}(Y)/A)_1.$$

For a morphism $f: Y \rightarrow Z$ in C_X , the forward span $Y \xleftarrow{\text{id}} Y \xrightarrow{f} Z$ acts through the counit of $(f/A)_{1,\sharp} \dashv (f/A)_1^*$. Since f belongs to both the left and the right class, Theorem B.14 identifies $(f/A)_{1,\sharp}$ with $(f/A)_{1,*}$; after evaluation at the unit, this is precisely the map $f_{0,!}: \mathcal{O}(Y)_1 \rightarrow \mathcal{O}(Z)_1$ constructed above. Forgetting along $A_2 = \mathcal{O}(X)_2 \rightarrow 1\text{Pr}$, we therefore obtain a lax symmetric monoidal functor

$$D_X: \text{Span}((E_{\text{tr}})_{/X}) \longrightarrow 1\text{Pr}, \quad (Y \rightarrow X) \longmapsto \mathcal{O}(Y)_1.$$

We now assemble these functors. For $Y \rightarrow X$ in $(E_{\text{tr}})_{/X}$ and $T \rightarrow X$ in $\text{Gest}_{/X}$, define

$$D_{/X}(Y)(T) := D_T(T \times_X Y) = \mathcal{O}(T \times_X Y)_1.$$

A span $Y \leftarrow W \rightarrow Z$ in $(E_{\text{tr}})_{/X}$ pulls back along $T \rightarrow X$ to a span in $(E_{\text{tr}})_{/T}$, and applying D_T defines the corresponding functor

$$D_{/X}(Y)(T) \longrightarrow D_{/X}(Z)(T).$$

The construction of D_T is natural under pullback, since both the relative algebra functor and the extension supplied by Theorem B.14 are natural. Hence these functors are natural in T , and we obtain functors

$$D_{/X}: \text{Span}((E_{\text{tr}})_{/X}) \longrightarrow \text{Fun}(\text{Gest}_{/X}^{\text{op}}, 1\text{Pr}).$$

The contravariant coefficient functor $O(-)_1: \text{Gest}^{\text{op}} \rightarrow \text{CAlg}(1\text{Pr})$ and the lax symmetric monoidal structures of the D_T supply natural external products

$$D_{/X}(Y)(T) \otimes D_{/X'}(Y')(T') \longrightarrow D_{/(X \times X')}(Y \times Y')(T \times T').$$

The naturality of the Künneth equivalences under pullback makes these products compatible with the span actions and with the following compatibility equivalences.

Their contravariant compatibility in X is the canonical base-change identification

$$T' \times_{X'} (X' \times_X Y) \simeq T' \times_X Y.$$

If $e: X' \rightarrow X$ lies in E_{tr} , the covariant compatibility required in Theorem B.27 is checked objectwise by the same pullback identification:

$$D_{/X}(e_{\#}Y)(T) = O(T \times_X Y)_1 \simeq O((T \times_X X') \times_{X'} Y)_1 = (e_* D_{/X'}(Y))(T).$$

The symmetric monoidal reconstruction result Corollary B.28 therefore assembles the D_X into a lax symmetric monoidal functor

$$O(-)_1: \text{Span}(\text{Gest}, E_{\text{tr}}) \longrightarrow 1\text{Pr}.$$

Its value at X is $D_{/X}(\text{id}_X)(\text{id}_X) = O(X)_1$, and the description of the exceptional pushforward in Theorem B.27 identifies the map attached to $f: Y \rightarrow X$ with the forward-span map in D_X , hence with $f_{0,!}$. All values are presentable, so the remaining right adjoints exist by Remark B.8. $\square$

Question 10.22. Can the six-functor formalism on Gest constructed in Proposition 10.21 be extended to a functor

$$O(-)_1: \text{Span}(\text{Gest}, \text{Gest}^!) \rightarrow 1\text{Pr},$$

defined on the span category of *all* $!$ -able maps of Gestalten (Definition 10.17)? Here $\text{Gest}^! \subseteq \text{Gest}$ denotes the wide subcategory of $!$ -able maps.

Question 10.23. If so, can the original six-functor formalism D be recovered from the six-functor formalism on Gest by precomposing with the transmutation, i.e. as the composite

$$\text{Span}(C, E) \xrightarrow{\text{Span}([_]_D)} \text{Span}(\text{Gest}, \text{Gest}^!) \xrightarrow{O(-)_1} 1\text{Pr}?$$

11 Universality of the motivic Gestalt

Talk given by Bastiaan.

Essentially all material in this chapter is taken from Aoki [Aok26].

In classical algebraic geometry, one studies smooth projective varieties over a field k by means of cohomology theories, like Betti cohomology, de Rham cohomology, and ℓ -adic cohomology. In order to understand the relation between these theories, Grothendieck envisioned the concept a universal abelian category $\text{Mot}(k)$ of motives, together with a “universal cohomology theory” of the form $\text{Sch} \rightarrow \text{Mot}(k)$; every other cohomology theory ought to factor through $\text{Mot}(k)$ by means of a *realization functor* from $\text{Mot}(k)$ to the category of vector spaces.

While Grothendieck’s original vision of $\text{Mot}(k)$ has remained conjectural, it was later substantially realized by Voevodsky [Voe00] through his triangulated category of motives $\text{DM}(k)$. Closely related is the stable motivic homotopy theory $\text{SH}(k)$ of Morel and Voevodsky [MV99], obtained from the category of smooth k -schemes by asking for presentability, Nisnevich descent, $\mathbb{A}^1$ -invariance and $\mathbb{P}^1$ -stability. For the purposes of this chapter, the categories $\text{SH}(k)$ are the main objects of interest.

The motivic stable category $\text{SH}(X)$ can be defined for any scheme X , and the assignment $X \mapsto \text{SH}(X)$ can be upgraded to a six-functor formalism

$$\text{SH}: \text{Span}(\text{Sch}_S) \rightarrow \mathbf{1Pr},$$

where Sch_S is the category of separated finite type schemes over some base scheme S . By a result of Drew–Gallauer [DG22], formulated in the closely related language of coefficient systems, this is the universal six-functor formalism on Sch_S satisfying four axioms: smooth base change and the projection formula, localization, $\mathbb{A}^1$ -invariance, and $\mathbb{P}^1$ -stability.

Applying the category-of-kernels construction provides a presentably symmetric monoidal 2-category

$$\text{SH}^{(2)}(S) := \mathbb{K}_{\text{SH}, S} \in \mathbf{CAlg}(\mathbf{2Pr}).$$

Its corepresentable objects give a symmetric monoidal functor

$$\text{Sch}_S^{\text{op}} \longrightarrow \text{SH}^{(2)}(S), \quad X \longmapsto [X],$$

and its defining property is the formula

$$\text{Hom}_{\text{SH}^{(2)}(S)}([X], [Y]) \simeq \text{SH}(X \times_S Y).$$

The goal of today’s talk is to go through the following results by Aoki [Aok26]:

- The functor $\mathrm{Sch}_S^{\mathrm{op}} \rightarrow \mathrm{SH}^{(2)}(S)$, $X \mapsto [X]$ is universal among symmetric monoidal functors from schemes into stable presentably symmetric monoidal 2-categories which satisfy the corresponding 2-categorical versions of the same four axioms.
- The Gestalt $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$ is the free Gestalt S equipped with a ring object $R \in \mathrm{CRing}(\mathrm{Gest}_S)$ which is 1-affine, 0-suave, homologically trivial and ‘sutured’ (roughly: $0 \hookrightarrow R$ and $R^\times \hookrightarrow R$ form a recollement).

11.1 Motivic spectra and the six operations

We first recall the motivic six-functor formalism that will be used throughout this chapter. We let $\mathrm{Sch} = \mathrm{Sch}_{\mathbb{Z}}^{\mathrm{sep}, \mathrm{f.t.}}$ denote the category of separated $\mathbb{Z}$ -schemes of finite type. For $X \in \mathrm{Sch}$, let Sm_X be the category of smooth X -schemes of finite presentation. The category of motivic spectra over X is the symmetric monoidal localization

$$\mathrm{SH}(X) := \mathrm{Shv}_{\mathrm{Nis}, \mathbb{A}^1}(\mathrm{Sm}_X; \mathrm{Sp})[(\Sigma^\infty \mathbb{P}_X^1)^{\otimes -1}].$$

Thus we first impose Nisnevich descent and $\mathbb{A}^1$ -invariance, and then invert the pointed projective line. This is a stable presentably symmetric monoidal category.

Base change of smooth schemes induces a contravariant functor

$$\mathrm{SH}^{*, \otimes}: \mathrm{Sch}^{\mathrm{op}} \longrightarrow \mathrm{CAlg}(\mathrm{1Pr}), \quad X \longmapsto \mathrm{SH}(X).$$

We generally suppress the superscript and write this functor simply as SH . Its basic properties follow directly from the construction. Every f^* is exact, colimit-preserving, and symmetric monoidal, hence has a right adjoint f_* . If f is smooth, then f^* also has a left adjoint $f_{\sharp}$; it satisfies smooth base change and the projection formula. The localization theorem gives a recollement

$$\mathrm{SH}(U) \xleftarrow{j^*} \mathrm{SH}(X) \xrightarrow{i^*} \mathrm{SH}(Z)$$

for every closed immersion $i: Z \hookrightarrow X$ with open complement $j: U \hookrightarrow X$. Finally, for the projection $\pi: \mathbb{A}_X^1 \rightarrow X$ and its zero section s , π^* is fully faithful and the Tate functor

$$\pi_{\sharp} s_*: \mathrm{SH}(X) \longrightarrow \mathrm{SH}(X)$$

is an equivalence. The first assertion is $\mathbb{A}^1$ -invariance, while the second is exactly the effect of inverting $\mathbb{P}^1$.

The non-formal input is the following theorem of Ayoub, in the extended form due to Cisinski–Déglise. We first isolate the axioms to which it applies.

Definition 11.1. Let S be noetherian and finite-dimensional. A *motivic coefficient system* over S is a lax symmetric monoidal functor

$$D: \mathrm{Sch}_S^{\mathrm{op}} \longrightarrow \mathrm{1Pr}_{\mathrm{st}}$$

satisfying the following four conditions.

- (1) Smooth maps admit left adjoints $f_{\sharp}$, and these left adjoints satisfy smooth base change and the projection formula.

(2) For every closed immersion $i: Z \hookrightarrow X$ with open complement $j: U \hookrightarrow X$, the diagram

$$D(U) \xleftarrow{j^*} D(X) \xrightarrow{i^*} D(Z)$$

is a recollement.

(3) For $\pi: \mathbb{A}_X^1 \rightarrow X$, the functor π^* is fully faithful.

(4) If $s: X \rightarrow \mathbb{A}_X^1$ is the zero section, then $\pi_{\#} s_*$ is an equivalence.

A morphism of motivic coefficient systems is a lax symmetric monoidal natural transformation which is compatible with the left adjoints for smooth maps. We write

$$\mathrm{Fun}_{\mathrm{mot}}^{\otimes\text{-lax}}(\mathrm{Sch}_S^{\mathrm{op}}, \mathrm{1Pr}_{\mathrm{st}})$$

for the category of motivic coefficient systems and their morphisms. Thus the objects are the functors satisfying the four conditions above, and the morphisms are the smooth-compatible lax symmetric monoidal natural transformations just described.

The discussion before the definition says exactly that $X \mapsto \mathrm{SH}(X)$ is a motivic coefficient system. It satisfies the following universal property:

Theorem 11.2 (Drew–Gallauer [DG22]). *The motivic coefficient system*

$$\mathrm{SH}: \mathrm{Sch}_S^{\mathrm{op}} \longrightarrow \mathrm{1Pr}_{\mathrm{st}}$$

is an initial object of $\mathrm{Fun}_{\mathrm{mot}}^{\otimes\text{-lax}}(\mathrm{Sch}_S^{\mathrm{op}}, \mathrm{1Pr}_{\mathrm{st}})$.

Theorem 11.3 (Ayoub–Cisinski–Déglise [Ayo07; CD19]). *Let $D: \mathrm{Sch}_S^{\mathrm{op}} \rightarrow \mathrm{1Pr}_{\mathrm{st}}$ be a motivic coefficient system. Then for every proper map $p: Y \rightarrow X$, the functor p^* admits a right adjoint p_* . These right adjoints satisfy proper base change and the projection formula. Moreover, if*

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ p' \downarrow & \lrcorner & \downarrow p \\ X' & \xrightarrow{g} & X \end{array}$$

is cartesian with p proper and g smooth, then the smooth–proper Beck–Chevalley map

$$g_{\#} p'_* \longrightarrow p_* g'_{\#}$$

is an equivalence.

Finally, if $\alpha: D \rightarrow D'$ is a morphism of motivic coefficient systems, then α is also compatible with the right adjoints for proper maps.

Remark 11.4. For the non-monoidal argument below we only need projective morphisms between quasiprojective schemes. In this setting, Aoki [Aok26, Corollary 3.11] proves projective right adjoints and projective base change for functors valued in an arbitrary stable presentable 2-category. No non-monoidal assertion for all proper morphisms is needed.

Remark 11.5. Ayoub proves the theorem in the quasiprojective setting, where proper morphisms are projective. Cisinski–Déglise remove this quasiprojectivity restriction and construct the exceptional functoriality for all separated finite type morphisms; in particular $p_! \simeq p_*$ for proper p , with the proper base-change and projection-formula compatibilities above.

Corollary 11.6. *The functor $\mathrm{SH}^{*,\otimes}$ extends to a presentable six-functor formalism*

$$\mathrm{SH}: \mathrm{Span}(\mathrm{Sch}) \longrightarrow \mathrm{1Pr}.$$

If $f: Y \rightarrow X$ factors as an open immersion $j: Y \hookrightarrow \bar{Y}$ followed by a proper map $p: \bar{Y} \rightarrow X$, then

$$f_! \simeq p_* j_\#.$$

Proof. Apply Theorem B.11 with I the open immersions, P the proper maps, and $E = \mathrm{Sch}$. Both classes are stable under base change and left cancellable. By Nagata compactification, every morphism in Sch factors as an open immersion followed by a proper map, and every morphism in $I \cap P$ is 0-truncated. The smooth functoriality recalled above supplies the left adjoints, Beck–Chevalley isomorphisms, and projection formula required for I . Theorem 11.3 gives the dual data for P and, applied to open immersions, gives the double Beck–Chevalley condition. The construction theorem therefore produces the asserted three-functor formalism and the formula for $f_!$. Since all its values are presentable, the needed right adjoints exist, so it is a six-functor formalism by Remark B.8. $\square$

11.2 The motivic Gestalt is 2-affine

Put $S := \mathrm{Spec}(\mathbb{Z})$ and write

$$\mathrm{SH}^{(2)}(X) := \mathbb{K}_{\mathrm{SH},X} \quad (X \in \mathrm{Sch}).$$

Thus $\mathrm{SH}^{(2)}(X)$ is the category of kernels of the motivic six-functor formalism on Sch_X . The following observation of Scholze is the first reason that the Gestalt attached to SH is unusually accessible.

Proposition 11.7 (Scholze [Sch26, Proposition 9.9]). *The Gestalt $[S]_{\mathrm{SH}}$ is 2-affine:*

$$[S]_{\mathrm{SH}} \simeq \mathrm{Gest}(\mathrm{SH}^{(2)}(S)).$$

Moreover, $\mathrm{SH}^{(2)}(S)$ is generated, under colimits and $\mathrm{1Pr}$ -tensors, by the corepresentable objects

$$[\mathbb{A}_{\mathbb{Z}}^n]_{\mathrm{SH}^{(2)}(S)} \quad (n \geq 0).$$

For every $X \in \mathrm{Sch}$, the structural map

$$[X]_{\mathrm{SH}} \longrightarrow [S]_{\mathrm{SH}}$$

is 1-affine.

Proof. Let $p: X \rightarrow S$ be the structural map. The adjunction of Proposition 10.7 gives a symmetric monoidal functor

$$p_1^*: \mathrm{SH}^{(2)}(S) \longrightarrow \mathrm{SH}^{(2)}(X)$$

with right adjoint $p_{1,*} \simeq p_{1,\#}$. Set

$$[X/S]_1 := p_{1,*}(\mathbb{1}_{\mathrm{SH}^{(2)}(X)}) \in \mathrm{CAlg}(\mathrm{SH}^{(2)}(S)).$$

This is the degree-1 relative algebra of the map $[X]_{\mathrm{SH}} \rightarrow [S]_{\mathrm{SH}}$. The functor $p_{1,*}$ induces a functor $\mathrm{SH}^{(2)}(X) \rightarrow \mathrm{Mod}_{[X/S]_1}(\mathrm{SH}^{(2)}(S))$, which admits a symmetric monoidal left adjoint

$$\Phi_X: \mathrm{Mod}_{[X/S]_1}(\mathrm{SH}^{(2)}(S)) \longrightarrow \mathrm{SH}^{(2)}(X)$$

carrying the free module $[X/S]_1 \otimes M$ to p_1^*M . It follows from the projection formula that the unit map $[X/S]_1 \otimes M \rightarrow p_{1,*}p_1^*M$ is an isomorphism on all free modules, and since the source of Φ_X is generated by free modules under colimits and 1Pr-tensors it follows that Φ_X is fully faithful.

It remains to prove that the image of p_1^* generates $\mathrm{SH}^{(2)}(X)$. The corepresentables associated to objects $Y \rightarrow X$ of Sch/X generate $\mathrm{SH}^{(2)}(X)$. A finite affine open cover of Y and the successive closed–open recollements express $[Y]$ in the localizing subcategory generated by the corepresentables of affine opens and their finite intersections. We may therefore assume that Y is affine. Choose a closed immersion $Y \hookrightarrow \mathbb{A}_{\mathbb{Z}}^n$. Since X is separated over S , the graph of $Y \rightarrow X$ is a closed immersion; composing it with the base change of $Y \hookrightarrow \mathbb{A}_{\mathbb{Z}}^n$ gives a closed immersion

$$i: Y \hookrightarrow \mathbb{A}_X^n.$$

Since i is a closed immersion, the recollement for the closed–open decomposition gives a fully faithful right adjoint i_* to i^* in the 2-category $\mathrm{SH}^{(2)}(X)$. Hence the composite

$$[Y] \xrightarrow{i_*} [\mathbb{A}_X^n] \xrightarrow{i^*} [Y]$$

is the identity, so $[Y]$ is a retract of $[\mathbb{A}_X^n]$. But

$$\mathbb{A}_X^n = X \times_S \mathbb{A}_{\mathbb{Z}}^n,$$

and base change identifies the latter corepresentable with the image under p_1^* of $[\mathbb{A}_{\mathbb{Z}}^n]_{\mathrm{SH}^{(2)}(S)}$. We conclude that the image of p_1^* generates $\mathrm{SH}^{(2)}(X)$, hence Φ_X is an equivalence. Taking $S = X$, this also proves the stated assertion about generators, and it identifies

$$\mathrm{SH}^{(2)}(X) \simeq \mathrm{Mod}_{[X/S]_1}(\mathrm{SH}^{(2)}(S))$$

for every X .

The projection formula and base change also give

$$[X \times_S Y/S]_1 \simeq [X/S]_1 \otimes [Y/S]_1.$$

Hence the preceding equivalence yields, for all $X, Y \in \mathrm{Sch}$,

$$\mathrm{SH}^{(2)}(X) \otimes_{\mathrm{SH}^{(2)}(S)} \mathrm{SH}^{(2)}(Y) \simeq \mathrm{Mod}_{[X/S]_1 \otimes [Y/S]_1}(\mathrm{SH}^{(2)}(S)) \simeq \mathrm{SH}^{(2)}(X \times_S Y).$$

In other words, the 2-categorical coefficient system $X \mapsto \mathrm{SH}^{(2)}(X)$ satisfies the categorical Künneth formula. The argument was relative in the base: applying it in the slice over any $T \in \mathrm{Sch}$ shows that $\mathrm{SH}^{(2)}$ satisfies the relative Künneth formula required in Corollary 10.12. That corollary, applied to $D = \mathrm{SH}$ and $n = 2$, proves the 2-affineness of $[S]_{\mathrm{SH}}$.

The same corollary applied in the slice over X shows that $[X]_{\text{SH}}$ is also 2-affine. Hence the structural map $[X]_{\text{SH}} \rightarrow [S]_{\text{SH}}$ is automatically 2-affine: in Stefanich-ring language, this follows from Proposition 6.3(4). To improve this to 1-affineness, only the first affine comparison remains. But this comparison is exactly the equivalence

$$\text{SH}^{(2)}(X) \simeq \text{Mod}_{[X/S]_1}(\text{SH}^{(2)}(S))$$

proved above. Hence $[X]_{\text{SH}} \rightarrow [S]_{\text{SH}}$ is 1-affine. $\square$

11.3 The universal property of the motivic 2-category

The result of the previous section shows that the Gestalt $[\text{Spec}(\mathbb{Z})]_{\text{SH}}$ is the 2-affine Gestalt associated to the presentably symmetric monoidal 2-category $\text{SH}^{(2)}(S)$. We now discuss the universal property established by Aoki [Aok26].

In this section S denotes a noetherian finite-dimensional base scheme, Sch_S denotes the category of separated finite type S -schemes, and $\text{QProj}_S \subseteq \text{Sch}_S$ denotes the full subcategory of quasiprojective schemes. Aoki proves the theorem using QProj_S , where proper morphisms are projective. His corrected thesis, Aoki [Aok26, Remark 1.3], states that replacing QProj_S by Sch_S gives the same universal property. We use the quasiprojective presentation in the proof. For the rest of this chapter one may keep $S = \text{Spec}(\mathbb{Z})$ in mind.

Let C be a stable presentably symmetric monoidal 2-category. We use the same word *motivic* for the 2-categorical analogue of Definition 11.1. Thus a symmetric monoidal functor

$$F: \text{QProj}_S^{\text{op}} \longrightarrow C$$

is motivic if smooth pullback admits left adjoints satisfying smooth base change, closed–open decompositions give recollements, the projection $\mathbb{A}_X^1 \rightarrow X$ is homotopy invariant, and the Tate operator $\pi_{\sharp} s_*$ is invertible. Morphisms between motivic functors are symmetric monoidal natural transformations which are compatible with the left adjoints for smooth maps.

Lemma 11.8. *Let C be a stable presentably symmetric monoidal 2-category, and let*

$$F: \text{QProj}_S^{\text{op}} \longrightarrow C$$

be motivic. Then F is biadjointable with respect to open immersions and projective maps. Consequently, by Theorem B.14, it extends canonically to a symmetric monoidal 2-functor

$$\text{Span}_2(\text{QProj}_S, \text{QProj}_S)_{\text{projective, open}} \longrightarrow C.$$

Proof. The motivic axioms already give the required left adjoints and Beck–Chevalley equivalences for open immersions, since open immersions are smooth. For projective maps, Aoki [Aok26, Corollary 3.11] gives right adjoints and projective base change in every stable presentable 2-category. Aoki’s complementary closed-immersion argument gives the Beck–Chevalley equivalence for a square whose horizontal maps are open and whose vertical maps are projective. This is precisely the double Beck–Chevalley condition required in Definition B.13.

It remains to explain the projection formulas. They are a formal consequence of symmetric monoidality together with the relevant Beck–Chevalley equivalence. We spell this out for a projective map

$p: X \rightarrow Y$; the argument for open immersions and left adjoints is the same, with left adjoints in place of right adjoints. Consider the pullback square

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_p} & Y \times X \\ p \downarrow & \lrcorner & \downarrow \text{id}_Y \times p \\ Y & \xrightarrow{\Delta_Y} & Y \times Y, \end{array}$$

where Γ_p is the graph of p . By symmetric monoidality, pullback along $\text{id}_Y \times p$ is the morphism $\text{id} \otimes p^*$, so its right adjoint is $\text{id} \otimes p_*$. For $N \in F(Y)$ and $M \in F(X)$, Beck–Chevalley for the displayed square gives

$$\Delta_Y^*(\text{id} \otimes p_*)(N \boxtimes M) \simeq p_* \Gamma_p^*(N \boxtimes M).$$

The left-hand side is $N \otimes p_* M$, while the right-hand side is $p_*(p^* N \otimes M)$. This is precisely the projection formula for p_* . For an open immersion j , one uses instead the pullback square

$$\begin{array}{ccc} U & \xrightarrow{\Gamma_j} & U \times X \\ j \downarrow & \lrcorner & \downarrow j \times \text{id}_X \\ X & \xrightarrow{\Delta_X} & X \times X \end{array}$$

and the left Beck–Chevalley equivalence to obtain

$$j_\#(M \otimes j^* N) \simeq j_\#(M) \otimes N.$$

Thus F is biadjointable with respect to open immersions and projective maps, and the final assertion is precisely Theorem B.14. $\square$

We will also use the following concrete form of the universal property of the kernel category. Write

$$\mathcal{P}_S := \text{Fun}(\text{Span}(\text{QProj}_S), \text{1Pr}_{\text{st}})$$

with its Day convolution symmetric monoidal structure. Aoki’s quasiprojective construction, compared with the separated finite type construction in Aoki [Aok26, Remark 1.3], gives an equivalence

$$\text{SH}^{(2)}(S) = \mathbb{K}_{\text{SH}, S} \simeq \text{Mod}_{\text{SH}}(\mathcal{P}_S).$$

Lemma 11.9. *Let C be a presentably symmetric monoidal 2-category. To construct a morphism*

$$H: \text{SH}^{(2)}(S) \longrightarrow C$$

in $\text{CAlg}(2\text{Pr})$ it is equivalent to construct the following two pieces of data:

(1) *a symmetric monoidal functor*

$$T: \text{Span}(\text{QProj}_S) \longrightarrow C;$$

(2) *a morphism of commutative algebra objects in $\text{Fun}(\text{Span}(\text{QProj}_S), \text{1Pr}_{\text{st}})$*

$$\alpha: \text{SH} \longrightarrow \text{Hom}_C(\mathbb{1}, T(-)).$$

Under this correspondence, H sends the corepresentable object $[X] \in \mathrm{SH}^{(2)}(S)$ to $T(X)$. Moreover, natural transformations between such functors H are determined by the corresponding transformations of the two pieces of data above.

Proof. This is just the definition of $\mathrm{SH}^{(2)}(S)$ as a category of modules, unwound. A symmetric monoidal colimit-preserving $1\mathrm{Pr}$ -linear functor out of the Day-convolution category $\mathcal{P}_S$ is determined by its restriction to the representables, hence by a symmetric monoidal functor from the span category. Passing from $\mathcal{P}_S$ to $\mathrm{Mod}_{\mathrm{SH}}(\mathcal{P}_S)$ imposes the additional datum that the commutative algebra $\mathrm{SH} \in \mathcal{P}_S$ acts on the unit of C after applying the chosen functor, i.e. the resulting commutative algebra in C comes with an augmentation. By adjunction, such an augmentation is equivalently a morphism

$$\mathrm{SH} \longrightarrow \mathrm{Hom}_C(\mathbb{1}, T(-))$$

of commutative algebra objects in $\mathrm{Fun}(\mathrm{Span}(\mathrm{QProj}_S), 1\mathrm{Pr}_{\mathrm{st}})$. Finally, $\mathrm{SH}^{(2)}(S)$ is generated under colimits by the free SH -modules on representables, which are the corepresentables $[X]$; this gives both the formula $H([X]) \simeq T(X)$ and the assertion about natural transformations. $\square$

Theorem 11.10 (Aoki [Aok26, Theorem A]). *The functor of corepresentables*

$$[-]: \mathrm{QProj}_S^{\mathrm{op}} \longrightarrow \mathrm{SH}^{(2)}(S), \quad X \longmapsto [X],$$

is universal among motivic functors. Equivalently, for every stable presentably symmetric monoidal 2-category C , restriction along $[-]$ induces an equivalence

$$\mathrm{Fun}^{\otimes, L}(\mathrm{SH}^{(2)}(S), C) \simeq \mathrm{Fun}_{\mathrm{mot}}^{\otimes}(\mathrm{QProj}_S^{\mathrm{op}}, C).$$

The same statement holds with QProj_S replaced throughout by Sch_S .

Proof. Let $\mathrm{Mot}^{(2)}(S)$ denote the universal stable presentably symmetric monoidal 2-category equipped with a motivic functor

$$[-]_{\mathrm{Mot}}: \mathrm{QProj}_S^{\mathrm{op}} \longrightarrow \mathrm{Mot}^{(2)}(S).$$

Since $X \mapsto [X] \in \mathrm{SH}^{(2)}(S)$ satisfies the four axioms by the properties of motivic spectra recalled in Section 11.1, the universal property of $\mathrm{Mot}^{(2)}(S)$ gives a symmetric monoidal colimit-preserving functor

$$F: \mathrm{Mot}^{(2)}(S) \longrightarrow \mathrm{SH}^{(2)}(S)$$

under $\mathrm{QProj}_S^{\mathrm{op}}$. We have to construct an inverse.

Applying Lemma 11.8 to $[-]_{\mathrm{Mot}}$ gives a canonical symmetric monoidal extension

$$\widetilde{[-]}_{\mathrm{Mot}}: \mathrm{Span}_2(\mathrm{QProj}_S, \mathrm{QProj}_S)_{\mathrm{projective, open}} \longrightarrow \mathrm{Mot}^{(2)}(S).$$

Restricting this extension to the ordinary span category gives the first input in Lemma 11.9; call it

$$T: \mathrm{Span}(\mathrm{QProj}_S) \longrightarrow \mathrm{Mot}^{(2)}(S).$$

For the second input, map out of the unit of $\mathrm{Mot}^{(2)}(S)$. This produces a lax symmetric monoidal coefficient system

$$D(X) := \mathrm{Hom}_{\mathrm{Mot}^{(2)}(S)}(\mathbb{1}, [X]_{\mathrm{Mot}}), \quad D: \mathrm{Span}(\mathrm{QProj}_S) \longrightarrow 1\mathrm{Pr}_{\mathrm{st}}.$$

Here D is lax symmetric monoidal because T is symmetric monoidal and $\mathrm{Hom}_{\mathrm{Mot}^{(2)}(S)}(\mathbb{1}, -)$ is lax symmetric monoidal. The contravariant part $X \mapsto D(X)$ is a motivic coefficient system in the sense of Definition 11.1. Indeed, the smooth left adjoints, closed–open recollements, $\mathbb{A}^1$ -invariance, and Tate invertibility are obtained by applying $\mathrm{Hom}_{\mathrm{Mot}^{(2)}(S)}(\mathbb{1}, -)$ to the corresponding structures for $[-]_{\mathrm{Mot}}$.

The only point worth spelling out is the projection formula. Let $f: Y \rightarrow X$ be smooth. The graph-square argument in the proof of Lemma 11.8, applied now to the smooth left adjoint $f_{\sharp}$, gives an equivalence in $\mathrm{Mot}^{(2)}(S)$

$$f_{\sharp}(M \otimes f^*N) \simeq f_{\sharp}(M) \otimes N$$

for $M: \mathbb{1} \rightarrow [Y]_{\mathrm{Mot}}$ and $N: \mathbb{1} \rightarrow [X]_{\mathrm{Mot}}$. Applying $\mathrm{Hom}_{\mathrm{Mot}^{(2)}(S)}(\mathbb{1}, -)$ gives precisely the projection formula for the induced left adjoint on $D(Y) \rightarrow D(X)$. Hence Theorem 11.2 gives a unique morphism of motivic coefficient systems

$$\alpha^{\mathrm{op}}: \mathrm{SH} \longrightarrow D|_{\mathrm{QProj}_S^{\mathrm{op}}}.$$

We must still upgrade this contravariant map to spans. By definition of morphisms of motivic coefficient systems, α^{op} is compatible with the smooth left adjoints, hence in particular with the left adjoints for open immersions. By the final clause of Theorem 11.3, it is also compatible with the right adjoints for projective maps. Therefore the same extension criterion of Cnossen, Lenz, and Linskens [CLL25] used in Lemma 11.8 extends α^{op} uniquely to a morphism of commutative algebra objects

$$\alpha: \mathrm{SH} \longrightarrow \mathrm{Hom}_{\mathrm{Mot}^{(2)}(S)}(\mathbb{1}, T(-))$$

in $\mathrm{CAlg}(\mathrm{Fun}(\mathrm{Span}(\mathrm{QProj}_S), \mathrm{1Pr}_{\mathrm{st}}))$, which is the second input required in Lemma 11.9.

Now Lemma 11.9, applied to the pair (T, α) , gives a symmetric monoidal colimit-preserving functor

$$G: \mathrm{SH}^{(2)}(S) \longrightarrow \mathrm{Mot}^{(2)}(S)$$

which sends the corepresentable $[X] \in \mathrm{SH}^{(2)}(S)$ to $[X]_{\mathrm{Mot}}$.

It remains only to check that F and G are inverse equivalences. The composite GF is a symmetric monoidal colimit-preserving endofunctor of $\mathrm{Mot}^{(2)}(S)$ whose restriction to $\mathrm{QProj}_S^{\mathrm{op}}$ is the identity. By the universal property defining $\mathrm{Mot}^{(2)}(S)$, this identifies GF with the identity. For the other composite, use the last assertion of Lemma 11.9. The endofunctor FG of $\mathrm{SH}^{(2)}(S)$ induces the identity on the span-valued functor of corepresentables, and its induced endomorphism of the coefficient system SH is the identity by Drew–Gallauer’s initiality. Thus the two pieces of data attached to FG agree with those attached to the identity functor of $\mathrm{SH}^{(2)}(S)$, so $FG \simeq \mathrm{id}$. This proves that F is an equivalence, which is precisely the asserted universal property of $\mathrm{SH}^{(2)}(S)$ for quasiprojective schemes. The separated finite type version is Aoki [Aok26, Remark 1.3]. $\square$

Remark 11.11. The use of quasiprojective schemes in the proof is essential to the stated projective-base-change input. The extension of the universal property to separated finite type schemes is a separate result, recorded in Aoki [Aok26, Remark 1.3]; it does not require a non-monoidal proper-base-change theorem in the generality excluded by Remark 11.4.

11.4 Ring Gestalten

We now translate Aoki’s universal property into the language of ring Gestalten. The guiding idea is very simple: a sufficiently geometric transmutation

$$X \longmapsto X_R$$

of schemes is controlled by the single object $R = (\mathbb{A}^1)_R$, and the motivic axioms become intrinsic conditions on this ring Gestalt.

The section is organized as follows.

- First we prove the affine-line dictionary: finite-limit-preserving transmutations on affine schemes are the same thing as 0-truncated commutative ring objects.
- Local ring Gestalten are the ones for which this affine construction satisfies Zariski gluing, and hence extends from affine schemes to all schemes.
- Sutured ring Gestalten are those for which the basic decomposition $\mathbb{A}^1 = \mathbb{G}_m \cup \{0\}$ gives a recollement. By descent, this single condition controls general closed–open decompositions.
- Homological triviality and stability are the ring-Gestalt shadows of $\mathbb{A}^1$ -invariance and invertibility of the Tate twist.
- Finally, weak suaveness and smoothness encode the smooth-adjoint part of the motivic formalism. Aoki’s key observation is that, in the sutured situation, weak suaveness plus stability is equivalent to smoothness.

The final result says that the motivic ring Gestalt $[\mathbb{A}_{\mathbb{Z}}^1]_{\text{SH}}$ is universal among ring Gestalten satisfying these conditions.

11.4.1 Transmutations from ring Gestalten

Write

$$\text{Aff} := \text{Aff}_{\mathbb{Z}}^{\text{f.t.}} \subseteq \text{Sch}$$

for the full subcategory of affine schemes. We begin in an arbitrary category C with finite limits. We write $\text{Ring}(C)_{\leq 0}$ for the category of commutative ring objects in C whose underlying object is 0-truncated. When C is a category of Gestalten, we will simply refer to such objects as 0-truncated ring Gestalten in C ; in particular, RingGest denotes the category of commutative ring objects in Gest .

Construction 11.12 (Evaluation at the affine line). Let $F : \text{Aff} \rightarrow C$ preserve finite limits. Since $\mathbb{A}^1$ is a commutative ring object and F preserves finite products, $F(\mathbb{A}^1)$ is a commutative ring object in C . Moreover, Aff is an ordinary category, so all its objects are 0-truncated, and a finite-limit-preserving functor preserves truncated objects. We therefore obtain a functor

$$\text{ev}_{\mathbb{A}^1} : \text{Fun}^{\text{lex}}(\text{Aff}, C) \longrightarrow \text{Ring}(C)_{\leq 0}, \quad F \longmapsto F(\mathbb{A}^1).$$

Proposition 11.13. *The functor $\text{ev}_{\mathbb{A}^1}$ of Construction 11.12 is an equivalence.*

In the Gestalten examples, this says that affine transmutations are the same thing as 0-truncated ring Gestalten. More explicitly, let $R \in \text{Ring}(C)_{\leq 0}$ and suppose that

$$X = \text{Spec}(\mathbb{Z}[x_1, \dots, x_k]/(f_1, \dots, f_l)).$$

Writing $f = (f_1, \dots, f_l): \mathbb{A}^k \rightarrow \mathbb{A}^l$, the functor corresponding to R sends X to the pullback

$$\begin{array}{ccc} X_R & \longrightarrow & R^k \\ \downarrow & & \downarrow f_R \\ * & \xrightarrow{0} & R^l, \end{array}$$

where f_R is obtained by evaluating the polynomials $f_1, \dots, f_l$ in R .

Proof. Let $\text{Poly}_{\mathbb{Z}} \subseteq \text{Aff}$ be the full subcategory spanned by the affine spaces $\mathbb{A}^k$. A finite-product-preserving functor $\text{Poly}_{\mathbb{Z}} \rightarrow C$ is precisely a commutative ring object in C : the polynomial maps between affine spaces encode the ring operations and all their identities.

The category Aff is the finite-limit completion of this algebraic theory in ordinary categories. Indeed, because $\mathbb{Z}$ is Noetherian, every finite type $\mathbb{Z}$ -algebra is finitely presented, and hence every object of Aff is a finite limit of affine spaces as in the displayed pullback. It follows that a finite-product-preserving functor on $\text{Poly}_{\mathbb{Z}}$ admits a finite-limit-preserving extension to Aff precisely when its underlying ring object is 0-truncated; in that case the extension is unique and is computed by these finite-limit presentations. This construction is inverse to evaluation at $\mathbb{A}^1$, proving the proposition. $\square$

11.4.2 Local ring Gestalten

For this subsection, suppose that C is countably toposic. Thus C has finite limits and countable colimits, and these satisfy the countable Giraud axioms: countable colimits are universal, coproducts are disjoint, and groupoid objects are effective. We have in mind especially the categories $\text{Gest}_{/S}^{n\text{-}\acute{e}t}$, which are countably toposic by Theorem 7.13.

We next formulate Zariski locality intrinsically in C . Let $R \in \text{Ring}(C)_{\leq 0}$. Its object of units $R^\times$ is defined by the cartesian square

$$\begin{array}{ccc} R^\times & \longrightarrow & R \times R \\ \downarrow & \lrcorner & \downarrow \mu \\ * & \xrightarrow{1} & R. \end{array}$$

Composing the upper horizontal map with the first projection defines $\iota: R^\times \rightarrow R$. Equivalently, $R^\times$ classifies a generalized element of R together with its multiplicative inverse. Define the locus on which $1 - a$ is invertible by

$$R_{1-}^\times := R \times_{1-(-), R, \iota} R^\times.$$

The two projections give a canonical map

$$R^\times \sqcup R_{1-}^\times \longrightarrow R.$$

Definition 11.14. A 0-truncated commutative ring object R in C is *local* if the map

$$R^\times \sqcup R_{1-}^\times \longrightarrow R$$

is a cover, i.e. if the geometric realization of its Čech nerve is R . We write $\text{Ring}^{\text{loc}}(C)_{\leq 0}$ for the full subcategory of local ring objects. In the Gestalten examples, its objects are the local ring Gestalten.

This is the usual internal local-ring condition. Indeed, a generalized element of R over $T \in C$ is a map $a: T \rightarrow R$. Pulling back the displayed cover gives the two loci in T on which a and $1 - a$, respectively, are invertible; these loci cover T precisely when R is local.

Let $\text{Fun}^{\text{lex,Zar}}(\text{Sch}, C)$ denote the full subcategory of $\text{Fun}^{\text{lex}}(\text{Sch}, C)$ spanned by the functors satisfying Zariski gluing: for every finite affine open cover $U = \coprod_i U_i \rightarrow X$, the canonical map

$$\text{colim}_{[p] \in \Delta^{\text{op}}} F(U_p) \longrightarrow F(X)$$

is an isomorphism, where $U_{\bullet}$ denotes the Čech nerve of $U \rightarrow X$.

Proposition 11.15. *Evaluation at the affine line induces an equivalence*

$$\text{ev}_{\mathbb{A}^1}: \text{Fun}^{\text{lex,Zar}}(\text{Sch}, C) \xrightarrow{\sim} \text{Ring}^{\text{loc}}(C)_{\leq 0}.$$

Proof. We first identify the effect of Zariski gluing on the affine restriction. Suppose that $F: \text{Sch} \rightarrow C$ preserves finite limits and satisfies Zariski gluing, and put $R := F(\mathbb{A}^1)$. Applying F to the standard cover

$$\mathbb{A}^1 = D(t) \cup D(1 - t)$$

identifies the two open subobjects with $R^{\times}$ and $R_{1-}^{\times}$. The gluing assumption says that their coproduct is a cover of R , so R is local.

Conversely, let $R \in \text{Ring}^{\text{loc}}(C)_{\leq 0}$ and let $F_R: \text{Aff} \rightarrow C$ be the finite-limit-preserving functor of Proposition 11.13. Consider a finite principal-open cover

$$\text{Spec}(A) = D(f_1) \cup \cdots \cup D(f_r).$$

Choose $g_1, \dots, g_r \in A$ with $\sum_i g_i f_i = 1$. The internal local-ring condition, applied inductively to this equality, implies that the loci on which one of the elements $g_i f_i$ is invertible cover $F_R(\text{Spec}(A))$. On such a locus f_i is invertible, so it factors through $F_R(D(f_i))$. Hence F_R carries the given principal-open cover to a cover in C . Every finite affine Zariski cover admits a refinement by a finite principal-open cover, and effectiveness of groupoid objects in C then shows that F_R carries its Čech nerve to a colimit diagram.

It remains to pass between affine schemes and arbitrary objects of Sch . Affine opens form a basis for the Zariski topology, and because the schemes in Sch are separated, this basis is closed under finite intersections. The usual basis-extension argument therefore identifies finite-limit-preserving Zariski-gluing functors on Sch with finite-limit-preserving functors on Aff which carry affine Zariski covers to colimit diagrams. Explicitly, if $X = \bigcup_i U_i$ is a finite affine open cover, the extension is forced to satisfy

$$F_R(X) \simeq \text{colim}_{[p] \in \Delta^{\text{op}}} F_R(U_p).$$

Universality of countable colimits in C shows that this construction is independent of the cover, functorial in X , and preserves finite limits. Thus restriction along $\text{Aff} \hookrightarrow \text{Sch}$ gives an equivalence onto the affine functors satisfying Zariski gluing. Combining this with the preceding paragraphs and Proposition 11.13 proves the proposition. $\square$

For $R \in \text{Ring}^{\text{loc}}(C)_{\leq 0}$, we denote the corresponding transmutation by

$$(-)_R: \text{Sch} \longrightarrow C, \quad X \longmapsto X_R.$$

11.4.3 Sutured ring Gestalten

The next condition is no longer only a condition on the underlying finite-limit-preserving functor. It uses the coefficient categories attached to the objects X_R . Aoki phrases this using stacks, by which he means commutative algebras in a symmetric monoidal 2-category. For us it is more natural to keep the Gestalten language throughout. Thus, from now on fix a presentably symmetric monoidal 2-category $\mathcal{C}$, and set

$$\mathcal{G}_C := \text{Gest}_{/\text{Gest}(C)}^{1\text{-}\acute{\text{e}}\text{t}}.$$

Here $\text{Gest}(C)$ denotes the 1-affine Gestalt corresponding to C . The basic input will be a commutative ring object $R \in \mathcal{G}_C$ whose underlying object $R = (\mathbb{A}^1)_R$ is 1-affine over $\text{Gest}(C)$. Then the affine-line dictionary gives X_R for every affine X , and these affine pieces are again 1-affine over $\text{Gest}(C)$. If R is local, this extends by Zariski gluing to a transmutation

$$(-)_R: \text{Sch} \longrightarrow \mathcal{G}_C, \quad X \longmapsto X_R$$

on all schemes; the resulting X_R need not be 1-affine for non-affine X . Whenever X_R is defined and $n \geq 0$, set

$$D_R^{(n)}(X) := (O(X_R)/O(\text{Gest}(C)))_n.$$

For affine X , the 1-affineness of X_R gives

$$D_R^{(1)}(X) \in \text{CAlg}(C), \quad D_R^{(2)}(X) \simeq \text{Mod}_{D_R^{(1)}(X)}(C) \in \text{Mod}_C(2\text{Pr}),$$

while for general X the object $D_R^{(2)}(X)$ is defined by descent from affine opens. This is the Gestalten version of Aoki's convention that a stack has an associated coefficient category.

We will use the following intrinsic form of recollement. Let $\mathcal{E}$ be a pointed 2-category with finite limits and finite colimits. A sequence

$$A \longrightarrow B \longrightarrow C$$

in $\mathcal{E}$ is a *bifiber sequence* if the square

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & C \end{array}$$

is both cartesian and cocartesian.

Definition 11.16 (Recollements in a pointed 2-category). Let $\mathcal{E}$ be as above. A diagram

$$C_U \xleftarrow{j^*} C_X \xrightarrow{i^*} C_Z$$

in $\mathcal{E}$ is a *recollement* if the following two conditions hold:

- (1) j^* admits a left adjoint $j_\#$, and

$$C_U \xrightarrow{j_\#} C_X \xrightarrow{i^*} C_Z$$

is a bifiber sequence.

(2) i^* admits a right adjoint i_* , and

$$C_Z \xrightarrow{i_*} C_X \xrightarrow{j^*} C_U$$

is a bifiber sequence.

Recall that $R^\times \rightarrow R$ denotes the object of units in the ring object R , and that the zero section is the map $0: * \rightarrow R$. More precisely, if R is the ring Gestalt, then in the following formulas R , $R^\times$, and $*$ stand for $(\mathbb{A}^1)_R$, $(\mathbb{G}_m)_R$, and $(\text{Spec}(\mathbb{Z}))_R$, respectively.

Definition 11.17 (Sutured ring Gestalten). Such a ring object $R \in \mathcal{G}_C$ is *sutured* if the map $R^\times \rightarrow R$ is an open immersion, the zero section $* \rightarrow R$ is a closed immersion, and the diagram

$$R^\times \longrightarrow R \longleftarrow *$$

induces a recollement

$$D_R^{(2)}(\mathbb{G}_m) \xleftarrow{j^*} D_R^{(2)}(\mathbb{A}^1) \xrightarrow{i^*} D_R^{(2)}(\text{Spec}(\mathbb{Z}))$$

in the pointed 2-category $\text{Mod}_C(2\text{Pr})$, where $j: R^\times \rightarrow R$ is the inclusion and $i: * \rightarrow R$ is the zero section.

The open and closed immersion hypotheses are part of the geometry of the word “sutured”. In particular, they imply that a sutured ring Gestalt is 0-truncated: the shear automorphism of $R \times R$ identifies the diagonal of R with the base change of the closed immersion $* \rightarrow R$, namely the inclusion $R \times \{0\} \rightarrow R \times R$. Thus the diagonal is a monomorphism.

Thus suturedness says that the basic decomposition

$$\mathbb{A}^1 = \mathbb{G}_m \cup \{0\}$$

behaves like an open–closed decomposition after transmutation. The following observation explains how, once the two pieces are sufficiently geometric, such a recollement also remembers the corresponding cover.

Lemma 11.18. *Let $j: U \rightarrow X$ and $i: Z \rightarrow X$ be 1-étale, 1-affine maps of Gestalten. Suppose that the diagram*

$$O(U)_2 \xleftarrow{j^*} O(X)_2 \xrightarrow{i^*} O(Z)_2$$

is a recollement. Then

$$U \sqcup Z \longrightarrow X$$

is a cover.

Proof. Set $f: U \sqcup Z \rightarrow X$. Since 1-étale maps are closed under coproducts in the slice over X , the map f is 1-étale and hence 1-suave. By the cover criterion of Proposition 8.22, it is therefore enough to show that

$$f_1^*: O(X)_2 \longrightarrow O(U \sqcup Z)_2$$

is conservative. Relative global sections carry coproducts of Gestalten to products, so

$$O(U \sqcup Z)_2 \simeq O(U)_2 \times O(Z)_2,$$

and under this identification f_1^* is the pair (j^*, i^*) .

It remains to recall the basic conservativity property of a recollement. If an object or morphism of the middle term becomes trivial after applying both j^* and i^* , then the two bifiber sequences in Definition 11.16 force it to be trivial already in the middle term. Equivalently, the pair (j^*, i^*) is jointly conservative. Thus f_1^* is conservative, and Proposition 8.22 shows that f is a cover. $\square$

In particular, suppose that R is sutured and that the two maps $R^\times \rightarrow R$ and $* \rightarrow R$ in the defining sutured diagram are 1-étale and 1-affine. Then Lemma 11.18 shows that $R^\times \sqcup * \rightarrow R$ is a cover. Since the zero section factors through $R_{1-}^\times \rightarrow R$, this cover refines the locality map

$$R^\times \sqcup R_{1-}^\times \longrightarrow R.$$

Thus, under these geometric hypotheses, suturedness implies the locality cover from Definition 11.14. By base change, the single recollement in the definition of suturedness controls all principal open–closed decompositions of affine schemes.

Proposition 11.19. *Let R be sutured. Let $X = \operatorname{Spec}(A)$ be affine of finite type over $\mathbb{Z}$, and let $a \in A$. Put*

$$U := D(a) \subseteq X, \quad Z := V(a) = \operatorname{Spec}(A/a) \subseteq X.$$

Then the diagram

$$D_R^{(2)}(U) \xleftarrow{j^*} D_R^{(2)}(X) \xrightarrow{i^*} D_R^{(2)}(Z)$$

is a recollement in $\operatorname{Mod}_C(2\operatorname{Pr})$.

Proof. The element $a \in A$ determines a morphism $a: X \rightarrow \mathbb{A}^1$. The principal open $D(a)$ and the principal closed subscheme $V(a)$ are the two pullbacks

$$\begin{array}{ccc} D(a) & \longrightarrow & \mathbb{G}_m \\ \downarrow & & \downarrow \\ X & \xrightarrow{a} & \mathbb{A}^1, \end{array} \quad \begin{array}{ccc} V(a) & \longrightarrow & \operatorname{Spec}(\mathbb{Z}) \\ \downarrow & & \downarrow 0 \\ X & \xrightarrow{a} & \mathbb{A}^1. \end{array}$$

Applying the transmutation attached to R , these become the corresponding pullbacks of

$$R^\times \longrightarrow R \longleftarrow *.$$

Suturedness gives the recollement for this universal diagram in degree 2. Base-changing this recollement along the map $X_R \rightarrow R$ induced by a gives precisely the displayed diagram

$$D_R^{(2)}(U) \longleftarrow D_R^{(2)}(X) \longrightarrow D_R^{(2)}(Z).$$

Hence the pulled-back diagram for $D(a) \subseteq X \supseteq V(a)$ is a recollement. $\square$

The previous proposition is the affine principal form of the localization axiom. The general closed–open case follows from it by two elementary operations: finite induction and Zariski descent.

Lemma 11.20. *Let R be local and sutured, and write $X \mapsto X_R$ for the associated Zariski-gluing transmutation. Then for every closed immersion $i: Z \hookrightarrow X$ in Sch with open complement $j: U \hookrightarrow X$, the diagram*

$$D_R^{(2)}(U) \xleftarrow{j^*} D_R^{(2)}(X) \xrightarrow{i^*} D_R^{(2)}(Z)$$

is a recollement.

Proof. First note that recollements may be checked after finite Zariski covers. If $V = \coprod_i V_i \rightarrow X$ is a finite Zariski cover with Čech nerve $V_\bullet$, then Zariski gluing gives

$$X_R \simeq \text{colim}_{[p] \in \Delta^{\text{op}}} (V_p)_R$$

in $\mathcal{G}_C = \text{Gest}_{/\text{Gest}(C)}^{1\text{-}\acute{\text{e}}\text{t}}$. Applying relative global sections over $\text{Gest}(C)$ and taking degree 2 gives

$$D_R^{(2)}(X) \simeq \lim_{[p] \in \Delta} D_R^{(2)}(V_p)$$

in $\text{Mod}_C(2\text{Pr})$, and the same assertion holds for U and Z . Thus the projections to the finite intersections of the cover jointly reflect the adjunctions and bifiber squares appearing in Definition 11.16.

We may therefore work affine-locally on X . Suppose first that $X = \text{Spec}(A)$ and that $Z = V(I)$ with $I = (a_1, \dots, a_r)$. The cases $U = \emptyset$ and $Z = \emptyset$ are tautological, so assume $r \geq 1$. We claim, by induction on r , that

$$D_R^{(2)}(\mathbb{A}^r \setminus \{0\}) \xleftarrow{} D_R^{(2)}(\mathbb{A}^r) \xrightarrow{} D_R^{(2)}(\text{Spec}(\mathbb{Z}))$$

is a recollement. For $r = 1$ this is precisely the definition of suturedness. For the induction step, decompose $\mathbb{A}^r$ first into the principal open $D(t_1)$ and the principal closed $\mathbb{A}^{r-1} = V(t_1)$. On the closed part, the complement of the origin is $\mathbb{A}^{r-1} \setminus \{0\}$. Hence the induction hypothesis gives the localization sequence inside $V(t_1)$, and Proposition 11.19 gives the principal localization sequence for $D(t_1) \subseteq \mathbb{A}^r$. Pasting the two recollements, or equivalently pasting the corresponding bifiber squares in Definition 11.16, gives the desired recollement for $\mathbb{A}^r \setminus \{0\} \subseteq \mathbb{A}^r$.

Now the tuple $(a_1, \dots, a_r)$ defines a map $a: X \rightarrow \mathbb{A}^r$, and the closed–open pair

$$U = X \setminus Z, \quad Z = V(I)$$

is the pullback of

$$\mathbb{A}^r \setminus \{0\} \subseteq \mathbb{A}^r \supseteq \{0\}.$$

Base-changing the recollement just constructed along $X_R \rightarrow (\mathbb{A}^r)_R$ gives the asserted recollement for the affine scheme X .

For general X , choose a finite affine Zariski cover. Since the schemes in Sch are separated, the finite intersections in this cover are again affine. On each such affine, the closed subscheme is cut out by finitely many functions, so the affine case applies. The initial descent observation then glues these affine recollements to the desired recollement on X . $\square$

Thus suturedness is precisely the ring-Gestalt translation of the closed–open localization axiom in Definition 11.1.

11.4.4 Homologically trivial and stable ring Gestalten

We now translate the two remaining motivic axioms which only involve the affine line: $\mathbb{A}^1$ -invariance and invertibility of the Tate twist. Continue to work in $\mathcal{G}_C$. Let

$$\pi: R = (\mathbb{A}^1)_R \longrightarrow * = (\mathrm{Spec}(\mathbb{Z}))_R$$

be the projection, and let

$$s: * \longrightarrow R$$

be the zero section.

Definition 11.21 (Homologically trivial ring Gestalten). A ring Gestalt R is *homologically trivial* if the pullback

$$\pi^*: D_R^{(2)}(\mathrm{Spec}(\mathbb{Z})) \longrightarrow D_R^{(2)}(\mathbb{A}^1)$$

admits a left adjoint $\pi_{\#}$, and the counit

$$\pi_{\#}\pi^* \longrightarrow \mathrm{id}_{D_R^{(2)}(\mathrm{Spec}(\mathbb{Z}))}$$

is an equivalence.

Thus homological triviality says that the affine line has the same homology as the point: pulling coefficients to R is fully faithful.

Definition 11.22 (Stable ring Gestalten). Let R be sutured, so that the zero-section pullback

$$s^*: D_R^{(2)}(\mathbb{A}^1) \longrightarrow D_R^{(2)}(\mathrm{Spec}(\mathbb{Z}))$$

admits a right adjoint s_* . We call R *stable* if π^* admits a left adjoint $\pi_{\#}$ and the composite

$$\pi_{\#}s_*: D_R^{(2)}(\mathrm{Spec}(\mathbb{Z})) \longrightarrow D_R^{(2)}(\mathrm{Spec}(\mathbb{Z}))$$

is an autoequivalence.

The autoequivalence $\pi_{\#}s_*$ is the *Tate twist*.

Proposition 11.23. *Let R be local. Then:*

- (1) *If R is homologically trivial, then the assignment $X \mapsto D_R^{(2)}(X)$ satisfies $\mathbb{A}^1$ -invariance: for every $X \in \mathrm{Sch}$, the projection*

$$\pi_X: (\mathbb{A}_X^1)_R \longrightarrow X_R$$

has the property that π_X^ admits a left adjoint $\pi_{X,\#}$ and the counit*

$$\pi_{X,\#}\pi_X^* \longrightarrow \mathrm{id}_{D_R^{(2)}(X)}$$

is an equivalence.

(2) If R is sutured and stable, then the assignment $X \mapsto D_R^{(2)}(X)$ satisfies Tate invertibility: for every $X \in \text{Sch}$, if

$$s_X: X_R \longrightarrow (\mathbb{A}_X^1)_R$$

is the zero section, then

$$\pi_{X,\#} s_{X,*}: D_R^{(2)}(X) \longrightarrow D_R^{(2)}(X)$$

is an autoequivalence.

Proof. The point is that all of these constructions are obtained from the corresponding construction over the point by base change. We first treat the case where X is affine. Then X_R is 1-affine over $\text{Gest}(C)$, so products over the base are computed on degree-2 coefficient categories by relative tensor product. In particular,

$$D_R^{(2)}(\mathbb{A}_X^1) \simeq D_R^{(2)}(X) \otimes_C D_R^{(2)}(\mathbb{A}^1),$$

and under this identification the pullback π_X^* is obtained from π^* by tensoring with $D_R^{(2)}(X)$.

If R is homologically trivial, tensoring the adjunction $\pi_{\#} \dashv \pi^*$ with $D_R^{(2)}(X)$ gives an adjunction $\pi_{X,\#} \dashv \pi_X^*$. The counit is the tensor product of the counit $\pi_{\#} \pi^* \rightarrow \text{id}_{D_R^{(2)}(\text{Spec}(\mathbb{Z}))}$ with the identity of $D_R^{(2)}(X)$, hence is an equivalence. This proves (1).

For (2), suturedness gives the right adjoint s_* . Base change along $X_R \rightarrow *$ gives the right adjoint $s_{X,*}$, and the same tensor-product description identifies

$$\pi_{X,\#} s_{X,*} \simeq \text{id}_{D_R^{(2)}(X)} \otimes (\pi_{\#} s_*).$$

If R is stable, the second tensor factor is an autoequivalence, hence so is the displayed functor.

For general X , choose a finite affine Zariski cover and pass to its Čech nerve. Since the transmutation $X \mapsto X_R$ was constructed by gluing in $\mathcal{G}_C$, the categories $D_R^{(2)}(-)$ are obtained as the corresponding limits over this nerve. Adjunctions, counits, and the assertion that a functor is an equivalence may therefore be checked after restriction to the finite intersections of the cover, where the affine argument applies. $\square$

11.4.5 Weakly suave and smooth ring Gestalten

The last condition in Aoki's theorem is smoothness. Here one has to be slightly careful about terminology. In the general theory of Stefanich rings, recalled in Section 6.5, maps come with a hierarchy of n -suaveness conditions. Aoki's word "suave" corresponds in our indexing to 0-suaveness. He first uses a weaker condition for stacks: only the visible pullback functor has to admit a linear left adjoint. In our notation this is the following.

Definition 11.24 (Weakly suave ring Gestalten). A ring Gestalt R is *weakly suave* if, for the projection

$$\pi: R = (\mathbb{A}^1)_R \longrightarrow * = (\text{Spec}(\mathbb{Z}))_R,$$

the pullback

$$\pi^*: D_R^{(2)}(\text{Spec}(\mathbb{Z})) \longrightarrow D_R^{(2)}(\mathbb{A}^1)$$

admits a $D_R^{(2)}(\text{Spec}(\mathbb{Z}))$ -linear left adjoint $\pi_{\#}$.

Thus weak suaveness is the degree-2 shadow of 0-suaveness for the map π . Full 0-suaveness asks for the corresponding higher adjointability and dualizability data from Definition 6.18. The point of Aoki's argument is that, for sutured ring Gestalten, stability supplies exactly the missing duality.

Definition 11.25 (Smooth ring Gestalten). A sutured ring Gestalt R is *smooth* if the projection

$$\pi: R \rightarrow *$$

is 0-suave in the sense of the Gestalten/Stefanich-ring formalism, and its dualizing object

$$\omega_\pi := \pi^! \mathbb{1}$$

is tensor-invertible.

This is the direct analogue of the usual criterion for cohomological smoothness: 0-suaveness gives the exceptional pullback, and invertibility of the dualizing object is the smoothness condition.

Proposition 11.26 (Aoki [Aok26]). *Let R be a sutured weakly suave ring Gestalt. Then R is smooth if and only if it is stable.*

Proof. First, note that suturedness makes the diagonal of R into a closed immersion. Indeed, the shearing automorphism

$$R \times R \longrightarrow R \times R, \quad (x, y) \longmapsto (x, x - y),$$

identifies the diagonal $\Delta: R \rightarrow R \times R$ with the map $\text{id}_R \times 0: R \times * \rightarrow R \times R$. But the zero section is the closed part of the sutured recollement, and closed immersions are stable under base change.

Now use the left adjoint $\pi_\#$ supplied by weak suaveness. The closed diagonal gives a right adjoint Δ_* to Δ^* , and hence maps

$$\eta: D_R^{(2)}(\text{Spec}(\mathbb{Z})) \xrightarrow{\pi^*} D_R^{(2)}(\mathbb{A}^1) \xrightarrow{\Delta_*} D_R^{(2)}(\mathbb{A}^1 \times \mathbb{A}^1)$$

and

$$\varepsilon: D_R^{(2)}(\mathbb{A}^1 \times \mathbb{A}^1) \xrightarrow{\Delta^*} D_R^{(2)}(\mathbb{A}^1) \xrightarrow{\pi_\#} D_R^{(2)}(\text{Spec}(\mathbb{Z})).$$

These are the expected coevaluation and evaluation maps for self-duality of R over the point. The triangle composites are not literally identities at first. After applying the same shearing trick as above, they identify with the Tate operator $\pi_\# s_*$, where $s: * \rightarrow R$ is the zero section. Stability says precisely that this operator is invertible. The elementary duality fact used by Aoki says that if the two triangle composites are equivalences, then one can twist the coevaluation to obtain an honest duality datum. Thus $R \rightarrow *$ is 0-suave: weak suaveness has been promoted to the dualizability required in the Gestalten sense.

It remains to identify the invertibility of the dualizing object. Let BR denote the classifying object of the additive group underlying R , and let

$$t: * \longrightarrow BR$$

be the universal point. The standard torsor square shows that

$$\omega_\pi = \pi^! \mathbb{1}$$

is pulled back from the object

$$L := t^! \mathbb{1}.$$

Consequently, smoothness is equivalent to invertibility of L . On the other hand, using $s^* \pi^! \mathbb{1} \simeq L$, the projection formula, and the fact that the composite $\pi \circ s$ is the identity, Aoki computes

$$L \simeq s^* \pi^! \mathbb{1} \simeq \pi_{\#} s_* \mathbb{1}.$$

Thus L is invertible exactly when the Tate operator $\pi_{\#} s_*$ is invertible. This is stability, and the proposition follows. $\square$

In slogan form: suturedness gives the closed diagonal, weak suaveness gives the left adjoint, and stability turns the resulting weak duality into an invertible one.

11.4.6 Universality of the motivic ring Gestalt

We can now assemble the preceding translations into Aoki's ring-Gestalt form of the universal property. The definitions above did not require stability of the ambient 2-category C . In Aoki's formulation, the category generated by the universal ring stack is stable, but the ambient presentably symmetric monoidal 2-category in which one places ring stacks need not itself be stable.

By Proposition 11.7, we may identify the base Gestalt $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$ with $\mathrm{Gest}(\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})))$. Under this identification the motivic affine line is a ring Gestalt

$$R_{\mathrm{SH}} := [\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}} \in \mathcal{G}_{\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))},$$

and this object is 1-affine over $\mathrm{Gest}(\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})))$. If C is a presentably symmetric monoidal 2-category, let

$$\mathrm{RingGest}_{\mathrm{mot}}(C)$$

denote the category of sutured, homologically trivial, and smooth ring Gestalten in $\mathcal{G}_C$; morphisms are morphisms of ring Gestalten. Such a ring Gestalt is automatically local: the open and closed maps in Definition 11.17 are 1-étale and 1-affine, so Lemma 11.18 applies.

Theorem 11.27 (Aoki's ring-Gestalt theorem). *For every presentably symmetric monoidal 2-category C , evaluation on the motivic affine line induces an equivalence*

$$\mathrm{Fun}^{\otimes, \mathrm{L}}(\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})), C) \xrightarrow{\sim} \mathrm{RingGest}_{\mathrm{mot}}(C), \quad H \mapsto H(R_{\mathrm{SH}}).$$

Equivalently, the stable presentably symmetric monoidal 2-category $\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))$ is freely generated by a homologically trivial, smooth, sutured ring Gestalt.

Proof. This is Aoki [Aok26, Theorem B], rewritten in the Gestalten language used here. Let us spell out how the dictionary works.

First suppose that

$$H: \mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})) \longrightarrow C$$

is symmetric monoidal and colimit-preserving. The notation $H(R_{\mathrm{SH}})$ means the following: the 1-affine Gestalt $R_{\mathrm{SH}} \rightarrow [\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$ is equivalently encoded by its algebra of functions, a commutative

algebra object of $\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))$, and we apply H to that commutative algebra. This gives a ring Gestalt over $\mathrm{Gest}(C)$.

By Theorem 11.10, the associated functor

$$X \mapsto H([X])$$

is motivic, where $[X]$ denotes the corepresentable object of $\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))$. Evaluating at $\mathbb{A}^1$ therefore gives a ring Gestalt with exactly the properties isolated above. The localization axiom for the decomposition $\mathbb{A}^1 = \mathbb{G}_m \cup \{0\}$ gives suturedness, and homotopy invariance gives homological triviality. The smooth axiom gives weak suaveness of the projection $H(R_{\mathrm{SH}}) \rightarrow *$, while the Tate axiom gives stability; by Proposition 11.26, this is the same as smoothness. Locality then follows from suturedness by Lemma 11.18.

Conversely, let R be a sutured, homologically trivial, smooth ring Gestalt over $\mathrm{Gest}(C)$. By the geometric clauses in the definition of suturedness and Lemma 11.18, R is local. Hence Proposition 11.15 gives the transmutation

$$(-)_R: \mathrm{Sch} \longrightarrow \mathcal{G}_C$$

by Proposition 11.15. Taking degree-1 functions gives a symmetric monoidal functor

$$F_R: \mathrm{Sch}^{\mathrm{op}} \longrightarrow C, \quad X \mapsto D_R^{(1)}(X),$$

where we forget the commutative algebra structure on $D_R^{(1)}(X)$ when viewing it as an object of C .

The preceding subsections translate the ring-Gestalt conditions into the motivic axioms for F_R . Suturedness gives closed–open recollements, by Lemma 11.20. Homological triviality gives $\mathbb{A}^1$ -invariance, by Proposition 11.23. Since smoothness implies weak suaveness, Proposition 11.26 gives stability, and hence the Tate operator is invertible. Finally, smoothness of $R \rightarrow *$ supplies the smooth-adjoint part of the motivic axioms; for general smooth morphisms this follows by the usual reduction to affine space, together with stability of 0-suaveness under products, base change, composition, and Zariski descent.

Thus F_R is a motivic functor. By Theorem 11.10, it is induced by a unique symmetric monoidal colimit-preserving functor

$$H_R: \mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})) \longrightarrow C.$$

The affine-line dictionary of Proposition 11.13 and Proposition 11.15 says that the resulting ring Gestalt $H_R(R_{\mathrm{SH}})$ is canonically identified with the original R .

The same argument also identifies morphisms. A natural transformation between functors out of $\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))$ is determined by its effect on the corepresentable motivic functor, by Lemma 11.9; under the affine-line dictionary, this is the same as a morphism of the corresponding local ring Gestalten. Hence the two constructions above are inverse equivalences. $\square$

For a Gestalt X , let $\mathrm{RingGest}_{\mathrm{mot}}(X)$ denote the category of sutured, homologically trivial, and smooth ring Gestalten over X , i.e. such ring Gestalten in $\mathrm{Gest}_{/X}^{1\text{-}\acute{\mathrm{e}}\mathrm{t}}$ whose affine-line object is 1-affine over X . If $X = \mathrm{Gest}(C)$, this recovers the notation $\mathrm{RingGest}_{\mathrm{mot}}(C)$ above. Locality is automatic by suturedness, as above.

Corollary 11.28. *The presheaf*

$$\mathrm{Gest}^{\mathrm{op}} \longrightarrow \mathrm{An}, \quad X \mapsto \mathrm{RingGest}_{\mathrm{mot}}(X)^{\simeq}$$

is represented by the motivic Gestalt $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$. Equivalently, for every Gestalt X there is a natural equivalence

$$\mathrm{Hom}_{\mathrm{Gest}}(X, [\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}) \simeq \mathrm{RingGest}_{\mathrm{mot}}(X)^{\simeq}.$$

Proof. By Proposition 11.7, the motivic Gestalt is 2-affine:

$$[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}} \simeq \mathrm{Gest}(\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z}))).$$

Therefore maps from X to $[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$ are the same thing as symmetric monoidal colimit-preserving functors

$$\mathrm{SH}^{(2)}(\mathrm{Spec}(\mathbb{Z})) \longrightarrow \mathcal{O}(X)_2.$$

Applying Theorem 11.27 with target $\mathcal{O}(X)_2$ identifies these functors with motivic ring Gestalten over X . Passing to maximal subgroupoids gives the displayed equivalence, naturally in X . $\square$

Using Proposition 11.26, one may equivalently describe $\mathrm{RingGest}_{\mathrm{mot}}$ as the moduli problem of sutured, homologically trivial, weakly suave, and stable ring Gestalten. This is often the more practical formulation: weak suaveness is the visible adjoint, while stability is the extra condition which upgrades it to smoothness.

12 Cartier duality

Talk given by Han-Ung.

Today we introduce the statement of a very general *Cartier duality* in the topos of Gestalten, following Scholze [Sch26, Lecture X], and illustrate it with many examples. This duality unifies and generalizes a host of classical “Fourier isomorphisms”. The proof is deferred to the following talks; here we concentrate on the statement, its corollaries, and the way in which the classical examples are recovered.

Most of the material here has been taken verbatim from Scholze’s notes.

12.1 Fourier isomorphisms

Cartier duality is a form of Fourier duality, so we begin by recalling various incarnations of the latter.

Example 12.1 (Fourier). We work with $\mathbb{C}$ -coefficients throughout. The most basic instance of a Fourier isomorphism is the isomorphism of Hilbert spaces

$$L^2(S^1) \xrightarrow{\sim} L^2(\mathbb{Z}),$$

where we regard $S^1 = \mathbb{R}/\mathbb{Z}$ with coordinate x . In one direction a sequence $(a_n)_{n \in \mathbb{Z}}$ is sent to the function $\sum_n a_n \exp(2\pi i n x)$; in the other, a function f is sent to the sequence

$$n \mapsto \int_{S^1} f(x) \exp(-2\pi i n x) dx.$$

If we instead regard $S^1 \subseteq \mathbb{C}^\times$ with coordinate t , the formulas become $(a_n)_n \mapsto \sum_n a_n t^n$ and $f \mapsto (n \mapsto \int_{S^1} f(t) t^{-n} \frac{dt}{t})$.

There are various “decompleted” refinements of this isomorphism; for instance

$$\begin{aligned} C^\infty(S^1) &\xrightarrow{\sim} \{\text{sequences } (a_n)_n \text{ of rapid decay}\}, \\ \mathcal{D}'(S^1) &\xrightarrow{\sim} \{\text{sequences } (a_n)_n \text{ of polynomial growth}\}, \end{aligned}$$

where $\mathcal{D}'(S^1)$ is the space of distributions on S^1 , namely the continuous strong dual of the nuclear Fréchet space $C^\infty(S^1)$; the Hilbert space isomorphism is recovered by passing to completions.

This phenomenon has a categorical shadow: continuous finite-dimensional representations of S^1 decompose into weight spaces, giving an equivalence

$$\mathrm{Rep}^{\mathrm{fd}, \mathrm{cont}}(S^1) \simeq \bigoplus_{n \in \mathbb{Z}} \mathrm{Vect}^{\mathrm{fd}}(\mathbb{C}),$$

sending V to $(V_n)_n$ with $V_n = \{v \in V \mid t \cdot v = t^n v \text{ for all } t \in S^1\}$. This is even a symmetric monoidal equivalence, for the pointwise tensor product on the left and the Day convolution on the right. Heuristically, the right-hand side is “finite sheaves of $\mathbb{C}$ -vector spaces on $\mathbb{Z}$ ” and the left-hand side is “finite sheaves of $\mathbb{C}$ -vector spaces on $*/S^1$ ”.

Example 12.2 (The multiplicative group). The previous example has an elementary algebraic analogue, valid over any (for the moment, ordinary) commutative ring R . Viewing $S^1 \subseteq \mathbb{C}^\times$ inside the multiplicative group, one can decomplete $L^2(S^1)$ all the way to Laurent polynomials. Namely, for the multiplicative group $\mathbb{G}_{m,R} = \text{Spec}(R[t^{\pm 1}])$ we have

$$O(\mathbb{G}_{m,R}) \cong \bigoplus_{n \in \mathbb{Z}} R = \{\text{sequences } (a_n)_n \text{ of finite support}\}, \quad (a_n)_n \mapsto \sum_{n \in \mathbb{Z}} a_n t^n.$$

The right-hand side may be understood as the compactly supported global sections (or homology) of the structure sheaf of the scheme $\mathbb{Z} \times \text{Spec}(R)$, a countable disjoint union of copies of $\text{Spec}(R)$.

There is a clean categorification. Since representations of $\mathbb{G}_{m,R}$ decompose according to weights, there is an equivalence

$$D_{\text{qc}}(*/\mathbb{G}_{m,R}) \simeq \prod_{n \in \mathbb{Z}} D_{\text{qc}}(\text{Spec}(R)) = D_{\text{qc}}(\mathbb{Z} \times \text{Spec}(R)).$$

Concretely, an R -module M with $\mathbb{G}_{m,R}$ -action decomposes as a $\mathbb{Z}$ -graded module $M = \bigoplus_n M_n$ into weight spaces. Dually, there is an equivalence

$$D_{\text{qc}}(\mathbb{G}_{m,R}) \simeq D_{\text{qc}}(*/\mathbb{Z} \times \text{Spec}(R)),$$

since the right-hand side is a complex of R -modules equipped with an automorphism, i.e. a complex of $R[t^{\pm 1}]$ -modules.

The equivalences are realized by categorified Fourier transforms. Abbreviate $\mathbb{G}_m = \mathbb{G}_{m,R}$ and $\mathbb{Z} = \mathbb{Z} \times \text{Spec}(R)$, and consider the correspondence

$$\begin{array}{ccc} & */\mathbb{G}_m \times \mathbb{Z} & \xrightarrow{m} */\mathbb{G}_m \\ & \swarrow p & \searrow q \\ */\mathbb{G}_m & & \mathbb{Z}, \end{array}$$

where m is induced by the pairing $\mathbb{G}_m \times \mathbb{Z} \rightarrow \mathbb{G}_m$, $(t, n) \mapsto t^n$, and p, q are the projections. Let $\mathcal{L} \in D_{\text{qc}}(*/\mathbb{G}_m)$ be the tautological line bundle, corresponding to the tautological representation of $\mathbb{G}_m$. Using $\mathcal{L}^{-1}$ as a Fourier kernel gives

$$D_{\text{qc}}(*/\mathbb{G}_m) \xrightarrow{\sim} D_{\text{qc}}(\mathbb{Z}), \quad M \mapsto q_!(p^* M \otimes m^* \mathcal{L}^{-1}),$$

with inverse $N \mapsto p_!(q^* N \otimes m^* \mathcal{L})$. Here $q_!$ plays the role of integration over $*/\mathbb{G}_m$, while $p^* M \otimes m^* \mathcal{L}^{-1}$ is the analogue of $f(t)t^{-n}$. Note that q is proper in the quasicohherent six-functor formalism, so $q_! \simeq q_*$, while p is étale, so $p_! \simeq p_\#$.

Symmetrically, the correspondence

$$\begin{array}{ccc} & \mathbb{G}_m \times */\mathbb{Z} & \xrightarrow{m} */\mathbb{G}_m \\ & \swarrow p & \searrow q \\ \mathbb{G}_m & & */\mathbb{Z} \end{array}$$

gives the equivalence $D_{\text{qc}}(\mathbb{G}_m) \simeq D_{\text{qc}}(*/\mathbb{Z})$. Under it the structure sheaf of $\mathbb{G}_m$ corresponds to the regular representation of $\mathbb{Z}$, and passing to endomorphisms recovers the Fourier isomorphism on the level of vector spaces. Working with analytic stacks over the liquid complex numbers, one obtains analogous equivalences for the full derived categories of quasicoherent sheaves on the various $(C^\infty, \text{real-analytic}, \dots)$ versions of S^1 , categorifying the isomorphisms for C^∞ -functions or distributions above.

Example 12.3 (Cartier). Let R be an ordinary commutative ring. Cartier introduced a duality on finite flat commutative group schemes $G = \text{Spec}(O(G))$ over R ; here $O(G)$ is a finite projective R -module carrying a commutative algebra structure together with a compatible cocommutative coalgebra structure encoding the addition on G . Basic examples are the constant group $\mathbb{Z}/n$, the group of n -th roots of unity

$$\mu_n = \ker(\mathbb{G}_m \xrightarrow{t \mapsto t^n} \mathbb{G}_m) = \text{Spec}(R[t]/(t^n - 1)),$$

and the n -torsion $A[n] = \ker(A \xrightarrow{\cdot n} A)$ of an abelian variety A . If R has characteristic 0, then any such G is étale-locally isomorphic to a constant commutative group (for μ_n , one adjoins a primitive n -th root of unity); in positive or mixed characteristic, finite flat group schemes are far more interesting.

As $O(G)$ is simultaneously a commutative algebra and a cocommutative coalgebra, the dual R -module $O(G)^*$ is again both, with the roles of algebra and coalgebra exchanged. Thus

$$G^* = \text{Spec}(O(G)^*)$$

is another finite flat group scheme, the *Cartier dual* of G . This has a geometric meaning: $G^* = \mathcal{H}om(G, \mathbb{G}_m)$, the internal hom in, say, fpqc sheaves. It is induced by a natural pairing $m: G \times G^* \rightarrow \mathbb{G}_m$, corresponding to the map $R[t^{\pm 1}] \rightarrow O(G) \otimes_R O(G)^*$ sending t to the image of 1 under the coevaluation $R \rightarrow O(G) \otimes_R O(G)^*$ (which turns out to be an invertible element). For instance $(\mathbb{Z}/n)^* = \mu_n$, while $A[n]^* = A^*[n]$ where A^* is the dual abelian variety. Using the correspondence

$$\begin{array}{ccc} */G \times G^* & \xrightarrow{m} & */\mathbb{G}_m \\ p \swarrow & & \searrow q \\ */G & & G^*, \end{array}$$

one again obtains a Fourier equivalence $D_{\text{qc}}(* / G) \simeq D_{\text{qc}}(G^*)$, and by symmetry the version exchanging G and G^* .

Example 12.4 (Mukai). A parallel story applies to abelian varieties themselves. For an abelian variety A/R there is a dual abelian variety

$$A^* = \mathcal{E}xt^1(A, \mathbb{G}_m),$$

the internal $\mathcal{E}xt^1$ in, say, fpqc sheaves. Note that as $\mathbb{G}_m$ is affine while $O(A) = R$, one has $\mathcal{H}om(A, \mathbb{G}_m) = 0$, so one must pass to $\mathcal{E}xt^1$ to see anything interesting. There is a tautological pairing $m: A \times A^* \rightarrow */\mathbb{G}_m$ and hence a line bundle

$$\mathcal{P} = m^* \mathcal{L} \in D_{\text{qc}}(A \times A^*),$$

the *Poincaré line bundle*. Mukai observed that the correspondence

$$\begin{array}{ccc} & A \times A^* & \xrightarrow{m} * / \mathbb{G}_m \\ & \swarrow p \quad \searrow q & \\ A & & A^* \end{array}$$

induces a Fourier equivalence $D_{\text{qc}}(A) \simeq D_{\text{qc}}(A^*)$, $M \mapsto q_*(p^*M \otimes \mathcal{P})$; see Huybrechts [Huy06]. In contrast to the previous examples, here one does *not* pass to classifying spaces.

Warning 12.5 (Breen). The preceding examples make it tempting to form duals in general by taking the derived internal $\mathcal{R}Hom(-, \mathbb{G}_m)$. However, the higher Ext-groups are (probably) very sensitive to the choice of Grothendieck topology, hard to compute in practice, and the naively expected vanishing in degrees > 0 (resp. > 1) already fails in the examples above. This makes it difficult to extend Cartier duality naively to “chain complexes of finite flat group schemes” or of abelian varieties. The theory of Gestalten circumvents this difficulty.

Example 12.6 (Deligne). Similar Fourier equivalences exist for other sheaf theories. A famous example is the ℓ -adic Fourier transform. Work over a field k of positive characteristic and fix a nontrivial additive character $\psi: \mathbb{F}_p \rightarrow \overline{\mathbb{Q}}_\ell^\times$. Via the Artin–Schreier sequence

$$0 \rightarrow \mathbb{F}_p \rightarrow \mathbb{A}_k^1 \xrightarrow{x \mapsto x^p - x} \mathbb{A}_k^1 \rightarrow 0,$$

ψ yields a rank-one local system $\mathcal{L}_\psi$ on $\mathbb{A}_k^1$, the *Artin–Schreier sheaf*. For a finite-dimensional k -vector space V with dual V^* , regarded as affine schemes, the correspondence

$$\begin{array}{ccc} & V \times V^* & \xrightarrow{m} \mathbb{A}_k^1 \\ & \swarrow p \quad \searrow q & \\ V & & V^* \end{array}$$

(with m the evaluation pairing) induces an equivalence $D_{\text{ét}}(V, \overline{\mathbb{Q}}_\ell) \simeq D_{\text{ét}}(V^*, \overline{\mathbb{Q}}_\ell)$, $M \mapsto q_!(p^*M \otimes m^*\mathcal{L}_\psi)$.

Example 12.7 (D -modules). The previous example has an analogue for D -modules over a field k of characteristic 0. Recall that via Simpson’s de Rham stack one may define $D\text{mod}(X) = D_{\text{qc}}(X_{\text{dR}})$, importing the six functors from the quasicoherent formalism. The role of the Artin–Schreier sheaf is played by the *exponential D -module* $\mathcal{L}$ on $\mathbb{A}_k^1$, corresponding to $k[T] \cdot e$ with $\partial(e) = e$ (“ $e = \exp(t)$ ”). This yields a map of commutative group stacks

$$\mathbb{A}_{\text{dR}}^1 \rightarrow * / \mathbb{G}_m,$$

which may also be described as the composite $\mathbb{A}_{\text{dR}}^1 = \mathbb{A}^1 / \widehat{\mathbb{G}}_a \rightarrow * / \widehat{\mathbb{G}}_a \xrightarrow{\exp} * / \mathbb{G}_m$. For V as above, the correspondence

$$\begin{array}{ccccc} & V_{\text{dR}} \times V_{\text{dR}}^* & \xrightarrow{m} & \mathbb{A}_{\text{dR}}^1 & \longrightarrow * / \mathbb{G}_m \\ & \swarrow p \quad \searrow q & & & \\ V_{\text{dR}} & & & & V_{\text{dR}}^* \end{array}$$

induces a Fourier equivalence $D_{qc}(V_{dR}) \simeq D_{qc}(V_{dR}^*)$, $M \mapsto q_!(p^*M \otimes m^*\mathcal{L})$. For $\mathbb{A}_k^1$ this has an elementary description in terms of actual D -modules (modules over the Weyl algebra $k\langle t, \partial \rangle / ([\partial, t] = 1)$) as the automorphism sending $t \mapsto \partial$ and $\partial \mapsto -t$.

Remark 12.8. There are many more dualities of this shape:

- (1) Over a field k of characteristic 0, the affine line $\mathbb{A}^1$ is Cartier dual to the formal additive group $\widehat{\mathbb{G}}_a$;
- (2) Over a nonarchimedean field k of characteristic 0, the analytic affine line $\mathbb{A}^{1,an}$ is Cartier dual to the overconvergent neighborhood of 0 in $\mathbb{A}^1$;
- (3) Over $\mathbb{Q}_p$, the open unit disc $D_{\mathbb{Q}_p}$ is Cartier dual to the locally analytic p -adic numbers $\mathbb{Z}_p^{la}$, a version of Amice duality;
- (4) Working directly with symmetric monoidal $(\infty, 1)$ -categories, Lurie has a generalization of Cartier duality.

All of these, except the very first using L^2 -functions, turn out to be special cases of the general Cartier duality for Gestalten established below.

12.2 Cartier duality for Gestalten

Fix a Gestalt X as base. The objects of the duality are *spectrum objects* in Gestalten over X . Working in a fully derived setting, commutative (E_∞) -groups in Gest_X give an analogue of the connective part $D_{\geq 0}(\mathbb{S}) \subseteq D(\mathbb{S})$; it is better to work stably and consider

$$\text{Sp}(\text{Gest}_X) := \{(M_0, M_1, \dots) \mid M_k \in \text{CGrp}(\text{Gest}_X), M_k \xrightarrow{\sim} \Omega M_{k+1}\}.$$

For each $n \geq 0$ we have the full subcategory $\text{Sp}(\text{Gest}_X^{n\text{-}\acute{e}t}) \subseteq \text{Sp}(\text{Gest}_X)$ on those sequences with all M_k being n -étale over X . By ambidexterity (Proposition 6.25) an n -étale map is $(n+1)$ -proper and vice versa, and so

$$\bigcup_n \text{Sp}(\text{Gest}_X^{n\text{-}\acute{e}t}) = \bigcup_n \text{Sp}(\text{Gest}_X^{n\text{-}prop}).$$

Remark 12.9. Equivalently, $\text{Sp}(\text{Gest}_X^{n\text{-}\acute{e}t})$ is the category of $\aleph_1$ -compact objects in the category of sheaves of $\mathbb{S}$ -modules on the $(\infty, 1)$ -topos $\text{Ind}_{\aleph_1}(\text{Gest}_X^{n\text{-}\acute{e}t})$.

Construction 12.10. There is a canonical object $\text{GL}_1 \in \text{Sp}(\text{Gest}_X)$ representing the functor

$$\text{Gest}_X^{\text{op}} \rightarrow \text{Sp}, \quad Y \mapsto (O(Y)_0^\times, O(Y)_1^\times, O(Y)_2^\times, \dots),$$

where $O(Y)_n^\times \subseteq O(Y)_n$ is the subanima of $\otimes$ -invertible objects, a commutative group under tensor product. This is indeed a spectrum object: $\Omega(O(Y)_{n+1}^\times)$ is the subanima of $\text{Hom}_{O(Y)_{n+1}}(\mathbb{1}, \mathbb{1}) \simeq O(Y)_n$ spanned by the invertible endomorphisms of $\mathbb{1}$, and the monoidal structure on $O(Y)_n$ coming from composition agrees with the one inherited from $O(Y)_{n+1}$, so an endomorphism is invertible precisely when the corresponding object of $O(Y)_n$ is $\otimes$ -invertible. We will in fact see that $\text{GL}_1 \in \text{Sp}(\text{Gest}_X^{0\text{-}prop})$.

Theorem 12.11 (Scholze–Stefanich). *For every $M \in \bigcup_n \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t})$, the functor*

$$\mathrm{Sp}(\mathrm{Gest}_X)^{\mathrm{op}} \rightarrow \mathbf{An}, \quad N \mapsto \mathrm{Hom}(N \otimes M, \mathrm{GL}_1)$$

is representable by an object

$$M^* := \mathcal{H}om(M, \mathrm{GL}_1) \in \bigcup_n \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t}).$$

For every $n \geq 0$, the assignment $M \mapsto M^$ induces an equivalence*

$$\mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t}) \simeq \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}prop})^{\mathrm{op}};$$

since n -proper maps are $(n+1)$ -étale and vice versa, $(-)^$ is thus an anti-self-equivalence of $\bigcup_n \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t})$. Moreover, for every such M , all $m \geq n$, and all $0 \leq k \leq m$, there is an equivalence*

$$(O(M_k)/O(X))_m^! \simeq (O(M_{m-k}^*)/O(X))_m,$$

given by an explicit Fourier transform. (If $m > n$, the left-hand side is just $(O(M_k)/O(X))_m$ by ambidexterity.)

Remark 12.12. All the dualities of Section 12.1, except the L^2 -example, are, suitably interpreted, instances of this theorem. For example, the connective spectrum objects $\mathbb{Z}$ and $\mathbb{G}_m = \mathrm{Gest}_X(\mathbb{S}[t^{\pm 1}])$ over $X = \mathrm{Gest}(\mathbb{D}(\mathbb{S}))$ are Cartier dual.

Remark 12.13. For $m > n$ the object $(O(M_k)/O(X))_m$ is dualizable, and the equivalence of Theorem 12.11 is induced by the canonical invertible object in

$$(O(M_{m-k}^*)/O(X))_m \otimes (O(M_k)/O(X))_m \xrightarrow{\sim} (O(M_{m-k}^* \times M_k)/O(X))_m$$

coming from the pairing $M_{m-k}^* \otimes M_k \rightarrow (\mathrm{GL}_1)_m$.

The proof of the theorem is given in the next talk. We now discuss some examples and consequences.

Corollary 12.14. *For every $M \in \bigcup_n \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t})$, the biduality map $M \rightarrow (M^*)^*$ is an isomorphism.*

Proof. Both sides lie in $\bigcup_n \mathrm{Sp}(\mathrm{Gest}_X^{n\text{-}\acute{e}t})$. For any N in this union,

$$\mathrm{Hom}(N, (M^*)^*) \simeq \mathrm{Hom}(N \otimes M^*, \mathrm{GL}_1) \simeq \mathrm{Hom}(M^*, N^*),$$

using the symmetry of the tensor pairing. Under these identifications, the biduality map corresponds to the map $\mathrm{Hom}(N, M) \rightarrow \mathrm{Hom}(M^*, N^*)$ given by contravariant functoriality of $(-)^*$. As $(-)^*$ is an anti-self-equivalence of the union (Theorem 12.11), this map of animae is an equivalence. The Yoneda lemma therefore identifies $M \rightarrow (M^*)^*$ as an isomorphism. $\square$

Strikingly, Theorem 12.11 imposes essentially no finiteness hypothesis on M . The following example makes the resulting bidualities concrete.

Example 12.15. Work over $X = \text{Gest}(\mathbf{D}(\mathbb{C}))$ (in fact $\mathbb{Z}[\frac{1}{p}, \zeta_p]$ would suffice) and fix a prime p . Then $\mathbb{Z}$ is Cartier dual to $\mathbb{G}_m = \text{Gest}_X(\mathbb{C}[t^{\pm 1}])$. As $\mathcal{H}om(-, \text{GL}_1)$ is exact, $\mathbb{Z}/p$ is Cartier dual to

$$\mu_p = \ker(\mathbb{G}_m \xrightarrow{t \mapsto t^p} \mathbb{G}_m) \cong \mathbb{Z}/p$$

(the isomorphism depending on a choice of p -th root of unity). On the derived category of sheaves of $\mathbb{F}_p$ -vector spaces, Cartier duality becomes the naive duality $\mathcal{H}om_{\mathbb{F}_p}(-, \mathbb{F}_p)$. Using biduality (with all sums and products taken in the derived category) we find:

- (1) $\bigoplus_{\mathbb{N}} \mathbb{F}_p$ is 0-étale, with dual the 0-proper $\prod_{\mathbb{N}} \mathbb{F}_p$, which is therefore 1-étale;
- (2) $\bigoplus_{\mathbb{N}} \prod_{\mathbb{N}} \mathbb{F}_p$ is 1-étale, with dual the 1-proper (hence 2-étale) $\prod_{\mathbb{N}} \bigoplus_{\mathbb{N}} \mathbb{F}_p$;
- (3) $\bigoplus_{\mathbb{N}} \prod_{\mathbb{N}} \bigoplus_{\mathbb{N}} \mathbb{F}_p$ is 2-étale, with dual the 2-proper (hence 3-étale) $\prod_{\mathbb{N}} \bigoplus_{\mathbb{N}} \prod_{\mathbb{N}} \mathbb{F}_p$; and so on.

It is remarkable that any topos supports these bidualities. In naive $\mathbb{F}_p$ -vector spaces the very first biduality for $\prod_{\mathbb{N}} \mathbb{F}_p$ already fails; it can be repaired in condensed $\mathbb{F}_p$ -vector spaces, where it is consistent that the derived biduality holds for $\prod_{\mathbb{N}} \bigoplus_{\mathbb{N}} \mathbb{F}_p$. This is independent of ZFC (it fails if the continuum is “too small”), the third biduality certainly fails there, and everything fails for uncountable index sets (already for $\aleph_1$). Putting a Gestalt structure on these vector-space objects thus yields a surprisingly clean duality theory. The restriction to countable sums and products reflects our foundational choice of $\aleph_1$ -presentable categories; a larger cutoff cardinal would permit larger index sets.

12.3 Recovering the classical examples

We now indicate how the internal hom into GL_1 recovers the classical duals. Consider $\text{GL}_1 \in \text{Sp}(\text{Gest}_X)$. For each $n \geq 0$, we denote by $(\text{GL}_1)_n \in \text{CGrp}(\text{Gest}_X)$ the commutative group representing the functor

$$\text{Gest}_X^{\text{op}} \rightarrow \text{CGrp}(\text{An}), \quad X' \mapsto \mathcal{O}(X')_n^{\times}.$$

We also denote by $(\mathbb{A}^1)_n \in \text{CMon}(\text{Gest}_X)$ the commutative monoid representing

$$\text{Gest}_X^{\text{op}} \rightarrow \text{CMon}(\text{An}), \quad X' \mapsto \mathcal{O}(X')_n^{\simeq, \aleph_1},$$

the monoid of $\aleph_1$ -compact objects of $\mathcal{O}(X')_n$ under tensor product. Note that we have an inclusion $(\text{GL}_1)_n \subseteq (\mathbb{A}^1)_n$ of commutative monoids. We then have the following result:

Theorem 12.16. *Let $X = \text{Gest}(A)$ and let $G \in \text{CGrp}(\text{Gest}_X)$ be n -suave. Then the internal hom*

$$\mathcal{H}om_{\text{CGrp}}(G, (\text{GL}_1)_n)$$

is representable by the n -affine Gestalt $\text{Gest}_X((\mathcal{O}(G)/A)_n^!)$. More generally, for every $G \in \text{CMon}(\text{Gest}_X^{n\text{-suave}})$ one has

$$\mathcal{H}om_{\text{CMon}(\text{Gest}_X)}(G, (\mathbb{A}^1)_n) \simeq \text{Gest}_X((\mathcal{O}(G)/A)_n^!).$$

Again, we will prove this result next week. Before continuing, let us provide more details about where the commutative algebra structure on $(\mathcal{O}(G)/A)_n^!$ comes from.

Construction 12.17. Put $C := \text{Gest}_{/X}^{n\text{-suave}}$. By Proposition 6.20, this full subcategory of $\text{Gest}_{/X}$ is closed under finite products. We construct a symmetric monoidal functor

$$(C, \times_X) \longrightarrow A_{n+1}, \quad Y \longmapsto (\mathcal{O}(Y)/A)_n^!.$$

We first record the coherent mate construction that we use. Let K be an $(\infty, 2)$ -category with a chosen object $\mathbb{1}$, and write $\text{RAdj}_{\mathbb{1}/}(K)$ for the category of right-adjoint 1-morphisms $u: \mathbb{1} \rightarrow B$, with commutative squares that are the identity on $\mathbb{1}$ as morphisms. Haugseng's mate equivalence and its compatibility with composition [Hau21, Theorem 1.1, Corollary 4.9, and Remark 4.11] give a functor

$$\Theta_K: \text{RAdj}_{\mathbb{1}/}(K)^{\text{op}} \longrightarrow \text{End}_K(\mathbb{1}), \quad (u: \mathbb{1} \rightarrow B) \longmapsto u^L u.$$

On a commutative square from u to v , the functor Θ_K takes the left mate and then whisks it with the appropriate right adjoint. Thus this formula specifies a functor, including its higher coherences, rather than only its values on objects and morphisms.

We now make the symmetric monoidality of this construction formal. Encode a symmetric monoidal $(\infty, 2)$ -category K by the finite-product-preserving functor [BH21, Appendix C]

$$K^{\otimes}: \text{Span}(\text{Fin}) \longrightarrow \text{Cat}_{(\infty, 2)}, \quad I \longmapsto K^I.$$

The assignments

$$K \longmapsto \text{RAdj}_{\mathbb{1}/}(K)^{\text{op}} \quad \text{and} \quad K \longmapsto \text{End}_K(\mathbb{1})$$

are functorial under unit-preserving $(\infty, 2)$ -functors and preserve finite products. Every structure functor of $K^{\otimes}$ preserves the tuple of unit objects, and the functors Θ_K are natural in K by Haugseng's result. Applying these two assignments and Θ to the entire functor $K^{\otimes}$, not separately to its individual operations, therefore gives a natural transformation between finite-product-preserving functors $\text{Span}(\text{Fin}) \rightarrow \text{Cat}$. Equivalently, Θ_K is a symmetric monoidal functor. This supplies in one construction the comparisons for every operation, their symmetric-group actions, and all compatibilities among them.

Write $\text{CAlg}(K)^{\text{radj}}$ for the full subcategory of commutative algebras whose unit is a right adjoint. It is closed under tensor products. The unit-arrow construction is natural in K and preserves finite products, so applying it to the entire $\text{Span}(\text{Fin})$ -diagram gives a morphism of finite-product-preserving functors and hence a symmetric monoidal functor

$$\eta: \text{CAlg}(K)^{\text{radj}} \longrightarrow \text{RAdj}_{\mathbb{1}/}(K), \quad B \longmapsto (\mathbb{1} \rightarrow B).$$

Take K to be the underlying symmetric monoidal $(\infty, 2)$ -category of A_{n+2} . The lax symmetric monoidal right adjoint

$$F_X := (\mathcal{O}(-)/A)_{n+1}: C^{\text{op}} \longrightarrow \text{CAlg}(K)$$

to the $(n+1)$ -affine inclusion lands in $\text{CAlg}(K)^{\text{radj}}$: for $f: Y \rightarrow X$, the unit of $F_X(Y)$ is the relative pullback

$$\eta_Y = (f/A)_n^*: \mathbb{1} \longrightarrow (\mathcal{O}(Y)/A)_{n+1}.$$

It admits the left adjoint $(f/A)_{n,\#}$ because f is n -suave. The composite of F_X with the unit-arrow functor is lax symmetric monoidal. Passing to opposites and composing with the symmetric monoidal functor Θ_K therefore gives an oplax symmetric monoidal functor

$$D_X: C \longrightarrow \text{End}_{A_{n+2}}(\mathbb{1}) \simeq A_{n+1}.$$

Here the structural equivalence $\text{End}_{A_{n+2}}(\mathbb{1}) \simeq A_{n+1}$ of the Stefanich ring A is symmetric monoidal. Under it, the value at Y is

$$D_X(Y) = (f/A)_{n,\#}(f/A)_n^* \mathbb{1} = (O(Y)/A)_n^!.$$

For orientation, if $r: Y \rightarrow Z$ is a morphism over X , naturality of η gives $\eta_Y \simeq (r/A)_n^* \eta_Z$. Its coherent left mate is

$$(f/A)_{n,\#}(r/A)_n^* \longrightarrow (g/A)_{n,\#},$$

where $g: Z \rightarrow X$ is the structure map. Whiskering with η_Z gives $D_X(Y) \rightarrow D_X(Z)$. This describes the resulting map, but its functoriality, including all higher coherences, comes from the mate equivalence above.

It remains to show that the oplax structure maps are isomorphisms. We first identify the already constructed binary comparison, without using this identification to define its coherences. Write

$$\mu_{Y,Z}: F_X(Y) \otimes F_X(Z) \longrightarrow F_X(Y \times_X Z)$$

for the lax product map. The binary oplax comparison is the composite

$$\begin{aligned} \eta_{Y \times_X Z}^L \eta_{Y \times_X Z} &\simeq \eta_{Y \times_X Z}^L \mu_{Y,Z}(\eta_Y \otimes \eta_Z) \\ &\longrightarrow (\eta_Y^L \otimes \eta_Z^L)(\eta_Y \otimes \eta_Z) \\ &\simeq D_X(Y) \otimes_{A_{n+1}} D_X(Z), \end{aligned}$$

where the middle map is the left mate of the square $\mu_{Y,Z}(\eta_Y \otimes \eta_Z) \simeq \eta_{Y \times_X Z}$. The preceding $\text{Span}(\text{Fin})$ -construction is what makes these maps coherent as Y and Z and the operation vary.

Let $p: Y \times_X Z \rightarrow Y$ be the base change of g , and put $h = fp$. Compatibility of mates with composition identifies this comparison with the pasted base-change and projection maps

$$\begin{aligned} D_X(Y \times_X Z) &= h_{n,\#}(\mathbb{1}) \\ &\simeq f_{n,\#} p_{n,\#}(\mathbb{1}) \\ &\simeq f_{n,\#} f_n^* D_X(Z) \\ &\simeq D_X(Y) \otimes_{A_{n+1}} D_X(Z). \end{aligned}$$

Here the second isomorphism is suave base change, and the third is the suave projection formula. This is precisely the Künneth isomorphism of Lemma 6.22, not merely some isomorphism between the same two objects. The unit comparison is an isomorphism since $D_X(X) = \mathbb{1}$. Hence D_X is symmetric monoidal.

Passing to commutative algebra objects now gives

$$\text{CMon}(C) \simeq \text{CAlg}(C, \times_X) \longrightarrow \text{CAlg}(A_{n+1}),$$

which endows $(O(G)/A)_n^!$ with the commutative algebra structure induced by the commutative monoid structure on G .

Proposition 12.18. *Let $G \in \text{CGrp}(\text{Gest}/X)$ be n -suave with $BG = */G$ being $(n+1)$ -affine. Then $G^* \in \text{Sp}(\text{Gest}/X)$ is $(-n)$ -connective, and is determined by*

$$G_n^* = \mathcal{H}om_{\text{CGrp}(\text{Gest}/X)}(G, (\text{GL}_1)_n) = \text{Gest}_X((O(G)/A)_n^!).$$

Example 12.19. Let $X = \text{Gest}(\text{D}(R))$ and let $G = \mathbb{Z} = \bigsqcup_{\mathbb{Z}} X$ with the commutative group structure inherited from $\mathbb{Z}$ (corresponding to the ring $\prod_{\mathbb{Z}} R$). This is 0-étale, in particular 0-suave, with

$$(O(G)/R)_0^! = \bigoplus_{\mathbb{Z}} R \in \text{D}(R).$$

Indeed, n -suave maps of Stefanich rings are stable under countable limits (Proposition 7.3), and the assignment $B \mapsto (B/A)_m^!$ carries limits to colimits, so the countable product $\prod_{\mathbb{Z}} R$ is sent to the countable coproduct $\bigoplus_{\mathbb{Z}} R$. The algebra structure is induced by the effect of $0: * \rightarrow \mathbb{Z}$ and $+: \mathbb{Z}^2 \rightarrow \mathbb{Z}$ on indexing sets,

$$R = \bigoplus_{*} R \rightarrow \bigoplus_{\mathbb{Z}} R, \quad \bigoplus_{\mathbb{Z}} R \otimes \bigoplus_{\mathbb{Z}} R \xrightarrow{\sim} \bigoplus_{\mathbb{Z}^2} R \rightarrow \bigoplus_{\mathbb{Z}} R,$$

so that $O(\mathbb{G}_{m,R})_0 = R[t^{\pm 1}]$, as expected.

Example 12.20. Let G be a finite flat group scheme over R . By Raynaud’s theorem, the structural morphism $G \rightarrow \text{Spec}(R)$ is a finite local complete intersection, hence G is 0-suave. Taking $n = 0$ and using that $O(G)$ is finite projective,

$$(O(G)/R)_0^! = O(G)^* = O(G^*),$$

recovering the classical Cartier dual.

12.4 Brown–Comenetz duality

Example 12.21. Let $X = \text{Gest}(\text{D}(\mathbb{C}))$ (the cyclotomic extension of $\mathbb{Q}$ would suffice for this discussion). The global sections of GL_1 form the spectrum

$$\text{GL}_1(X) = (\mathbb{C}^{\times}, \text{D}(\mathbb{C})^{\times}, 1\text{Pr}_{\mathbb{C}}^{\times}, \dots).$$

Its homotopy groups are hard to compute in general, but:

- $\pi_0 = \mathbb{C}^{\times}$;
- $\pi_{-1} = \mathbb{Z}$, since any $\otimes$ -invertible object of $\text{D}(\mathbb{C})$ is a shift $\mathbb{C}[n]$;
- $\pi_{-2} = 0$, by a theorem of Stefanich [Ste21];
- $\pi_{-3} = 0$ (working with “small” higher linear categories);
- π_{-4} is the “Witt group”, a highly nontrivial but known abelian group, countably but not finitely generated.

In the language of mathematical physics, these higher invertible linear categories give rise to *invertible topological quantum field theories*, and there is an expectation (Kitaev and others) that these are classified by the Brown–Comenetz dual $I_{\mathbb{C}^\times}$ of the sphere spectrum, with

$$\pi_{-n}I_{\mathbb{C}^\times} = \mathrm{Hom}(\pi_n \mathbb{S}, \mathbb{C}^\times).$$

The rationale is that an n -dimensional invertible TQFT should assign to a framed closed n -manifold an invertible number in $\mathbb{C}^\times$, cobordism-invariantly, and stably framed cobordism classes of closed n -manifolds are given by $\pi_n \mathbb{S}$.

Note that this expectation differs drastically from the actual homotopy groups of $\mathrm{GL}_1(X)$ given above. However, looking not at global sections but at GL_1 as a sheaf on Gestalten recovers the expected answer:

Corollary 12.22. *The spectrum object GL_1 in $\mathrm{Gest}/_{\mathrm{D}(\mathbb{C})}$ is an internal Brown–Comenetz dual $I_{\mathbb{G}_m}$, in the sense that*

$$\mathcal{H}om(\mathbb{Z}, \mathrm{GL}_1) = \mathbb{G}_m.$$

In particular, for all $n > 0$,

$$\pi_{-n} \mathrm{GL}_1 \cong \mathrm{Hom}(\pi_n \mathbb{S}, \pi_0 \mathbb{G}_m) = \mathrm{Hom}(\pi_n \mathbb{S}, \mathbb{C}^\times)$$

as 0-truncated abelian groups in $\mathrm{Gest}/_{\mathrm{D}(\mathbb{C})}$, via the pairing $\pi_n \mathbb{S} \otimes \pi_{-n} \mathrm{GL}_1 \rightarrow \pi_0 \mathbb{G}_m$.

Proof. The first assertion is the Cartier duality between $\mathbb{Z}$ and $\mathbb{G}_m$. For the negative homotopy groups, note that $\mathbb{G}_m$ is the connective cover of GL_1 , and its 1-connective cover $\tau_{\geq 1} \mathbb{G}_m$ is rational: for any $A \in \mathrm{CAlg}(\mathrm{D}(\mathbb{C}))$ one has $\tau_{>0} A^\times = \tau_{>0} A$ via the logarithm, and $\tau_{>0} A$ is rational. Hence

$$\mathcal{H}om(\mathbb{Z}, \tau_{\leq 0} \mathrm{GL}_1) \simeq \tau_{\leq 0} \mathbb{G}_m,$$

with the right-hand side concentrated in degree 0. The identification of the negative homotopy groups now follows from general Brown–Comenetz nonsense: in any countable $(\infty, 1)$ -topos, a coconnective spectrum object G with $\mathcal{H}om(\mathbb{Z}, G) = A$ concentrated in degree 0 and divisible satisfies $\pi_{-n} G = \mathrm{Hom}(\pi_n \mathbb{S}, A)$ via the pairing $\pi_n \mathbb{S} \otimes \pi_{-n} G \rightarrow \pi_0 G = A$. Since $\pi_n \mathbb{S}$ is a finite abelian group, so is $\mathrm{Hom}(\pi_n \mathbb{S}, \mathbb{C}^\times)$; this generalizes the earlier computation $\pi_0 \mathrm{Pic} = \mu_2$ of Proposition 9.24. $\square$

12.5 The motivic Fourier transform

Most of the examples of Section 12.1 fit transparently into the setting of Gestalten. The subtlest is Deligne’s ℓ -adic Fourier transform (Example 12.6); to accommodate it one uses the transmutation of Chapter 10, turning a six-functor formalism into a Gestalt. One could work with the ℓ -adic formalism directly, but a more refined discussion is available in the motivic six-functor formalism SH. This example also highlights genuinely new geometry made possible by Gestalten.

Example 12.23. Let $X = [\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}} \in \mathrm{Gest}$. Over it we have the transmutation $[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}}$ of the affine line, which carries a commutative group structure (via addition). In fact, it carries a commutative ring structure, and in particular a module structure over itself. Hence the Cartier dual $[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}}^*$ is again a $[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}}$ -module. If it were isomorphic to $[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}}$ we would obtain a Fourier self-duality. This is not quite true, but it is true locally.

Proposition 12.24. *There is a cover $\tilde{X} \rightarrow X$ such that*

$$[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}}^* \times_X \tilde{X} \simeq ([\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}} \times_X \tilde{X})[-1]$$

as $[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}} \times_X \tilde{X}$ -modules.

The universal such $\tilde{X}$ is a $[\mathbb{G}_{m,\mathbb{Z}}]_{\mathrm{SH}}$ -torsor $\tilde{X} \rightarrow X$. Roughly, $\tilde{X}$ is obtained by freely adjoining an exponential local system on $\mathbb{A}_{\mathbb{Z}}^1$ to the motivic six-functor formalism; the resulting six-functor formalism $S \mapsto \mathcal{O}([S]_{\mathrm{SH}} \times_X \tilde{X})_1$ is a theory of *exponential motives*. Over

$$[\mathrm{Spec}(\mathbb{F}_p)]_{\mathrm{SH}} \times_{\mathrm{Gest}(\mathbb{S})} \mathrm{Gest}(\mathbb{S}[\frac{1}{p}, \zeta_p]) \rightarrow X,$$

one obtains an additive character ψ , and the Artin–Schreier cover yields an exponential local system on $\mathbb{A}_{\mathbb{F}_p}^1$, splitting this map to $\tilde{X}$. Writing $[\mathbb{A}_{\mathbb{F}_p}^1]_{\mathrm{SH}}[p] := [\mathbb{A}_{\mathbb{F}_p}^1]_{\mathrm{SH}} \times_{\mathrm{Gest}(\mathbb{S})} \mathrm{Gest}(\mathbb{S}[\frac{1}{p}, \zeta_p])$, one gets an isomorphism

$$[\mathbb{A}_{\mathbb{F}_p}^1]_{\mathrm{SH}}^*[p] \simeq [\mathbb{A}_{\mathbb{F}_p}^1]_{\mathrm{SH}}[p][-1],$$

a motivic version of Deligne’s Fourier equivalence; base-changing to the Gestalt of the ℓ -adic six-functor formalism recovers Deligne’s equivalence itself.

Finally, there is a Betti realization

$$\mathrm{Gest}(\mathrm{D}(\mathbb{S})) \rightarrow [\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}$$

corresponding to the ring Gestalt $\underline{\mathbb{R}}$ over $\mathrm{Gest}(\mathrm{D}(\mathbb{S}))$. The pullback of the motivic affine line is

$$[\mathbb{A}_{\mathbb{Z}}^1]_{\mathrm{SH}} \times_{[\mathrm{Spec}(\mathbb{Z})]_{\mathrm{SH}}} \mathrm{Gest}(\mathrm{D}(\mathbb{S})) \simeq \underline{\mathbb{R}} = \mathrm{Gest}(\mathrm{Shv}(\mathbb{R}, \mathrm{D}(\mathbb{S}))).$$

Under this realization, the torsor $\tilde{X} \rightarrow X$ yields an $\underline{\mathbb{R}}^\times$ -torsor over $\mathrm{Gest}(\mathrm{D}(\mathbb{S}))$. After removing a trivial $\{\pm 1\}$ -torsor, this is the $\underline{\mathbb{R}}_{>0}$ -torsor $\mathrm{Gest}(W) \rightarrow \mathrm{Gest}(\mathrm{D}(\mathbb{S}))$ of Witt vectors discussed by Scholze [Sch25b].

13 The proof of Cartier duality

Talk given by Sil.

In this chapter we prove the Cartier duality theorem for Gestalten (Theorem 12.11), whose statement was discussed in the previous chapter.

13.1 Recollections

Throughout, fix a Gestalt $X = \text{Gest}(A)$ as base.

Definition 13.1. Recall that $(\text{GL}_1)_n$ is the Gestalt over X defined by

$$(\text{GL}_1)_n := \text{Gest}_X(\mathbb{S} \otimes \mathcal{O}(X)_n),$$

where $\mathbb{S}$ is the free commutative group on a single generator. In light of the adjunction $\text{CMon}(\text{An}) \rightleftarrows \text{CAlg}(\mathcal{O}(X)_{n+1})$, the Gestalt $(\text{GL}_1)_n$ represents the functor

$$\text{Gest}_{/X}^{\text{op}} \rightarrow \text{An}, \quad Y \mapsto \mathcal{O}(Y)_n^\times.$$

Since this canonically lifts to a functor valued in $\text{CGrp}(\text{An})$, the Gestalt $(\text{GL}_1)_n$ is canonically a commutative group in $\text{Gest}_{/X}$. We similarly define

$$(\mathbb{A}^1)_n := \text{Gest}_X(\text{Fin}^\approx \otimes \mathcal{O}(X)_n),$$

which represents the functor $Y \mapsto \mathcal{O}(Y)_n^{\approx, \mathbb{N}_1}$ and is canonically a commutative monoid in $\text{Gest}_{/X}$.

Lemma 13.2. *There are isomorphisms of commutative groups $\Omega(\text{GL}_1)_n \simeq (\text{GL}_1)_{n-1}$, so that the sequence $((\text{GL}_1)_0, (\text{GL}_1)_1, (\text{GL}_1)_2, \dots)$ assembles into a spectrum object $\text{GL}_1 \in \text{Sp}(\text{Gest}_{/X})$.*

Proof. Naturally in Y we have

$$\Omega \mathcal{O}(Y)_n^\times = \text{Aut}_{\mathcal{O}(Y)_n}(\mathbb{1}) = \text{End}_{\mathcal{O}(Y)_n}(\mathbb{1})^\times = (\mathcal{O}(Y)_{n-1})^\times,$$

which gives the claim. □

The goal of this talk is to prove the Cartier duality theorem of Scholze and Stefanich (Theorem 12.11). Recall its content: for $M \in \text{Sp}(\text{Gest}_{/X}^{n\text{-ét}})$, the functor

$$\text{Sp}(\text{Gest}_{/X})^{\text{op}} \rightarrow \text{An}, \quad N \mapsto \text{Hom}(N \otimes M, \text{GL}_1)$$

is representable by $M^* = \mathcal{H}om(M, \mathrm{GL}_1) \in \mathrm{Sp}(\mathrm{Gest}_{/X}^{n\text{-prop}})$; the assignment $M \mapsto M^*$ induces an equivalence

$$\mathrm{Sp}(\mathrm{Gest}_{/X}^{n\text{-ét}})^{\mathrm{op}} \simeq \mathrm{Sp}(\mathrm{Gest}_{/X}^{n\text{-prop}});$$

and for all $m, k \geq 0$ with $m + k \geq n$ there is an equivalence

$$(O(M_k^*)/O(X))_{m+k} \simeq (O(M_m)/O(X))_{m+k}^!,$$

induced by a form of the Fourier–Mukai transform.

13.2 The key computation

The heart of the proof is the following computation, which refines Theorem 12.16 to a statement over the slice.

Theorem 13.3. *Let $X = \mathrm{Gest}(A)$ and let $G \in \mathrm{CGrp}(\mathrm{Gest}_{/X}^{n\text{-suave}})$. Then the functor*

$$\mathrm{Gest}_{/X}^{\mathrm{op}} \rightarrow \mathrm{An}, \quad Y \mapsto \mathrm{Hom}_{\mathrm{CGrp}(\mathrm{Gest}_{/Y})}(G \times_X Y, (\mathrm{GL}_1)_n)$$

is representable by the n -affine Gestalt

$$\mathrm{Gest}_X((O(G)/A)_n^!)$$

over X . The analogous statement holds with $(\mathrm{GL}_1)_n$ replaced by $(\mathbb{A}^1)_n$ and $G \in \mathrm{CMon}(\mathrm{Gest}_{/X}^{n\text{-suave}})$.

Remark 13.4. By Construction 12.17, the functor

$$(O(-)/A)_n^! : \mathrm{Gest}_{/X}^{n\text{-suave}} \rightarrow A_{n+1}$$

is canonically symmetric monoidal for the cartesian structure on the source. In particular, for $G \in \mathrm{CMon}(\mathrm{Gest}_{/X}^{n\text{-suave}})$ the object $(O(G)/A)_n^!$ acquires a commutative algebra structure in A_{n+1} .

Definition 13.5. We write

$$\mathcal{H}om(G, (\mathrm{GL}_1)_n) := \mathrm{Gest}_X((O(G)/A)_n^!)$$

for the representing object of Theorem 13.3. It is canonically a commutative group, and as such represents the functor $G' \mapsto \mathrm{Hom}(G \otimes G', (\mathrm{GL}_1)_n)$; in other words, $\mathcal{H}om(G, (\mathrm{GL}_1)_n)$ is the internal hom of commutative groups in $\mathrm{Gest}_{/X}$.

Corollary 13.6. *If $G \in \mathrm{CGrp}(\mathrm{Gest}_{/X})$ is eventually suave (i.e. n -suave for some n), then $\mathcal{H}om(G, (\mathrm{GL}_1)_n)$ exists for all n . Likewise, if $M \in \mathrm{Sp}(\mathrm{Gest}_{/X})$ is such that each M_n is eventually suave, then*

$$M^* = \mathcal{H}om(M, \mathrm{GL}_1)$$

exists.

Proof. Assume first that G is n -suave. Then $\mathcal{H}om(G, (\mathrm{GL}_1)_m)$ exists for all $m \geq n$ by Theorem 13.3, and by taking loops we obtain the claim for $m < n$ as well. For the spectral statement, use that

$$\tau_{\geq 0} \mathcal{H}om(M, \mathrm{GL}_1) = \lim_n \mathcal{H}om(M_n, (\mathrm{GL}_1)_n). \quad \square$$

Proof of Theorem 13.3. The $(\mathbb{A}^1)_n$ -statement implies the $(\mathrm{GL}_1)_n$ -statement by passing to units, so we prove the former.

Let first H be an n -suave Gestalt over X . For a map $f: Y \rightarrow X$ we compute

$$\mathrm{Hom}_{\mathrm{Gest}_Y}(H \times_X Y, (\mathbb{A}^1)_n) \simeq \mathcal{O}(H \times_X Y)_n^{\simeq, \mathbf{S}_1} \simeq \mathrm{Hom}_{\mathcal{O}(Y)_{n+1}}(\mathbb{1}, (\mathcal{O}(H \times_X Y)/\mathcal{O}(Y))_n) \simeq \mathrm{Hom}((\mathcal{O}(H \times_X Y)/\mathcal{O}(Y))_n^!, \mathbb{1}).$$

Recall that the assignment $(\mathcal{O}(-)/\mathcal{O}(-))_n^!$ commutes with base change along n -suave maps, and so

$$\mathrm{Hom}((\mathcal{O}(H \times_X Y)/\mathcal{O}(Y))_n^!, \mathbb{1}) = \mathrm{Hom}(f_n^*(\mathcal{O}(H)/\mathcal{O}(X))_n^!, \mathbb{1}) = \mathrm{Hom}((\mathcal{O}(H)/\mathcal{O}(X))_n^!, (\mathcal{O}(Y)/\mathcal{O}(X))_n).$$

If now $H = G \in \mathrm{CMon}(\mathrm{Gest}_X^{n\text{-suave}})$ is a commutative monoid, one deduces

$$\mathrm{Hom}_{\mathrm{CMon}(\mathrm{Gest}_Y)}(G \times_X Y, (\mathbb{A}^1)_n) \simeq \mathrm{Hom}_{\mathrm{CAlg}(\mathcal{O}(X)_{n+1})}((\mathcal{O}(G)/\mathcal{O}(X))_n^!, (\mathcal{O}(Y)/\mathcal{O}(X))_n),$$

which is exactly representability by $\mathrm{Gest}_X((\mathcal{O}(G)/\mathcal{O}(X))_n^!)$. $\square$

We have thus constructed the spectral dual out of the non-spectral duals. It is not yet clear how to describe this spectral dual directly. We now show that the limit computing the spectral dual can be understood stage by stage, with the various stages iteratively fixing the n -affinizations of the spectral dual.

Theorem 13.7. *Let $X = \mathrm{Gest}(A)$ and $M \in \mathrm{Sp}(\mathrm{Gest}_X)$, and assume that for all $k \geq n$ the map $* \rightarrow M_{k+1}$ is k -suave. (Note that this implies that each M_k is k -suave.) Then M_0^* is n -prim, and the natural map*

$$M_0^* \rightarrow \mathcal{H}om(M_n, (\mathrm{GL}_1)_n) = \mathrm{Gest}_X((\mathcal{O}(M_n)/\mathcal{O}(A))_n^!)$$

exhibits its target as the n -affinization of its source.

This follows from the following lemma.

Lemma 13.8. *Let $G \in \mathrm{CGrp}(\mathrm{Gest}_X)$ and assume that G is $(n+1)$ -suave and that the unit $i: * \rightarrow G$ is n -suave, so that $G' := \Omega G$ is n -suave. Then $\mathcal{H}om(G, (\mathrm{GL}_1)_{n+1})$ is n -prim, and the map*

$$\mathcal{H}om(G, (\mathrm{GL}_1)_{n+1}) \xrightarrow{\Omega} \mathcal{H}om(G', (\mathrm{GL}_1)_n)$$

exhibits its target as an n -affinization of its source.

Proof. The target is the n -affine Gestalt $\mathrm{Gest}_X((\mathcal{O}(G')/\mathcal{O}(A))_n^!)$ and the source is the $(n+1)$ -affine Gestalt $\mathrm{Gest}_X((\mathcal{O}(G)/\mathcal{O}(A))_{n+1}^!)$. We must show the following two statements.

(1) The map $\mathbb{1} \rightarrow (\mathcal{O}(G)/\mathcal{O}(A))_{n+1}^!$ in A_{n+2} admits a right adjoint.

(2) The map $(\mathcal{O}(G')/\mathcal{O}(A))_n^! \rightarrow \mathrm{End}_{(\mathcal{O}(G)/\mathcal{O}(A))_{n+1}^!}(\mathbb{1})$ is an isomorphism.

For (1), the map $\mathbb{1} \rightarrow (\mathcal{O}(G)/\mathcal{O}(A))_{n+1}^!$ is obtained by applying $f_{n+1, \#}$ to the map $i_{n, !}: (\mathcal{O}(*)/\mathcal{O}(G))_{n+1}^! \rightarrow \mathbb{1}$ in $\mathcal{O}(G)_{n+2}$, where $f: G \rightarrow *$. Since i is n -suave, the map $i_{n, !}$ admits a right adjoint $i_n^!$; as $f_{n+1, \#}(-)$ preserves adjunctions, this produces the required right adjoint, hence primness.

For (2), we may use (1) to identify the endomorphism object $\text{End}_{(O(G)/A)_{n+1}^!}(\mathbb{1}) \in A_{n+1} = \text{End}_{A_{n+2}}(\mathbb{1})$ with the composite

$$\mathbb{1} \xrightarrow{f_{n+1,\#}(i_{n,!})} (O(G)/A)_{n+1}^! \xrightarrow{f_{n+1,\#}(i_n^*)} \mathbb{1}.$$

Since $\text{Hom}(-, \mathbb{1})$ acts as the identity on the unit, this composite may alternatively be computed as

$$\mathbb{1} \xrightarrow{f_{n+1,*}(i_{n,\#})} (O(G)/A)_{n+1} \xrightarrow{f_{n+1,*}(i_n^*)} \mathbb{1}.$$

Now $f_{n+1,*}(i_n^* i_{n,\#}) = f_{n+1,*}(g_{n,\#} g_n^*)$ by the pullback square

$$\begin{array}{ccc} G' & \xrightarrow{g} & * \\ g \downarrow & \lrcorner & \downarrow i \\ * & \xrightarrow{i} & G. \end{array}$$

But the map $g_{n,\#} g_n^*: \mathbb{1} \rightarrow \mathbb{1}$ in A_{n+2} is precisely $g_{n,\#}(\mathbb{1}) = (O(G')/A)_n^! \in A_{n+1} = \text{End}_{A_{n+2}}(\mathbb{1})$, which finishes the proof. $\square$

13.3 Continuing the proof

Now let $M \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}\acute{e}\text{t}})$. Applying Theorem 13.7 to $\Omega^k M$ for all k shows that each map $M_k^* \rightarrow X$ is n -prim. It remains to see that it is n -proper. The diagonal of M_k^* is a base change of the zero section $* \rightarrow M_k^*$: indeed, for any commutative group G the composite

$$G \xrightarrow{\Delta} G \times G \xrightarrow{\text{shear}} G \times G$$

equals (id, e) , which is a base change of $e: * \rightarrow G$. Consequently the higher diagonals are base changed from the zero sections of the iterated loop groups $\Omega^j G$. (Note that the diagonal of the zero section of G is the zero section of ΩG .)

Proposition 13.9. *In the situation of Theorem 13.7, the map $* \rightarrow M_1^*$ is $(n+1)$ -prim.*

Proof. Assume for simplicity that $n = 0$. For $m \geq 1$, consider the map $i_{m-1}^*: (O(M_1^*)/A)_m \rightarrow \mathbb{1}$ in A_{m+1} ; we show that it admits a right adjoint. As before, it is identified with the map $f_{m-1,!}: (O(M_{m-1})/A)_m^! \rightarrow \mathbb{1}$, where $f: M_{m-1} \rightarrow X$. Since f is $(m-1)$ -suave, the functor $f_{m-1,!}$ admits a right adjoint $f_{m-1}^!$, and hence i_{m-1}^* admits a right adjoint $i_{m-1,*}$ in A_{m+1} for each $m \geq 1$. However, we need this right adjoint in $O(M_1^*)_{m+1}$.

Consider the map $\text{id} \rightarrow i_{m,*} i_m^*$ in $\text{End}_{A_{m+2}}((O(M_1^*)/A)_{m+1})$. Applying $\text{Hom}(\mathbb{1}, -)$ sends it to the unit map $\mathbb{1} \rightarrow i_{m,*} \mathbb{1}$ in $O(M_1^*)_{m+1}$, so it suffices to show that this unit map admits a right adjoint.

Consider the map

$$O(M_m \times M_m)_{m+1} \rightarrow \text{End}_{A_{m+2}}((O(M_m)/A)_{m+1}^!)$$

sending K to the composite

$$(O(M_m)/A)_{m+1}^! \xrightarrow{p_m^!} (O(M_m \times M_m)/A)_{m+1}^! \xrightarrow{-\otimes K} (O(M_m \times M_m)/A)_{m+1}^! \xrightarrow{q_m,!} (O(M_m)/A)_{m+1}^!,$$

where $p, q: M_m \times M_m \rightarrow M_m$ are the two projections. The tensoring $- \otimes K$ exists because $(\mathcal{O}(M_m \times M_m)/A)_{m+1}^!$ is the image of the unit under the left adjoint $\mathcal{O}(M_m \times M_m)_{m+2} \rightarrow A_{m+2}$, so that the endomorphisms $\mathcal{O}(M_m \times M_m)_{m+1} = \text{End}_{\mathcal{O}(M_m \times M_m)_{m+2}}(\mathbb{1})$ act on this object.

One computes that the map $\Delta_{f, m-1, !}$ in $\mathcal{O}(M_m \times M_m)_{m+1}$ is sent to the unit $\text{id} \rightarrow f_m^! f_{m, !}$. So to prove the claim, it remains to show that $\Delta_{f, m-1, !}$ admits a right adjoint. But this follows from the fact that $* \rightarrow M_{m-1}$ is $(m-1)$ -suave. $\square$

Corollary 13.10. *Given $M \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}\acute{\text{e}}\text{t}})$, we have $M^* \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}prop})$.*

In fact, the argument dualizes, which implies the following.

Proposition 13.11. *Given $M \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}prop})$, we have $M^* \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}\acute{\text{e}}\text{t}})$.*

Proof of Theorem 12.11. Let $M \in \text{Sp}(\text{Gest}_{/X}^{n\text{-}\acute{\text{e}}\text{t}})$. Then M^* is representable and lies in $\text{Sp}(\text{Gest}_{/X}^{n\text{-}prop})$ by Corollary 13.10, and moreover

$$(\mathcal{O}(M_k)/\mathcal{O}(X))_{m+k}^! \cong (\mathcal{O}(M_m^*)/\mathcal{O}(X))_{m+k}.$$

To conclude, it suffices to show that the canonical map $M \rightarrow (M^*)^*$ is an isomorphism. But for $m+k > n+1$ we have

$$(\mathcal{O}(M_k^{**})/A)_{m+k} \cong (\mathcal{O}(M_m^*)/A)_{m+k} \cong (\mathcal{O}(M_k)/A)_{m+k},$$

which gives the claim. $\square$

A $\aleph_1$ -compact objects

The formalism developed in these notes heavily relies on manipulations with κ -compact objects for uncountable cardinals κ (which we mostly fix to be $\aleph_1$). In this appendix, we record various basic properties of κ -compact objects.

Bastiaan: I thank Peter Scholze for explaining the key steps in these arguments that I wasn't able to figure out.

A.1 Compact algebras and modules

We state and prove various basic properties of κ -compact algebras in a κ -presentable category.

Lemma A.1. *Let κ be an uncountable regular cardinal, let $C \in \mathrm{CAlg}(\mathrm{Pr}_\kappa^\mathrm{L})$, let $R \in \mathrm{CAlg}(C)$, and let $B \in \mathrm{CAlg}_R(C)$. Then B is κ -compact as an R -algebra if and only if its underlying R -module is κ -compact.*

Proof. The point is that the E_∞ -operad is countable. We write

$$\mathrm{Sym}_R : \mathrm{Mod}_R(C) \rightleftarrows \mathrm{CAlg}_R(C) : U$$

for the free-forgetful adjunction. We will use that κ -small colimits of κ -compact objects are again κ -compact: mapping out of such a colimit gives a κ -small limit of anima, and κ -filtered colimits of anima commute with κ -small limits.

First let $M \in \mathrm{Mod}_R(C)$ be κ -compact. The underlying R -module of the free commutative R -algebra on M is

$$U \mathrm{Sym}_R(M) \simeq \coprod_{d \geq 0} (M^{\otimes_R d})_{h\Sigma_d}.$$

Here each tensor power is κ -compact, the homotopy orbits are a countable colimit, and the coproduct is countable. Thus $U \mathrm{Sym}_R(M)$ is κ -compact. Also $\mathrm{Sym}_R(M)$ is κ -compact as an algebra, since U preserves κ -filtered colimits.

Since the monad $U \mathrm{Sym}_R$ is κ -accessible and preserves κ -compact objects by the formula above, the κ -compact objects in $\mathrm{CAlg}_R(C)$ are generated under κ -small colimits and retracts by free algebras on κ -compact modules. Indeed, for any κ -accessible monad on a κ -presentable category which preserves κ -compact objects, the free algebras on κ -compact objects are κ -compact and generate the algebra category under colimits, so the κ -compact algebras are the retracts of their κ -small colimits.

It remains to check that the class of R -algebras with κ -compact underlying R -module is closed under these operations. Closure under retracts is clear. For κ -small colimits, write such a colimit as the

geometric realization of its simplicial replacement; the n -simplices are κ -small coproducts of the algebras in the diagram. A κ -small coproduct of commutative R -algebras is the sifted colimit over its finite sub-coproducts, and the underlying R -module of a finite coproduct is the corresponding finite tensor product in $\text{Mod}_R(C)$. Since U preserves sifted colimits, and since κ -compact R -modules are closed under finite tensor products and κ -small colimits, the underlying module of each simplicial degree, and hence of the realization, is κ -compact. Thus a κ -compact R -algebra has κ -compact underlying R -module.

Conversely, suppose that $U(B)$ is κ -compact. By monadicity, B is the geometric realization of the standard simplicial resolution by free algebras,

$$\cdots \rightrightarrows \text{Sym}_R U \text{Sym}_R U(B) \rightrightarrows \text{Sym}_R U(B).$$

By the free-algebra case above, every term in this simplicial object is κ -compact as an algebra. Since the geometric realization is a countable colimit, B is κ -compact in $\text{CAlg}_R(C)$. $\square$

Lemma A.2. *Let κ be an uncountable regular cardinal, let $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$, and let $A \rightarrow B$ be κ -compact in $\text{CAlg}_A(C)$. Then an object $M \in \text{Mod}_B(C)$ is κ -compact as a B -module if and only if its underlying A -module is κ -compact.*

Proof. By Lemma A.1, the underlying A -module of B is κ -compact. We write

$$f^* = B \otimes_A -: \text{Mod}_A(C) \rightleftarrows \text{Mod}_B(C): f_*$$

for the extension-restriction adjunction. First suppose that M is κ -compact as a B -module. The κ -compact objects in $\text{Mod}_B(C)$ are generated under κ -small colimits and retracts by the free B -modules $B \otimes X$ with $X \in C^\kappa$. Since f_* preserves colimits and sends such a free module to $B \otimes X$, which is κ -compact as an A -module because the C -action on $\text{Mod}_A(C)$ preserves κ -compact objects, it follows that f_*M is κ -compact.

Conversely, suppose that f_*M is κ -compact as an A -module. The standard monadic resolution

$$M \simeq |\cdots \rightrightarrows f^* f_* f^* f_* M \rightrightarrows f^* f_* M|$$

expresses M as a countable colimit of B -modules of the form f^*N . Since $f_* f^* N \simeq B \otimes_A N$ and the tensor product on $\text{Mod}_A(C)$ preserves κ -compact objects in both variables, the functor $f_* f^*$ preserves κ -compact A -modules. Hence every term in this resolution is of the form f^*N with N κ -compact as an A -module, and is therefore κ -compact as a B -module. Since the realization is a countable, hence κ -small, colimit, M is κ -compact. $\square$

Corollary A.3. *Let κ be an uncountable regular cardinal, let $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$, and let $g: R \rightarrow S$ be κ -compact in $\text{CAlg}_R(C)$. Then the restriction functor*

$$g_*: \text{Mod}_S(C) \rightarrow \text{Mod}_R(C)$$

preserves κ -compact objects.

Proof. This is one direction of Lemma A.2. $\square$

Lemma A.4. *Let κ be an uncountable regular cardinal, let $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$, and let $A \in \text{CAlg}(C)$.*

- (1) The κ -compact objects in $\text{CAlg}_A(C)$ are closed under κ -small colimits and retracts. In particular, they are closed under finite colimits.
- (2) If $B \in \text{CAlg}_A(C)$ is κ -compact and $A \rightarrow A'$ is any map in $\text{CAlg}(C)$, then $A' \otimes_A B \in \text{CAlg}_{A'}(C)$ is κ -compact.
- (3) If $B \in \text{CAlg}_A(C)$ is κ -compact and $D \in \text{CAlg}_B(C)$ is κ -compact, then D is κ -compact as an object of $\text{CAlg}_A(C)$.
- (4) If $B, D \in \text{CAlg}_A(C)$ are κ -compact and $B \rightarrow D$ is a map in $\text{CAlg}_A(C)$, then D is κ -compact as an object of $\text{CAlg}_B(C)$.

Proof. Part (1) is the general closure property of κ -compact objects in a κ -presentable category. For (2), extension of scalars

$$A' \otimes_A -: \text{CAlg}_A(C) \rightarrow \text{CAlg}_{A'}(C)$$

is left adjoint to the forgetful functor, and this forgetful functor preserves κ -filtered colimits.

For (3), Lemma A.1 shows that D is κ -compact as a B -module. By Lemma A.2, its underlying A -module is κ -compact, so Lemma A.1 again shows that D is κ -compact as an A -algebra.

For (4), the underlying A -module of D is κ -compact by Lemma A.1. Since B is κ -compact over A , Lemma A.2 implies that D is κ -compact as a B -module. Applying Lemma A.1 once more, D is κ -compact as a B -algebra. $\square$

Lemma A.5. *Let κ be an uncountable regular cardinal, let $C \in \text{CAlg}(\text{Pr}_\kappa^{\text{L}})$, and let $A \rightarrow B$ be κ -compact in $\text{CAlg}_A(C)$. If $A \rightarrow B$ admits an A -algebra retraction $B \rightarrow A$, then A is κ -compact as an object of $\text{CAlg}_B(C)$.*

Proof. Apply Lemma A.4(4) to the map $B \rightarrow A$ in $\text{CAlg}_A(C)$: both B and the initial A -algebra A are κ -compact in $\text{CAlg}_A(C)$. $\square$

A.2 Compact objects in sheaf categories

We now collect the compactness facts for presheaves and left exact localizations which are used in the main text.

Lemma A.6. *Let κ be an uncountable regular cardinal, and let C be a small category with finite limits. Then finite limits of κ -small coproducts of representables in $\text{PSh}(C)$ are again κ -small coproducts of representables.*

Proof. The terminal object is representable, so it is enough to prove stability under pullbacks. Let

$$F = \coprod_{a \in A} y(c_a), \quad G = \coprod_{b \in B} y(d_b), \quad H = \coprod_{e \in E} y(h_e)$$

with A, B, E κ -small. Consider maps $F \rightarrow H$ and $G \rightarrow H$. For each $a \in A$, the restriction $y(c_a) \rightarrow H$ chooses a summand $e(a) \in E$ and a map $c_a \rightarrow h_{e(a)}$ in C . Similarly, for each $b \in B$, the restriction $y(d_b) \rightarrow H$ chooses a summand $e'(b) \in E$ and a map $d_b \rightarrow h_{e'(b)}$.

Coproducts in a presheaf ∞ -topos are disjoint and universal. Therefore the pullback $F \times_H G$ decomposes as

$$F \times_H G \simeq \coprod_{\substack{(a,b) \in A \times B \\ e(a)=e'(b)}} \left(y(c_a) \times_{y(h_{e(a)})} y(d_b) \right).$$

Since the Yoneda embedding preserves finite limits, the summand is represented by $c_a \times_{h_{e(a)}} d_b$. The displayed coproduct is κ -small, because κ is infinite and A and B are κ -small. This proves stability under pullbacks, and hence under all finite limits. $\square$

Lemma A.7. *Let C be a small category, and let $Y_\bullet \in \text{Fun}(\Delta^{\text{op}}, \text{PSh}(C))$ be a simplicial presheaf. For a finite simplicial set K , write*

$$Y(K) := \lim_{\Delta^n \rightarrow K} Y_n$$

for the right Kan extension of $Y_\bullet$ from simplices to finite simplicial sets. For $r \geq 0$, set

$$(\text{Ex}^r(Y))_q := Y(\text{sd}^r \Delta^q), \quad \text{Ex}^\infty(Y) := \text{colim}_{r \in \mathbb{N}} \text{Ex}^r(Y).$$

Then:

- (1) *Each $(\text{Ex}^r(Y))_q$ is a finite limit of values of $Y_\bullet$.*
- (2) *The natural map $Y_\bullet \rightarrow \text{Ex}^\infty(Y)$ induces an isomorphism on geometric realizations, and $(\text{Ex}^\infty(Y))_0 \simeq Y_0$.*
- (3) *The simplicial presheaf $\text{Ex}^\infty(Y)$ is a Kan complex in $\text{PSh}(C)$, i.e. for every $n \geq 1$ and $0 \leq i \leq n$ the maps*

$$\text{Ex}^\infty(Y)(\Delta^n) \rightarrow \text{Ex}^\infty(Y)(\Lambda_i^n)$$

are effective epimorphisms, equivalently pointwise surjective on π_0 .

- (4) *If every Y_n is a κ -small coproduct of representables, then each*

$$(\text{Ex}^\infty(Y))_{q+1} \times_{(\text{Ex}^\infty(Y))_q} (\text{Ex}^\infty(Y))_{q+1}$$

is the countable colimit, over r , of κ -small coproducts of representables.

Proof. The simplicial set $\text{sd}^r \Delta^q$ is finite. Although its full simplex category contains infinitely many degenerate simplices, the right Kan extension can be computed from the finite diagram of nondegenerate simplices and their face maps. Thus the formula defining $(\text{Ex}^r(Y))_q$ expresses it as a finite limit of the objects Y_n .

The remaining assertions are the usual internal properties of the Ex^∞ fibrant replacement in an ∞ -topos; this is the countability argument used in Scholze [Sch26, the proof of Proposition 7.3]. The degree-zero assertion follows directly from $\text{sd}^r \Delta^0 = \Delta^0$.

For (4), write $Y^{(r)} = \text{Ex}^r(Y)$. By (1) and Lemma A.6, every $Y_q^{(r)}$ is a κ -small coproduct of representables. Since filtered colimits commute with finite limits in presheaves of anima, we have

$$\begin{aligned} & (\text{Ex}^\infty(Y))_{q+1} \times_{(\text{Ex}^\infty(Y))_q} (\text{Ex}^\infty(Y))_{q+1} \\ & \simeq \text{colim}_{r \in \mathbb{N}} \left(Y_{q+1}^{(r)} \times_{Y_q^{(r)}} Y_{q+1}^{(r)} \right). \end{aligned}$$

Each term in the final colimit is again a κ -small coproduct of representables by Lemma A.6. $\square$

Proposition A.8. *Let C be a small category, and let $Y_\bullet \in \text{Fun}(\Delta^{\text{op}}, \text{PSh}(C))$ be a simplicial presheaf. There is a simplicial object $B_\bullet(Y)$, natural in $Y_\bullet$, with a natural isomorphism*

$$|B_\bullet(Y)| \simeq Y_0 \times_{|Y_\bullet|} Y_0.$$

Moreover, if $Y_\bullet^{\text{fib}} := \text{Ex}^\infty(Y)$, then

$$B_q(Y) \simeq Y_{q+1}^{\text{fib}} \times_{Y_q^{\text{fib}}} Y_{q+1}^{\text{fib}},$$

where the two maps $Y_{q+1}^{\text{fib}} \rightarrow Y_q^{\text{fib}}$ are the last and first face maps.

Proof. Set $Y_\bullet^{\text{fib}} := \text{Ex}^\infty(Y)$. By Lemma A.7, the map $Y_\bullet \rightarrow Y_\bullet^{\text{fib}}$ induces an isomorphism on geometric realizations and identifies degree zero. It is therefore enough to compute

$$Y_0^{\text{fib}} \times_{|Y_\bullet^{\text{fib}}|} Y_0^{\text{fib}}.$$

Let $\text{Dec}_\perp(Y^{\text{fib}})$ and $\text{Dec}_\top(Y^{\text{fib}})$ be the two decalages. Thus both have q -simplices Y_{q+1}^{fib} : the first forgets the last face and degeneracy, and the second forgets the first face and degeneracy. They carry natural maps to $Y_\bullet^{\text{fib}}$, given respectively by the last and first face maps. The augmented simplicial animae

$$\text{Dec}_\perp(Y^{\text{fib}}) \rightarrow Y_0^{\text{fib}}, \quad \text{Dec}_\top(Y^{\text{fib}}) \rightarrow Y_0^{\text{fib}}$$

have extra degeneracies, hence their realizations are both equivalent to Y_0^{fib} .

Since $Y_\bullet^{\text{fib}}$ is Kan in $\text{PSh}(C)$, the maps $\text{Dec}_\perp(Y^{\text{fib}}) \rightarrow Y_\bullet^{\text{fib}}$ and $\text{Dec}_\top(Y^{\text{fib}}) \rightarrow Y_\bullet^{\text{fib}}$ are Kan fibrations in $\text{PSh}(C)$.¹ Thus geometric realization preserves the pullback defining

$$B_\bullet(Y) := \text{Dec}_\perp(Y^{\text{fib}}) \times_{Y_\bullet^{\text{fib}}} \text{Dec}_\top(Y^{\text{fib}}).$$

We obtain

$$|B_\bullet(Y)| \simeq Y_0^{\text{fib}} \times_{|Y_\bullet^{\text{fib}}|} Y_0^{\text{fib}} \simeq Y_0 \times_{|Y_\bullet|} Y_0.$$

The formula for $B_q(Y)$ is immediate from the definition of the two decalages. $\square$

Corollary A.9. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let $Y_\bullet$ be a simplicial object of $\text{PSh}(C)$ such that every Y_n is a κ -small coproduct of representables. Then*

$$Y_0 \times_{|Y_\bullet|} Y_0$$

is a κ -small colimit of representables. More precisely, as an object over $Y_0 \times Y_0$, it is the geometric realization of a simplicial object whose q -simplices are countable colimits of κ -small coproducts of representables.

Proof. This follows from Proposition A.8 and part (4) of Lemma A.7. Since κ is uncountable regular, the countable colimits in the simplicial degrees and the countable geometric realization together form a κ -small colimit of representables. $\square$

¹For the decalage path fibration, see the nLab entry on decalage, <https://ncatlab.org/nlab/show/decalage>; for preservation of homotopy pullbacks along such realization fibrations, compare Rezk, *When are homotopy colimits compatible with homotopy base change?*, <https://rezk.web.illinois.edu/i-hate-the-pi-star-kan-condition.pdf>.

Proposition A.10. *Let κ be an uncountable regular cardinal, and let C be a small category with finite limits. Then the κ -compact objects of $\mathbf{PSh}(C)$ are closed under finite limits.*

Proof. The category $\mathbf{PSh}(C)$ is κ -compactly generated by representables, and its κ -compact objects are κ -small colimits of representables. Indeed, the only point is to absorb retracts: if X is a retract of Y in a category with countable colimits, represented by an idempotent $e : Y \rightarrow Y$, then

$$X \simeq \operatorname{colim} \left(Y \xrightarrow{e} Y \xrightarrow{e} Y \xrightarrow{e} \cdots \right).$$

The terminal object of $\mathbf{PSh}(C)$ is representable, hence κ -compact. We are thus reduced to pullbacks. Let

$$X \rightarrow Y \leftarrow Z$$

be a diagram in which all three objects are κ -small colimits of representables. We first reduce to the case where X and Z are representable. Write $X \simeq \operatorname{colim}_i X_i$ and $Z \simeq \operatorname{colim}_j Z_j$ as κ -small colimits of representables, with the induced maps to Y . Since colimits in a presheaf topos are universal, pullback in the slice over Y preserves colimits. Hence

$$X \times_Y Z \simeq \operatorname{colim}_{i,j} (X_i \times_Y Z_j).$$

The indexing category is still κ -small, so it remains to handle the case where X and Z are representable.

Write Y as a κ -small colimit of representables. The simplicial replacement expresses this colimit as the geometric realization of a simplicial object $Y_\bullet$ of $\mathbf{PSh}(C)$ such that each Y_n is a κ -small coproduct of representables. The map $Y_0 \rightarrow |Y_\bullet|$ is an effective epimorphism. Evaluating at the objects representing X and Z gives effective epimorphisms of anima, whose homotopy fibers are nonempty. We may therefore choose lifts of the two maps $X, Z \rightarrow |Y_\bullet|$ to Y_0 , together with the required homotopies over $|Y_\bullet|$.

Set

$$R := Y_0 \times_{|Y_\bullet|} Y_0.$$

Then

$$X \times_{|Y_\bullet|} Z \simeq (X \times Z) \times_{Y_0 \times Y_0} R,$$

where the map $X \times Z \rightarrow Y_0 \times Y_0$ is induced by the chosen lifts. By Corollary A.9, the map $R \rightarrow Y_0 \times Y_0$ is the geometric realization, in the slice over $Y_0 \times Y_0$, of a simplicial object $R_\bullet$ such that every R_q is a countable colimit $\operatorname{colim}_s R_{q,s}$ of κ -small coproducts of representables, compatibly over $Y_0 \times Y_0$. Since colimits in a presheaf topos are universal,

$$(X \times Z) \times_{Y_0 \times Y_0} R \simeq |(X \times Z) \times_{Y_0 \times Y_0} R_\bullet|.$$

The q -simplices on the right are

$$\operatorname{colim}_s ((X \times Z) \times_{Y_0 \times Y_0} R_{q,s}).$$

Here $X \times Z$ is representable, and $Y_0 \times Y_0$ is a κ -small coproduct of representables by Lemma A.6. Thus each term $(X \times Z) \times_{Y_0 \times Y_0} R_{q,s}$ is a finite limit of κ -small coproducts of representables, hence is itself a κ -small coproduct of representables. Since Δ^{op} and $\mathbb{N}$ are countable and κ is uncountable regular, the total realization is a κ -small colimit of representables. Therefore $X \times_Y Z$ is a κ -small colimit of representables. $\square$

Lemma A.11. *Let κ be an uncountable regular cardinal, and let*

$$L: C \rightleftarrows D : \iota$$

be an accessible localization of presentable categories. Assume that C is κ -compactly generated, that C^κ is closed under finite limits, that L preserves finite limits, and that ι preserves κ -filtered colimits. Then D is κ -compactly generated, and D^κ is closed under finite limits.

Proof. Since ι preserves κ -filtered colimits, the left adjoint L preserves κ -compact objects. Thus LK is κ -compact for every $K \in C^\kappa$.

We next show that these objects generate D under κ -filtered colimits. In fact, we will exhibit every object $d \in D$ as an explicit κ -filtered colimit of objects of this form. Observe that the category

$$(C^\kappa)_{/\iota d}$$

is κ -filtered: a κ -small diagram of maps $K_\alpha \rightarrow \iota d$ is dominated by the map $\text{colim}_\alpha K_\alpha \rightarrow \iota d$, and C^κ is closed under κ -small colimits. Since C is κ -compactly generated, the canonical map

$$\text{colim}_{(K \rightarrow \iota d)} K \rightarrow \iota d$$

is an isomorphism. Applying L gives

$$\text{colim}_{(K \rightarrow \iota d)} LK \simeq d.$$

Thus D is κ -compactly generated by the objects LK . Moreover, if d is κ -compact, then the identity of d factors through one of the structure maps $LK \rightarrow d$, so d is a retract of some LK .

Now let $F: J \rightarrow D^\kappa$ be a finite diagram. We apply the previous paragraph not to D itself, but to the induced localization

$$\text{Fun}(J, C) \rightleftarrows \text{Fun}(J, D).$$

Since J is finite, Lemma 1.5 shows that $\text{Fun}(J, C)$ is again κ -compactly generated, with κ -compact generators given by diagrams $J \rightarrow C^\kappa$; similarly, F is κ -compact as an object of $\text{Fun}(J, D)$. The right adjoint preserves κ -filtered colimits pointwise. Hence F is a retract, as a diagram, of $LK_\bullet$ for some diagram $K_\bullet: J \rightarrow C^\kappa$.

Taking limits, $\lim_J F$ is a retract of

$$\lim_J LK_\bullet \simeq L(\lim_J K_\bullet),$$

where we used that L preserves finite limits. Since C^κ is closed under finite limits, $\lim_J K_\bullet$ lies in C^κ . Therefore $L(\lim_J K_\bullet)$ is κ -compact in D , and so its retract $\lim_J F$ is κ -compact. $\square$

Lemma A.12. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let τ be a Grothendieck topology on C . Suppose that the τ -sheaf condition can be tested on a collection of covering sieves*

$$R \subseteq y(c)$$

which are κ -compact as objects of $\text{PSh}(C)$. Then τ -sheaves are closed under κ -filtered colimits in $\text{PSh}(C)$.

In particular, this applies if τ is generated by a pretopology whose covering families have fewer than κ morphisms. More generally, it applies whenever there is such a testing collection in which every sieve is generated by fewer than κ morphisms into the target.

Proof. Let $F \simeq \operatorname{colim}_i F_i$ be a κ -filtered colimit in $\operatorname{PSh}(C)$, where every F_i is a τ -sheaf. Let $R \subseteq y(c)$ be one of the covering sieves in the testing collection. Since $y(c)$ and R are κ -compact in $\operatorname{PSh}(C)$, we have

$$\begin{aligned} F(c) &\simeq \operatorname{Hom}_{\operatorname{PSh}(C)}(y(c), F) \\ &\simeq \operatorname{colim}_i \operatorname{Hom}_{\operatorname{PSh}(C)}(y(c), F_i) \\ &\simeq \operatorname{colim}_i \operatorname{Hom}_{\operatorname{PSh}(C)}(R, F_i) \\ &\simeq \operatorname{Hom}_{\operatorname{PSh}(C)}(R, F). \end{aligned}$$

The middle isomorphism uses that each F_i satisfies descent for the covering sieve R . Hence F satisfies descent for every sieve in the testing collection, and is therefore a τ -sheaf.

It remains to justify the final assertion. Let $R \subseteq y(c)$ be the sieve generated by a family

$$\{c_a \rightarrow c\}_{a \in A}$$

with A κ -small. Set $U := \coprod_{a \in A} y(c_a)$. The map $U \rightarrow y(c)$ factors through an effective epimorphism $U \rightarrow R$, and R is the geometric realization of its Čech nerve. Since $R \rightarrow y(c)$ is a monomorphism, the terms of this Čech nerve are the corresponding fiber products over $y(c)$:

$$U_n \simeq \coprod_{(a_0, \dots, a_n) \in A^{n+1}} y(c_{a_0} \times_c \cdots \times_c c_{a_n}).$$

Here the indicated finite limits exist in C . Each U_n is a κ -small coproduct of representables, and the simplicial indexing category is countable. Since κ is uncountable regular, $R \simeq |U_\bullet|$ is a κ -small colimit of representables. Such an object is κ -compact: mapping out of it gives a κ -small limit of evaluations, and κ -filtered colimits of animae commute with κ -small limits. $\square$

Corollary A.13. *Let κ be an uncountable regular cardinal, let C be a small category with finite limits, and let τ be a Grothendieck topology on C . Suppose that τ -sheaves are closed under κ -filtered colimits in $\operatorname{PSh}(C)$ (e.g. in the situation of Lemma A.12). Then $\operatorname{Shv}_\tau(C)$ is κ -compactly generated, and the κ -compact objects of $\operatorname{Shv}_\tau(C)$ are closed under finite limits.*

Moreover, sheafification sends κ -compact presheaves to κ -compact sheaves, and every κ -compact sheaf is a retract of a κ -small colimit of sheafified representables.

Proof. Let

$$L: \operatorname{PSh}(C) \rightleftarrows \operatorname{Shv}_\tau(C) : \iota$$

be sheafification and the inclusion. This is an accessible left exact localization, so L preserves finite limits. By assumption, ι preserves κ -filtered colimits. The category $\operatorname{PSh}(C)$ is κ -compactly generated by representables, and Proposition A.10 shows that its κ -compact objects are closed under finite limits. Applying Lemma A.11 gives the first claim.

The same adjunction shows directly that L sends κ -compact presheaves to κ -compact sheaves. The proof of Lemma A.11 also shows that every κ -compact sheaf is a retract of LK for some κ -compact presheaf K . Since κ -compact presheaves are retracts of κ -small colimits of representables, and since such retracts may be absorbed by the usual countable telescope, every κ -compact sheaf is a retract of a κ -small colimit of the objects $Ly(c)$. $\square$

B Six-functor formalisms

Some of the contents of this appendix were written by Ou Liu in preparation for Chapter 10.

In this appendix, we record the relevant definitions in the formalism of *six-functor formalisms* that have appeared throughout these notes. Following Mann [Man22] and Scholze [Sch25a], the six operations and all of their coherences are encoded by a single lax symmetric monoidal functor out of a category of spans. We then recall, following Mann [Man22, Proposition A.5.10], a criterion that produces such a formalism from the more accessible datum of a functor $C^{\text{op}} \rightarrow \text{CAlg}(\text{Cat})$ equipped with adjoints satisfying base change and the projection formula; we also record the 2-categorical enhancement from Cnossen, Lenz, and Linskens [CLL25]. Finally, we recall the definitions of cohomologically smooth/suave/prim/étale/proper morphisms with respect to a six-functor formalism.

B.1 The span category

We begin by recalling the construction of the category of spans. Throughout, C denotes a category.

Definition B.1. Let $E \subseteq C$ be a wide subcategory, i.e. a subcategory containing every object of C . We say that E is *closed under base change* if for every morphism $g: X \rightarrow Y$ in E and every morphism $f: Y' \rightarrow Y$ in C , the pullback $Y' \times_Y X$ exists in C and the projection $Y' \times_Y X \rightarrow Y'$ again lies in E .

Construction B.2 (The span category). Let $E \subseteq C$ be a wide subcategory closed under base change. The *span category* (or *category of correspondences*) $\text{Span}(C, E)$ is the category with the same objects as C , whose morphisms from X to Y are *spans*

$$\begin{array}{ccc} & U & \\ f \swarrow & & \searrow g \\ X & & Y \end{array}$$

in which the right leg g lies in E and the left leg f is arbitrary. The identity of X is the span $X \xleftarrow{=} X \xrightarrow{=} X$, and the composite of $X \xleftarrow{f} U \xrightarrow{g} Y$ with $Y \xleftarrow{f'} V \xrightarrow{g'} Z$ is the outer span of

$$\begin{array}{ccccc} & & U \times_Y V & & \\ & \swarrow & & \searrow & \\ & U & & V & \\ f \swarrow & & g \searrow & f' \swarrow & g' \searrow \\ X & & Y & & Z, \end{array}$$

formed by the pullback $U \times_Y V$. This pullback exists, and its right leg $U \times_Y V \rightarrow Z$ lies in E since E is closed under base change and composition.

Remark B.3. Making Construction B.2 into a genuine ∞ -category requires more care: one realizes $\text{Span}(C, E)$ as a complete Segal anima whose anima of n -simplices is the anima of functors $\text{Tw}[n] \rightarrow C$ out of the twisted arrow category sending forward maps into E and forming the relevant pullbacks. We refer to Barwick [Bar17] and Haugseng, Hebestreit, Linskens, and Nuiten [HHLN23] for the details.

Construction B.4 (Monoidal structure on spans). Suppose now that C has finite products. Then $\text{Span}(C, E)$ inherits a symmetric monoidal structure whose unit is the terminal object of C and whose underlying tensor product sends a pair (X, Y) to the product $X \times Y$.

Indeed, for $g: X \rightarrow Y$ and $g': X' \rightarrow Y'$ in E the product $g \times g': X \times X' \rightarrow Y \times Y'$ again lies in E , since it factors as

$$X \times X' \xrightarrow{g \times \text{id}} Y \times X' \xrightarrow{\text{id} \times g'} Y \times Y'$$

and each factor is a base change of g resp. g' along a projection. Hence the cartesian product equips the triple (C, C, E) with the structure of a commutative monoid among adequate triples. As the functor $\text{Span}(-)$ preserves finite products [HHLN23], this results in a symmetric monoidal structure on $\text{Span}(C, E)$.

B.2 Six-functor formalisms

Let C be a category with finite products and $E \subseteq C$ a wide subcategory closed under base change. We endow $\text{Span}(C, E)$ with the symmetric monoidal structure of Construction B.4 and Cat with the cartesian one.

Definition B.5 (Three-functor formalism). A *three-functor formalism* on (C, E) is a lax symmetric monoidal functor

$$D: \text{Span}(C, E) \rightarrow \text{Cat}.$$

Discussion B.6 (The six operations). Let us unwind the data of a three-functor formalism D .

- (1) Each object X of C is canonically a commutative algebra in $(\text{Span}(C, E), \times)$, with multiplication and unit given by the diagonal $X \rightarrow X \times X$ and the projection $X \rightarrow *$ read as backward maps. Since lax symmetric monoidal functors preserve commutative algebras, every $D(X)$ is itself a symmetric monoidal category, with tensor product $\otimes$ and unit $\mathbb{1}_X$.
- (2) Every morphism $f: X \rightarrow Y$ in C defines a backward span $Y \xleftarrow{f} X \xrightarrow{\text{id}} X$, and hence a *pullback* functor $f^* := D(Y \xleftarrow{f} X \xrightarrow{\text{id}} X): D(Y) \rightarrow D(X)$, which is symmetric monoidal and depends functorially on f .
- (3) Every morphism $g: X \rightarrow Y$ in E defines a forward span $X \xrightarrow{\text{id}} X \xrightarrow{g} Y$, and hence an *exceptional pushforward* $g_! := D(X \xrightarrow{\text{id}} X \xrightarrow{g} Y): D(X) \rightarrow D(Y)$, depending functorially on g . The lax structure makes $g_!$ into a $D(Y)$ -linear functor, which is the content of the *projection formula* $g_!(g^*(A) \otimes B) \xrightarrow{\sim} A \otimes g_!(B)$.

- (4) The two factorizations of a span $X \xleftarrow{f} U \xrightarrow{g} Y$ as a backward span followed by a forward span yield the *base change* isomorphism $g_! f^* \simeq f'^* g'_!$ associated to a pullback square, i.e. the assignment $(X \xleftarrow{f} U \xrightarrow{g} Y) \mapsto g_! f^*$ is functorial.

These are the three functors $(\otimes, f^*, f_!)$. When the relevant adjoints exist, one obtains the remaining three operations as their right adjoints: the *pushforward* f_* right adjoint to f^* , the *exceptional pullback* $g^!$ right adjoint to $g_!$, and the *internal hom* $\underline{\mathrm{Hom}}(M, -)$ right adjoint to $M \otimes -$. This accounts for the name: the six operations are $(f^*, f_*, f_!, f^!, \otimes, \underline{\mathrm{Hom}})$.

Definition B.7 (Six-functor formalism). A three-functor formalism $D: \mathrm{Span}(C, E) \rightarrow \mathrm{Cat}$ is a *six-functor formalism* if the right adjoints of Discussion B.6 all exist; that is, if each $D(X)$ is closed symmetric monoidal, each pullback f^* admits a right adjoint f_* , and each exceptional pushforward $g_!$ admits a right adjoint $g^!$.

Remark B.8. The right adjoints in Definition B.7 are a *property* of the three-functor formalism, not extra structure. In practice one frequently considers formalisms valued in presentable categories, e.g. a lax symmetric monoidal functor $\mathrm{Span}(C, E) \rightarrow \mathrm{Pr}^{\mathrm{L}}$ as in the constructions $\mathrm{Spec}(A) \mapsto n\mathrm{Pr}_A$ considered earlier in these notes; there the existence of these adjoints is automatic by the adjoint functor theorem, so that every such three-functor formalism is already a six-functor formalism.

B.3 The construction of six-functor formalisms

Three-functor formalisms encode their data in a highly coherent fashion, which makes them powerful but correspondingly difficult to construct directly. In practice one starts from the much more accessible datum of the pullbacks and tensor products alone, i.e. a functor

$$D_0: C^{\mathrm{op}} \rightarrow \mathrm{CAlg}(\mathrm{Cat}), \quad X \mapsto (D_0(X), \otimes), \quad f \mapsto f^*,$$

and asks for conditions guaranteeing that it extends to a three-functor formalism. The fundamental result in this direction is due to Liu–Zheng and Mann [Man22, Proposition A.5.10]. The hypotheses concern a factorization of the maps in E together with the interaction of the resulting adjoints. We first recall the Beck–Chevalley conditions and the projection formula.

Definition B.9 (Beck–Chevalley condition). Let $D_0: C^{\mathrm{op}} \rightarrow \mathrm{Cat}$ and consider a pullback square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

in C , giving a commutative square $f'^* g^* \simeq g'^* f^*$ of functors $D_0(Y) \rightarrow D_0(X')$.

- (1) Suppose f^* and f'^* admit left adjoints $f_{\#}$ and $f'_{\#}$. The square is *left adjointable*, or satisfies the *Beck–Chevalley condition*, if the mate

$$\mathrm{BC}_{\#}: f'_{\#} g'^* \rightarrow g^* f_{\#}$$

is an isomorphism.

- (2) Dually, suppose f^* and f'^* admit right adjoints f_* and f'_* . The square is *right adjointable*, or satisfies the *dual Beck–Chevalley condition*, if the mate

$$\mathrm{BC}_* : g^* f_* \rightarrow f'_* g'^*$$

is an isomorphism.

Definition B.10 (Projection formula). Let $D_0 : C^{\mathrm{op}} \rightarrow \mathrm{Cat}$, so that each $D_0(X)$ is symmetric monoidal and each f^* is symmetric monoidal.

- (1) If $f : X \rightarrow Y$ is such that f^* admits a left adjoint $f_{\#}$, then for $M \in D_0(X)$ and $N \in D_0(Y)$ there is a natural *projection map*

$$f_{\#}(M \otimes f^* N) \rightarrow f_{\#}(M) \otimes N,$$

adjoint to $M \otimes f^* N \rightarrow f^* f_{\#}(M) \otimes f^* N \simeq f^*(f_{\#}(M) \otimes N)$. We say $f_{\#}$ satisfies the *projection formula* if this map is an isomorphism for all M and N .

- (2) Dually, if p^* admits a right adjoint p_* , then the *projection map*

$$N \otimes p_*(M) \rightarrow p_*(p^* N \otimes M)$$

is required to be an isomorphism; we then say p_* satisfies the *projection formula*.

With these notions in place, we can state the construction theorem.

Theorem B.11 (Mann [Man22, Proposition A.5.10]). *Let C be a category with finite products, and let $I, P \subseteq E \subseteq C$ be wide subcategories closed under base change satisfying:*

- (1) *I and P are left cancellable (if $gf, g \in I$ then $f \in I$, and likewise for P);*
- (2) *every map in E factors as a map in I followed by a map in P ;*
- (3) *every map in $I \cap P$ is truncated.*

Let $D_0 : C^{\mathrm{op}} \rightarrow \mathrm{Cat}$ be a functor satisfying:

- (1) *for every $i : X \rightarrow Y$ in I , the pullback i^* admits a left adjoint $i_{\#}$ satisfying the Beck–Chevalley condition (Definition B.9) and the projection formula (Definition B.10);*
- (2) *for every $p : X \rightarrow Y$ in P , the pullback p^* admits a right adjoint p_* satisfying the dual Beck–Chevalley condition and the projection formula;*
- (3) *for every pullback square*

$$\begin{array}{ccc} X' & \xrightarrow{i} & X \\ p \downarrow & \lrcorner & \downarrow q \\ Y' & \xrightarrow{j} & Y \end{array}$$

in C with horizontal maps in I and vertical maps in P , the double Beck–Chevalley map $j_{\#} p_ \rightarrow q_* i_{\#}$ is an isomorphism.*

Then D_0 extends to a three-functor formalism

$$D: \text{Span}(C, E) \rightarrow \text{Cat}$$

with the property that for every $e \in E$ and every factorization $e = p \circ i$ with $i \in I$ and $p \in P$, the exceptional pushforward is given by $e_! \simeq p_* i_\#$.

B.4 A 2-categorical enhancement

The construction of Theorem B.11 is the shadow of a finer statement, in which $\text{Span}(C, E)$ is promoted to a 2-category and the extension is pinned down by a universal property. This refinement is due to Gaitsgory–Rozenblyum [GR17] in special cases and due to Cnossen–Lenz–Linskens [CLL25] in the present generality. It has two useful features that we will make use of: it characterizes the extension uniquely, and it works with an arbitrary symmetric monoidal 2-category as target. We give an informal account.

Construction B.12 (The span 2-category). Let $I, P \subseteq E \subseteq C$ be wide subcategories closed under base change. There is a 2-category $\text{Span}_2(C, E)_{P, I}$, a mild generalization of a construction of Haugseng, refining $\text{Span}(C, E)$ as follows:

- Its objects are the objects of C ;
- Its 1-morphisms $X \rightarrow Y$ are the spans $X \xleftarrow{f} U \xrightarrow{e} Y$ with $e \in E$, exactly as in $\text{Span}(C, E)$;
- A 2-morphism from $X \xleftarrow{f} U \xrightarrow{e} Y$ to $X \xleftarrow{f'} V \xrightarrow{e'} Y$ is a *span of spans*, i.e. an object W together with maps $p: W \rightarrow U$ in P and $i: W \rightarrow V$ in I making

$$\begin{array}{ccccc} & & U & & \\ & \swarrow & \uparrow p & \searrow e & \\ X & & W & & Y \\ & \swarrow & \downarrow i & \searrow e' & \\ & & V & & \end{array}$$

commute.

Composition of 2-cells is again by pullback, and the underlying 1-category is $\text{Span}(C, E)$. Making this precise (as a 2-fold complete Segal anima) is the technical heart of Cnossen, Lenz, and Linskens [CLL25], to which we refer for details.

Definition B.13 ((I, P) -biadjointable lax symmetric monoidal functor). Let $\mathbb{E}$ be a symmetric monoidal 2-category. A lax symmetric monoidal functor

$$F: C^{\text{op}} \rightarrow \mathbb{E}, \quad f \mapsto f^* := F(f),$$

is (I, P) -biadjointable if it satisfies the following conditions, interpreted via adjunctions internal to $\mathbb{E}$:

- (1) For every $i \in I$, the morphism i^* admits a left adjoint $i_\#$ satisfying the Beck–Chevalley condition and the projection formula.

- (2) For every $p \in P$, the morphism p^* admits a right adjoint p_* satisfying the dual Beck–Chevalley condition and the projection formula.
- (3) For every pullback square

$$\begin{array}{ccc} X' & \xrightarrow{i} & X \\ p \downarrow & \lrcorner & \downarrow q \\ Y' & \xrightarrow{j} & Y \end{array}$$

with horizontal maps in I and vertical maps in P , the double Beck–Chevalley map $j_{\#} p_* \rightarrow q_* i_{\#}$ is an isomorphism.

For $\mathbb{E} = \text{Cat}$, these are precisely the Beck–Chevalley and projection-formula conditions of Definitions B.9 and B.10.

Theorem B.14 (Cnossen, Lenz, and Linskens [CLL25, Theorem B]). *Let $I, P \subseteq E \subseteq C$ satisfy the hypotheses of Theorem B.11, and let $\mathbb{E}$ be a symmetric monoidal 2-category. Restriction along the inclusion $h: C^{\text{op}} \hookrightarrow \text{Span}_2(C, E)_{P, I}$ induces an equivalence of 2-categories*

$$h^*: \text{Fun}^{\text{lax}\otimes}(\text{Span}_2(C, E)_{P, I}, \mathbb{E}) \simeq \text{Fun}_{(I, P)\text{-biadj}}^{\text{lax}\otimes}(C^{\text{op}}, \mathbb{E}),$$

where the right-hand side is the locally full sub-2-category on the (I, P) -biadjointable lax symmetric monoidal functors and the lax monoidal natural transformations satisfying the evident Beck–Chevalley conditions. In particular, every such functor extends uniquely to a lax symmetric monoidal 2-functor on $\text{Span}_2(C, E)_{P, I}$. For every $e \in E$ and every factorization $e = p \circ i$ with $i \in I$ and $p \in P$, its exceptional pushforward is given by

$$e! \simeq p_* i_{\#}.$$

Specializing to $\mathbb{E} = \text{Cat}$ recovers Theorem B.11, now with the additional information that the extension is unique and characterized by this universal property.

Finally, we record an observation that we will need later, and which follows directly by inspecting Construction B.12.

Construction B.15 (Endomorphisms of the unit). Assume that $E = C$. We may then describe the endomorphism category of $*$ in $\text{Span}_2(C, E)_{P, I}$ as follows: an object of $\text{Hom}(*, *)$ is a span of the form $* \leftarrow X \rightarrow *$ corresponding to an object $X \in C$, while a morphism $X \rightarrow Y$ corresponds to a span $X \xleftarrow{p} W \xrightarrow{i} Y$ with $p \in P$ and $i \in I$. In other words, we have

$$\text{Hom}_{\text{Span}_2(C, E)_{P, I}}(*, *) \simeq \text{Span}(C, P, I),$$

the span category of C with backward maps in P and forward maps in I .

Now let $\mathbb{E}$ be symmetric monoidal and let $F: \text{Span}_2(C, E)_{P, I} \rightarrow \mathbb{E}$ be a lax symmetric monoidal 2-functor, e.g. one obtained from Theorem B.14. Then F carries the unit to a commutative algebra $F(*) \in \text{CAlg}(\mathbb{E})$, and its action on the endomorphisms of the units yields a functor (in fact lax symmetric monoidal)

$$\text{Span}(C, P, I) \simeq \text{Hom}(*, *) \xrightarrow{F} \text{Hom}_{\mathbb{E}}(F(*), F(*)).$$

B.5 Geometric properties of morphisms

The six operations make it possible to express, purely in terms of the six-functor formalism, several geometric properties of a morphism: cohomological smoothness, suaveness, primness, étaleness, and properness. We explicitly record this vocabulary here, so that we can compare it to the analogous notions in the Gestalten formalism under the transmutation functor $[-]_D$; see Section 10.3. Throughout, D denotes a six-functor formalism on (C, E) in the sense of Definition B.7. For the kernel constructions below, we additionally assume that E is closed under relative diagonals. Equivalently, the wide subcategory C_E admits fiber products and its inclusion into C preserves them, as in the notion of a geometric setup used by Heyer and Mann [HM24, Definition 2.1.1]. Morphisms are assumed to lie in E unless stated otherwise.

Definition B.16 (Cohomologically smooth morphism). A morphism $f: X \rightarrow Y$ in E is *cohomologically smooth* (or *D-smooth*) if it satisfies:

- (1) The natural transformation

$$f^!(\mathbb{1}_Y) \otimes f^*(-) \longrightarrow f^!(-)$$

is an isomorphism; we write $\omega_f := f^!(\mathbb{1}_Y)$ and call it the *dualizing complex* of f .

- (2) The dualizing complex ω_f is $\otimes$ -invertible in $D(X)$.

- (3) For every pullback square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & \lrcorner & \downarrow f \\ Y' & \xrightarrow{g} & Y, \end{array}$$

with g arbitrary, the morphism f' also satisfies conditions (1) and (2), and the canonical map $g'^* f^!(\mathbb{1}_Y) \rightarrow f'^!(\mathbb{1}_{Y'})$ is an isomorphism.

Cohomological smoothness can be conveniently formulated in terms of the 2-category of kernels K_D associated to D .

Definition B.17 (2-category of kernels). Let $Y \in C$. The 2-category of kernels $K_{D,Y}$ is the 2-category whose:

- *Objects* are objects of $(C_E)_{/Y}$, i.e. morphisms $X \rightarrow Y$ in E .
- The *hom category* from X_1 to X_2 is $\text{Hom}_{K_D}(X_1, X_2) := D(X_1 \times_Y X_2)$.
- The *identity* on X is $\Delta_{f!}(\mathbb{1}_X) \in D(X \times_Y X)$, where $\Delta_f: X \rightarrow X \times_Y X$ is the relative diagonal.
- *Composition* of $M \in \text{Hom}_{K_D}(X_1, X_2)$ and $N \in \text{Hom}_{K_D}(X_2, X_3)$ is

$$N \circ M := (\pi_{13})_!(\pi_{12}^* M \otimes \pi_{23}^* N) \in D(X_1 \times_Y X_3),$$

where the π_{ij} are the projections from $X_1 \times_Y X_2 \times_Y X_3$.

The extra diagonal hypothesis above ensures that the identity kernels and all exceptional pushforwards in the composition formula are defined. Formally, one equips $\text{Span}(E/Y)$ with the closed symmetric monoidal structure induced by the cartesian monoidal structure on E/Y , regards it as self-enriched via the internal homs $[X, X'] = X \times_Y X'$ supplied by self-duality in a span category, and transfers this enrichment along the lax symmetric monoidal functor $D_Y : \text{Span}(E/Y) \rightarrow \text{Cat}$.

An alternative approach to the 2-category of kernels is presented in Definition 10.1.

The origin of the name *2-category of kernels* is that the 1-morphisms $X_1 \rightarrow X_2$ of $K_{D,Y}$ are exactly the *kernels* $K \in D(X_1 \times_Y X_2)$ of Fourier–Mukai transforms

$$F_K := (\pi_2)_! (\pi_1^*(-) \otimes K) : D(X_1) \rightarrow D(X_2),$$

and the composition law of Definition B.17 corresponds to composing the associated transforms. In fact, there is a canonical 2-functor

$$\Psi_{D,Y} = \text{Hom}_{K_D}(Y, -) : K_{D,Y} \longrightarrow \text{Cat},$$

sending an object X to $D(X)$ and a 1-morphism $K \in \text{Hom}_{K_D}(X_1, X_2)$ to the Fourier–Mukai transform F_K .

Theorem B.18 (Smoothness via kernels, cf. Heyer and Mann [HM24, §4.5]). *A morphism $f : X \rightarrow Y$ in E is D -smooth if and only if*

- (1) *The 1-morphism $\mathbb{1}_X \in \text{Hom}_{K_D}(X, Y) = D(X)$ is a left adjoint in $K_{D,Y}$. Denote its right adjoint by $\omega_f \in \text{Hom}_{K_D}(Y, X) = D(X)$.*
- (2) *ω_f is $\otimes$ -invertible in $D(X)$.*

When these hold, $\omega_f \simeq f^!(\mathbb{1}_Y)$.

In light of this result, it makes sense to study more general adjunctions in $K_{D,Y}$. This leads to the following definition.

Definition B.19 (Suave and prim objects). *Let $f : X \rightarrow Y$ be in E and $A \in D(X)$.*

- (1) *A is f -suave if, viewed as a 1-morphism $X \rightarrow Y$ in $K_{D,Y}$, it is a left adjoint. Its right adjoint is the f -suave dual $SD_f(A) \in D(X)$. Write $\text{Suave}_f(X) \subseteq D(X)$ for the full subcategory of f -suave objects.*
- (2) *A is f -prim if, viewed as a 1-morphism $Y \rightarrow X$ in $K_{D,Y}$, it is a left adjoint. Its right adjoint is the f -prim dual $PD_f(A) \in D(X)$. Write $\text{Prim}_f(X) \subseteq D(X)$ for the full subcategory of f -prim objects.*

(Equivalently, via the self-duality $K_{D,Y} \simeq K_{D,Y}^{\text{op}}$, A is f -prim iff it is a *right* adjoint 1-morphism $X \rightarrow Y$; this is the form used by Heyer–Mann [HM24].)

Proposition B.20 (Heyer and Mann [HM24, §4.4]). *Let D be a six-functor formalism on (C, E) and $f : X \rightarrow Y$ in E .*

- (1) If $A \in \mathbf{D}(X)$ is f -suave, then $\mathrm{SD}_f(A)$ is also f -suave (with right adjoint A in $K_{\mathbf{D},Y}$), and there is a natural equivalence

$$\mathrm{SD}_f(A) \otimes f^*(-) \simeq \underline{\mathrm{Hom}}(A, f^!(-)) : \mathbf{D}(Y) \rightarrow \mathbf{D}(X).$$

Setting $- = \mathbb{1}_Y$ yields $\mathrm{SD}_f(A) \simeq \underline{\mathrm{Hom}}(A, f^!(\mathbb{1}_Y))$, exhibiting $\mathrm{SD}_f(A)$ as a relative Verdier dual of A . The bidual identity $\mathrm{SD}_f(\mathrm{SD}_f(A)) \simeq A$ then holds, and SD_f defines a contravariant equivalence

$$\mathrm{SD}_f : \mathrm{Suave}_f(X)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Suave}_f(X).$$

- (2) Dually, if $A \in \mathbf{D}(X)$ is f -prim, then $\mathrm{PD}_f(A)$ is also f -prim, with

$$f_!(\mathrm{PD}_f(A) \otimes -) \simeq f_* \underline{\mathrm{Hom}}(A, -) : \mathbf{D}(X) \rightarrow \mathbf{D}(Y),$$

and the bidual $\mathrm{PD}_f(\mathrm{PD}_f(A)) \simeq A$ holds. Consequently PD_f defines a contravariant equivalence

$$\mathrm{PD}_f : \mathrm{Prim}_f(X)^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Prim}_f(X).$$

We may specialize Definition B.19 to $A = \mathbb{1}_X$ to obtain two classes of morphisms.

Definition B.21 (Suave and prim morphisms). Let $f : X \rightarrow Y$ be in E .

- (1) f is $\mathbf{D}$ -suave if $\mathbb{1}_X \in \mathrm{Suave}_f(X)$. As before we write $\omega_f := \mathrm{SD}_f(\mathbb{1}_X) \in \mathbf{D}(X)$ for its *dualizing complex*.
- (2) f is $\mathbf{D}$ -prim if $\mathbb{1}_X \in \mathrm{Prim}_f(X)$. We then write $\delta_f := \mathrm{PD}_f(\mathbb{1}_X) \in \mathbf{D}(X)$ for the *co-dualizing complex* of f .

Remark B.22. Combining Theorem B.18 with Definition B.21 gives

$$f \text{ is } \mathbf{D}\text{-smooth} \iff f \text{ is } \mathbf{D}\text{-suave and } \omega_f \text{ is invertible.}$$

By Proposition B.20, the dualizing complex is the relative Verdier dual $\omega_f \simeq f^!(\mathbb{1}_Y)$. The co-dualizing complex may be computed by

$$\delta_f \simeq \pi_{2*} \Delta_{f!} \mathbb{1}_X,$$

where $\pi_2 : X \times_Y X \rightarrow X$ is the projection onto the second factor.

For $\mathbf{D}$ -suave f , the formula $\omega_f \otimes f^*(-) \xrightarrow{\sim} f^!(-)$ is equivalent (by adjunction) to $f^*(-) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\omega_f, f^!(-))$. Plugging in $\mathbb{1}_Y$ yields $\mathbb{1}_X \xrightarrow{\sim} \underline{\mathrm{Hom}}(\omega_f, \omega_f)$, and for dualizable ω_f this gives $\mathbb{1}_X \simeq \omega_f^\vee \otimes \omega_f$. We record this as an observation.

Observation B.23. For a $\mathbf{D}$ -suave morphism f , the dualizing complex ω_f is $\otimes$ -invertible if and only if it is dualizable. Consequently the verification of $\mathbf{D}$ -smoothness reduces to checking suaveness and dualizability of ω_f .

Suave and prim morphisms satisfy various base change properties.

Lemma B.24 (Base change for suave and prim morphisms, Heyer and Mann [HM24, Lemma 4.5.13]).
Let D be a six-functor formalism on (C, E) and consider a pullback square

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ g' \downarrow & \lrcorner & \downarrow g \\ X & \xrightarrow{f} & Y \end{array}$$

in C_E .

(1) If f is D -suave, the natural transformations

$$g'_* f'^* \simeq f^* g_*, \quad g'_! f'^! \simeq f^! g_!, \quad f'^* g^! \simeq g'^! f^*, \quad g'^* f^! \simeq f'^! g^*$$

are all isomorphisms.

(2) If f is D -prim, the natural transformations

$$g^* f_* \simeq f'_* g'^*, \quad g^! f_! \simeq f'_! g'^!, \quad f! g_* \simeq g_* f'_!, \quad f_* g'_! \simeq g'_! f_*$$

are all isomorphisms.

Passing to iterated diagonals gives notions of étaleness and properness.

Definition B.25 (Cohomologically étale and proper morphisms). Let D be a six-functor formalism on (C, E) , and let $f: X \rightarrow Y$ be a truncated morphism in E (i.e. n -truncated for some integer $n \geq -2$).

- f is D -étale (or *cohomologically étale*) if f is D -suave and Δ_f is either D -étale or an isomorphism.
- f is D -proper (or *cohomologically proper*) if f is D -prim and Δ_f is either D -proper or an isomorphism.

These definitions are well-founded because n -truncated morphisms have $(n-1)$ -truncated diagonals.

Suppose that Δ_f is D -étale. This hypothesis already implies that f is truncated: by definition Δ_f is truncated, and a morphism with truncated diagonal is truncated. We may therefore argue inductively on the truncation level of f . Write $\pi_1, \pi_2: X \times_Y X \rightarrow X$ for the projections, so that

$$\begin{array}{ccccc} X & & \xrightarrow{\text{id}_X} & & X \\ & \searrow \Delta_f & & \searrow \pi_2 & \\ & & X \times_Y X & \xrightarrow{\pi_2} & X \\ & \searrow \text{id}_X & \downarrow \pi_1 & \lrcorner & \downarrow f \\ & & X & \xrightarrow{f} & Y \end{array}$$

The base-change isomorphism $f^* f_! \xrightarrow{\sim} \pi_{1,!} \pi_2^*$ has a double mate

$$\text{Ex}^{*!}: \pi_1^! f^* \longrightarrow \pi_2^* f^!.$$

By induction, the étale comparison $c_{\Delta_f}: \Delta_f^* \rightarrow \Delta_f^!$ is already an isomorphism. We define the *étale comparison* for f to be the composite

$$c_f: f^* \xrightarrow{\sim} \Delta_f^! \pi_1^! f^* \xrightarrow{\Delta_f^! \text{Ex}^!} \Delta_f^! \pi_2^* f^! \xrightarrow{c_{\Delta_f}^{-1}} \Delta_f^* \pi_2^* f^! \xrightarrow{\sim} f^!.$$

Equivalently, under $f^* \dashv f_*$ this comparison is the mate

$$\text{id} \longrightarrow f_* f^!,$$

while under $f_! \dashv f^!$ it is the mate of the trace map

$$\text{tr}_f: f_! f^* \longrightarrow \text{id}.$$

In terms of the base-change description above, these mates are obtained by inserting the unit for the adjunction associated with Δ_f , after identifying Δ_f^* with $\Delta_f^!$ by c_{Δ_f} . Thus the construction uses only the six-functor structure and the comparison already constructed for the diagonal; no comparison is asserted when Δ_f is not known to be D-étale.

The following proposition shows that f is D-étale if and only if this inductively defined comparison is an isomorphism. A dual inductive construction produces a map $f_! \rightarrow f_*$ whenever Δ_f is known to be D-proper.

Proposition B.26 (Characterization of étaleness and properness, cf. Heyer and Mann [HM24, §4.6]). *Let $\mathbf{D}$ be a six-functor formalism on (C, E) and let $f: X \rightarrow Y$ be a morphism in E . In either part below, the hypothesis on Δ_f already implies that f is truncated.*

(1) *Suppose Δ_f is D-étale. Then the following are equivalent:*

- (a) *f is D-étale;*
- (b) *The comparison $f^* \mathbb{1}_Y \rightarrow f^! \mathbb{1}_Y$ is an isomorphism in $\mathbf{D}(X)$;*
- (c) *The comparison $f^* \rightarrow f^!$ is an isomorphism of functors.*

(2) *Suppose Δ_f is D-proper. Then the following are equivalent:*

- (a') *f is D-proper;*
- (b') *$f_! \mathbb{1}_X \simeq f_* \mathbb{1}_X$ in $\mathbf{D}(Y)$;*
- (c') *$f_! \xrightarrow{\sim} f_*$ as functors.*

In Section 10.3 we study how the notions of cohomological smoothness, suaveness, primness, étaleness and properness can be translated to the corresponding notions for Gestalten via the transmutation $[-]_{\mathbf{D}}: C \rightarrow \text{Gest}$.

B.6 Recovering six-functor formalisms from slices

Given a six-functor formalism on $\text{Span}(C, E)$, we may restrict it along the forgetful functor $E/X \rightarrow C$ for every $X \in C$ to obtain a six-functor formalism on the slice E/X . The point of this section is to

ask about the converse: when can the original six-functor formalism be recovered from all of these restrictions?

The reason it is useful to be able to do so is that in categorified situations it is sometimes easier to get a handle on the restricted formalisms, using the fact that $\text{Span}(E/X)$ is the endomorphism category of the span 2-category $\text{Span}_2(C, E)$ considered in Construction B.12. The localization argument proving the reconstruction result below admits a symmetric monoidal refinement, which assembles compatible slice-wise three-functor formalisms; see Corollary B.28.

Before giving a precise result, let us set the stage by making precise the restriction data and the compatibilities between them. Given $D: \text{Span}(C, E) \rightarrow \text{Cat}$, we may consider for every morphism $W \rightarrow X$ in C the functor

$$\text{Span}(E/X) \rightarrow \text{Cat}, \quad (Z \rightarrow X) \mapsto D(Z \times_X W).$$

Collecting these functors for all $W \in C/X$ results in a functor

$$D/X: \text{Span}(E/X) \rightarrow \text{Fun}(C/X^{\text{op}}, \text{Cat}).$$

Moreover, these functors are compatible with the two canonical functorialities in X on both sides:

- On the one hand, both sides are contravariantly functorial in maps $f: Y \rightarrow X$ in C . For the left-hand side, we may consider the pullback functor $f^*: E/X \rightarrow E/Y$, which induces a functor on span categories $\text{Span}(f^*): \text{Span}(E/X) \rightarrow \text{Span}(E/Y)$. For the right-hand side, we can restrict along the postcomposition functor $f_{\#}: C/Y \rightarrow C/X$ to obtain a functor $f^*: \text{Fun}(C/X^{\text{op}}, \text{Cat}) \rightarrow \text{Fun}(C/Y^{\text{op}}, \text{Cat})$. We then have the relation

$$f^* \circ D/X \simeq D/Y \circ \text{Span}(f^*), \quad (Z, W) \mapsto D(Z \times_X W) \simeq D((Z \times_X Y) \times_Y W)$$

of functors $\text{Span}(E/X) \rightarrow \text{Fun}(C/Y^{\text{op}}, \text{Cat})$.

- On the other hand, both sides are covariantly functorial in maps $e: Y \rightarrow X$ in E . For the left-hand side, the postcomposition functor $e_{\#}: E/Y \rightarrow E/X$ preserves pullbacks and thus induces a functor on spans. For the right-hand side, restricting along the pullback functor $e^*: C/Y \rightarrow C/X$ gives a functor $e_*: \text{Fun}(C/Y^{\text{op}}, \text{Cat}) \rightarrow \text{Fun}(C/X^{\text{op}}, \text{Cat})$. We then have the relation

$$D/X \circ \text{Span}(e_{\#}) \simeq e_* \circ D/Y, \quad (Z, W) \mapsto D(Z \times_X W) \simeq D(Z \times_Y (Y \times_X W))$$

of functors $\text{Span}(E/Y) \rightarrow \text{Fun}(C/X^{\text{op}}, \text{Cat})$.

These two compatibilities are closely related. The adjunction $e_{\#} \dashv e^*$ induces an adjunction

$$\text{Span}(e^*) \dashv \text{Span}(e_{\#}).$$

Similarly, restriction along e^* is right adjoint to restriction along $e_{\#}$. The contravariant compatibility is therefore the mate of the covariant compatibility. In particular, if we remember the covariant compatibility relations as coherent data, the contravariant compatibilities can be stated as properties.

We can now state the main result.

Theorem B.27 (Slicewise reconstruction, Cnossen, Lenz, and Linskens [CLL24, §8]). *Let C have finite limits and let $E \subseteq C$ be closed under base change and left cancellation. For any category A , there is an equivalence between the following two types of data:*

- *Functors $D: \text{Span}(C, E) \longrightarrow A$;*
- *Natural transformations*

$$D_{/\bullet}: \text{Span}(E_{/\bullet}) \rightarrow \text{Fun}(C_{/\bullet}^{\text{op}}, A)$$

of functors $C^{\text{op}} \rightarrow \text{Cat}$ satisfying the property that for each morphism $e: Y \rightarrow X$ in E the canonical map

$$D_{/X} \circ \text{Span}(e_{\#}) \rightarrow e_* \circ D_{/Y}$$

is a natural equivalence of functors $\text{Span}(E_{/Y}) \rightarrow \text{Fun}(C_{/Y}^{\text{op}}, A)$.

Proof. The proof is given in Cnossen, Lenz, and Linskens [CLL24, §8]; there, the data in the second bullet point occurs under the name *E -commutative monoid in the C -category $X \mapsto \text{Fun}(C_{/X}, A)$* .

The rough idea of the proof is as follows. Natural transformations between functors $C^{\text{op}} \rightarrow \widehat{\text{Cat}}$ can be alternatively encoded as morphisms between cocartesian fibrations over C^{op} . The cocartesian unstraightening of $X \mapsto \text{Span}(E_{/X})$ is the span category of $\text{Ar}_E(C)$, the full subcategory of the arrow category $\text{Ar}(C)$ spanned by the morphisms $Z \rightarrow X$ in E . The span category $\text{Span}(C, E)$ is a localization of this span category, and the class of maps at which one needs to localize is precisely the class of maps encoding the second compatibility relation. $\square$

We have already seen how to produce the data of the functor $D_{/X}$ from D . In the other direction, we may recover D from $D_{/\bullet}$ by setting

$$D(X) := D_{/X}(\text{id}_X)(\text{id}_X).$$

For an arbitrary map $f: U \rightarrow X$, the pullback functor $f^*: D(X) \rightarrow D(U)$ is obtained by applying the presheaf $D_{/X}(\text{id}_X): C_{/X}^{\text{op}} \rightarrow A$ to the morphism $f: U \rightarrow X$ in $C_{/X}$, using that $D_{/X}(\text{id}_X)(U) \simeq D_{/U}(\text{id}_U)(\text{id}_U)$. For a map $e: U \rightarrow Y$ in E , the exceptional pushforward $e_!: D(U) \rightarrow D(Y)$ is obtained by applying $D_{/Y}$ to the forward span

$$e \xleftarrow{\text{id}} e \xrightarrow{e} \text{id}_Y$$

in $E_{/Y}$ and then evaluating at id_Y . A general span $X \xleftarrow{f} U \xrightarrow{e} Y$ is sent to the composite $e_! f^*$. The covariant compatibility in Theorem B.27, whose symbol e_* denotes restriction between presheaf categories rather than a pushforward in a six-functor formalism, is precisely what makes this construction functorial under composition of spans.

Corollary B.28 (Symmetric monoidal slicewise reconstruction). *Let A be a symmetric monoidal category. In the situation of Theorem B.27, suppose that the slicewise data are equipped with coherent external products*

$$D_{/X}(Z)(W) \otimes D_{/X'}(Z')(W') \longrightarrow D_{/(X \times X')}(Z \times Z')(W \times W')$$

and a unit, satisfying the lax symmetric monoidal coherences and compatible with the span actions and with the contravariant and covariant compatibility equivalences. Then the reconstructed functor

$$D: \text{Span}(C, E) \longrightarrow A$$

admits a unique lax symmetric monoidal structure inducing the given slicewise structures. In particular, for $A = \text{Cat}$, compatible families of slicewise three-functor formalisms assemble into a three-functor formalism on (C, E) . The same applies to 1Pr and Pr^{L} .

Proof. We refine the localization proof of Theorem B.27. In the notation of Cnossen, Lenz, and Linskens [CLL24, Propositions 8.18 and 8.21], let

$$K := \text{Span}_{\text{fw}}(\text{Ar}_E(C))$$

be the span category obtained from the arrow category of E -maps. Slicewise data are equivalently functors $K \rightarrow A$ which invert the class W encoding the covariant compatibility, and the source functor induces a localization

$$s: K \longrightarrow \text{Span}(C, E)$$

at W .

Cartesian product, computed pointwise in the arrow category, makes the adequate triple defining K into a commutative algebra among adequate triples. Since the span construction preserves finite products, K is symmetric monoidal and s is symmetric monoidal. The class W is closed under tensoring with arbitrary objects: its generators arise from composable maps $f, q \in E$, and tensoring such a generator with an arrow in E gives the generator associated with the corresponding product maps. Consequently s is a symmetric monoidal localization. Thus precomposition with s identifies lax symmetric monoidal functors $\text{Span}(C, E) \rightarrow A$ with lax symmetric monoidal functors $K \rightarrow A$ whose underlying functors invert W .

Finally, the straightening and unstraightening equivalences used to identify functors $K \rightarrow A$ with the slicewise data are compatible with cartesian products on the indexing categories. Under this identification, a lax symmetric monoidal structure on a functor $K \rightarrow A$ is precisely a coherent system of external products as in the statement. Applying the symmetric monoidal universal property of s gives the asserted refinement and its uniqueness. $\square$

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